



RECENT ADVANCES IN FINANCIAL ENGINEERING

editors

Masaaki Kijima • Chiaki Hara
Keiichi Tanaka • Yukio Muromachi

2009

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Proceedings of the
KIER-TMU International Workshop
on Financial Engineering 2009

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Otemachi, Sankei Plaza, Tokyo
3 – 4 August 2009

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PREFACE

This book is the Proceedings of the *KIER-TMU International Workshop on Financial Engineering 2009* held in Summer 2009. The workshop is the successor of “Daiwa International Workshop on Financial Engineering” that was held in Tokyo every year since 2004 in order to exchange new ideas in financial engineering among workshop participants. Every year, various interesting and high quality studies were presented by many researchers from various countries, from both academia and industry. As such, this workshop served as a bridge between academic researchers in the field of financial engineering and practitioners.

We would like to mention that the workshop is jointly organized by the Institute of Economic Research, Kyoto University (KIER) and the Graduate School of Social Sciences, Tokyo Metropolitan University (TMU). Financial support from the Public Management Program, the Program for Enhancing Systematic Education in Graduate Schools, the Japan Society for Promotion of Science’s Program for Grants-in Aid for Scientific Research (A) #21241040, the Selective Research Fund of Tokyo Metropolitan University, and Credit Pricing Corporation are greatly appreciated.

We invited leading scholars including four keynote speakers, and various kinds of fruitful and active discussions were held during the KIER-TMU workshop. This book consists of eleven papers related to the topics presented at the workshop. These papers address state-of-the-art techniques and concepts in financial engineering, and have been selected through appropriate referees’ evaluation followed by the editors’ final decision in order to make this book a high quality one. The reader will be convinced of the contributions made by this research.

We would like to express our deep gratitude to those who submitted their papers to this proceedings and those who helped us kindly by refereeing these papers. We would also thank Mr. Satoshi Kanai for editing the manuscripts, and Ms. Kakarlapudi Shalini Raju and Ms. Grace Lu Huiru of World Scientific Publishing Co. for their kind assistance in publishing this book.

February, 2010

Masaaki Kijima, Tokyo Metropolitan University
Chiaki Hara, Institute of Economic Research, Kyoto University
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KIER-TMU International Workshop on Financial Engineering 2009

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Graduate School of Social Sciences, Tokyo Metropolitan University

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Program

August 3 (Monday)

Chair: Masaaki Kijima

10:00–10:10 Yasuyuki Kato, Nomura Securities/Kyoto University
Opening Address

Chair: Chiaki Hara

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Optimal and Robust Contracts for a Risk-Constrained Principal

10:55–11:25 Yumiharu Nakano, Tokyo Institute of Technology
Quantile Hedging for Defaultable Claims

11:25–12:45 Lunch

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12:45–13:30 Michael Gordy, Federal Reserve Board
Constant Proportion Debt Obligations: A Post-Mortem Analysis of Rating Models (with Soren Willemann)

13:30–14:00 Kyoko Yagi, University of Tokyo
An Optimal Investment Policy in Equity-Debt Financed Firms with Finite Maturities (with Ryuta Takashima and Katsushige Sawaki)

14:00–14:20 Afternoon Coffee I

Chair: Stéphane Crépey

14:20–14:50 Hidetoshi Nakagawa, Hitotsubashi University
Surrender Risk and Default Risk of Insurance Companies (with Olivier Le Courtois)

14:50–15:20 Kyo Yamamoto, University of Tokyo
Generating a Target Payoff Distribution with the Cheapest Dynamic Portfolio: An Application to Hedge Fund Replication (with Akihiko Takahashi)

15:20–15:50 Yasuo Taniguchi, Sumitomo Mitsui Banking Corporation/Tokyo Metropolitan University
Looping Default Model with Multiple Obligors

15:50–16:10 Afternoon Coffee II

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16:10–16:40 Stéphane Crépey, Evry University

Counterparty Credit Risk (with Samson Assefa, Tomasz R. Bielecki,
Monique Jeanblanc and Behnaz Zagari)

16:40–17:10 Kohta Takehara, University of Tokyo

Computation in an Asymptotic Expansion Method (with Akihiko Takahashi
and Masashi Toda)

August 4 (Tuesday)

Chair: Takashi Shibata

10:00–10:45 Chiaki Hara, Kyoto University
Heterogeneous Beliefs and Representative Consumer

10:45–11:15 Xue-Zhong He, University of Technology, Sydney
Boundedly Rational Equilibrium and Risk Premium (with Lei Shi)

11:15–11:45 Yuan Tian, Kyoto University/Tokyo Metropolitan University
Financial Synergy in M&A (with Michi Nishihara and Takashi Shibata)

11:45–13:15 Lunch

Chair: Andrea Macrina

13:15–14:00 Mark Davis, Imperial College London
Jump-Diffusion Risk-Sensitive Asset Management (with Sebastien Lleo)

14:00–14:30 Masahiko Egami, Kyoto University
A Game Options Approach to the Investment Problem with Convertible Debt Financing

14:30–15:00 Katsunori Ano
Optimal Stopping Problem with Uncertain Stopping and its Application to Discrete Options

15:00–15:30 Afternoon Coffee

Chair: Xue-Zhong He

15:30–16:00 Andrea Macrina, King's College London/Kyoto University
Information-Sensitive Pricing Kernels (with Lane Hughston)

16:00–16:30 Hiroki Masuda, Kyushu University
Explicit Estimators of a Skewed Stable Model Based on High-Frequency Data

16:30–17:00 Takayuki Morimoto, Kwansei Gakuin University
A Note on a Statistical Hypothesis Testing for Removing Noise by The Random Matrix Theory, and its Application to Co-Volatility Matrices (with Kanta Tachibana)

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Risk Sensitive Investment Management with Affine Processes: A Viscosity Approach*

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In this paper, we extend the jump-diffusion model proposed by Davis and Lleo to include jumps in asset prices as well as valuation factors. The criterion, following earlier work by Bielecki, Pliska, Nagai and others, is risk-sensitive optimization (equivalent to maximizing the expected growth rate subject to a constraint on variance). In this setting, the Hamilton-Jacobi-Bellman equation is a partial integro-differential PDE. The main result of the paper is to show that the value function of the control problem is the unique viscosity solution of the Hamilton-Jacobi-Bellman equation.

Keywords: Asset management, risk-sensitive stochastic control, jump diffusion processes, Poisson point processes, Lévy processes, HJB PDE, policy improvement.

1. Introduction

In this paper, we extend the jump diffusion risk-sensitive asset management model proposed by Davis and Lleo [19] to allow jumps in both asset prices and factor levels.

Risk-sensitive control generalizes classical stochastic control by parametrizing explicitly the degree of risk aversion or risk tolerance of the optimizing agent. In risk-sensitive control, the decision maker's objective is to select a control policy $h(t)$ to maximize the criterion

$$J(t, x, h; \theta) := -\frac{1}{\theta} \ln \mathbf{E} \left[e^{-\theta F(t, x, h)} \right] \quad (1)$$

*The authors are very grateful to the editors and an anonymous referees for a number of very helpful comments.

where t is the time, x is the state variable, F is a given reward function, and the risk sensitivity $\theta \in]-1, 0[\cup]0, \infty)$ is an exogenous parameter representing the decision maker's degree of risk aversion. A Taylor expansion of this criterion around $\theta = 0$ yields

$$J(t, x, h; \theta) = \mathbf{E}[F(t, x, h)] - \frac{\theta}{2} \mathbf{Var}[F(t, x, h)] + O(\theta^2) \quad (2)$$

which shows that the risk-sensitive criterion amounts to maximizing $\mathbf{E}[F(t, x, h)]$ subject to a penalty for variance. Jacobson [28], Whittle [35], Bensoussan and Van Schuppen [9] led the theoretical development of risk sensitive control while Lefebvre and Montulet [32], Fleming [25] and Bielecki and Pliska [11] pioneered the financial application of risk-sensitive control. In particular, Bielecki and Pliska proposed the logarithm of the investor's wealth as a reward function, so that the investor's objective is to maximize the risk-sensitive (log) return of his/her portfolio or alternatively to maximize a function of the power utility (HARA) of terminal wealth. Bielecki and Pliska brought an enormous contribution to the field by studying the economic properties of the risk-sensitive asset management criterion (see [13]), extending the asset management model into an intertemporal CAPM ([14]), working on transaction costs ([12]), numerical methods ([10]) and considering factors driven by a CIR model ([15]). Other main contributors include Kuroda and Nagai [31] who introduced an elegant solution method based on a change of measure argument. Davis and Lleo applied this change of measure technique to solve a benchmarked investment problem in which an investor selects an asset allocation to outperform a given financial benchmark (see [18]) and analyzed the link between optimal portfolios and fractional Kelly strategies (see [20]). More recently, Davis and Lleo [19] extended the risk-sensitive asset management model by allowing jumps in asset prices.

In this chapter, our contribution is to allow not only jumps in asset prices but also in the level of the underlying valuation factors. Once we introduce jumps in the factors, the Bellman equation becomes a nonlinear Partial Integro-Differential equation and an analytical or classical $C^{1,2}$ solutions may not exist. As a result, to give a sense to the relation between the value function and the risk sensitive Hamilton-Jacobi-Bellman Partial Integro Differential Equation (RS HJB PIDE), we consider a class of weak solutions called viscosity solutions, which have gained a widespread acceptance in control theory in recent years. The main results are a comparison theorem and the proof that the value function of the control problem under consideration is the unique continuous viscosity solution of the associated RS HJB PIDE. In particular, the proof of the comparison results uses non-standard arguments to circumvent difficulties linked to the highly nonlinear nature of the RS HJB PIDE and to the unboundedness of the instantaneous reward function g .

This chapter is organized as follows. Section 2 introduces the general setting of the model and defines the class of random Poisson measures which will be used to model the jump component of the asset and factor dynamics. In Section 3 we formulate the control problem and apply a change of measure to obtain a simpler auxiliary criterion. Section 4 outlines the properties of the value function. In Section 5 we show that the value function is a viscosity solution of the RS HJB PIDE before proving a comparison result in Section 6 which provides uniqueness.

2. Analytical Setting

Our analytical setting is based on that of [19]. The notable difference is that we allow the factor processes to experience jumps.

2.1 Overview

The growth rates of the assets are assumed to depend on n valuation factors $X_1(t), \dots, X_n(t)$ which follow the dynamics given in equation (4) below. The assets market comprises m risky securities S_i , $i = 1, \dots, m$. Let $M := n + m$. Let $(\Omega, \{\mathcal{F}_t\}, \mathcal{F}, \mathbb{P})$ be the underlying probability space. On this space is defined an $\mathbb{R}^M$ -valued $(\mathcal{F}_t)$ -Brownian motion $W(t)$ with components $W_k(t)$, $k = 1, \dots, M$. Moreover, let $(\mathbf{Z}, \mathcal{B}_{\mathbf{Z}})$ be a Borel space¹. Let $\mathbf{p}$ be an $(\mathcal{F}_t)$ -adapted σ -finite Poisson point process on $\mathbf{Z}$ whose underlying point functions are maps from a countable set $\mathbf{D}_{\mathbf{p}} \subset (0, \infty)$ into $\mathbf{Z}$. Define

$$\mathfrak{Z}_{\mathbf{p}} := \{U \in \mathcal{B}(\mathbf{Z}), \mathbb{E}[N_{\mathbf{p}}(t, U)] < \infty \forall t\} \quad (3)$$

Consider $N_{\mathbf{p}}(dt, dz)$, the Poisson random measure on $(0, \infty) \times \mathbf{Z}$ induced by $\mathbf{p}$. Following Davis and Lleo [19], we concentrate on stationary Poisson point processes of class (QL) with associated Poisson random measure $N_{\mathbf{p}}(dt, dx)$. The class (QL) is defined in [27] (Definition II.3.1, p. 59) as

Definition 2.1. An $(\mathcal{F}_t)$ -adapted point process $\mathbf{p}$ on $(\Omega, \mathcal{F}, \mathbb{P})$ is said to be of class (QL) with respect to $(\mathcal{F}_t)$ if it is σ -finite and there exists $\hat{N}_{\mathbf{p}} = (\hat{N}_{\mathbf{p}}(t, U))$ such that

- (i) for $U \in \mathfrak{Z}_{\mathbf{p}}$, $t \mapsto \hat{N}_{\mathbf{p}}(t, U)$ is a continuous $(\mathcal{F}_t)$ -adapted increasing process;
- (ii) for each t and a.a. $\omega \in \Omega$, $U \mapsto \hat{N}_{\mathbf{p}}(t, U)$ is a σ -finite measure on $(\mathbf{Z}, \mathcal{B}(\mathbf{Z}))$;
- (iii) for $U \in \mathfrak{Z}_{\mathbf{p}}$, $t \mapsto \tilde{N}_{\mathbf{p}}(t, U) = N_{\mathbf{p}}(t, U) - \hat{N}_{\mathbf{p}}(t, U)$ is an $(\mathcal{F}_t)$ -martingale;

The random measure $\{\hat{N}_{\mathbf{p}}(t, U)\}$ is called the *compensator* of the point process p .

¹ $\mathbf{Z}$ is a standard measurable (metric or topological) space and $\mathcal{B}_{\mathbf{Z}}$ is the Borel σ -field endowed to $\mathbf{Z}$.

Since the Poisson point processes we consider are stationary, then their compensators are of the form $\hat{N}_{\mathbf{p}}(t, U) = \nu(U)t$, where ν is the σ -finite characteristic measure of the Poisson point process $\mathbf{p}$. For notational convenience, we define the Poisson random measure $\bar{N}_{\mathbf{p}}(dt, dz)$ as

$$\begin{aligned} & \bar{N}_{\mathbf{p}}(dt, dz) \\ &= \begin{cases} N_{\mathbf{p}}(dt, dz) - \hat{N}_{\mathbf{p}}(dt, dz) = N_{\mathbf{p}}(dt, dz) - \nu(dz)dt =: \tilde{N}_{\mathbf{p}}(dt, dz) & \text{if } z \in \mathbf{Z}_0 \\ N_{\mathbf{p}}(dt, dz) & \text{if } z \in \mathbf{Z} \setminus \mathbf{Z}_0 \end{cases} \end{aligned}$$

where $\mathbf{Z}_0 \subset \mathcal{B}_{\mathbf{Z}}$ such that $\nu(\mathbf{Z} \setminus \mathbf{Z}_0) < \infty$.

2.2 Factor Dynamics

We model the dynamics of the n factors with an affine jump diffusion process

$$dX(t) = (b + BX(t^-))dt + \Lambda dW(t) + \int_{\mathbf{Z}} \xi(z) \bar{N}_{\mathbf{p}}(dt, dz), \quad X(0) = x \quad (4)$$

where $X(t)$ is the $\mathbb{R}^n$ -valued factor process with components $X_j(t)$ and $b \in \mathbb{R}^n$, $B \in \mathbb{R}^{n \times n}$, $\Lambda := [\Lambda_{ij}]$, $i = 1, \dots, n$, $j = 1, \dots, N$ and $\xi(z) \in \mathbb{R}^n$ with $-\infty < \xi_i^{\min} \leq \xi_i(z) \leq \xi_i^{\max} < \infty$ for $i = 1, \dots, n$. Moreover, the vector-valued function $\xi(z)$ satisfies:

$$\int_{\mathbf{Z}_0} |\xi(z)|^2 \nu(dz) < \infty$$

(See for example Definition II.4.1 in Ikeda and Watanabe [27] where $\mathbf{F}_{\mathbf{p}}$ and $\mathbf{F}_{\mathbf{p}}^{2,loc}$ are given in equations II(3.2) and II(3.5) respectively.)

2.3 Asset Market Dynamics

Let S_0 denote the wealth invested in the money market account with dynamics given by the equation:

$$\frac{dS_0(t)}{S_0(t)} = (a_0 + A_0' X(t)) dt, \quad S_0(0) = s_0 \quad (5)$$

where $a_0 \in \mathbb{R}$ is a scalar constant, $A_0 \in \mathbb{R}^n$ is a n -element column vector and where M' denotes the transposed matrix of M . Note that if we set $A_0 = 0$ and $a_0 = r$, then equation (5) can be interpreted as the dynamics of a globally risk-free asset. Let $S_i(t)$ denote the price at time t of the i th security, with $i = 1, \dots, m$. The dynamics of risky security i can be expressed as:

$$\begin{aligned} \frac{dS_i(t)}{S_i(t^-)} &= (a + AX(t))_i dt + \sum_{k=1}^N \sigma_{ik} dW_k(t) + \int_{\mathbf{Z}} \gamma_i(z) \bar{N}_{\mathbf{p}}(dt, dz), \\ S_i(0) &= s_i, \quad i = 1, \dots, m \end{aligned} \quad (6)$$

where $a \in \mathbb{R}^m$, $A \in \mathbb{R}^{m \times n}$, $\Sigma := [\sigma_{ij}]$, $i = 1, \dots, m$, $j = 1, \dots, M$ and $\gamma(z) \in \mathbb{R}^m$ satisfies Assumption 2.1.

Assumption 2.1. $\gamma(z) \in \mathbb{R}^m$ satisfies

$$-1 \leq \gamma_i^{\min} \leq \gamma_i(z) \leq \gamma_i^{\max} < +\infty, \quad i = 1, \dots, m$$

and

$$-1 \leq \gamma_i^{\min} < 0 < \gamma_i^{\max} < +\infty, \quad i = 1, \dots, m$$

for $i = 1, \dots, m$. Furthermore, define

$$\mathbf{S} := \text{supp}(\nu) \in \mathcal{B}_{\mathbf{Z}}$$

and

$$\tilde{\mathbf{S}} := \text{supp}(\nu \circ \gamma^{-1}) \in \mathcal{B}(\mathbb{R}^m)$$

where $\text{supp}(\cdot)$ denotes the measure's support, then we assume that $\prod_{i=1}^m [\gamma_i^{\min}, \gamma_i^{\max}]$ is the smallest closed hypercube containing $\tilde{\mathbf{S}}$.

In addition, the vector-valued function $\gamma(z)$ satisfies:

$$\int_{\mathbf{Z}_0} |\gamma(z)|^2 \nu(dz) < \infty$$

As noted in [19], Assumption 2.1 requires that each asset has, with positive probability, both upward and downward jumps and as a result bounds the space of controls.

Define the set $\mathcal{J}$ as

$$\mathcal{J} := \{h \in \mathbb{R}^m : -1 - h'\psi < 0 \quad \forall \psi \in \tilde{\mathbf{S}}\} \quad (7)$$

For a given z , the equation $h'\gamma(z) = -1$ describes a hyperplane in $\mathbb{R}^m$. Under Assumption 2.1 $\mathcal{J}$ is a convex subset of $\mathbb{R}^m$.

2.4 Portfolio Dynamics

We will assume that:

Assumption 2.2. The matrix $\Sigma\Sigma'$ is positive definite.

and

Assumption 2.3. The systematic (factor-driven) and idiosyncratic (asset-driven) jump risks are uncorrelated, i.e. $\forall z \in \mathbf{Z}$ and $i = 1, \dots, m$, $\gamma_i(z)\xi'(z) = 0$.

The second assumption implies that there cannot be simultaneous jumps in the factor process and any asset price process. This assumption, which will prove sufficient to show the existence of a unique optimal investment policy, may appear somewhat restrictive as it does not enable us to model a jump correlation structure across factors and assets, although we can model a jump correlation structure within the factors and within the assets.

Remark 2.1. Assumption (2.3) is automatically satisfied when jumps are only allowed in the security prices and the state variable $X(t)$ is modelled using a diffusion process (see [19] for a full treatment of this case).

Let $\mathcal{G}_t := \sigma((S(s), X(s)), 0 \leq s \leq t)$ be the sigma-field generated by the security and factor processes up to time t .

An *investment strategy* or *control process* is an $\mathbb{R}^m$ -valued process with the interpretation that $h_i(t)$ is the fraction of current portfolio value invested in the i th asset, $i = 1, \dots, m$. The fraction invested in the money market account is then $h_0(t) = 1 - \sum_{i=1}^m h_i(t)$.

Definition 2.2. An $\mathbb{R}^m$ -valued control process $h(t)$ is in class $\mathcal{H}$ if the following conditions are satisfied:

1. $h(t)$ is progressively measurable with respect to $\{\mathcal{B}([0, t]) \otimes \mathcal{G}_t\}_{t \geq 0}$ and is càdlàg;
2. $P\left(\int_0^T |h(s)|^2 ds < +\infty\right) = 1, \quad \forall T > 0;$
3. $h'(t)\gamma(z) > -1, \quad \forall t > 0, z \in \mathbf{Z}, \text{ a.s. } d\nu.$

Define the set $\mathcal{K}$ as

$$\mathcal{K} := \{h(t) \in \mathcal{H} : h(t) \in \mathcal{J} \quad \forall \text{ t.a.s.}\} \quad (8)$$

Lemma 2.1. Under Assumption 2.1, a control process $h(t)$ satisfying condition 3 in Definition 2.2 is bounded.

Proof. The proof of this result is immediate. □

Definition 2.3. A control process $h(t)$ is in class $\mathcal{A}(T)$ if the following conditions are satisfied:

1. $h(t) \in \mathcal{H} \quad \forall t \in [0, T];$

2. $\mathbf{E}\chi_T^h = 1$ where χ_t^h is the Doléans exponential defined as

$$\begin{aligned} \chi_t^h := & \exp \left\{ -\theta \int_0^t h(s)' \Sigma dW_s - \frac{1}{2} \theta^2 \int_0^t h(s)' \Sigma \Sigma' h(s) ds \right. \\ & + \int_0^t \int_{\mathbf{Z}} \ln(1 - G(z, h(s); \theta)) \tilde{N}_{\mathbf{p}}(ds, dz) \\ & \left. + \int_0^t \int_{\mathbf{Z}} \{ \ln(1 - G(z, h(s); \theta)) + G(z, h(s); \theta) \} \nu(dz) ds \right\}; \end{aligned} \quad (9)$$

and

$$G(z, h; \theta) = 1 - (1 + h' \gamma(z))^{-\theta} \quad (10)$$

Definition 2.4. We say that a control process $h(t)$ is *admissible* if $h(t) \in \mathcal{A}(T)$.

The proportion invested in the money market account is $h_0(t) = 1 - \sum_{i=1}^m h_i(t)$. Taking this budget equation into consideration, the wealth $V(t, x, h)$, or $V(t)$, of the investor in response to an investment strategy $h(t) \in \mathcal{H}$, follows the dynamics

$$\begin{aligned} \frac{dV(t)}{V(t^-)} = & (a_0 + A'_0 X(t)) dt + h'(t) (a - a_0 \mathbf{1} + (A - \mathbf{1} A'_0) X(t)) dt \\ & + h'(t) \Sigma dW_t + \int_{\mathbf{Z}} h'(t) \gamma(z) \tilde{N}_{\mathbf{p}}(dt, dz) \end{aligned}$$

where $\mathbf{1} \in \mathbf{R}^m$ denotes the m -element unit column vector and with $V(0) = v$. Defining $\hat{a} := a - a_0 \mathbf{1}$ and $\hat{A} := A - \mathbf{1} A'_0$, we can express the portfolio dynamics as

$$\frac{dV(t)}{V(t^-)} = (a_0 + A'_0 X(t)) dt + h'(t) (\hat{a} + \hat{A} X(t)) dt + h'(t) \Sigma dW_t + \int_{\mathbf{Z}} h'(t) \gamma(z) \tilde{N}_{\mathbf{p}}(dt, dz) \quad (11)$$

3. Problem Setup

3.1 Optimization Criterion

We will follow Bielecki and Pliska [11] and Kuroda and Nagai [31] and assume that the objective of the investor is to maximize the long-term risk adjusted growth of his/her portfolio of assets. In this context, the objective of the risk-sensitive management problem is to find $h^*(t) \in \mathcal{A}(T)$ that maximizes the control criterion

$$J(t, x, h; \theta) := -\frac{1}{\theta} \ln \mathbf{E} \left[e^{-\theta \ln V(t, x, h)} \right] \quad (12)$$

By Itô, the log of the portfolio value in response to a strategy h is

$$\begin{aligned}
\ln V(t) = \ln v &+ \int_0^t \left(a_0 + A'_0 X(s) \right) + h(s)' \left(\hat{a} + \hat{A}X(s) \right) ds - \frac{1}{2} \int_0^t h(s)' \Sigma \Sigma' h(s) ds \\
&+ \int_0^t h(s)' \Sigma dW(s) \\
&+ \int_0^t \int_{\mathbf{Z}_0} \{ \ln(1 + h(s)' \gamma(z)) - h(s)' \gamma(z) \} \nu(dz) ds \\
&+ \int_0^t \int_{\mathbf{Z}} \ln(1 + h(s)' \gamma(z)) \bar{N}_{\mathbf{p}}(ds, dz)
\end{aligned} \tag{13}$$

Hence,

$$e^{-\theta \ln V(t)} = v^{-\theta} \exp \left\{ \theta \int_0^t g(X_s, h(s); \theta) ds \right\} \chi_t^h \tag{14}$$

where

$$\begin{aligned}
g(x, h; \theta) = &\frac{1}{2} (\theta + 1) h' \Sigma \Sigma' h - a_0 - A'_0 x - h' (\hat{a} + \hat{A}x) \\
&+ \int_{\mathbf{Z}} \left\{ \frac{1}{\theta} \left[(1 + h' \gamma(z))^{-\theta} - 1 \right] + h' \gamma(z) 1_{\mathbf{Z}_0}(z) \right\} \nu(dz)
\end{aligned} \tag{15}$$

and the Doléans exponential χ_t^h is given by (9).

3.2 Change of Measure

Let $\mathbb{P}_h^\theta$ be the measure on $(\Omega, \mathcal{F})$ be defined as

$$\left. \frac{d\mathbb{P}_h^\theta}{d\mathbb{P}} \right|_{\mathcal{F}_t} := \chi_t \tag{16}$$

For a change of measure to be possible, we must ensure that the following technical condition holds:

$$G(z, h(s); \theta) < 1$$

for all $s \in [0, T]$ and z a.s. $d\nu$. This condition is satisfied iff

$$h'(s) \gamma(z) > -1 \tag{17}$$

a.s. $d\nu$, which was already one of the conditions required for h to be in class $\mathcal{H}$ (Condition 3 in Definition 2.2).

$\mathbb{P}_h^\theta$ is a probability measure for $h \in \mathcal{A}(T)$. For $h \in \mathcal{A}(T)$,

$$W_t^h = W_t + \theta \int_0^t \Sigma' h(s) ds$$

is a standard Brownian motion under the measure $\mathbb{P}_h^\theta$ and we define the $\mathbb{P}_h^\theta$ compensated Poisson measure as

$$\begin{aligned} \int_0^t \int_{\mathbf{Z}} \tilde{N}_{\mathbf{p}}^h(ds, dz) &= \int_0^t \int_{\mathbf{Z}} N_{\mathbf{p}}(ds, dz) - \int_0^t \int_{\mathbf{Z}} \{1 - G(z, h(s); \theta)\} \nu(dz) ds \\ &= \int_0^t \int_{\mathbf{Z}} N_{\mathbf{p}}(ds, dz) - \int_0^t \int_{\mathbf{Z}} \{(1 + h' \gamma(z))^{-\theta}\} \nu(dz) ds \end{aligned}$$

As a result, $X(s)$, $0 \leq s \leq t$ satisfies the SDE:

$$dX(s) = f(X(s^-), h(s); \theta) ds + \Lambda dW_s^h + \int_{\mathbf{Z}} \xi(z) \tilde{N}_{\mathbf{p}}^h(ds, dz) \quad (18)$$

where

$$f(x, h; \theta) := b + Bx - \theta \Lambda \Sigma' h + \int_{\mathbf{Z}} \xi(z) \left[(1 + h' \gamma(z))^{-\theta} - 1_{\mathbf{Z}_0}(z) \right] \nu(dz) \quad (19)$$

We will now introduce the following two auxiliary criterion functions under the measure $\mathbb{P}_h^\theta$:

- the auxiliary function directly associated with the risk-sensitive control problem:

$$I(v, x; h; t, T; \theta) = -\frac{1}{\theta} \ln \mathbf{E}_{t,x}^{h,\theta} \left[\exp \left\{ \theta \int_t^T g(X_s, h(s); \theta) ds - \theta \ln v \right\} \right] \quad (20)$$

where $\mathbf{E}_{t,x}^{h,\theta} [\cdot]$ denotes the expectation taken with respect to the measure $\mathbb{P}_h^\theta$ and with initial conditions (t, x) .

- the exponentially transformed criterion

$$\tilde{I}(v, x, h; t, T; \theta) := \mathbf{E}_{t,x}^{h,\theta} \left[\exp \left\{ \theta \int_t^T g(X_s, h(s); \theta) ds - \theta \ln v \right\} \right] \quad (21)$$

which we will find convenient to use in our derivations.

We have completed our reformulation of the problem under the measure $\mathbb{P}_h^\theta$. The state dynamics (18) is a jump-diffusion process and our objective is to maximize the criterion (20) or alternatively minimize (21).

3.3 The HJB Equation

In this section we derive the risk-sensitive Hamilton-Jacobi-Bellman partial integro differential equation (RS HJB PIDE) associated with the optimal control problem. Since we do not anticipate that a classical solution generally exists, we will not attempt to derive a verification theorem. Instead, we will show that the

value function Φ is a solution of the RS HJB PIDE in the viscosity sense. In fact, we will show that the value function is the unique continuous viscosity solution of the RS HJB PIDE. This result will in turn justify the association of the RS HJB PIDE with the control problem and replace the verification theorem we would derive if a classical solution existed.

Let Φ be the value function for the auxiliary criterion function $I(v, x; h; t, T)$ defined in (20). Then Φ is defined as

$$\Phi(t, x) = \sup_{h \in \mathcal{H}(T)} I(v, x; h; t, T) \quad (22)$$

We will show that Φ satisfies the HJB PDE

$$\frac{\partial \Phi}{\partial t}(t, x) + \sup_{h \in \mathcal{H}} L_t^h \Phi(t, X(t)) = 0 \quad (23)$$

where

$$\begin{aligned} L_t^h \Phi(t, x) = & f(x, h; \theta)' D\Phi + \frac{1}{2} \text{tr} \left(\Lambda \Lambda' D^2 \Phi \right) - \frac{\theta}{2} (D\Phi)' \Lambda \Lambda' D\Phi \\ & + \int_{\mathbf{Z}} \left\{ -\frac{1}{\theta} \left(e^{-\theta(\Phi(t, x + \xi(z)) - \Phi(t, x))} - 1 \right) - \xi'(z) D\Phi \right\} \nu(dz) \\ & - g(x, h; \theta) \end{aligned} \quad (24)$$

$D \cdot = \frac{\partial}{\partial x}$, and subject to terminal condition

$$\Phi(T, x) = \ln v \quad (25)$$

Similarly, let $\tilde{\Phi}$ be the value function for the auxiliary criterion function $\tilde{I}(v, x; h; t, T)$. Then $\tilde{\Phi}$ is defined as

$$\tilde{\Phi}(t, x) = \inf_{h \in \mathcal{H}(T)} \tilde{I}(v, x; h; t, T) \quad (26)$$

The corresponding HJB PDE is

$$\begin{aligned} \frac{\partial \tilde{\Phi}}{\partial t}(t, x) + \frac{1}{2} \text{tr} \left(\Lambda \Lambda' D^2 \tilde{\Phi}(t, x) \right) + H(x, \tilde{\Phi}, D\tilde{\Phi}) \\ + \int_{\mathbf{Z}} \left\{ \tilde{\Phi}(t, x + \xi(z)) - \tilde{\Phi}(t, x) - \xi'(z) D\tilde{\Phi}(t, x) \right\} \nu(dz) = 0 \end{aligned} \quad (27)$$

subject to terminal condition

$$\tilde{\Phi}(T, x) = v^{-\theta} \quad (28)$$

and where

$$H(s, x, r, p) = \inf_{h \in \mathcal{J}} \left\{ (b + Bx - \theta \Lambda \Sigma' h(s))' p + \theta g(x, h; \theta) r \right\} \quad (29)$$

for $r \in \mathbb{R}$, $p \in \mathbb{R}^n$ and in particular,

$$\tilde{\Phi}(t, x) = \exp \{-\theta \Phi(t, x)\} \quad (30)$$

The supremum in (23) can be expressed as:

$$\begin{aligned} & \sup_{h \in \mathcal{J}} L_t^h \Phi \\ &= (b + Bx)' D\Phi + \frac{1}{2} \text{tr} \left(\Lambda \Lambda' D^2 \Phi \right) - \frac{\theta}{2} (D\Phi)' \Lambda \Lambda' D\Phi + a_0 + A_0' x \\ &+ \int_{\mathbf{Z}} \left\{ -\frac{1}{\theta} \left(e^{-\theta(\Phi(t, x + \xi(z)) - \Phi(t, x))} - 1 \right) - \xi'(z) D\Phi 1_{\mathbf{Z}_0}(z) \right\} \nu(dz) \\ &+ \sup_{h \in \mathcal{J}} \left\{ -\frac{1}{2} (\theta + 1) h' \Sigma \Sigma' h - \theta h' \Sigma \Lambda' D\Phi + h' (\hat{a} + \hat{A}x) \right. \\ &\left. - \frac{1}{\theta} \int_{\mathbf{Z}} \left\{ (1 - \theta \xi'(z) D\Phi) \left[(1 + h' \gamma(z))^{-\theta} - 1 \right] + \theta h' \gamma(z) 1_{\mathbf{Z}_0}(z) \right\} \nu(dz) \right\} \quad (31) \end{aligned}$$

Under Assumption 2.2 the term

$$-\frac{1}{2} (\theta + 1) h' \Sigma \Sigma' h - \theta h' \Sigma \Lambda' D\Phi + h' (\hat{a} + \hat{A}x) - \int_{\mathbf{Z}} h' \gamma(z) 1_{\mathbf{Z}_0}(z) \nu(dz)$$

is strictly concave in h . Under Assumption 2.3, the nonlinear jump-related term

$$-\frac{1}{\theta} \int_{\mathbf{Z}} \left\{ (1 - \theta \xi'(z) D\Phi) \left[(1 + h' \gamma(z))^{-\theta} - 1 \right] \right\} \nu(dz)$$

simplifies to

$$-\frac{1}{\theta} \int_{\mathbf{Z}} \left\{ \left[(1 + h' \gamma(z))^{-\theta} - 1 \right] \right\} \nu(dz)$$

which is also concave in $h \forall z \in \mathbf{Z}$ a.s. $d\nu$. Therefore, the supremum is reached for a unique optimal control h^* , which is an interior point of the set $\mathcal{J}$ defined in equation (7), and the supremum, evaluated at h^* , is finite.

4. Properties of the Value Function

4.1 “Zero Beta” Policies

As in [19], we will use “zero beta” (0β) policies (initially introduced by Black [16])).

Definition 4.1. 1.20β -policy]By reference to the definition of the function g in equation (15), a ‘zero beta’ (0β) control policy $\check{h}(t)$ is an admissible control policy for which the function g is independent from the state variable x .

In our problem, the set $\mathcal{Z}$ of 0β -policies is the set of admissible policies $\check{h}$ which satisfy the equation

$$\check{h}'\hat{A} = -A_0$$

As $m > n$, there is potentially an infinite number of 0β -policies as long as the following assumption is satisfied

Assumption 4.1. *The matrix $\hat{A}$ has rank n .*

Without loss of generality, we fix a 0β control $\check{h}$ as a constant function of time so that

$$g(x, \check{h}; \theta) = \check{g}$$

where $\check{g}$ is a constant.

4.2 Convexity

Proposition 4.1. *The value function $\Phi(t, x)$ is convex in x .*

Proof. See the proof of Proposition 6.2 in [19]. □

Corollary 4.1. *The exponentially transformed value function $\tilde{\Phi}$ has the following property: $\forall (x_1, x_2) \in \mathbb{R}^2, \kappa \in (0, 1,)$,*

$$\tilde{\Phi}(t, \kappa x_1 + (1 - \kappa)x_2) \geq \tilde{\Phi}^\kappa(t, x_1)\tilde{\Phi}^{1-\kappa}(t, x_2) \quad (32)$$

Proof. The property follows immediately from the definition of $\Phi(t, x) = -\frac{1}{\theta} \ln \tilde{\Phi}(t, x)$. □

4.3 Boundedness

Proposition 4.2. *The exponentially transformed value function $\tilde{\Phi}$ is positive and bounded, i.e. there exists $M > 0$ such that*

$$0 \leq \tilde{\Phi}(t, x) \leq \check{M} \quad \forall (t, x) \in [0, T] \times \mathbb{R}^n$$

Proof. By definition,

$$\tilde{\Phi}(t, x) = \inf_{h \in \mathcal{A}(\mathcal{T})} \mathbf{E}_{t,x}^{h,\theta} \left[\exp \left\{ \theta \int_t^T g(X_s, h(s); \theta) ds - \theta \ln v \right\} \right] \geq 0$$

Consider the zero-beta policy $\check{h}$. By the Dynamic Programming Principle

$$\tilde{\Phi}(t, x) \leq e^{\theta \left[\int_t^T g(X(s), \check{h}; \theta) ds - \ln v \right]} = e^{\theta [\check{g}(T-t) - \ln v]}$$

which concludes the proof. $\square$

4.4 Growth

Assumption 4.2. *There exist $2n$ constant controls $\bar{h}^k, k = 1, \dots, 2n$ such that the $2n$ functions $\beta^k : [0, T] \rightarrow \mathbb{R}^n$ defined by*

$$\beta^k(t) = \theta B^{-1} \left(1 - e^{B(T-t)} \right) \left(A_0 + \bar{h}^k \hat{A} \right) \quad (33)$$

and $2n$ functions $\alpha^k : [0, T] \rightarrow \mathbb{R}$ defined by

$$\alpha(t) = - \int_t^T q(s) ds \quad (34)$$

where

$$\begin{aligned} q(t) := & \left(b - \theta \Lambda \Sigma' \bar{h} + \int_{\mathcal{Z}} \xi(z) \left[\left(1 + \bar{h}^{k'} \gamma(z) \right)^{-\theta} - 1_{\mathcal{Z}_0}(z) \right] v(dz) \right)' \beta^{k'}(t) \\ & + \frac{1}{2} \text{tr} \left(\Lambda \Lambda' \beta^{k'}(t) \beta^k(t) \right) + \int_{\mathcal{Z}} \left\{ e^{\beta^k \xi(z)} - 1 - \xi'(z) \beta^{k'}(t) \right\} v(dz) \\ & + \frac{1}{2} \theta (\theta + 1) \bar{h}^{k'} \Sigma \Sigma' \bar{h}^k - \theta a_0 - \theta \hat{a} \\ & + \theta \int_{\mathcal{Z}} \left\{ \frac{1}{\theta} \left[\left(1 + \bar{h}^{k'} \gamma(z) \right)^{-\theta} - 1 \right] + \bar{h}^{k'} \gamma(z) 1_{\mathcal{Z}_0}(z) \right\} v(dz) \end{aligned}$$

exist and for $i = 1, \dots, n$ satisfy:

$$\beta_i^i(t) < 0$$

$$\beta_i^{n+i}(t) > 0 \quad (35)$$

where $\beta_j^i(t)$ denotes the j th component of the vector $\beta^i(t)$.

Remark 4.1. Key to this assumption is the condition (35) which imposes a specific constraint on one element of each of the $2n$ vectors $\beta^k(t)$. To clarify the structure of this constraint, define M_β^- as the square $n \times n$ matrix whose i -th column (with $i = 1, \dots, n$) is the n -element column vector $\beta^i(t)$. Then all the elements m_{jj}^- , $j = 1, \dots, n$ on the diagonal of M_β^- are such that

$$m_{jj}^- = \beta_j^j(t) < 0$$

Similarly, define M_β^+ as the square $n \times n$ matrix whose i -th column (with $i = 1, \dots, n$) is the n -element column vector $\beta^{n+i}(t)$. Then all the elements m_{jj}^+ , $j = 1, \dots, n$ on the diagonal of M_β^+ are such that

$$m_{jj}^+ = \beta_j^{n+j}(t) > 0$$

Note that there is no requirement for either M_β^- or M_β^+ to have full rank. It would in fact be perfectly acceptable to have rank 1 as a result of column duplication.

Remark 4.2. For the function β^k in equation (33) to exist, B must be invertible. Moreover, the existence of $2n$ constant controls h^k , $k = 1, \dots, 2n$ such that (33) satisfies (35) is only guaranteed when $J = \mathbb{R}^n$. However, since finding the controls is equivalent to solving a system of at most n inequalities with m variables and $m > n$, it is likely that one could find constant controls after some adjustments to the elements of the matrices A_0, A, B or to the maximum jump size allowed.

Proposition 4.3. Suppose Assumption 4.2 holds and consider the $2n$ constant controls h^k , $k = 1, \dots, 2n$ parameterizing the $4n$ functions

$$\alpha^k : [0, T] \rightarrow \mathbb{R}, \quad k = 1, \dots, 2n$$

$$\beta^k : [0, T] \rightarrow \mathbb{R}^n, \quad k = 1, \dots, 2n$$

such that for $i = 1, \dots, n$,

$$\beta_i^i(t) < 0$$

$$\beta_i^{n+i}(t) > 0$$

where $\beta_j^i(t)$ denotes the j -th component of the vector $\beta^i(t)$. Then we have the following upper bounds:

$$\tilde{\Phi}(t, x) \leq e^{\alpha^k(t) + \beta^{k'}(t)x}$$

in each element x_i , $i = 1, \dots, n$ of x .

Proof. Setting $\mathbf{Z} = \mathbb{R}^n - \{0\}$ and recalling that the dynamics of the state variable $X(t)$ under the $\mathbb{P}_h^\theta$ -measure is given by

$$dX(t) = f(X(t^-), h(t); \theta) + \Lambda dW_t^h + \int_{\mathbb{R}^n} \xi(z) \tilde{N}_p^h(dt, dz)$$

we note that the associated Lévy measure $\tilde{\nu}$ can be defined via the map:

$$\tilde{\nu} = \nu \circ \xi^{-1} \quad (36)$$

We will now limit ourselves the class $\mathcal{H}^c$ of constant controls. By the optimality principle, for an arbitrary admissible constant control policy h , we have

$$\tilde{\Phi}(t, x) \leq \tilde{I}(x; \bar{h}; t, T) \leq \mathbf{E}_{t,x} \left[\exp \left\{ \theta \int_t^T g(X_s, \bar{h}) ds - \theta \ln \nu \right\} \right] := W(t, x) \quad (37)$$

In this setting, we note that the function g is an affine function of the affine process $X(t)$. Affine process theory (See Appendix A in Duffie and Singleton [24], Duffie, Pan and Singleton [23] or Duffie, Filipovic and Schachermayer [21] for more details on the properties of affine processes) leads us to expect that the expectation on the right-hand side of equation (37) takes the form

$$W(t, x) = \exp \{ \alpha(t) + \beta(t)x \} \quad (38)$$

where

$$\alpha : t \in [0, T] \rightarrow \mathbb{R}$$

$$\beta : t \in [0, T] \rightarrow \mathbb{R}^n$$

are functions solving two ODEs.

Indeed, applying the Feynman-Kac formula, we find that the function $W(t, x)$ satisfies the integro-differential PDE:

$$\begin{aligned} & \frac{\partial W}{\partial t} + \left(b + BX_s - \theta \Lambda \Sigma' \bar{h} + \int_{\mathbf{Z}} \xi(z) \left[\left(1 + \bar{h}' \gamma(z) \right)^{-\theta} - 1_{\mathbf{Z}_0}(z) \right] \nu(dz) \right)' DW(t, x) \\ & + \frac{1}{2} \text{tr} \left(\Lambda \Lambda' D^2 W(t, x) \right) + \int_{\mathbf{Z}} \{ W(t, x + \xi(z)) - W(t, x) - \xi'(z) DW(t, x) \} \nu(dz) \\ & + \theta g(x, \bar{h}; \theta) W(t, x) \\ & = 0 \end{aligned}$$

subject to terminal condition $\tilde{\Phi}(T, x) = \nu^{-\theta}$.

Now, taking a candidate solution of the form

$$W(t, x) = \exp \{ \alpha(t) + \beta(t)x \}$$

we have

$$\frac{\partial W}{\partial t} = (\alpha'(t) + \dot{\beta}(t)x) W(t, x)$$

$$DW = \beta'(t)W(t, x)$$

$$D^2W = \beta'(t)\beta(t)W(t, x)$$

Substituting into the PDE, we get

$$\begin{aligned} & (\alpha'(t) + \dot{\beta}(t)x) W(t, x) \\ & + \left(b + Bx - \theta \Lambda \Sigma' \bar{h} + \int_{\mathbf{Z}} \xi(z) \left[(1 + h' \gamma(z))^{-\theta} - 1_{\mathbf{Z}_0}(z) \right] \nu(dz) \right)' \beta'(t) W(t, x) \\ & + \frac{1}{2} \text{tr} (\Lambda \Lambda' \beta'(t) \beta(t)) W(t, x) \\ & + \int_{\mathbf{Z}} \{ W(t, x + \xi(z)) - W(t, x) - \xi'(z) \beta'(t) W(t, x) \} \nu(dz) \\ & + \theta \left(\frac{1}{2} (\theta + 1) \bar{h}' \Sigma \Sigma' \bar{h} - a_0 - A'_0 x - \bar{h}' (\hat{a} + \hat{A} x) \right. \\ & \left. + \int_{\mathbf{Z}} \left\{ \frac{1}{\theta} \left[(1 + h' \gamma(z))^{-\theta} - 1 \right] + \bar{h}' \gamma(z) 1_{\mathbf{Z}_0}(z) \right\} \nu(dz) \right) W(t, x) \\ & = 0 \end{aligned}$$

Dividing by $W(t, x)$ and rearranging, we get

$$\begin{aligned} & (\dot{\beta}(t) + B' \beta'(t) - \theta A'_0 - \theta \bar{h}' \hat{A}) x \\ & = - \left(\alpha'(t) + \left(b - \theta \Lambda \Sigma' \bar{h} + \int_{\mathbf{Z}} \xi(z) \left[(1 + \bar{h}' \gamma(z))^{-\theta} - 1_{\mathbf{Z}_0}(z) \right] \nu(dz) \right)' \beta'(t) \right. \\ & \quad + \frac{1}{2} \text{tr} (\Lambda \Lambda' \beta'(t) \beta(t)) + \int_{\mathbf{Z}} \{ e^{\beta \xi(z)} - 1 - \xi'(z) \beta'(t) \} \nu(dz) \\ & \quad \left. + \frac{1}{2} \theta (\theta + 1) \bar{h}' \Sigma \Sigma' \bar{h} - \theta a_0 - \theta \hat{a} + \theta \int_{\mathbf{Z}} \left\{ \frac{1}{\theta} \left[(1 + \bar{h}' \gamma(z))^{-\theta} - 1 \right] + \bar{h}' \gamma(z) 1_{\mathbf{Z}_0}(z) \right\} \nu(dz) \right) \end{aligned}$$

Since the left-hand side is independent from the right-hand side, then both sides are orthogonal. As a result we now only need to solve the two ODEs

$$\dot{\beta}(t) + B' \beta'(t) - \theta A'_0 - \theta \bar{h}' \hat{A} = 0 \quad (39)$$

and

$$\begin{aligned}
\dot{\alpha}(t) &+ \left(b - \theta \Lambda \Sigma' \bar{h} + \int_{\mathbf{z}} \xi(z) \left[\left(1 + \bar{h}' \gamma(z) \right)^{-\theta} - 1_{\mathbf{z}_0}(z) \right] \nu(dz) \right)' \beta'(t) \\
&+ \frac{1}{2} \text{tr} (\Lambda \Lambda' \beta'(t) \beta(t)) + \int_{\mathbf{z}} \left\{ e^{\beta \xi(z)} - 1 - \xi'(z) \beta'(t) \right\} \nu(dz) \\
&+ \frac{1}{2} \theta (\theta + 1) \bar{h}' \Sigma \Sigma' \bar{h} - \theta a_0 - \theta \hat{a} + \theta \int_{\mathbf{z}} \left\{ \frac{1}{\theta} \left[\left(1 + \bar{h}' \gamma(z) \right)^{-\theta} - 1 \right] + \bar{h}' \gamma(z) 1_{\mathbf{z}_0}(z) \right\} \nu(dz) \\
&= 0
\end{aligned} \tag{40}$$

to obtain the value of $W(t, x)$. The ODE (39) for β is linear and admits the solution

$$\beta(t) = \theta B^{-1} \left(1 - e^{B(T-t)} \right) \left(A_0 + \bar{h}^k \hat{A} \right) \tag{41}$$

As for the ODE (40) for α , we only need to integrate to get

$$\alpha(t) = - \int_t^T q(s) ds \tag{42}$$

where

$$\begin{aligned}
q(t) &:= \left(b - \theta \Lambda \Sigma' \bar{h} + \int_{\mathbf{z}} \xi(z) \left[\left(1 + \bar{h}' \gamma(z) \right)^{-\theta} - 1_{\mathbf{z}_0}(z) \right] \nu(dz) \right)' \beta'(t) \\
&+ \frac{1}{2} \text{tr} (\Lambda \Lambda' \beta'(t) \beta(t)) + \int_{\mathbf{z}} \left\{ e^{\beta \xi(z)} - 1 - \xi'(z) \beta'(t) \right\} \nu(dz) \\
&+ \frac{1}{2} \theta (\theta + 1) \bar{h}' \Sigma \Sigma' \bar{h} - \theta a_0 - \theta \hat{a} + \theta \int_{\mathbf{z}} \left\{ \frac{1}{\theta} \left[\left(1 + \bar{h}' \gamma(z) \right)^{-\theta} - 1 \right] + \bar{h}' \gamma(z) 1_{\mathbf{z}_0}(z) \right\} \nu(dz)
\end{aligned}$$

Observe that $W(t, x)$ is increasing in x_i , the i -th element of x , if $\beta_i > 0$, and conversely, $W(t, x)$ is decreasing in x_i if $\beta_i < 0$,

Equations (41) and (42) are respectively equations (33) and (34) from Assumption 4.2. By Assumption 4.2, there exists $2n$ constant controls $\bar{h}^k, k = 1, \dots, 2n$ such that for $i = 1, \dots, n$,

$$\beta_i^j(t) < 0$$

$$\beta_i^{n+i}(t) > 0$$

where $\beta_j^i(t)$ denotes the j th component of the vector $\beta^i(t)$. We can now conclude that we have the following upper bounds

$$\tilde{\Phi}(t, x) \leq e^{\alpha^k(t) + \beta^{k'}(t)x}$$

for each element $x_i, i = 1, \dots, n$ of x . □

Remark 4.3. To obtain the upper bounds and the asymptotic behaviour, we do not need the $2n$ constant controls to be pairwise different. In fact, we need at least 2 different controls and at most $2n$ different controls. Moreover, we could consider wider classes of controls extending beyond constant controls. This would require some modifications to the proof but would also alleviate the assumptions required for the result to hold.

Remark 4.4. For a given constant control $\bar{h}$, equation (39) is a linear n -dimensional ODE. However, if in the dynamics of the state variable $X(t)$, Λ and Ξ depended on X , the ODE would be nonlinear. Once ODE (39) is solved, obtaining $\alpha(t)$ from equation (40) is a simple matter of integration.

Remark 4.5. For a given constant control h , given $x \in \mathbb{R}^n$ and $t \in [0, T]$, the solution of ODE (39) is the same whether the dynamics of $S(t)$ and $X(t)$ is the jump diffusion considered here or the corresponding pure diffusion model. The converse is, however, not true since in the pure diffusion setting $h \in \mathbb{R}^m$, while in the jump diffusion case $h \in \mathcal{J} \subset \mathbb{R}^m$.

5. Viscosity Solution Approach

In recent years, viscosity solutions have gained a widespread acceptance as an effective technique to obtain a weak sense solution for HJB PDEs when no classical (i.e. $C^{1,2}$) solution can be shown to exist, which is the case for many stochastic control problems. Viscosity solutions also have a very practical interest. Indeed, once a solution has been interpreted in the viscosity sense and the uniqueness of this solution has been proved via a comparison result, the fundamental ‘stability’ result of Barles and Souganidis [8] opens the way to a numerical resolution of the problem through a wide range of schemes. Readers interested in an overview of viscosity solutions should refer to the classic article by Crandall, Ishii and Lions [17], the book by Fleming and Soner [26] and Øksendal and Sulem [30], as well as the notes by Barles [5] and Touzi [34].

While the use of viscosity solutions to solve classical diffusion-type stochastic control problems has been extensively studied and surveyed (see Fleming and Soner [26] and Touzi [34]), this introduction of a jump-related measure makes the jump-diffusion framework more complex. As a result, so far no general theory has been developed to solve jump-diffusion problems. Instead, the assumptions made to derive a comparison result are closely related to what the specific problem allows. Broadly speaking, the literature can be split along two lines of analysis, depending on whether the measure associated with the jumps is assumed to be finite.

In the case when the jump measure is finite, Alvarez and Tourin [1] consider a fairly general setting in which the jump term does not need to be linear in the function u which solves the integro-differential PDE. In this setting, Alvarez and

Tourin develop a comparison theorem that they apply to a stochastic differential utility problem. Amadori [3] extends Alvarez and Tourin's analysis to price European options. Barles, Buckdahn and Pardoux [6] study the viscosity solution of integro-differential equations associated with backward SDEs (BSDEs).

The Lévy measure is the most extensively studied measure with singularities. Pham [33] derives a comparison result for the variational inequality associated with an optimal stopping problem. Jakobsen and Karlsen [29] analyse in detail the impact of the Lévy measure's singularity and propose a maximum principle. Amadori, Karlsen and La Chioma [4] focus on geometric Lévy processes and the partial integro differential equations they generate before applying their results to BSDEs and to the pricing of European and American derivatives. A recent article by Barles and Imbert [7] takes a broader view of PDEs and their non-local operators. However, the authors assume that the nonlocal operator is broadly speaking linear in the solution which may prove overly restrictive in some cases, including our present problem.

As far as our jump diffusion risk-sensitive control problem is concerned, we will promote a general treatment and avoid restricting the class of the compensator ν . At some point, we will however need ν to be finite. This assumption will only be made for a purely technical reason arising in the proof of the comparison result (in Section 6). Since the rest of the story is still valid if ν is not finite, and in accordance with our goal of keeping the discussion as broad as possible, we will write the rest of the article in the spirit of a general compensator ν .

5.1 Definitions

Before proceeding further, we will introduce the following definition:

Definition 5.1. The upper semicontinuous envelope $u^*(x)$ of a function u at x is defined as

$$u^*(x) = \limsup_{y \rightarrow x} u(y)$$

and the lower semicontinuous envelope $u_*(x)$ of $u(x)$ is defined as

$$u_*(x) = \liminf_{y \rightarrow x} u(y)$$

Note in particular the fundamental inequality between a function and its upper and lower semicontinuous envelopes:

$$u_* \leq u \leq u^*$$

The theory of viscosity solutions was initially developed for elliptical PDEs of the form

$$H(x, u, Du, D^2u) = 0$$

and parabolic PDEs of the form

$$\frac{\partial u}{\partial t} + H(x, u, Du, D^2u) = 0$$

for what Crandall, Ishii and Lions [17] term a “proper” functional $H(x, r, p, A)$.

Definition 5.2. A functional $H(x, r, p, A)$ is said to be *proper* if it satisfies the following two properties:

1. (degenerate) ellipticity:

$$H(x, r, p, A) \leq H(x, r, p, B), \quad B \leq A$$

and

2. monotonicity

$$H(x, r, p, A) \leq H(x, s, p, A), \quad r \leq s$$

In our problem, the functional F defined as

$$\begin{aligned} F(x, p, A) := & -\sup_{h \in \mathcal{J}} \left\{ f(x, h)'p + \frac{1}{2} \text{tr}(\Lambda \Lambda' A) \right. \\ & - \frac{\theta}{2} p' \Lambda \Lambda' p \\ & + \int_{\mathbb{Z}} \left\{ -\frac{1}{\theta} \left(e^{-\theta(\Phi(t, x + \xi(z)) - \Phi(t, x))} - 1 \right) - \xi'(z)p \right\} \nu(dz) \\ & \left. - g(x, h) \right\} \end{aligned} \quad (43)$$

plays a similar role to the functional H in the general equation (43), and we note that it is indeed “proper”. As a result, we can develop a viscosity approach to show that the value function Φ is the unique solution of the associated RS HJB PIDE.

We now give two equivalent definitions of viscosity solutions adapted from Alvarez and Tourin [1]:

- a definition based on the notion of semijets;
- a definition based on the notion of test function

Before introducing these two definitions, we need to define parabolic semijet of upper semicontinuous and lower semicontinuous functions and to add two additional conditions.

Definition 5.3. Let $u \in USC([0, T] \times \mathbb{R}^n)$ and $(t, x) \in [0, T] \times \mathbb{R}^n$. We define:

- the Parabolic superjet $\mathcal{P}_u^{2,+}$ as

$$\begin{aligned} \mathcal{P}_u^{2,+} := \{ & (p, q, A) \in \mathbb{R} \times \mathbb{R}^n \times \mathcal{S}_n : \\ & u(s, y) \leq u(s, x) + p(s - t) + \langle q, y - x \rangle + \frac{1}{2} \langle A(y - x), y - x \rangle \\ & + o(|s - t| + |y - x|^2) \text{ as } (s, y) \rightarrow (t, x) \} \end{aligned}$$

- the closure of the Parabolic superjet $\overline{\mathcal{P}}_u^{2,+}$ as

$$\begin{aligned} \overline{\mathcal{P}}_u^{2,+} := \{ & (p, q, A) = \lim_{k \rightarrow \infty} (p_k, q_k, A_k) \text{ with } (p_k, q_k, A_k) \in \mathcal{P}_u^{2,+} \\ & \text{and } \lim_{k \rightarrow \infty} (t_k, x_k, u(t_k, x_k)) = (t, x, u(t, x)) \} \end{aligned}$$

Let $u \in LSC([0, T] \times \mathbb{R}^n)$ and $(t, x) \in [0, T] \times \mathbb{R}^n$. We define:

- the Parabolic subjet $\mathcal{P}_u^{2,-}$ as $\mathcal{P}_u^{2,-} := -\mathcal{P}_u^{2,+}$, and
- the closure of the Parabolic subjet $\overline{\mathcal{P}}_u^{2,-}$ as $\overline{\mathcal{P}}_u^{2,-} = -\overline{\mathcal{P}}_u^{2,+}$

Condition 5.1. Let $(t, x) \in [0, T] \times \mathbb{R}^n$ and $(p, q, A) \in \mathcal{P}_u^{2,+}u(t, x)$, there are $\varphi \in C(\mathbb{R}^n)$, $\varphi \geq 1$ and $R > 0$ such that for

$$((s, y), z) \in (\mathcal{B}_R(t, x) \cap ([0, T] \times \mathbb{R}^n)) \times \mathbf{Z},$$

$$\int_{\mathbf{Z}} \left\{ -\frac{1}{\theta} \left(e^{-\theta(u(s, y + \xi(z)) - u(s, y))} - 1 \right) - \xi'(z)q \right\} \nu(dz) \leq \varphi(y)$$

Condition 5.2. Let $(t, x) \in [0, T] \times \mathbb{R}^n$ and $(p, q, A) \in \mathcal{P}_u^{2,-}u(t, x)$, there are $\varphi \in C(\mathbb{R}^n)$, $\varphi \geq 1$ and $R > 0$ such that for

$$((s, y), z) \in (\mathcal{B}_R(t, x) \cap ([0, T] \times \mathbb{R}^n)) \times \mathbf{Z},$$

$$\int_{\mathbf{Z}} \left\{ -\frac{1}{\theta} \left(e^{-\theta(u(s, y + \xi(z)) - u(s, y))} - 1 \right) - \xi'(z)q \right\} \nu(dz) \geq -\varphi(y)$$

The purpose of these conditions on u and v is to ensure that the jump term is semicontinuous at any given point $(t, x) \in [0, T] \times \mathbb{R}^n$ (see Lemma 1 and Conditions

(6) and (7) in [1]). In our setting, we note that since the value function Φ and the function $x \mapsto e^x$ are locally bounded, these two conditions are satisfied.

Remark 5.1. Note that the jump-related integral term

$$\int_{\mathbf{Z}} \left\{ -\frac{1}{\theta} \left(e^{-\theta(u(s,y+\xi(z))-u(s,y))} - 1 \right) - \xi'(z)q \right\} \nu(dz)$$

is well defined when $(p, q, A) \in \mathcal{P}_u^{2,\pm}$. First, by Taylor,

$$\begin{aligned} & \int_{\mathbf{Z}} \left\{ -\frac{1}{\theta} \left(e^{-\theta(u(s,y+\xi(z))-u(s,y))} - 1 \right) - \xi'(z)q \right\} \nu(dz) \\ &= \int_{\mathbf{Z}} \left\{ (u(s,y+\xi(z)) - u(s,y)) - \frac{\theta}{2} (u(s,y+\xi(z)) - u(s,y))^2 \right. \\ & \quad \left. + \frac{\theta^2}{3!} (u(s,y+\xi(z)) - u(s,y))^3 + \dots - \xi'(z)q \right\} \nu(dz) \end{aligned}$$

By definition of the Parabolic superjet $\mathcal{P}_u^{2,+}$, for $t = s$, the pair (q, A) satisfies the inequality

$$u(s,y+\xi(z)) - u(s,y) - \xi'(z)q \leq \frac{1}{2}\xi'(z)A\xi(z) + o(|\xi(z)|^2)$$

Similarly, by definition of the Parabolic subjet $\mathcal{P}_u^{2,-}$, for $t = s$, the pair (q, A) satisfies the inequality

$$u(s,y+\xi(z)) - u(s,y) - \xi'(z)q \geq \frac{1}{2}\xi'(z)A\xi(z) + o(|\xi(z)|^2)$$

Thus, if u is a viscosity solution, we have

$$u(s,y+\xi(z)) - u(s,y) - \xi'(z)q = \frac{1}{2}\xi'(z)A\xi(z) + o(|\xi(z)|^2)$$

and the jump-related integral is equal to

$$\begin{aligned} & \int_{\mathbf{Z}} \left\{ -\frac{1}{\theta} \left(e^{-\theta(u(s,y+\xi(z))-u(s,y))} - 1 \right) - \xi'(z)q \right\} \nu(dz) \\ &= \int_{\mathbf{Z}} \left\{ -\frac{\theta}{2} (u(s,y+\xi(z)) - u(s,y))^2 + \frac{1}{2}\xi'(z)A\xi(z) + o(|\xi(z)|^2) \right\} \nu(dz) \end{aligned}$$

which is well-defined.

Definition 5.4. A locally bounded function $u \in USC([0, T] \times \mathbb{R}^n)$ satisfying Condition 5.1 is a viscosity subsolution of (23), if for all $x \in \mathbb{R}^n$, $u(T, x) \leq g_0(x)$, and for all $(t, x) \in [0, T] \times \mathbb{R}^n$, $(p, q, A) \in \mathcal{P}^{2,+}u(t, x)$, we have

$$-p + F(x, q, A) - \int_{\mathbf{Z}} \left\{ -\frac{1}{\theta} \left(e^{-\theta(u(t, x + \xi(z)) - u(t, x))} - 1 \right) - \xi'(z)q \right\} \nu(dz) \leq 0$$

A locally bounded function $u \in LSC([0, T] \times \mathbb{R}^n)$ satisfying Condition 5.2 is a viscosity supersolution of (23), if for all $x \in \mathbb{R}^n$, $u(T, x) \geq g_0(x)$, and for all $(t, x) \in [0, T] \times \mathbb{R}^n$, $(p, q, A) \in \mathcal{P}^{2,-}u(t, x)$, we have

$$-p + F(x, q, A) - \int_{\mathbf{Z}} \left\{ -\frac{1}{\theta} \left(e^{-\theta(u(t, x + \xi(z)) - u(t, x))} - 1 \right) - \xi'(z)q \right\} \nu(dz) \geq 0$$

A locally bounded function Φ whose upper semicontinuous and lower semicontinuous envelopes are a viscosity subsolution and a viscosity supersolution of (23) is a viscosity solution of (23).

Definition 5.5. A locally bounded function $u \in USC([0, T] \times \mathbb{R}^n)$ is a viscosity subsolution of (23), if for all $x \in \mathbb{R}^n$, $u(T, x) \leq g_0(x)$, and for all $(t, x) \in [0, T] \times \mathbb{R}^n$, $\psi \in C^2([0, T] \times \mathbb{R}^n)$ such that $u(t, x) = \psi(t, x)$, $u < \psi$ on $[0, T] \times \mathbb{R}^n \setminus \{(t, x)\}$, we have

$$-\frac{\partial \psi}{\partial t} + F(x, D\psi, D^2\psi) - \int_{\mathbf{Z}} \left\{ -\frac{1}{\theta} \left(e^{-\theta(\psi(t, x + \xi(z)) - \psi(t, x))} - 1 \right) - \xi'(z)D\psi \right\} \nu(dz) \leq 0$$

A locally bounded function $v \in LSC([0, T] \times \mathbb{R}^n)$ is a viscosity supersolution of (23), if for all $x \in \mathbb{R}^n$, $v(T, x) \geq g_0(x)$, and for all $(t, x) \in [0, T] \times \mathbb{R}^n$, $\psi \in C^2([0, T] \times \mathbb{R}^n)$ such that $v(t, x) = \psi(t, x)$, $v > \psi$ on $[0, T] \times \mathbb{R}^n \setminus \{(t, x)\}$, we have

$$-\frac{\partial \psi}{\partial t} + F(x, D\psi, D^2\psi) - \int_{\mathbf{Z}} \left\{ -\frac{1}{\theta} \left(e^{-\theta(\psi(t, x + \xi(z)) - \psi(t, x))} - 1 \right) - \xi'(z)D\psi \right\} \nu(dz) \geq 0$$

A locally bounded function Φ whose upper semicontinuous and lower semicontinuous envelopes are a viscosity subsolution and a viscosity supersolution of (23) is a viscosity solution of (23).

We would have similar definition for the viscosity supersolution, subsolution and solution of equation (27). Once again, the superjet and test function formulations are strictly equivalent (see Alvarez and Tourin [1] and Crandall, Ishii and Lions [17]).

Remark 5.2. A more classical but also more restrictive definition of viscosity solution is as the continuous function which is both a supersolution and a subsolution of (23) (see Definition 5.1 in Barles [5]). The line of reasoning we will follow will make full use of the latitude afforded by our definition and we will have to wait until the comparison result is established in Section 6 to prove the continuity of the viscosity solution.

5.2 Characterization of the Value Function as a Viscosity Solution

To show that the value function is a (discontinuous) viscosity solution of the associated RS HJB PIDE (23), we follow an argument by Touzi [34] which enables us to make a greater use of control theory in the derivation of the proof.

Theorem 5.1. *Φ is a (discontinuous) viscosity solution of the RS HJB PIDE (23) on $[0, T] \times \mathbb{R}^n$, subject to terminal condition (25).*

Proof.

Outline: This proof can be decomposed in five steps. First, we define $\tilde{\Phi}$ as a log transformation of Φ . In the next three steps, we prove that $\tilde{\Phi}$ is a viscosity solution of the exponentially transformed RS HJB PIDE by showing that it is 1) a viscosity subsolution; 2) a viscosity supersolution; and hence 3) a viscosity solution. Finally, applying a change of variable result, such as Proposition 2.2 in [34], we conclude that Φ is a viscosity solution of the RS HJB PIDE (23).

Step 1: Exponential Transformation

In order to prove that the value function Φ is a (discontinuous) viscosity solution of (23), we will start by proving that the exponentially transformed value function $\tilde{\Phi}$ is a (discontinuous) viscosity solution of (27).

Step 2: Viscosity Subsolution

Let $(t_0, x_0) \in Q := [0, t] \times \mathbb{R}^n$ and $u \in C^{1,2}(Q)$ satisfy

$$0 = (\tilde{\Phi}^* - u)(t_0, x_0) = \max_{(t,x) \in Q} (\tilde{\Phi}^*(t, x) - u(t, x)) \quad (44)$$

and hence

$$\tilde{\Phi} \leq \tilde{\Phi}^* \leq u \quad (45)$$

on Q .

Let (t_k, x_k) be a sequence in Q such that

$$\begin{aligned} \lim_{k \rightarrow \infty} (t_k, x_k) &= (t_0, x_0) \\ \lim_{k \rightarrow \infty} \tilde{\Phi}(t_k, x_k) &= \tilde{\Phi}^*(t_0, x_0) \end{aligned}$$

and define the sequence $\{\xi\}_k$ as $\xi_k := \tilde{\Phi}(t_k, x_k) - u(t_k, x_k)$. Since u is of class $C^{1,2}$, $\lim_{k \rightarrow \infty} \xi_k = 0$.

Fix $h \in \mathcal{J}$ and consider a constant control $\hat{h} = h$. Denote by X^k the state process with initial data $X_{t_k}^k = x_k$ and, for $k > 0$, define the stopping time

$$\tau_k := \inf \left\{ s > t_k : (s - t_k, X_s^k - x_k) \notin [0, \delta_k) \times \alpha \mathcal{B}_n \right\}$$

for a given constant $\alpha > 0$ and where $\mathcal{B}_n$ is the unit ball in $\mathbb{R}^n$ and

$$\delta_k := \sqrt{\xi_k} (1 - 1_{[0]}(\xi_k)) + k^{-1} 1_{[0]}(\xi_k)$$

From the definition of τ_k , we see that $\lim_{k \rightarrow \infty} \tau_k = t_0$.

By the Dynamic Programming Principle,

$$\tilde{\Phi}(t_k, x_k) \leq \mathbf{E}_{t_k, x_k} \left[\exp \left\{ \theta \int_{t_k}^{\tau_k} g(X_s, \hat{h}_s; \theta) ds \right\} \tilde{\Phi}(\tau_k, X_{\tau_k}^k) \right]$$

where $\mathbf{E}_{t_k, x_k} [\cdot]$ represents the expectation under the measure $\mathbb{P}$ given initial data (t_k, x_k) .

By inequality (45),

$$\tilde{\Phi}(t_k, x_k) \leq \mathbf{E}_{t_k, x_k} \left[\exp \left\{ \theta \int_{t_k}^{\tau_k} g(X_s, \hat{h}_s) ds \right\} u(\tau_k, X_{\tau_k}^k) \right]$$

and hence by definition of ξ_k ,

$$u(t_k, x_k) + \xi_k \leq \mathbf{E}_{t_k, x_k} \left[\exp \left\{ \theta \int_{t_k}^{\tau_k} g(X_s, \hat{h}_s) ds \right\} u(\tau_k, X_{\tau_k}^k) \right]$$

i.e.

$$\xi_k \leq \mathbf{E}_{t_k, x_k} \left[\exp \left\{ \theta \int_{t_k}^{\tau_k} g(X_s, \hat{h}_s) ds \right\} u(\tau_k, X_{\tau_k}^k) \right] - u(t_k, x_k)$$

Define $Z(t_k) = \theta \int_{t_k}^{\tau_k} g(X_s, \hat{h}_s) ds$, then

$$d(e^{Z_s}) := \theta g(X_s, \hat{h}_s) e^{Z_s} ds$$

Also, by Itô,

$$\begin{aligned} du_s &= \left\{ \frac{\partial u}{\partial s} + \mathcal{L}u \right\} ds + Du' \Lambda(s) dW_s \\ &\quad + \int_{\mathbf{Z}} \{u(s, X(s^-) + \xi(z)) - u(s, X(s^-))\} \tilde{N}_{\mathbf{p}}(ds, dz) \end{aligned}$$

for $s \in [t_k, \tau_k]$ and where the generator $\mathcal{L}$ of the state process $X(t)$ is defined as

$$\mathcal{L}u(t, x) := f(t, x, h; \theta)' Du + \frac{1}{2} \text{tr} \left(\Lambda \Lambda'(t, X) D^2 u \right) \quad (46)$$

By the Itô product rule, and since $dZ_s \cdot u_s = 0$, we get

$$d(u_s e^{Z_s}) = u_s d(e^{Z_s}) + e^{Z_s} du_s$$

and hence for $t \in [t_k, \tau_k]$

$$\begin{aligned} u(t, X_t^k) e^{Z_t} &= u(t_k, x_k) e^{Z_{t_k}} + \theta \int_{t_k}^t u(s, X_s^k) g(X_s^k, \hat{h}_s) e^{Z_s} ds \\ &\quad + \int_{t_k}^t \left(\frac{\partial u}{\partial s}(s, X_s^k) + \mathcal{L}u(s, X_s^k) e^{Z_s} \right) ds + \int_{t_k}^t Du'(s) dW_s \\ &\quad + \int_{t_k}^t \int_{\mathbf{Z}} \left\{ u(t, X^k(s^-) + \xi(z)) - u(t, X^k(s^-)) \right\} \tilde{N}_{\mathbf{p}}(dt, dz) \end{aligned}$$

Noting that $u(t_k, x_k) e^{Z_{t_k}} = u(t_k, x_k)$ and taking the expectation with respect to the initial data (t_k, x_k) , we get

$$\begin{aligned} \mathbf{E}_{t_k, x_k} [u(t, X_t) e^{Z_t}] \\ = u(t_k, x_k) e^{Z_{t_k}} + \mathbf{E}_{t_k, x_k} \left[\int_{t_k}^t \left(\frac{\partial u}{\partial s}(s, X_s) + \mathcal{L}u(s, X_s) + \theta u(s, X_s) g(X_s, \hat{h}_s) \right) e^{Z_s} ds \right] \end{aligned}$$

In particular, for $t = \tau_k$,

$$\begin{aligned} \xi_k &\leq \mathbf{E}_{t_k, x_k} [u(\tau_k, X_{\tau_k}) e^{Z_{\tau_k}}] - u(t_k, x_k) e^{Z_{t_k}} \\ &= + \mathbf{E}_{t_k, x_k} \left[\int_{t_k}^{\tau_k} \left(\frac{\partial u}{\partial s}(s, X_s) + \mathcal{L}u(s, X_s) + \theta u(s, X_s) g(X_s, \hat{h}_s) \right) e^{Z_s} ds \right] \end{aligned}$$

and thus

$$\begin{aligned} \frac{\xi_k}{\delta_k} &\leq \frac{1}{\delta_k} \left(\mathbf{E}_{t_k, x_k} [u(\tau_k, X_{\tau_k}) e^{Z_{\tau_k}}] - u(t_k, x_k) e^{Z_{t_k}} \right) \\ &= \frac{1}{\delta_k} \left(\mathbf{E}_{t_k, x_k} \left[\int_{t_k}^{\tau_k} \left(\frac{\partial u}{\partial s}(s, X_s) + \mathcal{L}u(s, X_s) + \theta u(s, X_s) g(X_s, \hat{h}_s) \right) e^{Z_s} ds \right] \right) \end{aligned}$$

As $k \rightarrow \infty$, $t_k \rightarrow t_0$, $\tau_k \rightarrow t_0$, $\frac{\xi_k}{\delta_k} \rightarrow 0$ and

$$\begin{aligned} &\frac{1}{\delta_k} \left(\mathbf{E}_{t_k, x_k} \left[\int_{t_k}^t \left(\frac{\partial u}{\partial s}(s, X_s) + \mathcal{L}u(s, X_s) + \theta u(s, X_s) g(X_s, \hat{h}_s) \right) e^{Z_s} ds \right] \right) \\ &\rightarrow \frac{\partial u}{\partial s}(s, X_s) + \mathcal{L}u(s, X_s) + \theta u(s, X_s) g(X_s, \hat{h}_s) \end{aligned}$$

a.s. by the Bounded Convergence Theorem, since the random variable

$$\frac{1}{\delta_k} \int_{t_k}^t \left(\frac{\partial u}{\partial s}(s, X_s) + \mathcal{L}u(s, X_s) + \theta u(s, X_s)g(X_s, \hat{h}_s) \right) e^{Z_s} ds$$

is bounded for large enough k .

Hence, we conclude that since $\hat{h}_s$ is arbitrary,

$$\frac{\partial u}{\partial s}(s, X_s) + \mathcal{L}u(s, X_s) + \theta u(s, X_s)g(X_s, \hat{h}_s) \geq 0$$

i.e.

$$-\frac{\partial u}{\partial s}(s, X_s) - \mathcal{L}u(s, X_s) - \theta u(s, X_s)g(X_s, \hat{h}_s) \leq 0$$

This argument proves that $\tilde{\Phi}$ is a (discontinuous) viscosity subsolution of the PDE (27) on $[0, t) \times \mathbb{R}^n$ subject to terminal condition $\tilde{\Phi}(T, x) = e^{g_0(x; T)}$.

Step 3: Viscosity Supersolution

This step in the proof is a slight adaptation of the proof for classical control problems in Touzi [34]. Let $(t_0, x_0) \in Q$ and $u \in C^{1,2}(Q)$ satisfy

$$0 = (\tilde{\Phi}_* - u)(t_0, x_0) < (\tilde{\Phi}_* - u)(t, x) \text{ for } Q \setminus (t_0, x_0) \quad (47)$$

We intend to prove that at (t_0, x_0)

$$\frac{\partial u}{\partial t}(t, x) + \inf_{h \in \mathcal{H}} \left\{ \mathcal{L}^h u(t, x) - \theta g(x, h) \right\} \leq 0$$

by contradiction. Thus, assume that

$$\frac{\partial u}{\partial t}(t, x) + \inf_{h \in \mathcal{H}} \left\{ \mathcal{L}^h u(t, x) - \theta g(x, h) \right\} > 0 \quad (48)$$

at (t_0, x_0) .

Since $\mathcal{L}^h u$ is continuous, there exists an open neighbourhood $\mathcal{N}_\delta$ of (t_0, x_0) defined for $\delta > 0$ as

$$\mathcal{N}_\delta := \{(t, x) : (t - t_0, x - x_0) \in (-\delta, \delta) \times \delta \mathcal{B}_n, \text{ and (48) holds}\} \quad (49)$$

Note that by (47) and since $\tilde{\Phi} > \tilde{\Phi}_* > u$,

$$\min_{Q \setminus \mathcal{N}_\delta} (\tilde{\Phi} - u) > 0$$

For $\rho > 0$, consider the set J^ρ of ρ -optimal controls h^ρ satisfying

$$\tilde{I}(t_0, x_0, h^\rho) \leq \tilde{\Phi}(t_0, x_0) + \rho \quad (50)$$

Also, let $\epsilon > 0$, $\epsilon \leq \gamma$ be such that

$$\min_{Q \setminus \mathcal{N}_\delta} (\tilde{\Phi} - u) \geq 3\epsilon e^{-\delta\theta M_\delta} > 0 \quad (51)$$

where M_δ is defined as

$$M_\delta := \max_{(t,x) \in \mathcal{N}_\delta^J, h \in \mathcal{J}^p} (-g(x, h), 0)$$

for

$$\mathcal{N}_\delta^J := \{(t, x) : (t - t_0, x - x_0) \in (-\delta, \delta) \times (\zeta + \delta)\mathcal{B}_n\} \quad (52)$$

and

$$\zeta := \max_{z \in \mathbb{Z}} \|\xi(z)\|$$

Note that $\zeta < \infty$ by boundedness of $\xi(z)$ and thus $M_\delta < \infty$.

Now let (t_k, x_k) be a sequence in $\mathcal{N}_\delta$ such that

$$\lim_{k \rightarrow \infty} (t_k, x_k) = (t_0, x_0)$$

and

$$\lim_{k \rightarrow \infty} \tilde{\Phi}(t_k, x_k) = \tilde{\Phi}_*(t_0, x_0)$$

Since $(\tilde{\Phi} - u)(t_k, x_k) \rightarrow 0$, we can assume that the sequence (t_k, x_k) satisfies

$$|(\tilde{\Phi} - u)(t_k, x_k)| \leq \epsilon, \quad \text{for } k \geq 1 \quad (53)$$

for ϵ defined by (51).

Consider the ϵ -optimal control h_k^ϵ , denote by $\tilde{X}_k^\epsilon$ the controlled process defined by the control process h_k^ϵ and introduce the stopping time

$$\tau_k := \inf \{s > \tau_k : (s, \tilde{X}_k^\epsilon(s)) \notin \mathcal{N}_\delta\}$$

Note that since we assumed that $-\infty \leq \xi_i^{\min} \leq \xi_i \leq \xi_i^{\max} < \infty$ for $i = 1, \dots, n$ and since v is assumed to be bounded then $X(\tau)$ is also finite and in particular,

$$(\tilde{\Phi} - u)(\tau_k, \tilde{X}_k^\epsilon(\tau_k)) \geq (\tilde{\Phi}_* - u)(\tau_k, \tilde{X}_k^\epsilon(\tau_k)) \geq 3\epsilon e^{-\delta\theta M_\delta} \quad (54)$$

Choose $\mathcal{N}_\delta^J$ so that $(\tau, \tilde{X}^\epsilon(\tau)) \in \mathcal{N}_\delta^J$. In particular, since $X^\epsilon(\tau)$ is finite then $\mathcal{N}_\delta^J$ can be defined to be a strict subset of Q and we can effectively use the local boundedness of g to establish M_δ .

Let $Z(t_k) = \theta \int_{t_k}^{\tau_k} g(\tilde{X}_s^\epsilon, h_s^\epsilon) ds$, since $\tilde{\Phi} \geq \tilde{\Phi}_*$ and by (53) and (54),

$$\begin{aligned} & \tilde{\Phi}(\tau_k, \tilde{X}_k^\epsilon(\tau_k))e^{Z(\tau_k)} - \tilde{\Phi}(t_k, x_k)e^{Z(t_k)} \\ & \geq u(\tau_k, \tilde{X}_k^\epsilon(\tau_k))e^{Z(\tau_k)} - \tilde{\Phi}(t_k, x_k)e^{Z(t_k)} + 3\epsilon e^{-\delta\theta M_\delta} e^{Z(\tau_k)} - \epsilon \\ & \geq \int_{t_k}^{\tau_k} d\left(u(s, \tilde{X}_k^\epsilon(s))e^{Z_s}\right) + 2\epsilon \end{aligned}$$

i.e.

$$\tilde{\Phi}(t_k, x_k) \leq \tilde{\Phi}(\tau_k, \tilde{X}_k^\epsilon(\tau_k))e^{Z(\tau_k)} - \int_{t_k}^{\tau_k} d\left(u(s, \tilde{X}_k^\epsilon(s))e^{Z_s}\right) - 2\epsilon$$

Taking expectation with respect to the initial data (t_k, x_k) ,

$$\tilde{\Phi}(t_k, x_k) \leq \mathbf{E}_{t_k, x_k} \left[\tilde{\Phi}(\tau_k, \tilde{X}_k^\epsilon(\tau_k))e^{Z(\tau_k)} - \int_{t_k}^{\tau_k} d\left(u(s, \tilde{X}_k^\epsilon(s))e^{Z_s}\right) \right] - 2\epsilon$$

Note that by the Itô product rule,

$$\begin{aligned} & d\left(u(s, \tilde{X}_k^\epsilon(s))e^{Z_s}\right) \\ & = u_s d\left(e^{Z_s}\right) + e^{Z_s} du_s \\ & = \frac{\partial u}{\partial t}(t, x) + \mathcal{L}^h u(t, x) + \theta g(x, h) \end{aligned}$$

Since we assumed that

$$-\frac{\partial u}{\partial t}(t, x) - \mathcal{L}^h u(t, x) - \theta g(x, h) < 0$$

then

$$-\int_{t_k}^{\tau_k} d\left(u(s, \tilde{X}_k^\epsilon(s))e^{Z_s}\right) < 0$$

and therefore

$$\begin{aligned} \tilde{\Phi}(t_k, x_k) & \leq \mathbf{E}_{t_k, x_k} \left[\tilde{\Phi}(\tau_k, \tilde{X}_k^\epsilon(\tau_k))e^{Z(\tau_k)} - \int_{t_k}^{\tau_k} d\left(u(s, \tilde{X}_k^\epsilon(s))e^{Z_s}\right) \right] - 2\epsilon \\ & \leq -2\epsilon + \mathbf{E} \left[\exp \left\{ \theta \int_{t_k}^{\tau_k} g(X_s, h_k^\epsilon(s)) ds \right\} \tilde{\Phi}(\tau_k, \tilde{X}_k^\epsilon(\tau_k)) \right] \\ & \leq -2\epsilon + \tilde{I}(t_k, x_k, h_k^\epsilon) \\ & \leq \tilde{\Phi}(t_k, x_k) - \epsilon \end{aligned}$$

where the third inequality follows from the Dynamic Programming Principle and the last inequality follows from the definition of ϵ -optimal controls (see equation (50)).

Hence, equation (48),

$$\frac{\partial u}{\partial t}(t, x) + \inf_{h \in \mathcal{H}} \left\{ \mathcal{L}^h u(t, x) - \theta g(x, h) \right\} > 0$$

is false and we have shown that

$$\frac{\partial u}{\partial t}(t, x) + \inf_{h \in \mathcal{H}} \left\{ \mathcal{L}^h u(t, x) - \theta g(x, h) \right\} \leq 0$$

This argument therefore proves that $\tilde{\Phi}$ is a (discontinuous) viscosity supersolution of the PDE (27) on $[0, t) \times \mathbb{R}^n$ subject to terminal condition $\tilde{\Phi}(T, x) = e^{g_0(x; T)}$.

Step 4: Viscosity Solution

Since $\tilde{\Phi}$ is both a (discontinuous) viscosity subsolution and a supersolution of (27), it is a (discontinuous) viscosity.

Step 5: Conclusion

Since by assumption Φ is locally bounded, so is $\tilde{\Phi}$. In addition, $\varphi(x) = e^{-\theta x}$ is of class $C_1(\mathbb{R})$. Also we note that $\frac{d\varphi}{dx} < 0$. By the change of variable property (see for example Proposition 2.2 in Touzi [34]), we see that

1. since $\tilde{\Phi}$ is a (discontinuous) viscosity subsolution of (27), $\Phi = \varphi^{-1} \circ \tilde{\Phi}$ is a (discontinuous) viscosity supersolution of (23);
2. since $\tilde{\Phi}$ is a (discontinuous) viscosity supersolution of (27), $\Phi = \varphi^{-1} \circ \tilde{\Phi}$ is a (discontinuous) viscosity subsolution of (23).

and therefore Φ is a (discontinuous) viscosity solution of (23) on $[0, t) \times \mathbb{R}^n$ subject to terminal condition $\tilde{\Phi}(T, x) = e^{g_0(x; T)}$. $\square$

We also note the following corollary:

Corollary 5.1.

- (i) Φ^* is a upper semicontinuous viscosity subsolution, and;
- (ii) Φ_* is a lower semicontinuous viscosity supersolution of the RS HJB PIDE (23) on $[0, T] \times \mathbb{R}^n$, subject to terminal condition (25).

As a result of this corollary, we note that Φ^* , Φ_* and Φ are respectively a viscosity subsolution, supersolution, and solution in the sense of Definitions 5.4 and 5.5.

6. Comparison Result

Once we have characterized the class of viscosity solutions associated with a given problem, the next task is to prove that the problem actually admits a unique viscosity solution by establishing a comparison theorem. Comparison theorems are the cornerstone of the application of viscosity theory. Their main use is to prove uniqueness, and in our case continuity, of the viscosity solution. Although a set of, by now fairly standard, techniques can be applied in the proof, the comparison theorem *per se* is generally customized to address both the specificities of the PDE and the requirements of the general problem.

We face three main difficulties in establishing a comparison result for our risk-sensitive control problem. The first obstacle is the behaviour of the value function Φ at infinity. In the pure diffusion case or LEQR case solved by Kuroda and Nagai [31], the value function is quadratic in the state and is therefore not bounded for $x \in \mathbb{R}^n$. Consequently, there is no reason to expect the solution to the integro-differential RS HJB PIDE (23) to be bounded. The second hurdle is the presence of an extra non-linearity: the quadratic growth term $(D\Phi)' \Lambda \Lambda' D\Phi$. This extra non-linearity could, in particular, increase the complexity of the derivation of a comparison result for an unbounded value function. Before dealing with the asymptotic growth condition we will therefore need to address this non-linear term. The traditional solution, an exponential change of variable such as the one proposed by Duffie and Lions [22], is equivalent to the log transformation we used to derive the RS HJB PIDE and again to prove that the value function is a viscosity solution of the RS HJB PIDE. However, the drawback of this method is that, by creating a new zeroth order term equal to the solution multiplied by the cost function g , it imposes a severe restriction on g for the PDE to satisfy the monotonicity property required to talk about viscosity solutions. The final difficulty lies in the presence of the jump term and of the compensator ν . If we assume that the measure is finite, this can be addressed following the general argument proposed by Alvarez and Tourin [1] and Amadori [2].

To address these difficulties, we will need to adopt a slightly different strategy from the classical argument used to proof comparison results as set out in Crandall, Ishii and Lions [17]. In particular, we will exploit the properties of the exponentially transformed value function $\tilde{\Phi}$ resulting from Assumption 4.2 and alternate between the log transformed RS HJB PIDE and the quadratic growth RS HJB PIDE (23) through the proof.

Theorem 6.1. *Let $\tilde{u} = e^{-\theta v} \in USC([0, T] \times \mathbb{R}^n)$ be a bounded from above viscosity subsolution of (23) and $\tilde{v} = e^{-\theta u} \in LSC([0, T] \times \mathbb{R}^n)$ be a bounded from below viscosity supersolution of (23). If the measure ν is bounded and Assumption 4.2 holds then*

$$u \leq v \quad \text{on } [0, T] \times \mathbb{R}^n$$

Proof outline: This proof can be decomposed in seven steps. In the first step, we perform the usual exponential transformation to rewrite the problem for the value function Φ into a problem for the value function $\tilde{\Phi}$. The rest of the proof is done by contradiction. In step 2, we state the assumption we are planning to disprove. The properties of the value function $\tilde{\Phi}$ related to Assumption 4.2 are used in Step 3 to deduce that it is enough to prove the comparison result for Φ on a bounded state space to reach our conclusion. We then double variables in step 4 before finding moduli of continuity for the diffusion and the jump components respectively in steps 5 and 6. Finally, we reach a contradiction in step 7 and conclude the proof.

Step 1: Exponential Transformation

Let $u \in USC([0, T] \times \mathbb{R}^n)$ be a viscosity subsolution of (23) and $v \in LSC([0, T] \times \mathbb{R}^n)$ be a viscosity supersolution of (23). Define:

$$\tilde{u} := e^{-\theta v}$$

$$\tilde{v} := e^{-\theta u}$$

By the change of variable property (see for example Proposition 2.2 in Touzi [34]), $\tilde{u}$ and $\tilde{v}$ are respectively a viscosity subsolution and a viscosity supersolution of the RS HJB PIDE (27) for the exponentially transformed value function $\tilde{\Phi}$.

Thus, to prove that

$$u \leq v \quad \text{on } [0, T] \times \mathbb{R}^n$$

it is sufficient to prove that

$$\tilde{u} \leq \tilde{v} \quad \text{on } [0, T] \times \mathbb{R}^n$$

Step 2: Setting the Problem

As is usual in the derivation of comparison results, we argue by contradiction and assume that

$$\sup_{(t,x) \in [0,T] \times \mathbb{R}^n} [\tilde{u}(t, x) - \tilde{v}(t, x)] > 0 \tag{55}$$

Step 3: Taking the Behaviour of the Value Function into Consideration

The assertion of this theorem is that the comparison result holds in the class of functions satisfying Assumption 4.2. As a result Proposition 4.3 holds and we can concentrate our analysis on subsolutions and supersolutions sharing the

same growth properties as the exponentially transformed value function $\tilde{\Phi}$. By Propositions 4.3 and 4.2,

$$0 < \tilde{u}(t, x) \leq e^{\alpha^k(t) + \beta^{k'}(t)x} \quad \forall (t, x) \in [0, T] \times \mathbb{R}^n$$

$$0 < \tilde{v}(t, x) \leq e^{\alpha^k(t) + \beta^{k'}(t)x} \quad \forall (t, x) \in [0, T] \times \mathbb{R}^n$$

and

$$\lim_{|x| \rightarrow \infty} \tilde{u}(t, x) = \lim_{|x| \rightarrow \infty} \tilde{v}(t, x) = 0 \quad \forall t \in [0, T] \quad (56)$$

for $k = 1, \dots, 2n$ where α^k and β^k are the functions given in Assumption 4.2. Since (56) holds at an exponential rate, then by Assumption (55) there exists $R > 0$, such that

$$\sup_{(t,x) \in [0,T] \times \mathbb{R}^n} [\tilde{u}(t, x) - \tilde{v}(t, x)] = \sup_{(t,x) \in [0,T] \times \mathcal{B}_R} [\tilde{u}(t, x) - \tilde{v}(t, x)]$$

Hence, it is enough to show a contradiction with respect to the hypothesis

$$\sup_{(t,x) \in Q} [\tilde{u}(t, x) - \tilde{v}(t, x)] > 0 \quad (57)$$

established on the set $Q := [0, T] \times \mathcal{B}_R$. Before proceeding to the next step, we will restate assumption (57) now needs to be restated in terms of u and v as

$$\sup_{(t,x) \in Q} [u(t, x) - v(t, x)] > 0 \quad (58)$$

Step 4: Doubling of Variables on the Set Q

Let $\eta > 0$ be such that

$$N := \sup_{(t,x) \in Q} [u(t, x) - v(t, x) - \varphi(t)] > 0$$

where $\varphi(t) := \frac{\eta}{t}$.

We will now double variables, a technique commonly used in viscosity solutions literature (see e.g. Crandall, Ishii and Lions [17]). Consider a global maximum point $(t_\epsilon, x_\epsilon, y_\epsilon) \in (0, T] \times \mathcal{B}_R \times \mathcal{B}_R =: Q_d$ of

$$u(t, x) - v(t, y) - \varphi(t) - \epsilon|x - y|^2$$

and define

$$N_\epsilon := \sup_{(t,x,y) \in Q_d} [u(t, x) - v(t, y) - \varphi(t) - \epsilon|x - y|^2] > 0$$

Note that $N_\epsilon > 0$ for ϵ large enough. Moreover, $N_\epsilon \geq N$ and $N_\epsilon \downarrow 0$ as $\epsilon \rightarrow \infty$.

It is well established (see Lemma 3.1 and Proposition 3.7 in [17]) that along a subsequence

$$\lim_{\epsilon \rightarrow \infty} (t_\epsilon, x_\epsilon, y_\epsilon) = (\hat{t}, \hat{x}, \hat{y})$$

for some $(\hat{t}, \hat{x}) \in [0, T] \times \mathbb{R}^n$ which is a maximum point of

$$u(t, x) - v(t, x) - \varphi(t)$$

Via the same argument, we also have

$$\lim_{\epsilon \rightarrow \infty} \epsilon |x_\epsilon - y_\epsilon|^2 = 0$$

as well as

$$\lim_{\epsilon \rightarrow \infty} u(t_\epsilon, x_\epsilon) = u(\hat{t}, \hat{x})$$

and

$$\lim_{\epsilon \rightarrow \infty} v(t_\epsilon, x_\epsilon) = v(\hat{t}, \hat{x})$$

In addition, we note that

$$\lim_{\epsilon \rightarrow \infty} N_\epsilon = N$$

Applying Theorem 8.3 in Crandall, Ishii and Lions [17] at $(t_\epsilon, x_\epsilon, y_\epsilon)$, we see that there exists $a_\epsilon, b_\epsilon \in \mathbb{R}$ and $A_\epsilon, B_\epsilon \in \mathcal{S}_n$ such that

$$(a_\epsilon, \epsilon(x_\epsilon - y_\epsilon), A_\epsilon) \in \overline{\mathcal{P}}_u^{2,+}$$

$$(b_\epsilon, \epsilon(x_\epsilon - y_\epsilon), B_\epsilon) \in \overline{\mathcal{P}}_v^{2,-}$$

$$a_\epsilon - b_\epsilon = \varphi'(t_\epsilon)$$

and

$$-3\epsilon \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} \leq \begin{bmatrix} A_\epsilon & 0 \\ 0 & -B_\epsilon \end{bmatrix} \leq 3\epsilon \begin{bmatrix} I & -I \\ -I & I \end{bmatrix}$$

Thus, we have for the subsolution u

$$\begin{aligned} & -a_\epsilon + F(x_\epsilon, \epsilon(x_\epsilon - y_\epsilon), A_\epsilon) \\ & + \int_{\mathbb{Z}} \left\{ \frac{1}{\theta} \left(e^{-\theta(u(t_\epsilon, x_\epsilon + \xi(z)) - u(t_\epsilon, x_\epsilon))} - 1 \right) + \epsilon \xi'(z)(x_\epsilon - y_\epsilon) \right\} \nu(dz) \\ & \leq 0 \end{aligned}$$

and for the supersolution v ,

$$\begin{aligned} & -b_\epsilon + F(y_\epsilon, \epsilon(x_\epsilon - y_\epsilon), B_\epsilon) \\ & + \int_Z \left\{ \frac{1}{\theta} \left(e^{-\theta(v(t_\epsilon, y_\epsilon + \xi(z)) - v(t_\epsilon, y_\epsilon))} - 1 \right) + \epsilon \xi'(z)(x_\epsilon - y_\epsilon) \right\} v(dz) \\ & \geq 0 \end{aligned}$$

Subtracting these two inequalities,

$$\begin{aligned} -\varphi'(t_\epsilon) &= b_\epsilon - a_\epsilon \\ &\leq F(y_\epsilon, \epsilon(x_\epsilon - y_\epsilon), B_\epsilon) - F(x_\epsilon, \epsilon(x_\epsilon - y_\epsilon), A_\epsilon) \\ &\quad + \int_Z \left\{ \frac{1}{\theta} \left(e^{-\theta(v(t_\epsilon, y_\epsilon + \xi(z)) - v(t_\epsilon, y_\epsilon))} - 1 \right) + \epsilon \xi'(z)(x_\epsilon - y_\epsilon) \right\} v(dz) \\ &\quad - \int_Z \left\{ \frac{1}{\theta} \left(e^{-\theta(u(t_\epsilon, x_\epsilon + \xi(z)) - u(t_\epsilon, x_\epsilon))} - 1 \right) + \epsilon \xi'(z)(x_\epsilon - y_\epsilon) \right\} v(dz) \\ &= F(y_\epsilon, \epsilon(x_\epsilon - y_\epsilon), B_\epsilon) - F(x_\epsilon, \epsilon(x_\epsilon - y_\epsilon), A_\epsilon) \\ &\quad + \frac{1}{\theta} \int_Z \left\{ e^{-\theta(v(t_\epsilon, y_\epsilon + \xi(z)) - v(t_\epsilon, y_\epsilon))} \right\} v(dz) \\ &\quad - \frac{1}{\theta} \int_Z \left\{ e^{-\theta(u(t_\epsilon, x_\epsilon + \xi(z)) - u(t_\epsilon, x_\epsilon))} \right\} v(dz) \end{aligned} \tag{59}$$

Step 5: Modulus of Continuity

In this step, we focus on the (diffusion) operator F .

$$\begin{aligned} & F(y_\epsilon, \epsilon(x_\epsilon - y_\epsilon), B_\epsilon) - F(x_\epsilon, \epsilon(x_\epsilon - y_\epsilon), A_\epsilon) \\ &= \sup_{h \in \mathcal{J}} \left\{ \epsilon f(t_\epsilon, y_\epsilon, h)'(x_\epsilon - y_\epsilon) + \frac{1}{2} \text{tr}(\Lambda \Lambda' B_\epsilon) - \frac{\theta}{2} \epsilon^2 (x_\epsilon - y_\epsilon)' \Lambda \Lambda' (x_\epsilon - y_\epsilon) - g(y_\epsilon, h) \right\} \\ &\quad - \sup_{h \in \mathcal{J}} \left\{ \epsilon f(t_\epsilon, x_\epsilon, h)'(x_\epsilon - y_\epsilon) + \frac{1}{2} \text{tr}(\Lambda \Lambda' A_\epsilon + \delta I_n) \right. \\ &\quad \left. - \frac{\theta}{2} \epsilon^2 (x_\epsilon - y_\epsilon)' \Lambda \Lambda' (x_\epsilon - y_\epsilon) - g(x_\epsilon, h) \right\} \\ &\leq \frac{1}{2} |\text{tr}(\Lambda \Lambda' B_\epsilon - \Lambda \Lambda' A_\epsilon)| + \sup_{h \in \mathcal{J}} \{ \epsilon |f(t_\epsilon, y_\epsilon, h) - f(t_\epsilon, x_\epsilon, h)| |x_\epsilon - y_\epsilon| \} \\ &\quad + \sup_{h \in \mathcal{J}} \{ |g(x_\epsilon, h) - g(y_\epsilon, h)| \} \\ &\leq \frac{1}{2} |\text{tr}(\Lambda \Lambda' A_\epsilon - \Lambda \Lambda' B_\epsilon)| + \sup_{h \in \mathcal{J}} \{ \epsilon |f(t_\epsilon, y_\epsilon, h) - f(t_\epsilon, x_\epsilon, h)| |x_\epsilon - y_\epsilon| \} \\ &\quad + \sup_{h \in \mathcal{J}} \{ |g(x_\epsilon, h) - g(y_\epsilon, h)| \} \end{aligned}$$

Note that the functional f defined in (19) satisfies

$$|f(t_\epsilon, y_\epsilon, h) - f(t_\epsilon, x_\epsilon, h)| \leq C_f |y_\epsilon - x_\epsilon|$$

for some constant $C_f > 0$. In addition,

$$\begin{aligned} & \text{tr}(\Lambda \Lambda' A_\epsilon - \Lambda \Lambda' B_\epsilon) \\ &= \text{tr} \left(\begin{bmatrix} \Lambda \Lambda' & \Lambda \Lambda' \\ \Lambda \Lambda' & \Lambda \Lambda' \end{bmatrix} \begin{bmatrix} A_\epsilon & 0 \\ 0 & -B_\epsilon \end{bmatrix} \right) \\ &\leq 3\epsilon \text{tr} \left(\begin{bmatrix} \Lambda \Lambda' & \Lambda \Lambda' \\ \Lambda \Lambda' & \Lambda \Lambda' \end{bmatrix} \begin{bmatrix} I & -I \\ -I & I \end{bmatrix} \right) \\ &= 0 \end{aligned}$$

Finally, by definition of g ,

$$|g(y_\epsilon, h) - g(x_\epsilon, h)| \leq C_g |y_\epsilon - x_\epsilon|$$

for some constant $C_g > 0$. Combining these estimates, we get

$$\begin{aligned} & F(y_\epsilon, \epsilon(x_\epsilon - y_\epsilon), B_\epsilon) - F(x_\epsilon, \epsilon(x_\epsilon - y_\epsilon), A_\epsilon) \\ &\leq \omega(\epsilon |y_\epsilon - x_\epsilon|^2 + |y_\epsilon - x_\epsilon|) \end{aligned} \quad (60)$$

for a function $\omega(\zeta) = C\zeta$, with $C = \max[C_f, C_g]$. The function $\omega : [0, \infty) \rightarrow [0, \infty)$, which satisfies the condition $\omega(0^+) = 0$, is called a modulus of continuity.

Step 6: The Jump Term

We now consider the jump term

$$\begin{aligned} & \frac{1}{\theta} \int_{\mathbf{Z}} \left\{ e^{-\theta(v(t_\epsilon, y_\epsilon + \xi(z)) - v(t_\epsilon, y_\epsilon))} - e^{-\theta(u(t_\epsilon, x_\epsilon + \xi(z)) - u(t_\epsilon, x_\epsilon))} \right\} \nu(dz) \\ &= \frac{1}{\theta} \int_{\mathbf{Z}} \left\{ e^{-\theta(v(t_\epsilon, y_\epsilon + \xi(z)) - v(t_\epsilon, y_\epsilon))} - e^{-\theta(u(t_\epsilon, x_\epsilon + \xi(z)) - u(t_\epsilon, x_\epsilon) + v(t_\epsilon, x_\delta) - v(t_\epsilon, x_\delta))} \right\} \nu(dz) \end{aligned} \quad (61)$$

Since for $\epsilon > 0$ large enough, $u(t, x) - v(t, y) \geq 0$ then

$$\begin{aligned} & u(t_\epsilon, x_\epsilon + \xi(z)) - u(t_\epsilon, x_\epsilon) + v(t_\epsilon, y_\epsilon) - v(t_\epsilon, y_\epsilon + \xi(z)) \\ &\leq -(u(t_\epsilon, x_\epsilon) - v(t_\epsilon, y_\epsilon)) + N \end{aligned}$$

by definition of N . Moreover, since $N_\epsilon = \sup_{(t, x, y) \in Q_d} [u(t, x) - v(t, y) - \varphi(t) - \epsilon|x - y|^2] > 0$, then $N_\epsilon \leq u(t_\epsilon, x_\epsilon) - v(t_\epsilon, y_\epsilon)$ and therefore

$$u(t_\epsilon, x_\epsilon + \xi(z)) - u(t_\epsilon, x_\epsilon) + v(t_\epsilon, y_\epsilon) - v(t_\epsilon, y_\epsilon + \xi(z)) \leq N - N_\epsilon$$

for $z \in \mathbf{Z}$. Thus,

$$e^{-\theta(u(t_\epsilon, x_\epsilon + \xi(z)) - u(t_\epsilon, x_\epsilon) + v(t_\epsilon, y_\epsilon) - v(t_\epsilon, y_\epsilon))} \geq e^{-\theta(v(t_\epsilon, y_\epsilon + \xi(z)) - v(t_\epsilon, y_\epsilon) + N - N_\epsilon)}$$

and equation (61) can be bounded from above by:

$$\begin{aligned} & \frac{1}{\theta} \int_{\mathbf{Z}} \left\{ e^{-\theta(v(t_\epsilon, y_\epsilon + \xi(z)) - v(t_\epsilon, y_\epsilon))} - e^{-\theta(u(t_\epsilon, x_\epsilon + \xi(z)) - u(t_\epsilon, x_\epsilon) + v(t_\epsilon, y_\epsilon) - v(t_\epsilon, y_\epsilon))} \right\} \nu(dz) \\ & \leq \frac{1}{\theta} \int_{\mathbf{Z}} \left\{ e^{-\theta(v(t_\epsilon, y_\epsilon + \xi(z)) - v(t_\epsilon, y_\epsilon))} - e^{-\theta(v(t_\epsilon, y_\epsilon + \xi(z)) - v(t_\epsilon, y_\epsilon) + N - N_\epsilon)} \right\} \nu(dz) \\ & = \frac{1}{\theta} \int_{\mathbf{Z}} \left\{ e^{-\theta(v(t_\epsilon, y_\epsilon + \xi(z)) - v(t_\epsilon, y_\epsilon))} (1 - e^{-\theta(N - N_\epsilon)}) \right\} \nu(dz) \\ & = \frac{1}{\theta} \int_{\mathbf{Z}} \left\{ e^{-\theta(-\frac{1}{\theta} [\ln \tilde{v}(t_\epsilon, y_\epsilon + \xi(z)) - \ln \tilde{v}(t_\epsilon, y_\epsilon)])} (1 - e^{-\theta(N - N_\epsilon)}) \right\} \nu(dz) \\ & = \frac{1}{\theta} \int_{\mathbf{Z}} \left\{ \frac{\tilde{v}(t_\epsilon, y_\epsilon + \xi(z))}{\tilde{v}(t_\epsilon, y_\epsilon)} (1 - e^{-\theta(N - N_\epsilon)}) \right\} \nu(dz) \end{aligned} \quad (62)$$

By Proposition 4.2 and since $\tilde{v}$ is LSC, then $\exists \lambda > 0 : 0 < \lambda \leq \tilde{v}(t, x) \leq C_\Phi \forall (t, x) \in Q$. As a result,

$$\frac{\tilde{v}(t_\epsilon, y_\epsilon + \xi(z))}{\tilde{v}(t_\epsilon, y_\epsilon)} \leq K$$

for some constant $K > 0$. In addition, since the measure ν is assumed to be finite and the function $\zeta \mapsto e^\zeta$ is continuous, we can establish the following upper bound for the right-hand side of (62):

$$\begin{aligned} & \frac{1}{\theta} \int_{\mathbf{Z}} \left\{ \frac{\tilde{v}(t_\epsilon, y_\epsilon + \xi(z))}{\tilde{v}(t_\epsilon, y_\epsilon)} (1 - e^{-\theta(N - N_\epsilon)}) \right\} \nu(dz) \\ & \leq \frac{K}{\theta} \int_{\mathbf{Z}} \{1 - e^{-\theta(N - N_\epsilon)}\} \nu(dz) \\ & \leq \omega_R(N - N_\epsilon) \sup_{(t, y) \in [0, T] \times \mathbb{R}^n} \nu(\mathbf{Z}) \end{aligned} \quad (63)$$

for some modulus of continuity ω_R related to the function $\zeta \mapsto 1 - e^\zeta$ and parameterized by the radius $R > 0$ of the Ball $\mathcal{B}_R$ introduced in Step 3. Note that this parametrization is implicitly due to the dependence of N and N_ϵ on R . The term $\sup_{(t, y) \in [0, T] \times \mathbb{R}^n} \nu(\mathbf{Z})$ is the upper bound for the measure ν .

Step 7: Conclusion

We now substitute the upper bound obtained in inequalities (60) and (63) in (59) to obtain:

$$-\varphi'(t_\epsilon) \leq \omega(\epsilon |y_\epsilon - x_\epsilon|^2 + |y_\epsilon - x_\epsilon|) + \omega_R(N - N_\epsilon) \sup_{(t,x) \in [0,T] \times \mathbb{R}^n} \nu(\mathbf{Z}) \quad (64)$$

Taking the limit superior in inequality (64) as $\epsilon \rightarrow \infty$ and recalling that

1. the measure ν is finite;
2. $\xi_i(z), i = 1, \dots, m$ is bounded $\forall z \in \mathbf{Z}$ a.s. $d\nu$

we see that

$$\nu(\mathbf{Z}) < \infty$$

Then

$$\lim_{\epsilon \rightarrow 0} \omega_R(N - N_\epsilon) \nu(\mathbf{Z}) = 0$$

which leads to the contradiction

$$-\varphi'(t) = \frac{\eta}{t^2} \leq 0$$

We conclude from this that Assumption 58 is false and therefore

$$\sup_{(t,x) \in Q} [\nu(t, x) - u(t, x)] \geq 0 \quad (65)$$

Stated differently, we conclude that

$$u \leq \nu \quad \text{on } [0, T] \times \mathbb{R}^n \quad \square$$

6.1 Uniqueness

Uniqueness is a direct consequence of Theorem 6.1. Another important corollary is the fact that the (discontinuous) locally bounded viscosity solution Φ is in fact continuous on $[0, T] \times \mathbb{R}^n$.

Corollary 6.1. *The function $\Phi(t, x)$ defined on $[0, T] \times \mathbb{R}^n$ is the unique continuous viscosity solution of the RS HJB PIDE (23) subject to terminal condition (25).*

Proof. Uniqueness is a standard by-product of Theorem 6.1. Continuity can be proved as follows. By definition of the upper and lower semicontinuous envelopes,

recall that

$$\Phi_* \leq \Phi \leq \Phi^*$$

By Corollary 5.1, Φ_* and Φ^* respectively are semicontinuous supersolution and subsolution of the RS HJB PIDE (23) subject to terminal condition (25).

We note that as a consequence of Theorem 6.1 is that

$$\Phi_* \geq \Phi^*$$

and hence

$$\Phi_* = \Phi^*$$

is a continuous viscosity solution of the RS HJB PIDE (23) subject to terminal condition (25).

Hence, $\Phi = \Phi_* = \Phi^*$ and it is the unique continuous viscosity solution of the RS HJB PIDE (23) subject to terminal condition (25). $\square$

Now that we have proved uniqueness and continuity of the viscosity solution Φ to the RS HJB PIDE (23) subject to terminal condition (25), we can deduce that the RS HJB PIDE (27) subject to terminal condition (28) also has a unique continuous viscosity solution. We formalize the uniqueness and continuity of $\tilde{\Phi}$ in the following corollary:

Corollary 6.2. *The function $\tilde{\Phi}(t, x)$ defined on $[0, T] \times \mathbb{R}^n$ is the unique continuous viscosity solution of the RS HJB PIDE (27) subject to terminal condition (28).*

7. Conclusion

In this chapter, we considered a risk-sensitive asset management model with assets and factors modelled using affine jump-diffusion processes. This apparently simple setting conceals a number of difficulties, such as the unboundedness of the instantaneous reward function g and the high nonlinearity of the HJB PIDE, which make the existence of classical $C^{1,2}$ solution unlikely barring the introduction of significant assumptions. As a result, we considered a wider class of weak solutions, namely viscosity solutions. We proved that the value function of a class of risk sensitive control problems and established uniqueness by proving a non-standard comparison result. The viscosity approach has proved remarkably useful at solving difficult control problems for which the classical approach may fail. However, it is limited by the fact that it only provides continuity of the value function and by its focus on the PDE in relative isolation from the actual optimization problem. The question is where to go from there? A possible avenue of research would be to look for a method to establish smoothness of the value function, for example through a connection between viscosity solutions and classical solutions.

Achieving this objective may also require changes to the analytic setting in order to remove some of the difficulties inherent in manipulating unbounded functions.

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Small-Sample Estimation of Models of Portfolio Credit Risk*

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This paper explores the small sample properties of the most commonly used estimators of ratings-based portfolio credit models. We consider both method of moments and maximum likelihood estimators, and show that unrestricted estimators are subject to large biases in realistic sample sizes. We demonstrate large potential gains in precision and bias-reduction from imposing parametric restrictions across rating buckets. The restrictions we consider are based on economically meaningful hypotheses on the structure of systematic risk.

Keywords: Portfolio credit risk, maximum likelihood, method of moments, small sample bias.

1. Introduction

Models of portfolio credit risk have widespread application in bank risk-management, the credit rating of structured credit products, and the assessment of regulatory capital requirements. At the level of the individual position, credit risk depends most importantly on obligor default and rating migration probabilities. At the portfolio level, aggregate risk-measures (such as value-at-risk) depend also on the correlation (or, more generally, the *dependence*) across obligors in credit events. In practice and in academic work, the most widely used models are constructed as multi-firm generalizations of the structural model of Merton [20]. The return on firm asset value determines the outcome for the obligor at the model horizon. Dependence across obligors is generated through a factor struc-

*This paper is drawn from an earlier working paper by the title “Estimating Default Correlations from Short Panels of Credit Rating Performance Data,” dated January 2002. The opinions expressed here are those of the authors, and do not reflect the views of the Board of Governors or its staff.

ture in which the obligor asset return is modeled as a weighted sum of systematic and idiosyncratic risk factors.

Calibration of these models often draws upon historical ratings performance data. These panel datasets may provide performance data on large numbers of rated obligors, but in the time-series dimension they invariably span just a few decades at most. As shown by Gagliardini and Gouriéroux [7], large n in the cross-sectional dimension is not sufficient for consistency of the parameter estimates. Rather, it is large T in the time-series dimension that is needed. Thus, for the foreseeable future, large sample asymptotics may not be an adequate guide to the performance of the estimators on available data. Furthermore, even if the asymptotics were reliable and the estimators unbiased, parameter uncertainty matters. Value-at-risk is a non-linear function of the model parameters, so the estimated VaR under parameter uncertainty is biased [15, 25]. Heitfield [14] draws similar conclusions in the context of model-based rating of collateralized debt obligations.

This paper explores the small sample properties of the most commonly used estimators of ratings-based portfolio credit models. We consider both method of moments and maximum likelihood estimators. Our main purpose is to measure the potential gain in precision and bias-reduction from imposing parametric restrictions across rating buckets. The restrictions we consider are based on economically meaningful hypotheses on the nature of the rating system and the structure of systematic risk.

The literature on estimation of portfolio credit risk models has grown enormously over the last decade. Method of moment estimators were introduced to this literature by Gordy [9] and Nagpal and Bahar [21], and refined by Frey and McNeil [6]. Early applications included [13] and [3]. Gagliardini and Gouriéroux [7] extend the method to models of rating migration. Maximum likelihood estimation of these models was considered by Frey and McNeil [6], and has since been extended by Feng, Gouriéroux and Jasiak [5] to models with rating migration. Gagliardini and Gouriéroux [8] and Gouriéroux and Jasiak [11] develop approximate maximum likelihood approaches that exploit the large cross-sectional dimension to reduce the computational burden of the estimator. A promising new development has been the introduction by McNeil and Wendin [18, 19] of Bayesian MCMC estimators of portfolio credit models. These methods are flexible and powerful, though their computational requirements are non-trivial. For a recent application and extension of the Bayesian approach, see [24].

The portfolio credit model is presented in Section 2. We work within a two-state (default/no-default) setting, and so do not consider rating migrations of surviving obligors. In many cases, the ratings performance data include information on rating migrations as well as on default. In principle, transition data can and should be exploited to increase the precision of the estimators. We restrict ourselves to the two-state case partly for simplicity in exposition, but also for two

practical reasons. First, some datasets might not contain information on rating migrations. Default information is the “least common denominator” in the credit risk world. Second, estimation of a model of rating migration requires stronger assumptions on the nature and objectives of the rating process. The “through-the-cycle” rating philosophies of the leading rating agencies are open to varied interpretations, some of which may be difficult to formalize in a statistical model.¹

Section 3 shows how model parameters can be estimated from ratings performance data using the method of moments or maximum likelihood. The method of moments estimator has a closed-form solution, so it is especially convenient. The maximum likelihood estimators are somewhat more computationally demanding, but are also more efficient. Furthermore, the ML estimators lend themselves to imposing structural parameter restrictions.

Section 4 presents results for a Monte Carlo study of the small sample properties of three different maximum likelihood estimators as well as the method of moments estimator. We find that the method of moments and the least-restricted maximum likelihood estimator are subject to large biases in realistic sample sizes. The restricted maximum likelihood estimators offer large improvements in performance. In Section 5, we explain the source of the bias in the method of moments estimator. Implications are discussed in the Conclusion.

2. A Structural Default Model

We adopt a two-state version of the popular CreditMetrics model [12]. Assume we have a set of obligors, indexed by i . Associated with each obligor is a latent variable R_i which represents the normalized return on an obligor’s assets. R_i is given by

$$R_i = \mathbf{Z}\eta_i + \xi_i\varepsilon_i. \quad (1)$$

where $\mathbf{Z}$ is a K -vector of systematic risk factors. These factors capture unanticipated changes in economy-wide variables such as interest rates and commodity prices. We assume that $\mathbf{Z}$ is a mean-zero normal random vector with variance matrix Ω . We measure the sensitivity of obligor i to $\mathbf{Z}$ by a vector of *factor loadings*, η_i . Obligor-specific risk is represented by ε_i . Each ε_i is assumed to have a standard normal distribution and is independent across obligors and independent of $\mathbf{Z}$. Without loss of generality, the covariance matrix Ω is assumed to have ones on the main diagonal (so each Z_k has a standard normal marginal distribution), and the weights η_i and ξ_i are scaled so that R_i has a mean of zero and a variance of one. The obligor defaults if R_i falls below the default threshold γ_i . By construction, then, the unconditional probability of default (“PD”) of obligor i is equal to the standard normal CDF evaluated at γ_i .

¹Alternative interpretations of “through-the-cycle” can be found in [2], [26], [1], and [16, 17].

To allow the model to be calibrated using historical data of the sort available from the rating agencies, we group the obligors into G homogeneous “buckets” indexed by g . In most applications, the buckets comprise an ordered set of rating grades. In principle, however, a bucketing system can be defined along multiple dimensions. For example, a bucket might be composed of obligors of a given rating in a particular industry and country. Within a bucket, each obligor has the same default threshold γ_g so that the PD of any obligor in grade g is

$$\bar{p}_g = \Phi(\gamma_g), \quad (2)$$

where $\Phi(z)$ is the standard normal CDF.

The vector of factor loadings is assumed to be constant across all obligors in a bucket. so we can re-write the equation for R_i as

$$R_i = X_g w_g + \varepsilon_i \sqrt{1 - w_g^2}. \quad (3)$$

where

$$X_g = \frac{\sum_k Z_k \eta_{g,k}}{\sqrt{\eta_g' \Omega \eta_g}}$$

is a univariate bucket-specific common risk factor. By construction, each X_g is normally distributed with mean zero and unit variance. The G -vector $\mathbf{X} = (X_1, \dots, X_G)$ has a multivariate normal distribution. Let σ_{gh} denote the covariance between X_g and X_h . The factor loading on X_g for obligors in bucket g is

$$w_g = \sqrt{\eta_g' \Omega \eta_g},$$

which is bounded between zero and one. We eliminate ξ_i from equation (1) by imposing the scaling convention that the variance of R_i is one.

The advantage of writing R_i in terms of X_g and w_g rather than $\mathbf{Z}$ and η_g is that we then only need to keep track of one risk factor per bucket. We can think of X_g as summarizing the total effect of $\mathbf{Z}$ on obligors in bucket g , and w_g as describing the sensitivity of those obligors to the bucket-specific common risk factor. In the discussion that follows, the term *risk factors* should be taken to refer to X_g . The term *structural risk factors* will be used to identify the elements of $\mathbf{Z}$ because they reflect underlying economic variables. Likewise *factor loadings* will refer to w_g and *structural factor loadings* will refer to η_g .

In this model, dependence across obligors i and j is summarized by their *asset correlation*, which is the correlation between the latent variables R_i and R_j . If i and j are in buckets g and h , respectively, then the asset correlation is $\rho_{gh} = w_g w_h \sigma_{gh}$. For two distinct obligors in the same bucket g , we have $\rho_{gg} = w_g^2$. In the Gaussian framework of the standard structural model, the matrix of asset correlations is a complete characterization of the dependence structure. As observed by Embrechts, McNeil and Straumann [4], linear correlations need not be sufficient under more general distributional assumptions.

In some applications, there is interest in the correlation between default event indicators $1[Y_i < \gamma_i]$ and $1[Y_j < \gamma_j]$. For obligor i in bucket g and obligor j in bucket h , the *default correlation* is

$$C_{gh} = \frac{\Phi_2(\gamma_g, \gamma_h, \rho_{gh}) - \bar{p}_g \bar{p}_h}{\sqrt{\bar{p}_g(1 - \bar{p}_g)} \sqrt{\bar{p}_h(1 - \bar{p}_h)}} \quad (4)$$

where $\Phi_2(z_1, z_2, \rho)$ is the bivariate normal cdf for standard normal marginals and correlation ρ . The same formula holds in the special case where the two (distinct) obligors lie in the same bucket.

Given sufficient data, one can estimate all $G(G + 1)/2$ asset correlations. When data are scarce, however, many of these parameters may be unidentified or poorly identified. To reduce the number of parameters to be estimated, we impose *ex ante* restrictions on the factor loadings and risk factor variance matrix. The most commonly applied restriction is

Restriction $\mathcal{R}1$. One Risk Factor: $\sigma_{gh} = 1$ for all (g, h) bucket pairs.

$\mathcal{R}1$ is equivalent to requiring that $X_1 = X_2 = \dots = X_G$. A sufficient condition for $\mathcal{R}1$ is that there is exactly one structural risk factor (i.e., $K = 1$). As shown by Gordy [10], $\mathcal{R}1$ is necessary assumption in the model underpinnings for the Basel II internal ratings-based capital standard and, indeed, is unavoidable (implicitly if not explicitly) in any system of ratings-based capital charges. Empirically, $\mathcal{R}1$ may be an overly strong assumption, as casual observation suggests that industry and country business cycles are not perfectly synchronized. Nonetheless, if a portfolio is relatively homogeneous, or if sectoral distinctions among obligors cannot be observed from available data, a single-factor representation can serve as a reasonable approximation.

While $\mathcal{R}1$ imposes a restriction on the correlation among reduced form risk factors, it does nothing to restrict the sensitivity of each obligor's asset return to those factors. A different reduced form factor loading is associated with each bucket, and no restrictions are imposed on how these loadings vary. In practice it may be reasonable to assume that factor loadings vary smoothly with obligor default probabilities (or equivalently with obligor default thresholds). This assumption can be imposed by expressing factor loadings as a continuous function of default thresholds.

Restriction $\mathcal{R}2$. Smooth Factor Loadings: $w_g = \Lambda(\lambda(\gamma_g))$ for all g , where $\Lambda(\cdot)$ is a continuous, strictly monotonic link function that maps real numbers onto the interval $(-1, 1)$ and $\lambda(\cdot)$ is a continuous index function that maps default thresholds onto the real line.

The choice of the link function is rather arbitrary. In the analysis that follows

we use the simple arctangent transformation

$$\Lambda(\lambda) = \frac{2}{\pi} \arctan(\lambda).$$

This function is linear with unit slope in a neighborhood of $\lambda = 0$ and asymptotes smoothly toward positive (negative) one as λ approaches positive (negative) infinity. The specification of the index function is more important than the choice of the link because it can be used to restrict the way w varies with γ . If the index function is monotonic in γ , then mapping from γ to w will be monotonic as well. The more parsimonious is the index function, the more restrictive is the implied relationship between the default thresholds and the factor loadings.

The strongest restriction one can impose on the factor loadings is to assume that they are constant across all obligors.

Restriction $\mathcal{R}3$. Constant Factor Loading: $w_g = w_h$ for all (g, h) bucket pairs.

Together, $\mathcal{R}1$ and $\mathcal{R}3$ imply that the structural factor loadings are constant across buckets. Note that $\mathcal{R}3$ is a special case of $\mathcal{R}2$ in which the index function $\lambda(g)$ is a constant.

3. Moment and Maximum Likelihood Estimators

In this section, we develop method of moments and maximum likelihood estimators for the structural model. The estimation framework assumes that we have access to historical performance data for a credit ratings system. For each of T years and G rating buckets, we observe the number of obligors in bucket g at the beginning of year t (a “bucket-cohort”), and the number of members of the bucket-cohort who default by year-end. We assume that the default threshold γ_g and the factor loading w_g are constant across time for each bucket, and that the vector of risk factors $\{\mathbf{X}, \varepsilon\}$ is serially independent. The task at hand is to estimate γ_g and w_g for each rating bucket and (in the full-information MLE case) the variance matrix Σ . Given these parameter estimates we can recover PDs and default correlations using equations (2) and (4).

Let n_g and d_g denote the number of obligors and the number of defaults in bucket g . Throughout this paper, we take n_g as exogenous, and so can treat it as a fixed parameter in moment conditions and likelihood functions.² Conditional on X_g , defaults in bucket g are independent, and each default event can be viewed as the outcome of a Bernoulli trial with success probability

$$p_g(X_g) = p(X_g; \gamma_g, w_g) = \Phi \left(\frac{\gamma_g - w_g X_g}{\sqrt{1 - w_g^2}} \right). \quad (5)$$

²It is clear that the number of obligors in each bucket is stochastic. We assume the random process that generates the vector $\mathbf{n}$ is independent of the process that generates defaults.

Thus, the total number of defaults in the bucket is conditionally binomial with parameters n_g and $p_g(X_g)$.

From the factorial moment of the binomial distribution, we have

$$E[d_g(d_g - 1)|X_g] = n_g(n_g - 1)p_g(X_g)^2.$$

Taking expectations, we obtain the unconditional second factorial moment

$$E[d_g(d_g - 1)] = n_g(n_g - 1)E[p_g(X_g)^2] = n_g(n_g - 1)\Phi_2(\gamma_g, \gamma_g, w_g^2) \quad (6)$$

where the last equality follows from Proposition 1 in [9]. This leads to the simple method of moments estimator for bucket parameters γ_g, w_g . Let $Y_{g,1}$ and $Y_{g,2}$ be the sample moments

$$Y_{g,1} = \frac{1}{T} \sum_{t=1}^T \frac{d_{g,t}}{n_{g,t}}$$

$$Y_{g,2} = \frac{1}{T} \sum_{t=1}^T \frac{d_{g,t}}{n_{g,t}} \frac{(d_{g,t} - 1)}{(n_{g,t} - 1)}$$

From equation (2), we have the moment restriction

$$E[Y_{g,1}] = \bar{p}_g = \Phi(\gamma_g) \quad (7)$$

which implies the MM estimator $\hat{\gamma}_g = \Phi^{-1}(Y_{g,1})$. From equation (6), we have

$$E[Y_{g,2}] = \Phi_2(\gamma_g, \gamma_g, w_g^2). \quad (8)$$

The Frey and McNeil [6] MM estimator of w_g is the value $\hat{w}_g$ that satisfies

$$Y_{g,2} = \Phi_2(\hat{\gamma}_g, \hat{\gamma}_g, \hat{w}_g^2).$$

Note that the sign of w_g is not identified. Without loss of generality, we impose $w_g \geq 0$ for the MM estimator.

We now develop the full and restricted maximum likelihood estimators for the model. The conditional binomial distribution for d_g implies the likelihood function

$$L(\gamma_g, w_g | d_g, X_g) = \binom{n_g}{d_g} p(X_g; \gamma_g, w_g)^{d_g} (1 - p(X_g; \gamma_g, w_g))^{n_g - d_g}. \quad (9)$$

Since defaults are conditionally independent across buckets, the joint likelihood of the vector $\mathbf{d}$ conditional on $\mathbf{X}$ is simply the product of the G conditional likelihoods defined in (9). The unconditional likelihood for $\mathbf{d}$ is thus

$$L(\gamma, \mathbf{w}, \Sigma | \mathbf{d}) = \int \prod_{g=1}^G \left(\binom{n_g}{d_g} p(x_g; \gamma_g, w_g)^{d_g} (1 - p(x_g; \gamma_g, w_g))^{n_g - d_g} \right) dF(\mathbf{x}; \Sigma) \quad (10)$$

where $F(\mathbf{x}; \Sigma)$ is the multivariate normal CDF of $\mathbf{X}$.

In principle, we could maximize the product of (10) across T observations with respect to all $2G + (G - 1)G/2$ free parameters simultaneously. This would provide unrestricted full information maximum likelihood estimates of the parameters. In practice, however, this strategy is computationally feasible only when G is small. To reduce the dimensionality of the optimization problem, we can integrate X_g out of equation (9) to yield the marginal likelihood

$$L(\gamma_g, \mathbf{w}_g | d_g) = \int_{\Re} \binom{n_g}{d_g} p(x; \gamma_g, \mathbf{w}_g)^{d_g} (1 - p(x; \gamma_g, \mathbf{w}_g))^{n_g - d_g} d\Phi(x). \quad (11)$$

This function depends only on the two parameters $\mathbf{w}_g$ and γ_g , so estimates of $\mathbf{w}$ and γ can be obtained by maximizing the marginal likelihood for each bucket, one bucket at a time.³ This procedure yields our least restrictive maximum likelihood estimator that imposes no restrictions in the parameters of the default model described in Section 2. Because this estimator does not utilize information about the potential correlation in default rates across buckets, it is not asymptotically efficient, except in the unrealistic special case where $\sigma_{gh} = 0$ for all $g \neq h$. It also provides no estimate of the variance matrix Σ , which is needed to calculate value-at-risk. In practical application, Σ is sometimes obtained from other data sources. For example, in CreditMetrics, Σ is estimated by taking pairwise correlations in stock market indices [12].

$\mathcal{R}1$ implies that the effect of $\mathbf{X}$ on all obligors can be represented by a single standard normal *scalar* variable X . Under this restriction we can re-write (10) as

$$L(\gamma, \mathbf{w} | \mathbf{d}) = \int_{\Re} \prod_{g=1}^G \left(\binom{n_g}{d_g} p(x; \gamma_g, \mathbf{w}_g)^{d_g} (1 - p(x; \gamma_g, \mathbf{w}_g))^{n_g - d_g} \right) d\Phi(x). \quad (12)$$

Maximizing this likelihood over $\mathbf{w}$ and γ yields a full information likelihood estimator that imposes the one risk factor restriction.

Rather than estimate the elements of $\mathbf{w}$ directly one can substitute the formula in $\mathcal{R}2$ into equation (12) and maximize the resulting equation over γ and the parameters of the index function $\lambda(\gamma)$. This procedure yields a FIML estimator that imposes both the one risk factor and the smooth factor loading restrictions. Similarly, $\mathcal{R}1$ and $\mathcal{R}3$ can be imposed by replacing the vector $\mathbf{w}$ in equation (12) with a single loading $w \geq 0$ and maximizing the resulting likelihood with respect to γ and the scalar w .

If both $\mathcal{R}1$ and $\mathcal{R}3$ hold, then all the maximum likelihood estimators described in this section are consistent for $T \rightarrow \infty$. Furthermore, the estimator that imposes

³As was the case for the MM estimator, the sign of $\mathbf{w}_g$ is not identified by the marginal likelihood estimator.

$\mathcal{R}1$ and $\mathcal{R}3$ is efficient in the sense that it achieves the lowest possible asymptotic variance among consistent estimators. It is important to emphasize, however, that in finite samples some or all of these maximum likelihood estimators may be biased. In the next section we use Monte Carlo simulations to investigate the small sample properties of these estimators.

4. Monte Carlo Simulations

If many decades of ratings performance data were available, the asymptotic results of the previous section would pose a clear trade-off. On one hand, the more restrictive maximum likelihood estimators yield more precise estimates if the restrictions they impose are valid; on the other hand, the less restrictive estimators are more robust to specification errors. When ratings performance data are in short supply (i.e., T is small) the tradeoff becomes more complicated because the less restrictive estimators may also be most biased. We use Monte Carlo simulations to study the small sample biases in our estimators.

The following four estimators are examined in this analysis.

MM: unrestricted method of moments estimator.

MLE1: limited information maximum likelihood estimator.

MLE2: full information maximum likelihood estimator that imposes $\mathcal{R}1$.

MLE3: full information maximum likelihood estimator that imposes $\mathcal{R}1$ and $\mathcal{R}3$.

In each Monte Carlo simulation, we constructed a synthetic dataset intended to represent the type of historical data available from the major rating agencies. Data were simulated for three rating grades. Grade “A” corresponds to medium to low investment grade (S&P A/BBB), grade “B” corresponds to high speculative grade (S&P BB), and grade “C” corresponds to medium speculative grade (S&P B). Table 1 summarizes characteristics of these three grades.⁴ Simulated defaults in each grade were generated according to the stochastic model described in Section 2 with $\mathcal{R}1$ and $\mathcal{R}3$ imposed.

Two sets of Monte Carlo simulations were undertaken. In the first, 500 synthetic datasets were generated for four different values of T : 20, 40, 80, and 160. In each case a “true” factor loading of 0.45 was assumed. These simulations were intended to shed light on the properties of our estimators as the number of years of default data increases. Though estimates of both factor loadings and default thresholds were obtained for each simulated dataset, we will postpone discussing

⁴S&P grade-cohorts are somewhat larger than we have assumed, but are similar in the relative preponderance of higher grade obligors.

Table 1. Characteristics of simulated rating grades.

Grade	PD	Default Threshold	No. of Obligors
A	0.0015	-2.9677	400
B	0.0100	-2.3263	250
C	0.0500	-1.6449	100

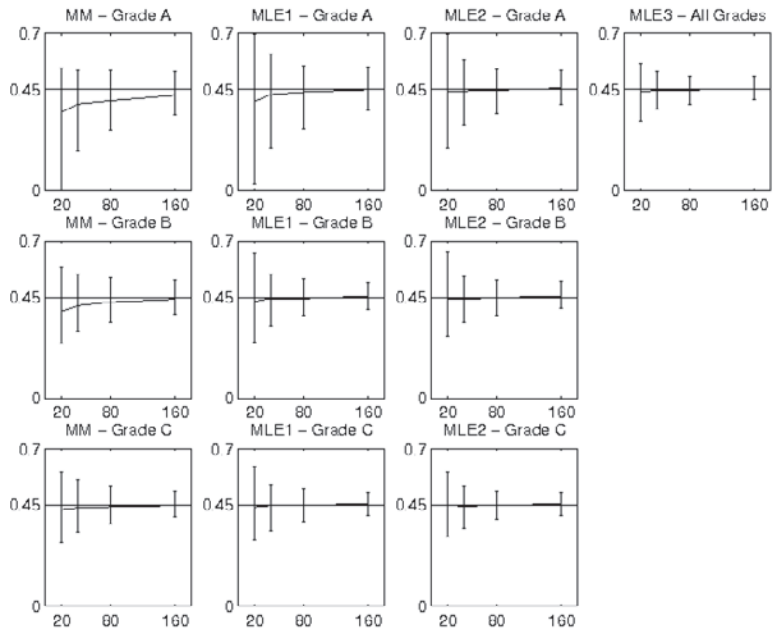


Figure 1. Median estimated factor loadings by sample size (error bars show 5th and 95th percentiles).

default thresholds for the time being. Table 2 summarizes the means, standard deviations, and root mean squared errors (“RMSE”) for the estimates of w given each of the four sample sizes. Figure 1 displays the median and the 5th and 95th percentiles of the estimated parameter values.

Not surprisingly, properties of all four estimators improve as T increases. The means become closer to 0.45 and the variances and RMSEs decrease. Also as expected, for large values of T the more restrictive estimators are more tightly clustered around 0.45 than the less restrictive estimators. More surprising is the rather poor performance of MM and MLE1 when T is small. Though all four estimators appear to be downward-biased in small samples, the bias of MM and

MLE1 is substantially worse than that of MLE2 and MLE3.

In real-world applications, we could never hope to observe 80 or 160 years of default data. S&P historical performance data currently cover 28 annual cohorts [28]. Moody's performance data go back to 1970, but there is believed to be an important break in the time-series at 1983 due to a change in Moody's rating methods. Banks' internal rating systems typically contain even shorter time-series, though larger grade-cohorts. For the vast majority of these internal systems, we would observe less than 20 years of data. To explore the small-sample properties of our estimators in greater detail, a second set of Monte Carlo simulations was run with T fixed at 20. Four groups of 1,500 synthetic datasets were simulated for a grid of "true" factor loadings from 0.15 to 0.60. For a small minority of trials, the simulated data did not permit identification of all model parameters. In other trials, the optimization routines used to calculate the maximum likelihood estimators failed to converge. In Appendix A, we provide details on the incidence and treatment of identification and convergence problems.

Tables 3 and 4 show the distributions of estimated default thresholds and implied default probabilities. Even when T is small, all four estimators generally produce minimally biased and reasonably precise estimates of default thresholds and, therefore, of the corresponding PDs. Although the direct estimator of the PD is unbiased, we favor estimation of default thresholds, because the distribution of $\hat{\gamma}$ is approximately symmetric. PDs, by contrast, are bounded at zero, so estimated PDs for the higher quality grades have highly asymmetric distributions. Therefore, standard test statistics should be better behaved for estimated default thresholds.

Tables 5(a) through 5(d) describe the distributions of estimated factor loadings. Several strong patterns can be seen in these tables, of which the most striking is the large downward bias associated with MM and MLE1. This problem is particularly significant for high quality grades when the true factor loadings are high. MLE2 and MLE3 are also biased downward, but the magnitude of the bias is less severe. In contrast to the results for MM and MLE1, the magnitude of the bias for MLE2 does not appear to depend on the grade in any systematic way.

Based on the root mean squared error criterion, MLE3 clearly outperforms the other three estimators; and more generally, the more restrictive estimators outperform the less restrictive estimators. The greatest gain in efficiency appears to occur when the single factor assumption ($\mathcal{R}1$) is imposed. Because it incorporates information on cross-grade default correlations, MLE2 produces substantially more accurate estimates of high-grade factor loadings than MLE1 or MM.

5. Bias in Method of Moments

Finite-sample bias in moment estimators arises when the moment restrictions are nonlinear functions of the parameters. In this section, we show why the MM estimator for factor loading w is subject to a large downward bias in realistic

Table 2. Distribution of estimated factor loadings by sample size for $w = 0.45$.

T		MM			MLE1			MLE2			MLE3
		A	B	C	A	B	C	A	B	C	All
20	Mean	0.3020	0.3816	0.4105	0.3748	0.4272	0.4390	0.4356	0.4389	0.4427	0.4374
	Std. Dev.	0.1504	0.0984	0.0923	0.1762	0.1053	0.0852	0.1319	0.0907	0.0773	0.0743
	RMSE	0.2110	0.1198	0.1004	0.1914	0.1077	0.0858	0.1325	0.0913	0.0775	0.0753
40	Mean	0.3660	0.4098	0.4293	0.4151	0.4429	0.4462	0.4418	0.4426	0.4444	0.4454
	Std. Dev.	0.1074	0.0768	0.0692	0.1269	0.0717	0.0609	0.0885	0.0610	0.0550	0.0529
	RMSE	0.1363	0.0867	0.0722	0.1315	0.0720	0.0610	0.0888	0.0614	0.0553	0.0530
80	Mean	0.3999	0.4272	0.4381	0.4329	0.4468	0.4499	0.4455	0.4481	0.4497	0.4486
	Std. Dev.	0.0799	0.0620	0.0503	0.0837	0.0512	0.0432	0.0619	0.0454	0.0401	0.0396
	RMSE	0.0943	0.0661	0.0516	0.0853	0.0513	0.0432	0.0620	0.0454	0.0400	0.0396
160	Mean	0.4208	0.4383	0.4444	0.4527	0.4536	0.4548	0.4573	0.4563	0.4571	0.4548
	Std. Dev.	0.0621	0.0483	0.0367	0.0595	0.0368	0.0320	0.0467	0.0369	0.0333	0.0311
	RMSE	0.0686	0.0497	0.0371	0.0595	0.0369	0.0324	0.0472	0.0374	0.0341	0.0315

Table 3. Distribution of estimated default thresholds by “true” factor loadings ($T = 20$).

w		MM			MLE1			MLE2			MLE3		
		A	B	C	A	B	C	A	B	C	A	B	C
0.15	Mean	-2.980	-2.333	-1.649	-2.982	-2.331	-1.647	-2.981	-2.331	-1.647	-2.982	-2.331	-1.647
	Std. Dev.	0.101	0.064	0.060	0.096	0.063	0.060	0.097	0.063	0.060	0.096	0.063	0.060
	RMSE	0.101	0.065	0.060	0.097	0.063	0.060	0.097	0.064	0.060	0.097	0.063	0.060
0.30	Mean	-2.995	-2.337	-1.656	-2.987	-2.334	-1.651	-2.985	-2.335	-1.653	-2.988	-2.334	-1.651
	Std. Dev.	0.134	0.092	0.084	0.125	0.094	0.088	0.125	0.093	0.088	0.124	0.094	0.088
	RMSE	0.136	0.093	0.085	0.126	0.094	0.088	0.126	0.094	0.088	0.126	0.094	0.088
0.45	Mean	-3.016	-2.352	-1.655	-3.008	-2.343	-1.657	-2.995	-2.345	-1.661	-3.005	-2.350	-1.664
	Std. Dev.	0.190	0.139	0.124	0.173	0.137	0.123	0.163	0.127	0.117	0.165	0.132	0.120
	RMSE	0.196	0.141	0.124	0.177	0.138	0.124	0.165	0.129	0.118	0.170	0.134	0.122
0.60	Mean	-3.088	-2.378	-1.670	-3.046	-2.360	-1.653	-3.009	-2.345	-1.652	-3.014	-2.360	-1.667
	Std. Dev.	0.311	0.214	0.176	0.213	0.186	0.155	0.181	0.155	0.133	0.185	0.159	0.137
	RMSE	0.334	0.220	0.177	0.227	0.189	0.155	0.186	0.156	0.133	0.191	0.163	0.139
True Value		-2.968	-2.326	-1.645	-2.968	-2.326	-1.645	-2.968	-2.326	-1.645	-2.968	-2.326	-1.645

settings. For clarity in exposition, we make a number of simplifying assumptions to reduce notation.

We fix a bucket with threshold γ and factor loading w . Assume that the cohort size n is constant across time. For now, let us assume that γ is known, so does not need to be estimated, and that we wish to estimate the asset correlation $\rho = w^2$. The MM estimator of ρ is the value $\hat{\rho}$ that satisfies

$$Y_2 = \Phi_2(\gamma, \gamma, \hat{\rho}).$$

To emphasize that this gives $\hat{\rho}$ as an implicit function, let us write $\bar{\Phi}_2(\hat{\rho}; \gamma)$ for

Table 4. Distribution of estimated default probabilities by “true” factor loadings (in percentage points).

w		MM			MLE1			MLE2			MLE3		
		A	B	C	A	B	C	A	B	C	A	B	C
0.15	Mean	0.151	0.996	4.987	0.149	1.000	5.008	0.150	0.999	5.008	0.149	1.000	5.008
	Std. Dev.	0.047	0.168	0.607	0.045	0.167	0.616	0.045	0.167	0.618	0.045	0.167	0.620
	RMSE	0.047	0.168	0.607	0.045	0.166	0.615	0.045	0.167	0.618	0.045	0.167	0.619
0.30	Mean	0.149	0.999	4.951	0.151	1.007	5.003	0.152	1.003	4.986	0.151	1.008	5.006
	Std. Dev.	0.059	0.243	0.855	0.059	0.252	0.909	0.061	0.248	0.908	0.059	0.249	0.911
	RMSE	0.059	0.243	0.856	0.059	0.252	0.909	0.061	0.248	0.908	0.059	0.249	0.911
0.45	Mean	0.149	0.991	5.025	0.152	1.014	5.006	0.155	1.000	4.953	0.151	0.990	4.923
	Std. Dev.	0.083	0.360	1.259	0.089	0.381	1.264	0.085	0.338	1.198	0.083	0.351	1.222
	RMSE	0.083	0.360	1.259	0.089	0.381	1.263	0.085	0.338	1.198	0.083	0.351	1.224
0.60	Mean	0.149	0.999	4.998	0.145	1.015	5.121	0.152	1.020	5.078	0.150	0.984	4.926
	Std. Dev.	0.142	0.565	1.769	0.108	0.525	1.646	0.085	0.399	1.372	0.085	0.387	1.369
	RMSE	0.142	0.565	1.769	0.108	0.525	1.650	0.085	0.400	1.374	0.085	0.387	1.370
True Value		0.150	1.000	5.000	0.150	1.000	5.000	0.150	1.000	5.000	0.150	1.000	5.000

the bivariate normal in the above equation. We denote Υ_γ as the inverse of this function, so that

$$\Upsilon_\gamma(\bar{\Phi}_2(\rho; \gamma)) = \rho. \quad (13)$$

The empirical moment Y_2 is a noisy but unbiased estimator of the quantity $y_2^* = \bar{\Phi}_2(\rho; \gamma)$ for the true parameter value ρ . As in [22], we take a Taylor series approximation for $\hat{\rho}$ as

$$\hat{\rho} = \Upsilon_\gamma(Y_2) \approx \Upsilon_\gamma(y_2^*) + (Y_2 - y_2^*)\Upsilon'_\gamma(y_2^*) + \frac{1}{2}(Y_2 - y_2^*)^2\Upsilon''_\gamma(y_2^*).$$

Taking expectations of both sides, and noting that $\Upsilon_\gamma(y_2^*) = \rho$, $E[Y_2 - y_2^*] = 0$ and $E[(Y_2 - y_2^*)^2]$ is the variance $V[Y_2]$, the bias is approximated as

$$E[\hat{\rho}] - \rho \approx \frac{1}{2}V[Y_2]\Upsilon''_\gamma(y_2^*). \quad (14)$$

By twice differentiating both sides of identity (13), we find

$$\Upsilon''_\gamma(y_2^*) = -\bar{\Phi}_2''(\rho; \gamma)/\bar{\Phi}_2'(\rho; \gamma)^3$$

As noted by Vasicek [27],

$$\bar{\Phi}_2'(\rho; \gamma) = \frac{\partial}{\partial \rho}\Phi_2(\gamma, \gamma, \rho) = \phi_2(\gamma, \gamma, \rho)$$

where ϕ_2 is the bivariate normal density. From this, it is straightforward to show that

$$\bar{\Phi}_2''(\rho; \gamma) = \frac{\partial}{\partial \rho}\phi_2(\gamma, \gamma, \rho) = \left(\left(\frac{\gamma}{1+\rho} \right)^2 + \frac{\rho}{1-\rho^2} \right) \phi_2(\gamma, \gamma, \rho).$$

Thus, we arrive at

$$\Upsilon''_{\gamma}(y_2^*) = - \left(\left(\frac{\gamma}{1+\rho} \right)^2 + \frac{\rho}{1-\rho^2} \right) \frac{1}{\phi_2(\gamma, \gamma, \rho)^2}. \quad (15)$$

In Appendix B, we derive the variance of Y_2 . Scaling by T , we have

$$\begin{aligned} T \cdot \text{V}[Y_2] &= \bar{\Phi}_4(\rho; \gamma) - \bar{\Phi}_2(\rho; \gamma)^2 \\ &\quad + \frac{1}{n(n-1)} (-2(2n-3)\bar{\Phi}_4(\rho; \gamma) \\ &\quad + 4(n-2)\bar{\Phi}_3(\rho; \gamma) + 2\bar{\Phi}_2(\rho; \gamma)) \end{aligned} \quad (16)$$

where $\bar{\Phi}_m$ is the m -variate normal cdf such that

$$\bar{\Phi}_m(\rho; \gamma) = \Pr(Z_1 \leq \gamma, \dots, Z_m \leq \gamma)$$

for Z_i that are standard normal variables with equal correlations $E[Z_i Z_j] = \rho$ for $i \neq j$. When $\rho = 0$, $\bar{\Phi}_4(\rho; \gamma) = \bar{\Phi}_2(\rho; \gamma)^2$, so that only the sampling variation term remains in the variance. In this case, the bias in $\hat{\rho}$ is $O(1/n)$. When $\rho > 0$, $\bar{\Phi}_4(\rho; \gamma) > \bar{\Phi}_2(\rho; \gamma)^2$, so the bias does not vanish even as the number of obligors increases to infinity.

A minor extension of these calculations gives us the bias for the factor loading w . The moment condition is $\hat{w} = \sqrt{\Upsilon_{\gamma}(Y_2)}$, so the Taylor series approximation to the bias is

$$E[\hat{w}] - w \approx \frac{1}{2} \text{V}[Y_2] \frac{d^2}{dy^2} \sqrt{\Upsilon_{\gamma}(y)} \Big|_{y=y_2^*} \quad (17)$$

Taking derivatives of $\sqrt{\Upsilon(y)}$,

$$\frac{d^2}{dy^2} \sqrt{\Upsilon(y)} = \frac{1}{2} \frac{\Upsilon''(y)}{\Upsilon(y)^{1/2}} - \frac{1}{4} \frac{\Upsilon'(y)^2}{\Upsilon(y)^{3/2}}$$

and substituting as before for $\Upsilon_{\gamma}(y_2^*)$ and its derivatives, we obtain

$$\frac{d^2}{dy^2} \sqrt{\Upsilon_{\gamma}(y)} \Big|_{y=y_2^*} = \frac{-1}{2w} \left(\left(\frac{\gamma}{1+w^2} \right)^2 + \frac{w^2}{1-w^4} + \frac{1}{2w^2} \right) \frac{1}{\phi_2(\gamma, \gamma, w^2)^2} \quad (18)$$

This expression is negative for $w > 0$, so it is clear that the bias in $\hat{w}$ must be towards zero.

Table 6 displays the approximate bias in the factor loading estimator as given by equation (17) for the three hypothetical buckets in Table 1. As in the previous section, we vary w from 0.15 to 0.60. The bias is expressed as a multiple of $1/T$ so, for example, if $T = 100$ and $w = 0.45$, then the approximate bias in $\hat{w}$ grade

Table 5(a). Distribution of estimated factor loadings for $w = 0.15$ and $T = 20$.

	MM			MLE1			MLE2			MLE3
	A	B	C	A	B	C	A	B	C	All
Mean	0.0956	0.1091	0.1201	0.1220	0.1180	0.1257	0.1643	0.1383	0.1412	0.1341
Std. Dev.	0.1090	0.0819	0.0765	0.1174	0.0813	0.0736	0.1032	0.0703	0.0638	0.0533
RMSE	0.1218	0.0915	0.0821	0.1206	0.0874	0.0775	0.1042	0.0712	0.0644	0.0556
Percentile										
2.5	0.0000	0.0000	0.0000	0.0013	0.0010	0.0019	0.0095	0.0099	0.0142	0.0111
5.0	0.0000	0.0000	0.0000	0.0021	0.0028	0.0042	0.0200	0.0194	0.0273	0.0318
50.0 (Med.)	0.0000	0.1170	0.1300	0.0955	0.1212	0.1314	0.1544	0.1379	0.1431	0.1387
95.0	0.2900	0.2384	0.2350	0.3390	0.2540	0.2430	0.3450	0.2570	0.2438	0.2173
97.5	0.3124	0.2590	0.2555	0.3814	0.2772	0.2622	0.3817	0.2793	0.2595	0.2305

Table 5(b). Distribution of estimated factor loadings for $w = 0.30$ and $T = 20$.

	MM			MLE1			MLE2			MLE3
	A	B	C	A	B	C	A	B	C	All
Mean	0.1960	0.2570	0.2718	0.2354	0.2723	0.2779	0.2898	0.2847	0.2850	0.2849
Std. Dev.	0.1372	0.0874	0.0776	0.1519	0.0863	0.0757	0.1173	0.0773	0.0707	0.0621
RMSE	0.1721	0.0973	0.0826	0.1650	0.0906	0.0788	0.1177	0.0788	0.0723	0.0639
Percentile										
2.5	0.0000	0.0531	0.1011	0.0040	0.0820	0.1179	0.0476	0.1307	0.1377	0.1605
5.0	0.0000	0.1171	0.1402	0.0081	0.1216	0.1498	0.0892	0.1529	0.1660	0.1793
50.0 (Med.)	0.2201	0.2583	0.2738	0.2393	0.2750	0.2813	0.2900	0.2841	0.2876	0.2871
95.0	0.4043	0.3994	0.3975	0.4784	0.4072	0.3968	0.4862	0.4115	0.3961	0.3858
97.5	0.4343	0.4297	0.4170	0.5429	0.4246	0.4199	0.5280	0.4333	0.4171	0.4018

Table 5(c). Distribution of estimated factor loadings for $w = 0.45$ and $T = 20$.

	MM			MLE1			MLE2			MLE3
	A	B	C	A	B	C	A	B	C	All
Mean	0.3020	0.3816	0.4105	0.3591	0.4209	0.4251	0.4289	0.4319	0.4278	0.4280
Std. Dev.	0.1504	0.0984	0.0923	0.1732	0.1022	0.0865	0.1255	0.0880	0.0796	0.0753
RMSE	0.2110	0.1198	0.1004	0.1955	0.1062	0.0900	0.1272	0.0898	0.0826	0.0784
Percentile										
2.5	0.0000	0.1948	0.2479	0.0119	0.2026	0.2553	0.1485	0.2475	0.2685	0.2813
5.0	0.0000	0.2302	0.2696	0.0238	0.2484	0.2838	0.2061	0.2816	0.2924	0.2986
50.0 (Med.)	0.3274	0.3788	0.4053	0.3849	0.4258	0.4277	0.4362	0.4354	0.4318	0.4309
95.0	0.5074	0.5463	0.5688	0.6184	0.5780	0.5598	0.6215	0.5677	0.5527	0.5495
97.5	0.5449	0.5782	0.6028	0.6493	0.6127	0.5788	0.6499	0.5968	0.5777	0.5702

Table 5(d). Distribution of estimated factor loadings for $w = 0.60$ and $T = 20$.

	MM			MLE1			MLE2			MLE3
	A	B	C	A	B	C	A	B	C	All
Mean	0.3675	0.4857	0.5388	0.4374	0.5517	0.5740	0.5384	0.5721	0.5767	0.5733
Std. Dev.	0.1802	0.1159	0.1106	0.2004	0.1095	0.0842	0.1193	0.0891	0.0749	0.0721
RMSE	0.2941	0.1628	0.1264	0.2580	0.1196	0.0881	0.1342	0.0933	0.0784	0.0768
Percentile										
2.5	0.0000	0.2718	0.3362	0.0107	0.3160	0.3930	0.2548	0.3755	0.4081	0.4004
5.0	0.0000	0.3078	0.3642	0.0248	0.3610	0.4267	0.3112	0.4122	0.4377	0.4368
50.0 (Med.)	0.4048	0.4776	0.5355	0.4769	0.5680	0.5747	0.5606	0.5843	0.5795	0.5822
95.0	0.6007	0.6862	0.7262	0.6803	0.7037	0.7020	0.6890	0.7008	0.6917	0.6742
97.5	0.6432	0.7372	0.7651	0.6970	0.7214	0.7294	0.7037	0.7244	0.7150	0.6895

Table 6. Bias (times T) in MM estimator for factor loading.

Grade	Bias			
	$w = 0.15$	$w = 0.30$	$w = 0.45$	$w = 0.60$
A	-6.95	-4.13	-10.59	-27.77
B	-2.83	-1.89	-3.50	-6.56
C	-3.21	-1.52	-1.77	-2.36

Table 7. Bias (times T) in MM estimator for default threshold.

Grade	γ	Bias			
		$w = 0.15$	$w = 0.30$	$w = 0.45$	$w = 0.60$
A	-2.968	-0.0013	-0.0021	-0.0047	-0.0131
B	-2.326	-0.0025	-0.0053	-0.0123	-0.0291
C	-1.645	-0.0057	-0.0123	-0.0258	-0.0508

B is -0.035 . As it derives from a Taylor series in powers of $1/\sqrt{T}$, the accuracy of the approximate bias may be poor for low values of T . For higher values of T , the results of Table 6 are comparable to the simulation results of Section 4.

Thus far, we have assumed that γ is known. If not, then the bias in $\hat{w}$ has components associated with the variance of Y_1 and the covariance between Y_1 and Y_2 , as well as the term (analyzed above) associated with the variance of Y_2 . The MM estimator $\hat{\gamma}$ is biased too. Arguments parallel to above show that

$$E[\hat{\gamma}] - \gamma \approx \frac{1}{2} V[Y_1] \frac{d^2}{dy^2} \Phi^{-1}(y) \Big|_{y=y_1^*} \quad (19)$$

Proceeding as before, we find the variance of Y_1 is given by

$$T \cdot V[Y_1] = \bar{\Phi}_2(\rho; \gamma) - \bar{p}^2 + \frac{1}{n} (\bar{p} - \bar{\Phi}_2(\rho; \gamma)). \quad (20)$$

For the second derivative of Φ^{-1} , we have

$$\frac{d^2}{dy^2} \Phi^{-1}(y) \Big|_{y=y_1^*} = \frac{-\Phi''(\gamma)}{\Phi'(\gamma)^3} = \frac{\gamma}{\phi(\gamma)^2}$$

The bias in $\hat{\gamma}$ is away from zero.

Table 7 displays the approximate bias in the default threshold estimator as given by equation (19) for the three hypothetical buckets in Table 1. As above, we vary w from 0.15 to 0.60, and express the bias as a multiple of $1/T$. We see that

the bias is negligible in realistic sample sizes. For example, with only ten years of data, the bias for grade B is -0.0012 when $w = 0.45$, so $E[\hat{\gamma}] = -2.3276$ against $\gamma = -2.3263$.

Conclusion

We have examined the small sample properties of method of moments and maximum likelihood estimators of portfolio credit risk models. We show that estimates of default thresholds are reasonably robust to the choice of estimators, but estimates of factor loadings (w) can differ markedly. The unrestricted estimators for w are subject to large bias towards zero and high mean square error in realistic sample sizes. The downward bias is most severe for higher quality grades.

The performance of the method of moments (MM) estimator for w is particularly dismal, as $E[\hat{w}]$ is roughly one-third less than the true value when we have $T = 20$ years of data. The virtue of the MM estimator, and indeed the main source of its relative popularity in practical application, is its tractability. The cost of this tractability is bias and inefficiency. In realistic sample sizes, the costs to MM clearly should outweigh the benefits. Work in progress will determinate whether we can improve the performance of moment estimators without too much sacrifice in computational facility. One possibility is to use cross-bucket moments as over-identifying information in generalized method of moments (GMM) estimation.

The three maximum likelihood estimators we study can be ordered by the restrictiveness of the assumptions they impose. The least restrictive (MLE1) allows for the possibility that obligors in different rating grades may be sensitive to different risk factors. The second (MLE2) imposes the restriction that obligors in all grades are sensitive to a *single* systematic risk factor, but allows factor loadings to vary across grades. Finally, the most restrictive (MLE3) requires that factor loadings be constant across rating grades.

If the restrictions imposed by the last estimator are correct, all three ML estimators are consistent. We find that all three estimators for w are downward biased in small sample, but the biases for MLE2 and MLE3 are much smaller than the bias for MLE1. The gap between MLE2 and MLE3 is relatively modest in terms of bias, though for higher quality grades MLE3 has a much smaller variance.

In applied work, an intermediate approach between MLE2 and MLE3 could be preferred. Such an estimator would allow for the possibility that highly-rated obligors have systematically higher or lower factor loadings than lower-rated obligors, while still capturing the benefits of imposing structure on the relationship between PDs and factor loadings. Instead of fixing a single common value for all factor loading as in MLE3, factor loadings would be expressed as a simple parametric function of the default threshold. This approach would permit greater flexibility in fitting data than MLE3, but afford greater efficiency than MLE2.

Finally, MLE3 or a blended version of MLE2 and MLE3 provides two practical advantages over the less restrictive estimators. First, by limiting the number of

parameters that must be estimated, cross-bucket restrictions on factor loadings go a long way toward solving identification problem that arise when the number of obligors in a bucket is small or when defaults are infrequent. When very few defaults are observed in a bucket, estimating all the parameters of the more general default models becomes difficult or impossible. Such circumstances may arise, for example, when buckets consist of a large number of narrowly-defined rating grades. Second, and perhaps more important, making factor loadings a (possibly constant) parametric function of default thresholds ensures that a bucket’s factor loading can be calculated directly from its PD. This provides a natural means for assigning factor loadings to bank rating grades that straddle or fall between rating agency grades.

Appendices

A. Identification and Convergence Problems

In the main Monte Carlo study four sets of 1,500 synthetic datasets were constructed with w set to 0.15, 0.30, 0.45, and 0.60. For some of these datasets, one or more of the estimators described in Section 3 failed to generate a full set of model parameters. The table below shows the fraction of simulations for which one or more parameters could not be estimated.

w	MM	MLE1	MLE2	MLE3
0.15	0.005	0.000	0.000	0.000
0.30	0.005	0.003	0.003	0.000
0.45	0.007	0.038	0.043	0.007
0.60	0.061	0.281	0.311	0.121

For grades where the PD implied by γ_g is small, a simulated dataset may contain a very small number of defaults. This outcome is particularly likely when w is large. When no defaults are observed in a bucket, the unrestricted model parameters (MM, MLE1) are not identified. When fewer than two defaults are observed, the MM asset correlation ($\hat{\rho}$) is negative. In this case, we impose a lower bound of $\hat{w} = 0$.

Even when model parameters are strictly identified by the data, the optimization algorithm used to obtain maximum likelihood estimators may fail to converge to a solution. Often such convergence problems arise when the matrix of second partial derivatives of the log-likelihood function (the Hessian matrix) is nearly singular. Rothenberg [23] shows that such singularity may result when model parameters are “nearly” unidentified. In general, highly correlated observations contain less information that is helpful in identifying model parameters than independent data. For this reason, it is perhaps not surprising that convergence problems are greater for higher values of w . Identification problems can be overcome by imposing parametric restrictions such as $\mathcal{R}3$. This helps explain why MLE3 is more likely to converge to a solution than MLE1 or MLE2.

B. Variance of the Second Factorial Moment Estimator

For the variance of Y_2 , we write

$$V[Y_2] = (n(n-1))^{-2} V[n(n-1)Y_2] = (n(n-1))^{-2} V[(1/T) \sum_t d_t(d_t-1)]. \quad (21)$$

As the d_t are identically and independently distributed across time,

$$V[(1/T) \sum_t d_t(d_t-1)] = \frac{1}{T} V[d_1(d_1-1)] = \frac{1}{T} (E[d_1^2(d_1-1)^2] - E[d_1(d_1-1)]^2).$$

We want to exploit the factorial moment rule

$$E[d_1(d_1-1) \cdots (d_1-j+1)] = n(n-1) \cdots (n-j+1) \bar{\Phi}_j(\rho; \gamma).$$

Straightforward algebra shows that

$$d^2(d-1)^2 = d(d-1)(d-2)(d-3) + 4d(d-1)(d-2) + 2d(d-1)$$

and from this we obtain

$$\begin{aligned} E[d_1^2(d_1-1)^2] &= n(n-1)(n-2)(n-3) \bar{\Phi}_4(\rho; \gamma) \\ &\quad + 4n(n-1)(n-2) \bar{\Phi}_3(\rho; \gamma) + 2n(n-1) \bar{\Phi}_2(\rho; \gamma). \end{aligned}$$

Substituting into equation (21), we arrive at

$$\begin{aligned} T \cdot V[Y_2] &= \frac{(n-2)(n-3)}{n(n-1)} \bar{\Phi}_4(\rho; \gamma) + \frac{4(n-2)}{n(n-1)} \bar{\Phi}_3(\rho; \gamma) \\ &\quad + \frac{2}{n(n-1)} \bar{\Phi}_2(\rho; \gamma) - \bar{\Phi}_2(\rho; \gamma)^2 \\ &= \bar{\Phi}_4(\rho; \gamma) - \bar{\Phi}_2(\rho; \gamma)^2 \\ &\quad + \frac{1}{n(n-1)} (-2(2n-3) \bar{\Phi}_4(\rho; \gamma) + 4(n-2) \bar{\Phi}_3(\rho; \gamma) + 2 \bar{\Phi}_2(\rho; \gamma)) \end{aligned}$$

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Heterogeneous Beliefs with Mortal Agents*

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This paper will examine a model with many agents, each of whom has a different belief about the dynamics of a risky asset. The agents are Bayesian and so learn about the asset over time. All agents are assumed to have a finite (but random) lifetime. When an agent dies, he passes his wealth (but not his knowledge) onto his heir. As a result, the agents never become sure of the dynamics of the risky asset. We derive expressions for the stock price and riskless rate. We then use numerical examples to exhibit their behaviour.

1. Introduction

This paper will look at a model of agents with heterogeneous beliefs. We assume that there is a single risky asset that produces a dividend process. Agents are unsure of the dynamics of the dividend process. Specifically, they do not know one of the parameters that governs its dynamics. Agents therefore form beliefs about this parameter and update these over time. To avoid agents eventually determining the true value of the parameter, we assume that agents are finite lived.

The paper will build on previous work of Brown & Rogers (2009). That paper explained the general theory of how to incorporate heterogeneous beliefs into a dynamic equilibrium model. However, in the case in which the agents were Bayesian, it was seen that the agents would eventually determine the true drift of the dividend process. The purpose of this paper is therefore to investigate a model in which there is a non-trivial steady state. This is done through the assumption that the different agents are in fact dynasties. Each member of the dynasty has a finite but random lifetime and when that member dies, he will pass on his wealth, but not his knowledge, to his heir. The paper will explain how to construct and solve this model and will lead to a stationary distribution for the stock price.

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As in Brown & Rogers (2009), we assume that there is a single risky asset which pays a dividend continuously in time. In addition there is a riskless asset in zero net supply. The dividend process of the stock is now assumed to be a quadratic function of an Ornstein-Uhlenbeck (OU) process. All the agents know all the parameters of the OU process except the mean to which it reverts. All the agents observe the OU process as it evolves and so as time progresses they update their beliefs about the unknown parameter. However, since they are finite lived, they will never find its true value.

The model described is quite simple, yet already there is enough to make the asset pricing non-trivial. Just as in Brown & Rogers (2009), the agents maximize their expected utilities subject to their budget constraints and we use these optimisation problems to derive a state price density. Using this state price density we can then price the risky asset as the net present value of future dividends. Comparative statics allow us to see how the stock price depends on the parameters of our model. We also produce a volatility surface for the stock, which behaves very reasonably.

The structure of the paper is as follows. We give a brief literature review below. Section 2 introduces the model and solves the equilibrium to determine a state price density. Section 3 then uses this state price density to calculate the prices of the stock and bond; these calculations are non-trivial. Section 4 looks at comparative statics of the model and Section 5 concludes.

1.1 Literature Review

There is a large literature on heterogeneous beliefs, which has been discussed in detail in Brown & Rogers (2009). Work includes Kurz (2008b), Kurz (1994), Kurz (1997), Kurz & Motolese (2006), Kurz (2008a), Kurz *et al.* (2005), Fan (2006), Harrison & Kreps (1978), Morris (1996), Wu & Guo (2003), Wu & Guo (2004), Harris & Raviv (1993), Kandel & Pearson (1995), Buraschi & Jiltsov (2006), Jouini & Napp (2007). Closer to the work presented here are the papers that assume that there is a parameter of the economy that is unknown to the agents. We briefly review such models here.

Basak (2000) considers a two-agent model in which each agent receives an endowment process. There is also an extraneous process that agents believe may effect the economy. The endowment process and all its parameters are observed. The extraneous process is observed, but the parameters of the stochastic differential equation (SDE) that drives it are not known to the agents. They form beliefs about the drift term in this SDE and update their beliefs in a Bayesian manner. The paper analyses this problem and derives quantities such as the consumption, the state price density and riskless rate. Basak also explains how to generalise the model to multiple agents and multiple extraneous processes.

Basak (2005) also considers a model with two agents, who each receive an endowment process. The aggregate endowment process is observed by the agents.

They also observe its volatility, but not its drift; they use filtering to determine this drift. There is assumed to be a bond and risky security, both in zero net supply. Again, agents do not know the drift of the stock price. Agents maximize the expected utility of consumption. He then solves for the equilibrium and uses it to derive interest rates and perceived market risk of the agents. He also gives a number of generalisations to the model. For example, he considers the case in which there is a process which does not directly affect the asset prices. However, each agent thinks that this process does affect the dynamics of the asset prices and so this changes the equilibrium. He also looks at the case of multiple agents and again derives the riskless rate and perceived market prices of risk. The final part of his paper looks at further extensions to his model; for example, he explores a monetary model in which there is a money supply that is stochastic and agents disagree on its drift.

Gallmeyer & Hollifield (2008) have considered the effects of adding a short-sale constraint to a model with heterogeneous beliefs. They consider a model with two agents. These agents are unsure about the drift of the output process of the economy. They start with initial beliefs about the drift and use filtering to update these. The agent who is initially more pessimistic is assumed to have logarithmic utility and a short sale constraint. The optimistic agent is assumed to have general CRRA utility and does not have a short sale constraint. The authors examine this model and derive expressions for the state price densities, stock price and consumption. In particular, they examine the effects of the imposition of the short sale constraint on the stock price.

The paper of Zapatero (1998) considers a model in which there is an aggregate endowment process that obeys an SDE driven by two independent Brownian motions. The constant drift of the process is unknown to the agents. There are 2 groups of agents and they each have a different Gaussian prior for this drift. Zapatero also considers the case in which as well as observing the endowment process, the agents also see a signal, which again is driven by the two Brownian motions, but has unknown drift. Again, agents have prior beliefs about this drift, which they update. He derives an equilibrium and shows that volatility of the interest rate is higher in an economy with the additional information source.

Li (2007) considers a model with 2 groups of agents. There is a dividend process which obeys some SDE, but the drift of this SDE is unknown. The drift can satisfy one of two different SDEs. Each group of agents attaches a different probability to the drift obeying the two different SDEs. They update this probability as they observe more data. Agents are assumed to have log utility and Li derives the stock price, wealth and consumption of agents in this model. He also analyses the volatility of the stock price.

Turning to the Bayesian learning side of our story, we remark that there is an extensive literature on Bayesian learning in finance and economics in which agents update their beliefs as they observe data. Work includes Hautsch & Hess

(2004), Kandel & Pearson (1995), Schinkel *et al.* (2002) and Kalai & Lehrer (1993), each of whom uses this Bayesian learning in quite different setups. For example, Schinkel *et al.* (2002) apply Bayesian learning to n competitive firms who set prices but do not know the demand function. They observe demand at each step and use this to update their posterior belief for the state of the world, which then impacts their perceived demand function. The authors show that prices converge. Kalai & Lehrer (1993) applies Bayesian learning to an n -person game in which agents do not know the payoff matrices of their competitors. They show that the equilibrium will approach the Nash equilibrium of the system. Hautsch & Hess (2004) apply Bayesian learning to explain why more precise data has a larger impact on market prices. They test this by looking at the behaviour of T-bond futures when unemployment data is announced.

Closer to our work, Guidolin & Timmermann (2001) look at a discrete time model in which the dividend process can have one of two different growth rates over each time period and the probability of each growth rate is unknown to the agents. The agents are learning, so they update their estimate for the unknown probability at each time step. In order to avoid the problem of agents discovering the true probability, they also consider agents who only look at a rolling window of data.

2. The Model

The setup of our model is similar to Brown & Rogers (2009). There is a single productive asset, which we refer to as the stock, which pays dividends continuously in time. The dividend at time t is δ_t . The dividend process is assumed to be a quadratic function of a stationary Ornstein Uhlenbeck (OU) process.

Since we are interested in obtaining a stationary distribution for the stock price, the construction of the probability space requires slightly more care than in Brown & Rogers (2009). Let Ω denote the sample space. We set $\Omega = C(\mathbb{R}, \mathbb{R})$, the space of continuous functions from $\mathbb{R}$ to $\mathbb{R}$. Let $X_t(\omega) \equiv \omega(t)$ denote the canonical process. Furthermore, let $\mathcal{F}_t = \sigma(X_s : -\infty \leq s \leq t)$.

As before, the reference measure is denoted by $\mathbb{P}_0$. We assume that under this measure X is a stationary OU process which reverts to mean zero and has reversion rate λ .¹

Next, we define:

$$W_t = X_t - X_0 + \int_0^t \lambda X_s ds \quad (1)$$

¹An Ornstein Uhlenbeck process which reverts to mean a' with reversion rate λ satisfies the SDE $dX_t = d\tilde{W}_t + \lambda(a' - X_t)dt$ where $\tilde{W}$ is a standard Brownian motion under the reference measure. While it is common to allow a non-unit volatility in the definition of the OU process, this can always be scaled to 1, and in view of the form (2) of the dividend process, this scaling can be absorbed into the constants a_0, a_1, a_2 . process which reverts to mean zero and has reversion rate λ .

for all $t \in \mathbb{R}$. Since X is an OU process, we observe that the process $(W_t)_{t \geq 0}$ is a standard Brownian motion².

2.1 The Dividend Process

We now define the dividend process by:

$$\delta_t = a_0 + a_1 X_t + a_2 X_t^2 \quad (2)$$

for some constants a_0, a_1, a_2 , where a_0 and a_2 are non-negative.

The simplest non-trivial setup is that in which $a_0 = a_2 = 0$, in which case the dividend process will simply be an OU process. However, choosing such values of a_0 and a_2 means that there is a positive probability that the dividend process will become negative, which is unrealistic. To overcome this problem, the constants can be chosen so that $a_0 \geq a_1^2/4a_2$, in which case the dividend process will always be non-negative. Furthermore, it will transpire that considering the case in which the dividend process is a quadratic function of X is no more difficult than the case in which δ is simply a scaling of X .³

2.2 The Agents

In our model there are N agents at all times. We assume that each person has a random lifetime. When this person dies, their wealth is immediately passed onto their (ignorant) child. Thus we are viewing each agent as a dynasty rather than a person⁴.

Formally, there exist times $(T_k^i)_{k \in \mathbb{Z}}$ which are the jump times of a stationary renewal process. At each of these times T_k^i , agent i will die and be replaced by his child. Thus, the wealth of the agent will be maintained, but their beliefs will not; the child will start with his own ignorant beliefs which will not depend on any historical data.

Turning now to the beliefs of the agents, first recall that, under the reference measure, $(X_t)_{t \in \mathbb{R}}$ is an OU process with zero mean. However, under the true measure, X will revert to level a , which will not necessarily be zero. The agents do not know this level. They will use Bayesian updating to deduce it.

²It will transpire that we are only interested in the increments of W ; thus it does not matter that W_0 is known before time 0.

³The case in which δ is a quadratic function of X is slightly more complicated, since two different values of X can give the same value of δ . Hence, $\sigma(X_s : t_0 \leq s \leq t) \neq \sigma(\delta_s : t_0 \leq s \leq t)$. Thus, we must assume that the agents observe the process X , rather than just observing the process δ .

⁴This idea of dynasties has been used by Nakata (2007), who considers an economy in which at any time point there are H young and H old agents. Each agent lives for 2 periods. Young agent $h \in \{1, \dots, H\}$ has the same preferences and beliefs as the old agent h . He then considers a Rational Beliefs Equilibrium as explained by Kurz. However, all agents in his model live for exactly two units of time, in contrast to our assumptions.

We need to determine the measure that each agent works under. First note that if we restrict to the time interval $[s, t]$, we may define a new measure by:

$$\frac{d\mathbb{P}_a}{d\mathbb{P}_0} = \exp\left(\lambda a(W_t - W_s) - \frac{1}{2}(\lambda a)^2(t - s)\right) \quad (3)$$

It follows from the Cameron-Martin-Girsanov theorem⁵ that a standard Brownian motion under $\mathbb{P}_0$ becomes a Brownian motion with drift λa under $\mathbb{P}_a$. Formally, $W_r = \bar{W}_r + \lambda ar$, for $s \leq r \leq t$ where $\bar{W}$ is a standard Brownian motion under $\mathbb{P}_a$. Thus,

$$dX_t = d\bar{W}_t + \lambda(a - X_t)dt$$

so we see that, under $\mathbb{P}_a$, X is an OU process which reverts to mean a .

Since agents do not know a , the beliefs of each agent simply consist of their distribution function for the parameter a . When a member of the i th dynasty is born, he gives λa a prior distribution⁶. We make the reasonable modelling assumption that this child's prior for the parameter $\alpha \equiv \lambda a$ is Normal with mean α_i and precision⁷ ε . Hence, all members of dynasty i begin life with the same prior precision ε . The agent then updates his prior according to his observation of $(X_s)_{t_k^i \leq s \leq t}$, where t_k^i denotes the time of birth of the current child and t is the current time.

If the agent knew the value of a , he would simply use a change of measure of the form (3). However, a is unknown, so the agent must weight each of the changes of measure according to his prior distribution for a . Hence at time t , agent i has posterior density

$$\begin{aligned} \pi_t^i(\alpha) &= \sqrt{\frac{\varepsilon}{2\pi}} \exp\left(-\frac{\varepsilon}{2}(\alpha - \alpha^i)^2 + \alpha(W_t - W_{t_k^i}) - \frac{1}{2}\alpha^2(t - t_k^i)\right) \\ &= \sqrt{\frac{\varepsilon}{2\pi}} \exp\left(-\frac{\varepsilon}{2}(\alpha - \alpha^i)^2 + \alpha\Delta W - \frac{1}{2}\alpha^2\Delta t\right), \end{aligned} \quad (4)$$

for α , where we use the abbreviations

$$\Delta t \equiv t - t_k^i \quad \Delta W \equiv W_t - W_{t_k^i}.$$

Notice that this posterior for α is of course Gaussian; when we maximize over α , we find the posterior mean to be

$$\hat{\alpha}_t = \frac{\Delta W + \varepsilon\alpha^i}{\varepsilon + \Delta t}, \quad (5)$$

⁵See Rogers & Williams (2000), IV.38 for an account.

⁶This is equivalent to having a prior distribution for a , since λ is known.

⁷Equivalently, the prior has variance ε^{-1} .

which summarizes the way that agent i learns from the observations.

Hence agent i 's law for the path has density with respect to the reference measure given by:

$$\begin{aligned}\Lambda_t^i &= \int_{-\infty}^{\infty} \pi_t^i(\alpha) d\alpha \\ &= \sqrt{\frac{\varepsilon}{\varepsilon + \Delta t}} \exp\left(\frac{(\Delta W)^2 + 2\alpha^i \varepsilon \Delta W - \varepsilon(\alpha^i)^2 \Delta t}{2(\varepsilon + \Delta t)}\right)\end{aligned}\quad (6)$$

2.3 Deriving the State Price Density

Associated with agent (or dynasty) i is a utility function, which we take to be CARA: $U_i(t, x) = -\frac{1}{\gamma_i} e^{-\gamma_i x} e^{-\rho t}$. Here, ρ is the discount factor, assumed to be the same for all agents. The agents seek to maximize the expected discounted utility of their consumption. Thus, agent i 's objective is:

$$\max \quad \mathbb{E}_0 \left[\int_{t_0}^{\infty} U_i(t, c_t^i) \Lambda_t^i dt \right] \quad (7)$$

where t_0 is some start value, which we will later allow to go to $-\infty$. Λ_t^i is the density derived in (6), which jumps at each of the times T_k^i .

The objectives of the agents have the same form as the previous Brown & Rogers (2009), so its theory can be used to derive a state price density. In particular, by looking at the price of an arbitrary contingent claim we can deduce that:

$$\zeta_s v_i = U_i'(s, c_s^i) \Lambda_s^i$$

where v_i is some $\mathcal{F}_{t_0}$ random variable⁸, and U_i' denotes the derivative of U_i with respect to its second argument. Recalling the CARA form of U_i and taking logs, we obtain:

$$\frac{\log \zeta_t}{\gamma_i} + \frac{\log v_i}{\gamma_i} = -\frac{\rho t}{\gamma_i} - c_t^i + \frac{\log \Lambda_t^i}{\gamma_i} \quad (8)$$

Summing (8) over i and using market clearing gives:

$$\log \zeta_t \frac{1}{N} \sum \frac{1}{\gamma_i} + \frac{1}{N} \sum \frac{\log v_i}{\gamma_i} = -\frac{1}{N} \sum \frac{\rho t}{\gamma_i} - \frac{\delta_t}{N} + \frac{1}{N} \sum \frac{\log \Lambda_t^i}{\gamma_i}$$

⁸We will shortly let t_0 tend to negative infinity and when this occurs, the $\mathcal{F}_{t_0}$ will be trivial, thus v_i will just be a constant.

2.4 A Continuum of Agents

Recall that there are N different agents in our model. We will now let N tend to infinity so that we can examine the case in which there is a continuum⁹ of agents. We assume that $\frac{1}{N} \sum \frac{1}{\gamma_i}$ has a finite limit and denote this limit by:

$$\Gamma^{-1} \equiv \lim_N \frac{1}{N} \sum \frac{1}{\gamma_i}$$

Abusing notation slightly, we use a_i to denote the $\lim_{N \rightarrow \infty} \frac{a_i}{N}$. Hence:

$$\log \zeta_t + G' = -\rho t - \Gamma(a_1 X_t + a_2 X_t^2) + \Gamma \lim_{N \rightarrow \infty} \sum \frac{1}{N \gamma_i} \log \Lambda_t^i \quad (9)$$

where G' is some $\mathcal{F}_{t_0}$ -measurable function. We now let t_0 tend to negative infinity; $\mathcal{F}_{t_0}$ then becomes trivial, so G' becomes a simple constant¹⁰.

Only the last term in (9) requires further development. Writing u^i for the time since the last person died in the i th dynasty, we obtain:

$$\begin{aligned} \Gamma \lim_{N \rightarrow \infty} \sum \frac{1}{N \gamma_i} \log \Lambda_t^i &= \Gamma \lim_{N \rightarrow \infty} \sum \frac{1}{N \gamma_i} \left[\frac{1}{2} \log \left(\frac{\varepsilon}{\varepsilon + u^i} \right) \right. \\ &\quad \left. + \left(\frac{(W_t - W_{t-u^i})^2 + 2\alpha^i \varepsilon (W_t - W_{t-u^i}) - \varepsilon (\alpha^i)^2 u^i}{2(\varepsilon + u^i)} \right) \right] \quad (10) \end{aligned}$$

We assume that the mean of α^i is given by $\langle \alpha \rangle$ and further that the distribution of u^i, α^i and γ_i are all independent¹¹. We further make the assumption that u has a density $\varphi(\cdot)$, given by:

$$\varphi(u) = A(\varepsilon + u)\lambda e^{-\lambda u} \quad (11)$$

where $A = \frac{\lambda}{1+\varepsilon\lambda}$ is chosen so that $\int_0^\infty \varphi(u) du = 1$. Since $\varphi(u)$ represents the probability of someone who is currently alive having age u , it follows that $\varphi(\cdot)$ must be decreasing. This gives the inequality $\lambda\varepsilon \geq 1$. The assumed form (11) of φ is restrictive; in particular, it confounds the effect of the mean reversion parameter λ and prior precision ε with the lifetimes of the individual members of the dynasties,

⁹Why do we not begin with a continuum of agents, then? We find the derivation of the state-price density and the evolution of beliefs easier to understand in the finite- N description, though it should be possible to derive these directly in a continuum model.

¹⁰We note that as $t_0 \rightarrow \infty$, the expression on the right of (9) is almost surely finite, so the left hand side must be as well. Since our ζ and $(v_i)_{1 \leq i \leq N}$ were only chosen up to a multiplicative constant, we may choose them to depend on t_0 in such a way that as $t_0 \rightarrow \infty$ both ζ and G' are a.s. finite.

¹¹The assumed independence of the α^i and γ^i is a substantive structural assumption made for tractability; that these are independent of the u^i is a consequence of the renewal process structure of the death times, and the fact that the renewal process — which has been running for infinite time — will have reached steady-state.

and this makes it impossible to give a clean interpretation of our later investigation of the effects of varying λ and ε . Nevertheless, we proceed with this assumption, as it would be difficult to make further progress without it.

Using our expression for φ , equation (10) becomes:

$$\begin{aligned} \log \zeta_t = & -G - \Gamma(a_1 X_t + a_2 X_t^2) - \rho t + \frac{1}{2} \int \frac{(W_t - W_{t-u})^2}{\varepsilon + u} \varphi(u) du \\ & + \langle \alpha \rangle \varepsilon \int \frac{(W_t - W_{t-u})}{\varepsilon + u} \varphi(u) du \end{aligned}$$

where G is some new constant. This then gives us:

$$\log \zeta_t = -G - \Gamma(a_1 X_t + a_2 X_t^2) - \rho t + \frac{A}{2} \eta_t + \langle \alpha \rangle \varepsilon A \xi_t$$

where

$$\begin{aligned} \xi_t &= \int_0^\infty (W_t - W_{t-u}) \lambda e^{-\lambda u} du \\ \eta_t &= \int_0^\infty (W_t - W_{t-u})^2 \lambda e^{-\lambda u} du \end{aligned}$$

By rearrangement and use of Fubini (see appendix), we are able to show that:

$$\begin{aligned} \xi_t &= X_t \\ \eta_t &= X_t^2 + e^{-\lambda t} \int_{-\infty}^t \lambda e^{\lambda s} X_s^2 ds \end{aligned}$$

Our final expression for the state price density is then given by:

$$\begin{aligned} \log \zeta_t = & -G - \Gamma(a_1 X_t + a_2 X_t^2) - \rho t \\ & + \frac{A}{2} [(X_t)^2 + e^{-\lambda t} \int_{-\infty}^t \lambda e^{\lambda s} X_s^2 ds] + \langle \alpha \rangle \varepsilon A X_t \end{aligned} \quad (12)$$

$$= -G + B X_t + C X_t^2 + U_t - \rho t \quad (13)$$

where:

$$B = \langle \alpha \rangle \varepsilon A - \Gamma a_1 \quad C = \frac{A}{2} - \Gamma a_2$$

and

$$U_t = \frac{1}{2} A e^{-\lambda t} \int_{-\infty}^t \lambda e^{\lambda s} X_s^2 ds$$

3. Asset Prices

3.1 The Interest Rate Process

We will use our state price density to derive the interest rate process. From Itô's formula, we have:

$$\begin{aligned} \frac{d\zeta_t}{\zeta_t} &= (B + 2CX_t)dW_t \\ &\quad + \left(C + \frac{\lambda A}{2}X_t^2 - \lambda U_t - \rho - B\lambda X_t - 2\lambda CX_t^2 + \frac{1}{2}(B + 2CX_t)^2\right)dt \\ &\doteq \left[(-\rho + C + \frac{1}{2}B^2) + (-\lambda B + 2CB)X_t + (-2\lambda C + \frac{\lambda A}{2} + 2C^2)X_t^2 - \lambda U_t\right]dt \end{aligned}$$

where the symbol $\doteq$ signifies that the two sides differ by a local martingale. The interest rate is equal to minus the coefficient of dt in the above expansion, hence:

$$r_t = r(X_t, U_t) \equiv \left(\rho - C - \frac{1}{2}B^2\right) + B(\lambda - 2C)X_t + \left(2\lambda C - \frac{\lambda A}{2} - 2C^2\right)X_t^2 + \lambda U_t \quad (14)$$

Thus, our model gives us an interest rate process of the form:

$$r_t = \alpha_0 + \alpha_1 X_t + \alpha_2 X_t^2 + \lambda U_t$$

for some constants $\alpha_i, i = 0, 1, 2$. Note that the interest rate process will depend on the behaviour of the dividend process in the past (via U_t) as well as on the current value of the dividend process. We therefore see that in some sense, high historical volatility generates high values of the riskless rate.

3.2 The Stock Price

We will now calculate the stock price. We have:

$$\begin{aligned} S_t &= \mathbb{E}_t^0 \left[\int_t^\infty \frac{\zeta_u \delta_u}{\zeta_t} du \right] \\ &= \frac{1}{\zeta_t} \int_t^\infty \mathbb{E}_t^0 [\zeta_u \delta_u] du \end{aligned} \quad (15)$$

3.2.1 A PDE for the stock price

From the form of ζ_t and the Markovian structure, we will have that:

$$\zeta_t S_t = \zeta_t h(X_t, U_t) \quad (16)$$

for some function h . This function will satisfy a PDE which we may determine by observing that $\zeta_t S_t + \int_0^t \zeta_s \delta_s ds$ is a martingale and applying Itô's formula. After a few calculations, we obtain the PDE:

$$0 = \frac{1}{2}h_{xx} + (B + (2C - \lambda)x)h_x + \lambda\left(\frac{A}{2}x^2 - u\right)h_u - r(x, u)h + (a_0 + a_1x + a_2x^2) \quad (17)$$

Unfortunately, it does not appear to be possible to solve this equation in closed form, so we will resort to another approach. However, before we do this, let us look at some of the consequences of (16) and (17). Suppose that under the real-world probability, $\mathbb{P}_*$, the OU process reverts to level a^* , then we have that:

$$dS_t = h_x dW_t^* + h_x \lambda (a^* - X_t) dt + h_u \left(\frac{\lambda A}{2} X_t^2 - \lambda U_t \right) dt + \frac{1}{2} h_{xx} dt$$

where W^* denotes a Brownian motion under measure $\mathbb{P}^*$. After using (17) we get that:

$$dS_t = h_x dW_t^* + h_x ((\lambda a^* - B) - 2CX_t) dt + r(X_t, U_t) h dt - (a_0 + a_1 X_t + a_2 X_t^2) dt$$

Hence, we see that the volatility and drift of the stock price are given by:

$$\Sigma_t = \frac{h_x(X_t, U_t)}{h(X_t, U_t)} \quad (18)$$

$$\mu_t^* = \frac{r(X_t, U_t) h(X_t, U_t) - (a_0 + a_1 X_t + a_2 X_t^2) + (\lambda a^* - 2CX_t - B) h_x(X_t, U_t)}{h(X_t, U_t)} \quad (19)$$

We shall use these expressions later.

3.2.2 Calculation of stock price via computation of conditional expectation

We will now proceed to determine the stock price via another method. Substituting the state price density from (13) into (15), we obtain:

$$S_t = \exp\{-BX_t - CX_t^2 - U_t + \rho t\} \\ \times \int_t^\infty \mathbb{E}_t^0[(a_0 + a_1 X_T + a_2 X_T^2) \exp\{BX_T + CX_T^2 + U_T - \rho T\}] dT$$

On first sight it may appear that it is very difficult to get any further with this expression. However, if we can calculate:

$$V^T(t, X_t; \theta) := \mathbb{E}_t^0[\exp\{\theta(a_0 + a_1 X_T + a_2 X_T^2) + BX_T + CX_T^2 + \int_t^T \frac{A}{2} \lambda e^{\lambda(s-T)} X_s^2 ds\}]$$

then we may differentiate with respect to θ to and set $\theta = 0$ to give:

$$S_t = \exp\{-BX_t - CX_t^2\} \int_t^\infty \exp\{(e^{-\lambda(T-t)} - 1)U_t - \rho(T-t)\} \frac{\partial}{\partial \theta} \Big|_{\theta=0} V^T(t, X_t; \theta) dT$$

We also define $\tau \equiv T - t$. We will show that:

$$V^T(t, X_t; \theta) = \exp\{\frac{1}{2}a(\tau)X_t^2 + b(\tau)X_t + c(\tau)\}$$

where a, b and c are functions which we will shortly deduce. To deduce these functions, we will use a martingale argument. For $t \leq T$ we define:

$$\begin{aligned} M_t^T &\equiv \mathbb{E}_t^0[\exp\{\theta(a_0 + a_1 X_T + a_2 X_T^2) + B X_T + C X_T^2 + \int_{-\infty}^T \frac{A}{2} \lambda e^{\lambda(s-T)} X_s^2 ds\}] \\ &= V^T(t, X_t; \theta) \exp\left\{\int_{-\infty}^t \frac{A}{2} \lambda e^{\lambda(s-T)} X_s^2 ds\right\} \end{aligned}$$

Now apply Itô's formula:

$$\begin{aligned} dM_t^T &= \exp\left\{\int_{-\infty}^t \frac{A}{2} \lambda e^{\lambda(s-T)} X_s^2 ds\right\} [V_t dt + V_x dX_t + \frac{1}{2} V_{xx} dX_t dX_t + \frac{\lambda A}{2} e^{\lambda(t-T)} X_t^2 V dt] \\ &= M_t^T \left[\frac{\lambda A}{2} e^{\lambda(t-T)} X_t^2 dt - \left(\frac{1}{2} \dot{a}(\tau) X_t^2 + \dot{b}(\tau) X_t + \dot{c}(\tau)\right) dt \right. \\ &\quad \left. + (a(\tau) X_t + b(\tau))(dW_t - \lambda X_t dt) + \frac{1}{2} (a(\tau) + (a(\tau) X_t + b(\tau))^2) dt \right] \end{aligned}$$

But $(M_t^T)_{t \leq T}$ is a martingale under $\mathbb{P}_0$, so the coefficient of dt in the above expression must be zero. Thus we obtain:

$$\begin{aligned} \frac{1}{2} \dot{a} &= \frac{\lambda A}{2} e^{-\lambda \tau} - \lambda a + \frac{1}{2} a^2 \\ \dot{b} &= ab - \lambda b \\ \dot{c} &= \frac{1}{2} (a + b^2) \end{aligned}$$

The boundary conditions are given by:

$$a(0) = 2(C + \theta a_2) \quad b(0) = B + \theta a_1 \quad c(0) = \theta a_0$$

3.2.3 Solving the ODEs

We now solve the ODEs. The first equation is a Riccati equation, so in order to solve we make the usual substitution:

$$a(\tau) = -\frac{\dot{g}(\tau)}{g(\tau)}$$

Substituting this into the ODE for a gives:

$$\frac{1}{2} \ddot{g} + \lambda \dot{g} + \frac{\lambda A}{2} e^{-\lambda \tau} g = 0$$

and the boundary condition becomes:

$$-\dot{g}(0) = 2(C + \theta a_2)g(0)$$

We can solve this equation using Maple to obtain:

$$g(u) = e^{-\lambda u} \left[(\sqrt{\lambda A} Y_1(2\sqrt{A/\lambda}) - 2(C + \theta a_2) Y_2(2\sqrt{A/\lambda})) J_2(2e^{-\lambda u/2} \sqrt{A/\lambda}) \right. \\ \left. - (\sqrt{\lambda A} J_1(2\sqrt{A/\lambda}) - 2(C + \theta a_2) J_2(2\sqrt{A/\lambda})) Y_2(2e^{-\lambda u/2} \sqrt{A/\lambda}) \right]$$

where J_i and Y_i are Bessel functions of order i of the first and second kind respectively. Turning now to the ODE for b , we may use our solution for a to deduce:

$$\dot{b} + \frac{\dot{g}}{g} b + \lambda b = 0$$

Rearranging gives:

$$\frac{d}{d\tau}(b g e^{\lambda \tau}) = 0$$

which we can solve subject to $b(0) = B + \theta a_1$ to give:

$$b(\tau) = \frac{(B + \theta a_1) g(0)}{e^{\lambda \tau} g(\tau)}$$

Finally, we obtain:

$$c(\tau) = \theta a_0 + \int_0^\tau \frac{1}{2} (a(\tau') + b(\tau')^2) d\tau'$$

Thus we have completely solved the ODEs. In order to calculate the stock price, we need to find $\frac{\partial V}{\partial \theta}$. We therefore need:

$$\frac{\partial g}{\partial \theta} = e^{-\lambda u} [-2a_2 Y_2(2\sqrt{A/\lambda}) J_2(2e^{-\lambda u/2} \sqrt{A/\lambda}) + 2a_2 J_2(2\sqrt{A/\lambda}) Y_2(2e^{-\lambda u/2} \sqrt{A/\lambda})]$$

and also:

$$\frac{\partial \dot{g}}{\partial \theta} = -\lambda \frac{\partial g}{\partial \theta} + e^{-\lambda u} [-2a_2 Y_2(2\sqrt{A/\lambda}) (\lambda J_2(2\sqrt{A/\lambda} e^{-\lambda u/2}) - \sqrt{A/\lambda} e^{-\lambda u/2} J_1(2\sqrt{A/\lambda} e^{-\lambda u/2})) \\ + 2a_2 J_2(2\sqrt{A/\lambda}) (\lambda Y_2(2\sqrt{A/\lambda} e^{-\lambda u/2}) - \sqrt{A/\lambda} e^{-\lambda u/2} Y_1(2\sqrt{A/\lambda} e^{-\lambda u/2}))]$$

We may then calculate expressions for $\frac{\partial V}{\partial \theta}$. First note that:

$$\frac{\partial V}{\partial \theta} = \left(\frac{1}{2} \frac{\partial a}{\partial \theta} X_t^2 + \frac{\partial b}{\partial \theta} X_t + \frac{\partial c}{\partial \theta} \right) \exp\left\{ \frac{1}{2} a(\tau) X_t^2 + b(\tau) X_t + c(\tau) \right\}$$

But:

$$\begin{aligned} \frac{\partial c}{\partial \theta}(\tau) &= a_0 + \int_0^\tau \frac{1}{2} \left(\frac{\partial a}{\partial \theta}(\tau') + 2b(\tau') \frac{\partial b}{\partial \theta}(\tau') \right) d\tau' \\ \frac{\partial b}{\partial \theta}(\tau) &= a_1 \frac{g(0)}{e^{\lambda \tau} g(\tau)} + \frac{(B + \theta a_1)}{e^{\lambda \tau}} \frac{\frac{\partial g}{\partial \theta}(0)}{g(\tau)} - \frac{(B + \theta a_1)}{e^{\lambda \tau}} \frac{g(0)}{g(\tau)^2} \frac{\partial g}{\partial \theta}(\tau) \\ \frac{\partial a}{\partial \theta}(\tau) &= -\frac{\frac{\partial \dot{g}}{\partial \theta}(\tau)}{g(\tau)} + \frac{g(\tau)}{g(\tau)^2} \frac{\partial g}{\partial \theta}(\tau) \end{aligned}$$

So finally we have:

$$S_t = \exp\{-BX_t - CX_t^2\} \int_0^\infty \exp\{-\rho\tau - (1 - e^{-\lambda\tau})U_t\} \\ \left(\frac{1}{2} \frac{\partial a}{\partial \theta} X_t^2 + \frac{\partial b}{\partial \theta} X_t + \frac{\partial c}{\partial \theta}\right) \exp\{\frac{1}{2}a(\tau)X_t^2 + b(\tau)X_t + c(\tau)\}d\tau \quad (20)$$

This is as far as we can get with the expression for the stock price. We see that the stock price depends not only on the dividend at time t , but also on U_t , a term reflecting the behaviour of $(X_s)_{-\infty \leq s \leq t}$. This is as we would expect, since agents need to use information from the whole of their lifetimes to make better estimates of the mean to which X is reverting. From properties of the OU process, we see that if X_t reverts to mean a then, since X is stationary, we have $X_t \sim N(a, \frac{1}{2\lambda})$. Hence,

$$\mathbb{E}U_t = \int_{-\infty}^t \frac{\lambda A}{2} e^{\lambda(r-t)} \left(\frac{1}{2\lambda} + a^2\right) dr = \frac{A}{2} \left(\frac{1}{2\lambda} + a^2\right)$$

This indicates a sensible value for U_t , which will be helpful for when we begin to look at numerical examples later on.

3.3 The Bond Price

The time- t price of a zero-coupon bond which has unit payoff at time T is given by:

$$\mathbb{E}^0\left[\frac{\zeta_T}{\zeta_t} \middle| \mathcal{F}_t\right] = \exp\left[-BX_t - CX_t^2 - U_t(1 - e^{-\lambda\tau}) - \rho\tau\right] V^T(t, X_t; \theta = 0)$$

Using our expression for $V^T(t, X_t; \theta = 0)$, we obtain:

$$\exp\left[\left(\frac{1}{2}a(\tau) - C\right)X_t^2 + (b(\tau) - B)X_t + c(\tau) - \rho\tau - (1 - e^{-\lambda\tau})U_t\right] \quad (21)$$

where the functions a, b and c are all evaluated using $\theta = 0$.

3.4 Remarks on the Case in which a is Known

Note that if we let $\varepsilon \rightarrow \infty$, then this corresponds to the case in which all the agents are certain that they know the value of a . By taking the limit in our expressions for the stock price, bond price and riskless rate, we can deduce expressions for these quantities in this limit. We note further that if the agents are sure about the value of a and this value corresponds to the true value, a^* , then the expressions we obtain will be the same as those for the model in which the true value of a was known to all the agents.

4. Numerical Results

The aim of this Section is to investigate how the stock price varies as the different parameters of the model are varied. We do not intend here to discuss the extent to which this model might fit actual prices; this would be an econometric study taking us some distance from the theoretical aims of this paper. However, we want to work with parameter values which are plausible, and choosing these requires some care.

We will restrict to the case in which $a_0 = a_1 = 0$, so that we have simply $\delta_t = a_2 X_t^2$. This ensures that the dividend process remains positive. Note further that the state price density (13) only depends on the *product* Γa_2 rather than the individual Γ and a_2 . Although the dividend process does depend on a_2 , changing a_2 simply corresponds to the changing the units in which we measure the dividend process. Hence, we may choose $a_2 = 1$.

Some of the parameters are relatively easy to choose, such as λ and ρ , for which we choose $\lambda = 2$ and $\rho = 0.04$; the impatience rate of the agents is 4%, so they have a mean time horizon of 25 years, reasonable for a human agent, and the mean reversion of the OU process for the dividend has a half-life of 6 months, again a plausible value. However, other parameters, such as Γ are much harder to determine. We are only interested in ensuring that the parameters are of the correct order. For this, we abbreviate $\langle \alpha \rangle = a$, and consider the thought experiment where $\varepsilon \rightarrow \infty$, which corresponds to the case in which agents are sure that they know the true value of a . This leaves the parameters a and Γ which we still need to determine.

One way to determine these parameters would be to choose them in order to match various moments from empirical data, such as the mean price-dividend ratio; this was the strategy employed in Brown & Rogers (2009) when we considered the equity premium puzzle. Ideally, we would use the same method here, but unfortunately our stock price is much more complicated. Thus, computing a given stock price requires the numerical computation of an integral. To work out the mean price dividend ratio, we would then need to compute a further integral as we averaged over the values of the driving Brownian motion. We would then vary the parameters and calculate the expected price dividend ratio each time in an attempt to find a realistic set of parameters. Given the additional complexity of this problem and the fact that we are only interested in determining parameters that are of the correct order, we will proceed in a different manner.

We first note that the interest rate process has a particularly simple form, which we can use to get a simple expression for the expected riskless rate. We can match this with the mean riskless rate from the Shiller data set.

Note that we are considering the case in which $a_0 = a_1 = 0, a_2 = 1$ and the limit as $\varepsilon \rightarrow \infty$ and hence $A\varepsilon \rightarrow 1, B \rightarrow \langle \alpha \rangle = a, C \rightarrow -\Gamma$. Substituting into expression (14) gives:

$$r = (\rho + \Gamma - \frac{1}{2}a^2) + a(\lambda + 2\Gamma)X_t - 2\Gamma(\lambda + \Gamma)X_t^2$$

Thus, the expected riskless rate is given by:

$$\mathbb{E}r = (\rho + \Gamma - \frac{1}{2}a^2) + a^2(\lambda + 2\Gamma) - 2\Gamma(\lambda + \Gamma)(a^2 + \frac{1}{2\lambda})$$

To determine Γ , we compare a CRRA agent (where we know a reasonable value for the constant of relative risk aversion¹²) with a CARA agent. If we consider a single agent model in which the value of a is known, the stock price will be given by:

$$S_0 = \mathbb{E} \int_0^\infty \frac{U'(\delta_t)}{U'(\delta_0)} \delta_t dt$$

Since we just want our parameters to be of the correct order, it is sufficient to check that the behaviour of

$$\frac{U'(\delta_t)}{U'(\delta_0)} \delta_t \quad (22)$$

when X is near to its mean value a is the same for both the CRRA and CARA case. If we set $X_0 = X_t = a$ then clearly (22) will be the same in both the CRRA and CARA case. We therefore impose the requirement that a small change in X_t from $X_t = a$ has the same effect in both cases, leading to the condition:

$$\frac{U''_{CRRA}(a^2)}{U'_{CRRA}(a^2)} = \frac{U''_{CARA}(a^2)}{U'_{CARA}(a^2)}$$

which leads us to the condition:

$$\Gamma = \frac{R}{a^2} \quad (23)$$

Since we know a sensible value for the coefficient of relative risk aversion R is $R = 2$, this equation gives us an equation from which we can determine Γ and a . Substituting in our expression for the riskless rate yields the cubic equation:

$$l(\Gamma) \equiv \frac{\Gamma^3}{\lambda} + 2R\Gamma^2 + (\mathbb{E}r - \rho + 2R(\lambda - 1))\Gamma + \frac{1}{2}R - R\lambda = 0$$

We will choose $R = 2$. We also choose $\mathbb{E}r = 0.01$, as given by the Shiller data set. We may then note that $l(0) < 0$ and $\frac{dl}{d\Gamma} > 0$ for $\Gamma > 0$, hence there is a unique positive solution to the above equation, which we can easily compute. Computation shows that the correct Γ to choose is $\Gamma = 0.49$ which we take as our default value. This gives $a = 2.01$.

¹²Ideally, we would have worked with CRRA agents throughout, but the combination of the individual agents' first-order conditions to specify the state-price density is intractable.

This concludes the thought experiment we used to find a reasonable value for Γ . We now use this with a more interesting value for ε which does not imply that agents know α with certainty. To summarise, the default parameters we choose are: $a_0 = a_1 = 0; a_2 = 1; \lambda = 2; \rho = 0.04; \varepsilon = 1.0; \Gamma = 0.49; \langle \alpha \rangle = a = 2.01$. We also choose $X_t = a, U_t = \frac{A}{2}(a^2 + \frac{1}{2\lambda})$. We then vary the parameters and examine the behaviour.

4.1 Comments on Results

Figure 1 shows that the stock price is decreasing in λ . Recall that λ is the parameter which tells us how quickly the dividend process returns to its mean. Hence, a lower value of λ means that the dividend process is more likely to reach high values, so is worth more to the agents. However, λ is also a parameter used in specifying the distribution of the lifetime of the agents. Increasing λ therefore decreases the expected lifetime of the agents. Each child in the dynasty therefore has less time to learn about the unknown parameter a and this increased uncertainty amongst the agents also means that the stock price decreases as λ increases.

Figure 2 shows that as ε increases, so does the stock price, which is to be expected since if the agents know more about the dividend process (i.e. their beliefs have a higher precision), the stock should be worth more to them. Once again, the effect of varying ε is confounded with the distribution of the agents' lifetimes.

Similarly, Figure 3 shows that the larger the value of ρ , the less the stock is worth. A large ρ indicates that the agents are impatient and want to consume their wealth in the near future, making the stock less attractive.

Figure 4 exhibits the dependence of the stock price on $\langle \alpha \rangle$. Recall that X_t and U_t are kept fixed as we vary $\langle \alpha \rangle$. A small $\langle \alpha \rangle$ indicates that the agents think the level to which X reverts is low. Thus, since we do not change X_t , a low value of $\langle \alpha \rangle$ relative to X indicates that X is currently abnormally high and so the dividends are abnormally high. Thus, the agents are keen to hold this stock. Furthermore, the relatively high level of X means that the agents have a large amount of dividend with which to buy the stock.

Figure 5 may at first seem surprising, since it shows that the stock price is increasing in the risk aversion, Γ . However, we recall that all agents have a CARA utility and furthermore, the parameters of our model are chosen so that the dividend process is non-negative. On the one hand, a larger value of Γ means that the value of the dividend process becoming larger are valued more highly than before. The downside of holding the stock is limited, since the dividend process is always non-negative. This explains the behaviour shown in Figure 5.

The volatility surface¹³ in Figure 6 shows that the volatility appears to be increasing in both X_t and U_t . This seems reasonable: if the dividend process has

¹³Note that the plot shows h_x/S_t ; the absolute value of this would give the volatility.

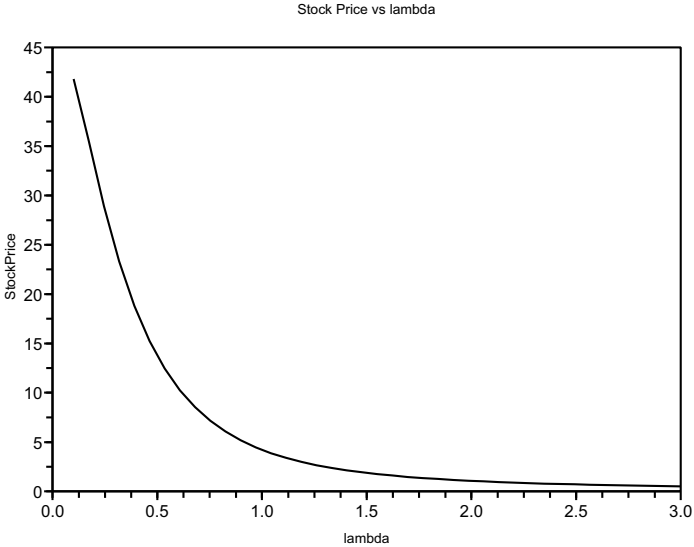


Figure 1. Graph of S_t against λ .

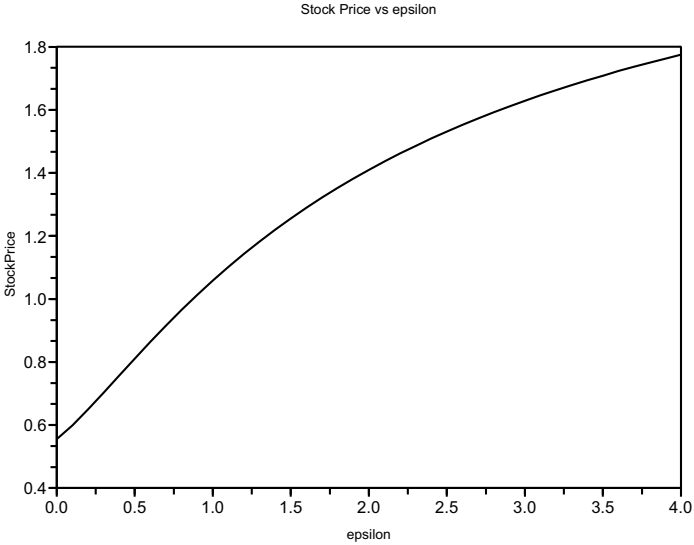


Figure 2. Graph of S_t against ϵ .

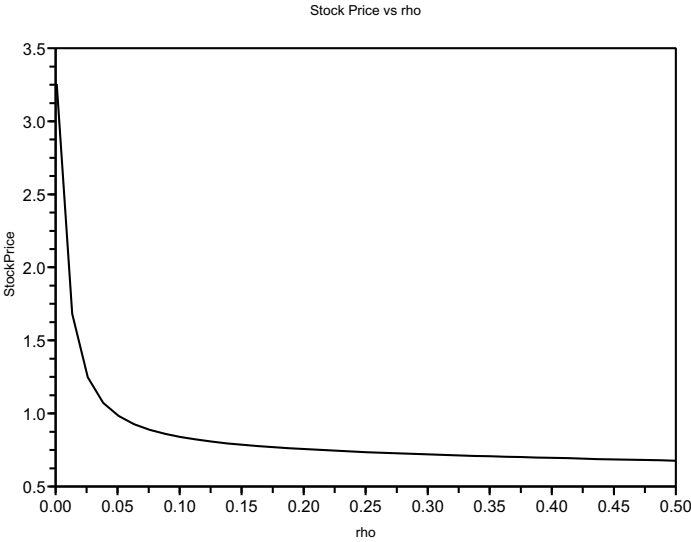


Figure 3. Graph of S_t against ρ .

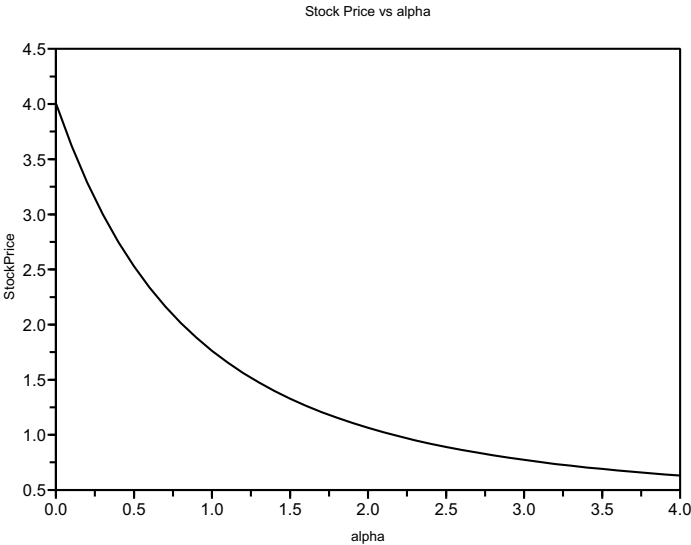


Figure 4. Graph of S_t against $\langle \alpha \rangle$.

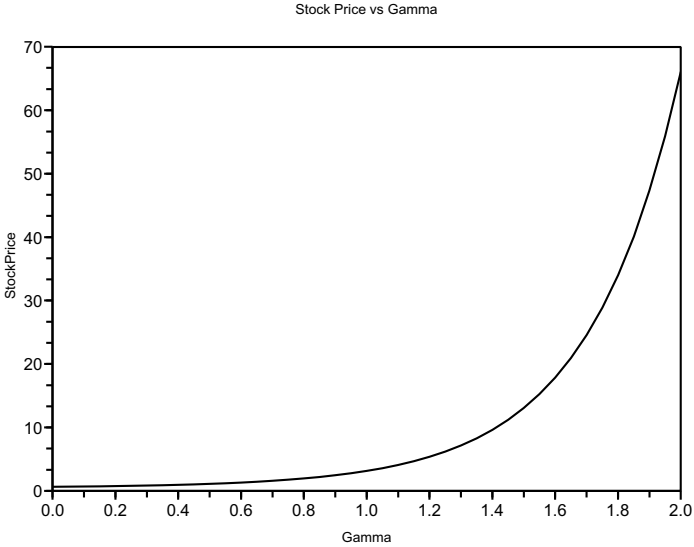


Figure 5. Graph of S_t against Γ .

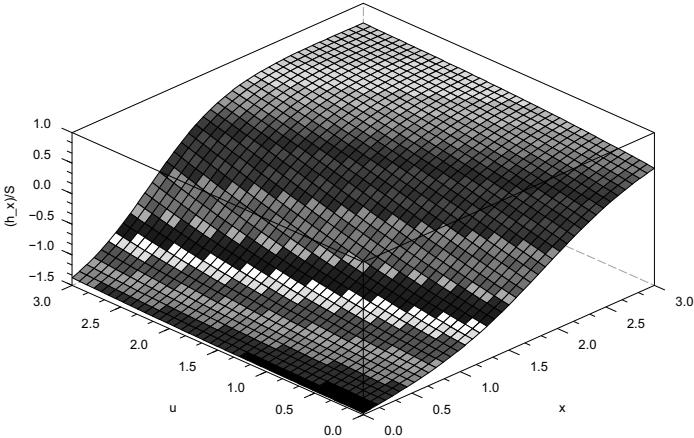


Figure 6. Volatility surface.

been varying greatly in the past, then U_t will be large, and in this case we would expect the stock to have a larger volatility.

5. Conclusions

We have introduced a new model in which the dividend of the stock obeys an OU process for which none of the agents know the mean. We derived a state price density and were able to use this to price the stock and a bond. We also were able to deduce an interest rate model. We produced graphs which illustrated the dependence of the stock price on the various parameters. The behaviour shown in these graphs seemed very reasonable. We also looked at how the parameter certainty case could be viewed as a special limit of the parameter uncertainty case.

Extensions to this work include using a different utility function for the agents; a CRRA utility would be a natural choice. In section 2.4 we also had to assume a quite specific form for the distribution of the lifetimes of the agents. An obvious improvement would be to consider the problem with a different distribution of lifetimes, in particular one that did not depend on the parameters of the dividend process. Unfortunately both these generalisations appear to make the calculations intractable.

Appendices. Stochastic Integrals

A.1. Calculating ξ_t

Recall that ξ_t is given by:

$$\xi_t = \int_0^\infty (W_t - W_{t-u})\lambda e^{-\lambda u} du$$

By change of variables,

$$\xi_t = W_t - e^{-\lambda t} \int_{-\infty}^t \lambda e^{\lambda s} W_s ds$$

So substituting from (1) gives:

$$\xi_t = W_t + X_0 - e^{-\lambda t} \left[\int_{-\infty}^t X_s \lambda e^{\lambda s} ds + \int_{-\infty}^t \lambda e^{\lambda s} \int_0^s \lambda X_r dr ds \right] \quad (24)$$

But the final term in the above expression is:

$$\begin{aligned} & -e^{-\lambda t} \int_{-\infty}^t \lambda e^{\lambda s} \int_0^s \lambda X_r dr ds \\ & = e^{-\lambda t} \int_{s=-\infty}^0 \int_{r=s}^0 \lambda e^{\lambda s} \lambda X_r dr ds - e^{-\lambda t} \int_{s=0}^t \int_{r=0}^s \lambda e^{\lambda s} \lambda X_r dr ds \end{aligned}$$

Applying Fubini, we obtain:

$$e^{-\lambda t} \int_{r=-\infty}^0 \int_{s=-\infty}^r \lambda e^{\lambda s} \lambda X_r ds dr - e^{-\lambda t} \int_{r=0}^t \int_{s=r}^t \lambda e^{\lambda s} \lambda X_r ds dr$$

Computing the integral with respect to s gives:

$$\begin{aligned} & e^{-\lambda t} \left[\int_{-\infty}^0 \lambda e^{\lambda r} X_r dr - e^{\lambda t} \int_0^t X_r \lambda dr + \int_0^t \lambda e^{\lambda r} X_r dr \right] \\ &= e^{\lambda t} \int_{-\infty}^t \lambda e^{\lambda r} X_r dr - \int_0^t \lambda X_r dr \end{aligned}$$

Substituting this into (24) gives:

$$\xi_t = W_t + X_0 - \int_0^t \lambda X_r dr$$

But recalling (1), we obtain:

$$\boxed{\xi_t = X_t}$$

A.2. Calculating η_t

Recall that η_t is given by:

$$\eta_t = \int_0^\infty (W_t - W_{t-u})^2 \lambda e^{-\lambda u} du$$

Changing variables we obtain:

$$\eta_t = e^{-\lambda t} \int_{-\infty}^t (W_t - W_r)^2 \lambda e^{\lambda r} dr$$

Substituting from (1) gives:

$$\begin{aligned} \eta_t &= e^{-\lambda t} \int_{-\infty}^t [(X_t - X_r) + \int_r^t \lambda X_s ds]^2 \lambda e^{\lambda r} dr \\ &= e^{-\lambda t} \int_{-\infty}^t (X_t - X_r)^2 \lambda e^{\lambda r} dr + 2e^{-\lambda t} \int_{-\infty}^t (X_t - X_r) \left(\int_r^t \lambda X_s ds \right) \lambda e^{\lambda r} dr \\ &\quad + e^{-\lambda t} \int_{-\infty}^t \left(\int_r^t \lambda X_s ds \right)^2 \lambda e^{\lambda r} dr \end{aligned} \tag{25}$$

We will now apply Fubini to two of these terms to deduce an expression for η_t . Firstly, we work on:

$$\int_{r=-\infty}^t X_t \int_{s=r}^t \lambda X_s ds \lambda e^{\lambda r} dr$$

By applying Fubini, we obtain:

$$\begin{aligned} & \int_{s=-\infty}^t X_t X_s \int_{r=-\infty}^s \lambda^2 e^{\lambda r} dr ds \\ &= \int_{-\infty}^t X_t X_s \lambda e^{\lambda s} ds \end{aligned}$$

Putting this into (25) gives:

$$\begin{aligned} \eta_t &= X_t^2 + e^{-\lambda t} \int_{-\infty}^t \lambda e^{\lambda r} X_r^2 dr \\ &\quad - 2e^{-\lambda t} \int_{-\infty}^t X_r \left(\int_r^t \lambda X_s ds \right) \lambda e^{\lambda r} dr \\ &\quad + e^{-\lambda t} \int_{-\infty}^t \int_r^t \lambda X_s ds \int_r^t \lambda X_v dv \lambda e^{\lambda r} dr \end{aligned} \quad (26)$$

The final term is:

$$2e^{-\lambda t} \int_{r=-\infty}^t \int_{s=r}^t \int_{v=s}^t \lambda X_s \lambda X_v \lambda e^{\lambda r} dv ds dr$$

where we have halved the area of integration in the $dvds$ integral. Applying Fubini yields:

$$\begin{aligned} & 2e^{-\lambda t} \int_{s=-\infty}^t \int_{v=s}^t \int_{r=-\infty}^s \lambda X_s \lambda X_v \lambda e^{\lambda r} dr dv ds \\ &= 2e^{-\lambda t} \int_{s=-\infty}^t \int_{v=s}^t \lambda X_s \lambda X_v e^{\lambda s} dv ds \\ &= 2e^{-\lambda t} \int_{r=-\infty}^t \lambda X_r e^{\lambda r} \int_{s=r}^t \lambda X_s ds dr \end{aligned}$$

Substituting this into (26) gives:

$$\boxed{\eta_t = X_t^2 + e^{-\lambda t} \int_{-\infty}^t \lambda e^{\lambda s} X_s^2 ds}$$

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Counterparty Risk on a CDS in a Markov Chain Copula Model with Joint Defaults*

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In this paper we study the counterparty risk on a payer CDS in a Markov chain model of two reference credits, the firm underlying the CDS and the protection seller in the CDS. We first state few preliminary results about pricing and CVA of a CDS with counterparty risk in a general set-up. We then introduce a Markov chain copula model in which wrong way risk is represented by the possibility of joint defaults between the counterpart and the firm underlying the CDS. In the set-up thus specified we derive semi-explicit formulas for most quantities of interest with regard to CDS counterparty risk such as price, CVA, EPE or hedging strategies. Model calibration is made simple by the copula property of the model. Numerical results show adequacy of the behavior of EPE and CVA in the model with stylized features.

Keywords: Counterparty credit risk, CDS, wrong way risk, CVA, EPE.

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[†]See <http://www.cris-creditrisk.com>.

1. Introduction

Since the sub-prime crisis, counterparty risk is a crucial issue in connection with valuation and risk management of credit derivatives. Counterparty risk in general is ‘the risk that a party to an OTC derivative contract may fail to perform on its contractual obligations, causing losses to the other party’ (cf. Canabarro and Duffie [13]). A major issue in this regard is the so-called *wrong way risk*, namely the risk that the value of the contract is particularly high from the perspective of the other party at the moment of default of the counterparty. As classic examples of wrong way risk, one can mention the situations of selling a put option to a company on its own stock, or entering a forward contract in which oil is bought by an airline company (see Redon [24]).

Among papers dealing with general counterparty risk, one can mention, apart from the abovementioned references, Canabarro *et al.* [14], Zhu and Pykhtin [26], and the series of papers by Brigo *et al.* [7, 9, 10, 8, 11, 12]. From the point of view of measurement and management of counterparty risk, two important notions emerge:

- The Credit Value Adjustment process (CVA), which measures the depreciation of a contract due to counterparty risk. So, in rough terms, $CVA_t = P_t - \Pi_t$, where Π and P denote the price process of a contract depending on whether one accounts or not for counterparty risk.
- The Expected Positive Exposure function (EPE), where $EPE(t)$ is the risk-neutral expectation of the loss on a contract conditional on a default of the counterparty occurring at time t .

Note that the CVA can be given an option-theoretic interpretation, so that counterparty risk can, in principle, be managed dynamically.

1.1 Counterparty Credit Risk

Wrong way risk is particularly important in the case of *credit derivatives* transactions, at least from the perspective of a credit protection buyer. Indeed, via economic cycle and default contagion effects, the time of default of a counterparty selling credit protection is typically a time of higher value of credit protection.

We consider in this paper a **Credit Default Swap with counterparty risk** (‘risky CDS’ in the sequel, as opposed to ‘risk-free CDS’, without counterparty risk). Note that this topic already received a lot of attention in the literature. It can thus be considered as a benchmark problem of counterparty credit risk. To quote but a few:

- Huge and Lando [17] propose a rating-based approach,
- Hull and White [18] study this problem in the set-up of a static copula model,

- Jarrow and Yu [19] use an intensity contagion model, further considered in Leung and Kwok [21],
- Brigo and Chourdakis [7] work in the set-up of their Gaussian copula and CIR++ intensity model, extended to the issue of bilateral counterparty credit risk in Brigo and Capponi [6],
- Blanchet-Scalliet and Patras [5] or Lipton and Sepp [22] develop structural approaches.

1.2 A Markov Copula Approach

We shall consider a Markovian model of credit risk in which simultaneous defaults are possible. Wrong way risk is thus represented in the model by the fact that at the time of default of the counterparty, there is a positive probability that the firm on which the CDS is written defaults too, in which case the loss incurred to the investor (Exposure at Default ED, cf. (3)) is the loss given default of the firm (up to the recovery on the counterparty), that is a very large amount. Of course, this simple model should not be taken too literally. We are not claiming here that simultaneous defaults can happen in actual practice. The rationale and financial interpretation of our model is rather that at the time of default of the counterparty, there is a positive probability of a high defaults spreads environment, in which case, the value of the CDS for a protection buyer is close to the loss given default of the firm.

More specifically, we shall be considering a four-state Markov Chain model of two obligors, so that all the computations are straightforward, either that there are explicit formulas for all the quantities of interest, or, in case less elementary parametrizations of the model are used, that these quantities can be easily and quickly computed by solving numerically the related Kolmogorov ODEs.

This Markovian set-up makes it possible to address in a dynamic and consistent way the issues of valuing (and also hedging) the CDS, and/or, if wished, the CVA, interpreted as an option as evoked above.

To make this even more practical, we shall work in a *Markovian copula* set-up in the sense of Bielecki *et al.* [3], in which calibration of the model marginals to the related CDS curves is straightforward. The only really free model parameters are thus the few dependence parameters, which can be calibrated or estimated in ways that we shall explain in the paper.

1.3 Outline of the Paper

In Section 2 we first describe the mechanism and cash flows of a payer CDS with counterparty credit risk. We then state a few preliminary results about pricing and CVA of this CDS in a general set-up. In Section 3 we introduce our Markov chain copula model, in which we derive explicit formulas for most quantities of interest in regard to a risky CDS, like price, EPE, CVA or hedging ratios. Section 4

is about implementation of the model. Alternative model parametrizations and related calibration or estimation procedures are proposed and analyzed. Numerical results are presented and discussed, showing good agreement of model's EPE and CVA with expected features. Section 5 recapitulates our model's main properties and presents some directions for possible extensions of the previous results.

2. General Set-Up

2.1 Cash Flows

As is well known, a CDS contract involves three entities: A reference credit (firm), a buyer of default protection on the firm, and a seller of default protection on the firm. The issue of counterparty risk on a CDS is:

- Primarily, the fact that the seller of protection may fail to pay the protection cash flows to the buyer in case of a default of the firm;
- Also, the symmetric concern that the buyer may fail to pay the contractual CDS spread to the seller.

We shall focus in this paper on the so-called *unilateral counterparty credit risk* involved in a payer CDS contract, namely the risk corresponding to the first bullet point above; however it should be noted that the approach of this paper could be extended to the issue of bilateral credit risk.

We shall refer to the buyer and the seller of protection on the firm as the risk-free *investor* and the defaultable *counterpart*, respectively. Indices 1 and 2 will refer to quantities related to the firm and to the counterpart. The default time of the firm and of the counterpart are denoted by τ_1 and τ_2 .

Under a risky CDS (payer CDS with counterparty credit risk), the investor pays to the counterpart a stream of premia with spread κ , or *Fees Cash Flows*, from the inception date (time 0 henceforth) until the occurrence of a credit event (default of the counterpart or the firm) or the maturity T of the contract, whichever comes first.

Let us denote by R_1 and R_2 the recovery of the firm and the counterpart, supposed to be adapted to the information available at time τ_1 and τ_2 , respectively. If the firm defaults prior to the expiration of the contract, the *Protection Cash Flows* paid by the counterpart to the investor depends on the situation of the counterpart:

- If the counterpart is still alive, she can fully compensate the loss of investor, i.e., she pays $(1 - R_1)$ times the face value of the CDS to the investor;
- If the counterpart defaults at the same time as the firm (note that it is important to take this case into account in the perspective of the model with simultaneous defaults to be introduced later in this paper), she will only be able to pay to the investor a fraction of this amount, namely $R_2(1 - R_1)$ times the face value of the CDS.

Finally, there is a *Close-Out Cash Flow* which is associated to clearing the positions in the case of early default of the counterpart. As of today, CDSs are sold over-the-counter (OTC), meaning that the two parties have to negotiate and agree on the terms of the contract. In particular the two parties can agree on one of the following three possibilities to exit (unwind) a trade:

- *Termination*: The contract is stopped after a terminal cash flow (positive or negative) has been paid to the investor;
- *Offsetting*: The counterpart takes the opposite protection position. This new contract should have virtually the same terms as the original CDS except for the premium which is fixed at the prevailing market level, and for the tenor which is set at the remaining time to maturity of the original CDS. So the counterpart leaves the original transaction in place but effectively cancels out its economic effect;
- *Novation (or Assignment)*: The original CDS is assigned to a new counterpart, settling the amount of gain or loss with him. In this assignment the original counterpart (or *transferor*), the new counterpart (*transferee*) and the investor agree to transfer all the rights and obligations of the transferor to transferee. So the transferor thereby ends his involvement in the contract and the investor thereafter deals with the default risk of the transferee.

In this paper we shall focus on **termination**. More precisely, if the counterpart defaults in the life-time of the CDS while the firm is still alive, a ‘fair value’ $\chi_{(\tau_2)}$ of the CDS is computed at time τ_2 according to a methodology specified in the CDS contract at inception. If this value (from the perspective of the investor) is negative, $(-\chi_{(\tau_2)})$ is paid by the investor to the counterpart, whereas if it is positive, the counterpart is assumed to pay to the investor a portion R_2 of $\chi_{(\tau_2)}$.

Remark 2.1. A typical specification is $\chi_{(\tau_2)} = P_{\tau_2}$, where P_t is the value at time t of a risk-free CDS on the same reference name, with the same contractual maturity T and spread κ as the original risky CDS. The consistency of this rather standard way of specifying $\chi_{(\tau_2)}$ is, in a sense, questionable. Given a pricing model accounting for the major risks in the product at hand, including, if appropriate, counterparty credit risk, with a related price process of the risky CDS denoted by Π , it could be argued that a more consistent specification would be $\chi_{(\tau_2)} = \Pi_{\tau_2}$ (or, more precisely, $\chi_{(\tau_2)} = \Pi_{\tau_2-}$, since $\Pi_{\tau_2} = 0$ in view of the usual conventions regarding the definition of ex-dividend prices). We shall see in section 4 that, at least in the specific model of this paper, adopting either convention makes little difference in practice.

2.2 Pricing

Let us be given a risk-neutral pricing model $(\Omega, \mathbb{F}, \mathbb{P})$, where $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ is a given filtration making the τ_i ’s stopping times. In absence of further precision, all

the *processes*, first of which, the *discount factor* process β , are supposed to be $\mathbb{F}$ -adapted, and all the *random variables* are assumed to be $\mathcal{F}_T$ -measurable. The fair value $\chi_{(\tau_2)}$ is supposed to be an $\mathcal{F}_{\tau_2}$ -measurable random variable. The recoveries R_1 and R_2 are assumed to be $\mathcal{F}_{\tau_1}$ - and $\mathcal{F}_{\tau_2}$ -measurable random variables. Let $\mathbb{E}_\tau$ stand for the conditional expectation under $\mathbb{P}$ given $\mathcal{F}_\tau$, for any stopping time τ .

We assume for simplicity that the face value of all the CDSs under consideration (risky or not) is equal to one monetary unit and that the spreads are paid continuously in time. All the cash flows and prices are considered from the perspective of the investor. In accordance with the usual convention regarding the definition of *ex-dividend* prices, the integrals in this paper are taken open on the left and closed on the right of the interval of integration. In view of the description of the cash-flows in subsection 2.1, one then has

Definition 2.2. (i) The model *price* process of a risky CDS is given by $\Pi_t = \mathbb{E}_t[\pi_T(t)]$, where $\pi_T(t)$ corresponds to the *risky CDS cumulative discounted cash flows* on the time interval $(t, T]$, so,

$$\begin{aligned} \beta_t \pi_T(t) = & -\kappa \int_{t \wedge \tau_1 \wedge \tau_2 \wedge T}^{\tau_1 \wedge \tau_2 \wedge T} \beta_s ds + \beta_{\tau_1} (1 - R_1) \mathbb{1}_{t < \tau_1 \leq T} [\mathbb{1}_{\tau_1 < \tau_2} + R_2 \mathbb{1}_{\tau_1 = \tau_2}] \\ & + \beta_{\tau_2} \mathbb{1}_{t < \tau_2 \leq T} \mathbb{1}_{\tau_2 < \tau_1} [R_2 \chi_{(\tau_2)}^+ - \chi_{(\tau_2)}^-] . \end{aligned} \quad (1)$$

(ii) The model *price* process of a risk-free CDS is given by $P_t = \mathbb{E}_t[p_T(t)]$, where $p_T(t)$ corresponds to the *risk-free CDS cumulative discounted cash flows* on the time interval $(t, T]$, so,

$$\beta_t p_T(t) = -\kappa \int_{t \wedge \tau_1 \wedge T}^{\tau_1 \wedge T} \beta_s ds + (1 - R_1) \beta_{\tau_1} \mathbb{1}_{t < \tau_1 \leq T} . \quad (2)$$

The first, second and third term on the right-hand side of (1) correspond to the fees, protection and close-out cash flows of a risky CDS, respectively. Note that there are no cash flows of any kind after $\tau_1 \wedge \tau_2 \wedge T$ (in the case of the risky CDS) or $\tau_1 \wedge T$ (in the case of the risk-free CDS), so $\pi_T(t) = 0$ for $t \geq \tau_1 \wedge \tau_2 \wedge T$ and $p_T(t) = 0$ for $t \geq \tau_1 \wedge T$.

Remark 2.3. In these definitions it is implicitly assumed that, consistent with the now standard theory of no-arbitrage (cf. Delbaen and Schachermayer [15]), a primary market of financial instruments (along with the risk-free asset β^{-1}) has been defined, with price processes given as locally bounded $(\Omega, \mathbb{F}, \mathbb{P})$ -local martingales. No-arbitrage on the extended market consisting of the primary assets and a further CDS then motivates the previous definitions. Since the precise specification of the primary market is irrelevant until the question of hedging is dealt with, we postpone it to section 3.3.

Definition 2.4. (i) The *Exposure at Default (ED)* is the $\mathcal{F}_{\tau_2}$ -measurable random variable $\xi_{(\tau_2)}$ defined by,

$$\xi_{(\tau_2)} = \begin{cases} (1 - R_2)(1 - R_1), & \tau_2 = \tau_1 \leq T, \\ P_{\tau_2} - (R_2\chi_{(\tau_2)}^+ - \chi_{(\tau_2)}^-) & \tau_2 < \tau_1, \tau_2 \leq T, \\ 0, & \text{otherwise} . \end{cases} \quad (3)$$

(ii) The *Credit Valuation Adjustment (CVA)* is the process killed at $\tau_1 \wedge \tau_2 \wedge T$ defined by, for $t \in [0, T]$,

$$\beta_t \text{CVA}_t = \mathbb{1}_{\{t < \tau_2\}} \mathbb{E}_t[\beta_{\tau_2} \xi_{(\tau_2)}] . \quad (4)$$

(iii) The *Expected Positive Exposure (EPE)* is the function of time defined by, for $t \in [0, T]$,

$$\text{EPE}(t) = \mathbb{E}[\xi_{(\tau_2)} | \tau_2 = t] . \quad (5)$$

The following proposition justifies the name of Credit Valuation Adjustment which is used for the CVA process defined by (4). In case $\chi_{(\tau_2)} = P_{\tau_2}$ (see Remark 2.1) then

$$\xi_{(\tau_2)} = \xi_{(\tau_2)}^0 := (1 - R_2) \times \begin{cases} (1 - R_1), & \tau_2 = \tau_1 \leq T, \\ P_{\tau_2}^+, & \tau_2 < \tau_1, \tau_2 \leq T, \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

and we essentially recover the basic result that has been established in Brigo and Masetti [8]. Note that as opposed to [8] we do not exclude simultaneous defaults in our set-up, whence further terms in $\mathbb{1}_{t < \tau_1 = \tau_2 \leq T}$ in the proof of Proposition 2.1.

Proposition 2.1. *One has $\text{CVA}_t = P_t - \Pi_t$ on $\{t < \tau_2\}$.*

Proof. If $\tau_1 \leq t < \tau_2$, then $\Pi_t = P_t = \text{CVA}_t = 0$ in view of (1), (2) and (4).

Assume $t < \tau_1 \wedge \tau_2$. Subtracting $\pi_T(t)$ from $p_T(t)$ yields,

$$\begin{aligned} \beta_t (p_T(t) - \pi_T(t)) &= -\kappa \int_{\tau_1 \wedge \tau_2 \wedge T}^{\tau_1 \wedge T} \beta_s ds + \beta_{\tau_1} (1 - R_1) \mathbb{1}_{\tau_1 \leq T} \mathbb{1}_{\tau_1 \geq \tau_2} \\ &\quad - \beta_{\tau_1} R_2 (1 - R_1) \mathbb{1}_{\tau_1 \leq T} \mathbb{1}_{\tau_1 = \tau_2} - \beta_{\tau_2} \mathbb{1}_{\tau_2 < \tau_1} \mathbb{1}_{\tau_2 \leq T} (R_2 \chi_{(\tau_2)}^+ - \chi_{(\tau_2)}^-) . \end{aligned} \quad (7)$$

Moreover, in view of (2), one has,

$$\beta_{\tau_2} p_T(\tau_2) \mathbb{1}_{\tau_2 < \tau_1} \mathbb{1}_{\tau_2 \leq T} = -\kappa \int_{\tau_1 \wedge \tau_2 \wedge T}^{\tau_1 \wedge T} \beta_s ds + (1 - R_1) \beta_{\tau_1} \mathbb{1}_{\tau_2 < \tau_1 \leq T} . \quad (8)$$

Now, using the following identity in the second term on the right-hand-side of (7):

$$\mathbb{1}_{\tau_1 \leq T} \mathbb{1}_{\tau_1 \geq \tau_2} = \mathbb{1}_{\tau_1 \leq T} \mathbb{1}_{\tau_2 < \tau_1} + \mathbb{1}_{\tau_1 = \tau_2 \leq T} ,$$

and plugging (8) into (7), we obtain (recall $t < \tau_1 \wedge \tau_2$),

$$\begin{aligned} \beta_t(p_T(t) - \pi_T(t)) &= \beta_{\tau_2} \mathbb{1}_{\tau_2 < \tau_1, \tau_2 \leq T} p_T(\tau_2) \\ &\quad + \beta_{\tau_2} \mathbb{1}_{\tau_2 = \tau_1 \leq T} (1 - R_2)(1 - R_1) - \beta_{\tau_2} \mathbb{1}_{\tau_2 < \tau_1, \tau_2 \leq T} (R_2 \chi_{(\tau_2)}^+ - \chi_{(\tau_2)}^-). \end{aligned}$$

Thus:

- On the set $\{\tau_2 < \tau_1, \tau_2 \leq T\}$,

$$\beta_t(p_T(t) - \pi_T(t)) = \beta_{\tau_2} p_T(\tau_2) - \beta_{\tau_2} (R_2 \chi_{(\tau_2)}^+ - \chi_{(\tau_2)}^-)$$

As $P_{\tau_2} = \mathbb{E}_{\tau_2}[p_T(\tau_2)]$, we then have, since R_2 and $\chi_{(\tau_2)}$ are $\mathcal{F}_{\tau_2}$ -measurable,

$$\beta_t \mathbb{E}_{\tau_2}[p_T(t) - \pi_T(t)] = \beta_{\tau_2} (P_{\tau_2} - (R_2 \chi_{(\tau_2)}^+ - \chi_{(\tau_2)}^-)) ; \quad (9)$$

- On the set $\{\tau_1 = \tau_2 \leq T\}$,

$$\beta_t(p_T(t) - \pi_T(t)) = \beta_{\tau_2} (1 - R_1)(1 - R_2)$$

and thus

$$\beta_t \mathbb{E}_{\tau_2}[p_T(t) - \pi_T(t)] = \mathbb{E}_{\tau_2}[\beta_{\tau_2} (1 - R_1)(1 - R_2)]. \quad (10)$$

Using the fact that $\tau_2 < \tau_1, \tau_2 \leq T$ and $\tau_2 = \tau_1 \leq T$ are $\mathcal{F}_{\tau_2}$ -measurable, it follows,

$$\begin{aligned} \beta_t P_t - \beta_t \Pi_t &= \beta_t \mathbb{E}_t[\mathbb{E}_{\tau_2}[p_T(t) - \pi_T(t)]] \\ &= \beta_t \mathbb{E}_t[\mathbb{E}_{\tau_2}[p_T(t) - \pi_T(t)] \mathbb{1}_{\tau_2 < \tau_1, \tau_2 \leq T} \\ &\quad + \mathbb{E}_{\tau_2}[p_T(t) - \pi_T(t)] \mathbb{1}_{\tau_2 = \tau_1 \leq T}] \\ &= \mathbb{E}_t[\beta_{\tau_2} \xi_{(\tau_2)}] = \beta_t \text{CVA}_t. \end{aligned}$$

□

2.3 Special Case $\mathbb{F} = \mathbb{H}$

Let $H = (H^1, H^2)$ denote the pair of the default indicator processes of the firm and the counterpart, so $H_t^i = \mathbb{1}_{\tau_i \leq t}$. The following proposition gathers a few useful results that can be established in the special case of a model filtration $\mathbb{F}$ given as

$$\mathbb{F} = \mathbb{H} = (\mathcal{H}_t^1 \vee \mathcal{H}_t^2)_{t \in [0, T]},$$

with $\mathcal{H}_t^i = \sigma(H_s^i; 0 \leq s \leq t)$.

Proposition 2.2. (i) For $t \in [0, T]$, any $\mathcal{H}_t$ -measurable random variable Y_t can be written as

$$Y_t = y_0(t) \mathbb{1}_{t < \tau_1 \wedge \tau_2} + y_1(t, \tau_1) \mathbb{1}_{\tau_1 \leq t < \tau_2} + y_2(t, \tau_2) \mathbb{1}_{\tau_2 \leq t < \tau_1} + y_3(t, \tau_1, \tau_2) \mathbb{1}_{\tau_2 \vee \tau_1 \leq t}$$

where $y_0(t), y_1(t, u), y_2(t, v), y_3(t, u, v)$ are deterministic functions.

(ii) For any integrable random variable Z , one has,

$$\mathbb{1}_{t < \tau_1 \wedge \tau_2} \mathbb{E}_t Z = \mathbb{1}_{t < \tau_1 \wedge \tau_2} \frac{\mathbb{E}(Z \mathbb{1}_{t < \tau_1 \wedge \tau_2})}{\mathbb{P}(t < \tau_1 \wedge \tau_2)}. \quad (11)$$

(iii) The price process of the risky CDS is given by $\Pi_t = \Pi(t, H_t)$, for a pricing function Π defined on $\mathbb{R}^+ \times E_1 \times E_1$ with $E_1 = \{0, 1\}$, such that $\Pi(t, e) = 0$ for $e \neq (0, 0)$. On the set $\{t < \tau_1 \wedge \tau_2\}$, Π_t is given by the deterministic function

$$\Pi(t, 0, 0) = u(t) := \frac{\mathbb{E}[\pi_T(t)]}{\mathbb{P}(\tau_1 \wedge \tau_2 > t)}. \quad (12)$$

(iv) One has, for suitable functions $\tilde{\chi}(\cdot), v(\cdot), \tilde{\xi}(\cdot, \cdot)$ and $\text{CVA}(\cdot)$,

$$\mathbb{1}_{\{\tau_2 < \tau_1\}} \chi(\tau_2) = \mathbb{1}_{\{\tau_2 < \tau_1\}} \tilde{\chi}(\tau_2), \quad \mathbb{1}_{\{\tau_2 < \tau_1\}} P_{\tau_2} = \mathbb{1}_{\{\tau_2 < \tau_1\}} v(\tau_2) \quad (13)$$

$$\xi_{(\tau_2)} = \tilde{\xi}(\tau_1, \tau_2) := (\mathbb{1}_{\tau_2 = \tau_1 \leq T} (1 - R_2)(1 - R_1) + \mathbb{1}_{\tau_2 < \tau_1, \tau_2 \leq T} (v(\tau_2) - (R_2 \tilde{\chi}^+(\tau_2) - \tilde{\chi}^-(\tau_2)))) \quad (14)$$

$$\text{CVA}_t = \mathbb{1}_{t < \tau_1 \wedge \tau_2} \text{CVA}(t). \quad (15)$$

(v) A function $\text{CVA}(\cdot)$ satisfying (15) is defined by, for $t \in [0, T]$,

$$\beta_t \text{CVA}(t) := \int_t^T \beta_s \text{EPE}(s) \frac{\mathbb{P}(\tau_2 \in ds)}{\mathbb{P}(t < \tau_1 \wedge \tau_2)}. \quad (16)$$

Proof. (i) and (ii) are standard (see, e.g., Bielecki and Rutkowski [4]; (ii) in particular is the so-called *Key Lemma*).

(iii) Since there are no cash flows of a risky CDS beyond the first default (cf. (1)), one has $\pi_T(t) = \pi_T(t) \mathbb{1}_{t < \tau_1 \wedge \tau_2}$. The Key Lemma then yields,

$$\Pi_t = \mathbb{E}_t[\mathbb{1}_{t < \tau_1 \wedge \tau_2} \pi_T(t)] = (1 - H_t^1)(1 - H_t^2) \frac{\mathbb{E}[\pi_T(t)]}{\mathbb{P}(\tau_1 \wedge \tau_2 > t)}.$$

Thus $\Pi_t = \Pi(t, H_t^1, H_t^2)$, for a pricing function Π defined by

$$\Pi(t, e_1, e_2) = (1 - e_1)(1 - e_2)u(t),$$

where $u(t)$ is defined by the right-hand-side of (12).

(iv) follows directly from part (i), given the definition of $P_{\tau_2}, \chi(\tau_2), \xi_{(\tau_2)}$ and of the CVA process.

(v) By (iv) and using (ii) again, one has, on the set $\{t < \tau_1 \wedge \tau_2\}$,

$$\begin{aligned}
 \beta_t \text{CVA}_t &= \mathbb{E}_t[\beta_{\tau_2} \xi_{(\tau_2)}] = \mathbb{E}_t[\beta_{\tau_2} \tilde{\xi}(\tau_1, \tau_2)] \\
 &= \frac{\mathbb{E}[\beta_{\tau_2} \tilde{\xi}(\tau_1, \tau_2) \mathbb{1}_{t < \tau_1 \wedge \tau_2}]}{\mathbb{P}(t < \tau_1 \wedge \tau_2)} = \frac{\mathbb{E}[\mathbb{E}(\beta_{\tau_2} \tilde{\xi}(\tau_1, \tau_2) \mathbb{1}_{t < \tau_1 \wedge \tau_2} | \tau_2)]}{\mathbb{P}(t < \tau_1 \wedge \tau_2)} \\
 &= \frac{\mathbb{E}[\mathbb{E}(\beta_{\tau_2} \tilde{\xi}(\tau_1, \tau_2) \mathbb{1}_{t < \tau_2 \leq T} | \tau_2)]}{\mathbb{P}(t < \tau_1 \wedge \tau_2)} = \frac{\mathbb{E}[\beta_{\tau_2} \mathbb{E}(\tilde{\xi}(\tau_1, \tau_2) | \tau_2) \mathbb{1}_{t < \tau_2 \leq T}]}{\mathbb{P}(t < \tau_1 \wedge \tau_2)} \\
 &= \frac{\mathbb{E}[\beta_{\tau_2} \text{EPE}(\tau_2) \mathbb{1}_{t < \tau_2 \leq T}]}{\mathbb{P}(t < \tau_1 \wedge \tau_2)} = \int_t^T \beta_s \text{EPE}(s) \frac{\mathbb{P}(\tau_2 \in ds)}{\mathbb{P}(t < \tau_1 \wedge \tau_2)},
 \end{aligned}$$

whence (v). □

3. Markov Copula Factor Set-Up

3.1 Factor Process Model

We shall now introduce a suitable *Markovian Copula Model* for the pair of default indicator processes $H = (H^1, H^2)$ of the firm and the counterparty. The name ‘Markovian Copula’ refers to the fact that the model will have prescribed marginals for the laws of H^1 and H^2 , respectively (see Bielecki *et al.* [2, 3] for a general theory). The practical interest of a Markovian copula model is clear in view of the task of model calibration, since the copula property allows one to decouple the calibration of the marginal and of the dependence parameters in the model (see again section 4.1). More fundamentally, the opinion developed in this paper is that it is also a virtue for a model to ‘take the right inputs to generate the right outputs’, namely taking as basic inputs the individual default probabilities (individual CDS curves), which correspond to the more reliable information on the market, and are then ‘coupled together’ in a suitable way (see section 4.1).

An apparent shortcoming of the Markov copula approach is that it does not allow for default contagion effects in the usual sense (default of a name impacting the default intensities of the other ones). Note also that in this work we assume that the underlying filtration is $\mathbb{H}$ and thus the default intensities are deterministic between defaults. The interest of this admittedly simplified set-up is that one is able to derive explicit formulas for most quantities of interest with regard to CDS counterparty risk like price, CVA, EPE or hedging ratios. In a forthcoming paper, we will generalize this setting to take into account the spread risk.

The way we shall introduce dependence between τ_1 and τ_2 is by relaxing the standard assumption of no simultaneous defaults. As we shall see, allowing for simultaneous defaults is a powerful way of modeling defaults dependence.

Specifically, we model the pair $H = (H^1, H^2)$ as an inhomogeneous Markov chain relative to its own filtration $\mathbb{H}$ on a probability space $(\Omega, \mathbb{P})$ (for the σ -algebra $\mathcal{H}_T$), with state space $E = \{(0, 0), (1, 0), (0, 1), (1, 1)\}$, and generator matrix at time t given by the following 4×4 matrix $A(t)$, where the first to fourth rows (or

columns) correspond to the four possible states $(0, 0)$, $(1, 0)$, $(0, 1)$ and $(1, 1)$ of H_t :

$$A(t) = \begin{bmatrix} -l(t) & l_1(t) & l_2(t) & l_3(t) \\ 0 & -q_2(t) & 0 & q_2(t) \\ 0 & 0 & -q_1(t) & q_1(t) \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (17)$$

In (17) the l 's and q 's denote deterministic functions of time integrable over $[0, T]$, with in particular $l(t) = l_1(t) + l_2(t) + l_3(t)$.

Remark 3.1. The intuitive meaning of '(17) being the generator matrix of H ' is the following (see, e.g., Rogers and Williams [25], Vol. I, Chap. III, Sec. 2, for standard definitions and results on Markov Chains):

- *First line:* Conditional on the pair $H_t = (H_t^1, H_t^2)$ being in state $(0, 0)$ (firm and counterpart still alive at time t), there is a probability $l_1(t)dt$, (resp. $l_2(t)dt$; resp. $l_3(t)dt$) of a default of the firm alone (resp. of the counterpart alone; resp. of a simultaneous default of the firm and the counterpart) in the infinitesimal time interval $(t, t + dt)$;
- *Second line:* Conditional on the pair $H_t = (H_t^1, H_t^2)$ being in state $(1, 0)$ (firm defaulted but counterpart still alive at time t), there is a probability $q_2(t)dt$ of a further default of the counterpart in the time interval $(t, t + dt)$;
- *Third line:* Conditional on the pair $H_t = (H_t^1, H_t^2)$ being in state $(0, 1)$ (firm still alive but counterpart defaulted at time t), there is a probability $q_1(t)dt$ of a further default of the firm in the time interval $(t, t + dt)$.

On each line the diagonal term is then set as minus the sum of the off-diagonal terms, so that the sum of the entries of each line is equal to zero, as should be for $A(t)$ to represent the generator of a Markov process. Moreover, for the sake of the desired **Markov copula property** (Proposition 3.1(iii) below), we impose the following relations between the l 's and the q 's.

Assumption 3.2. $q_1(t) = l_1(t) + l_3(t)$, $q_2(t) = l_2(t) + l_3(t)$.

Observe that in virtue of these relations:

- Conditional on H_t^1 being in state 0, and whatever the state of H_t^2 may be (that is, in the state $(0, 0)$ or $(0, 1)$ for H_t), there is a probability $q_1(t)dt$ of a default of the firm (alone or jointly with the counterpart) in the next time interval $(t, t + dt)$;

- Conditional on H_t^2 being in state 0, and whatever the state of H_t^1 may be (that is, in the states (0, 0) or (1, 0) for H_t), there is a probability $q_2(t)dt$ of a default of the counterpart (alone or jointly with the firm) in the next time interval $(t, t + dt)$.

In mathematical terms the default indicator processes H^1 and H^2 are $\mathbb{H}$ -Markov processes on the state space $E_1 = \{0, 1\}$ with time t generators respectively given by

$$A_1(t) = \begin{bmatrix} -q_1(t) & q_1(t) \\ 0 & 0 \end{bmatrix}, \quad A_2(t) = \begin{bmatrix} -q_2(t) & q_2(t) \\ 0 & 0 \end{bmatrix}. \quad (18)$$

To formalize the previous statements, and in view of the study of simultaneous jumps, let us further introduce the processes $H^{(1)}$, $H^{(2)}$ and $H^{(1,2)}$ standing for the indicator processes of a default of the firm alone, of the counterpart alone, and of a simultaneous default of the firm and the counterpart, respectively. So

$$H^{(1,2)} = [H^1, H^2], \quad H^{(1)} = H^1 - H^{(1,2)}, \quad H^{(2)} = H^2 - H^{(1,2)}, \quad (19)$$

where $[\cdot, \cdot]$ stands for the quadratic covariation. Equivalently, for $t \in [0, T]$,

$$H_t^{(1)} = \mathbb{1}_{\tau_1 \leq t, \tau_1 \neq \tau_2}, \quad H_t^{(2)} = \mathbb{1}_{\tau_2 \leq t, \tau_1 \neq \tau_2}, \quad H_t^{(1,2)} = \mathbb{1}_{\tau_1 = \tau_2 \leq t}.$$

Note that the natural filtration of $(H^i)_{i \in I}$, with here and henceforth $I = \{\{1\}, \{2\}, \{1, 2\}\}$, is equal to $\mathbb{H}$. The proof of the following Proposition is deferred to Appendix 5.

Proposition 3.1. (i) *The $\mathbb{H}$ -intensity of H^i is of the form $q_i(t, H_t)$ for a suitable function $q_i(t, e)$ for every $i \in I$, namely,*

$$\begin{aligned} q_{(1)}(t, e) &= \mathbb{1}_{e_1=0} (\mathbb{1}_{e_2=0} l_1(t) + \mathbb{1}_{e_2=1} q_1(t)) \\ q_{(2)}(t, e) &= \mathbb{1}_{e_2=0} (\mathbb{1}_{e_1=0} l_2(t) + \mathbb{1}_{e_1=1} q_2(t)) \\ q_{(1,2)}(t, e) &= \mathbb{1}_{e=(0,0)} l_3(t). \end{aligned}$$

Put another way, the processes M^i defined by, for every $i \in I$,

$$M_t^i = H_t^i - \int_0^t q_i(s, H_s) ds, \quad (20)$$

with

$$\begin{aligned} q_{(1)}(t, H_t) &= (1 - H_t^1) \left((1 - H_t^2) l_1(t) + H_t^2 q_1(t) \right) \\ q_{(2)}(t, H_t) &= (1 - H_t^2) \left((1 - H_t^1) l_2(t) + H_t^1 q_2(t) \right) \\ q_{(1,2)}(t, H_t) &= (1 - H_t^1)(1 - H_t^2) l_3(t), \end{aligned} \quad (21)$$

are $\mathbb{H}$ -martingales.

(ii) The $\mathbb{H}$ -intensity process of H^i is given by $(1 - H_t^i)q_i(t)$. In other words, the processes M^i defined by, for $i = 1, 2$,

$$M_t^i = H_t^i - \int_0^t (1 - H_s^i)q_i(s)ds, \quad (22)$$

are $\mathbb{H}$ -martingales.

(iii) The processes H^1 and H^2 are $\mathbb{H}$ -Markov processes with generator matrix at time t given by $A_1(t)$ and $A_2(t)$ (cf. (18)).

(iv) One has,

$$\mathbb{P}(\tau_1 > s, \tau_2 > t) = \exp\left(-\int_0^s l_1(u)du - \int_0^t l_2(u)du - \int_0^{s \vee t} l_3(u)du\right) \quad (23)$$

and therefore

$$\begin{aligned} \mathbb{P}(\tau_1 > t) &= e^{-\int_0^t q_1(u)du}, \quad \mathbb{P}(\tau_2 > t) = e^{-\int_0^t q_2(u)du}, \\ \mathbb{P}(\tau_1 \wedge \tau_2 > t) &= e^{-\int_0^t l(u)du} \\ \mathbb{P}(\tau_1 > s, \tau_2 \in dt) &= q_2(t)e^{-\int_0^s l(u)du}e^{-\int_s^t q_2(u)du}dt \\ \mathbb{P}(\tau_1 \in dt, \tau_2 > s) &= q_1(t)e^{-\int_0^s l(u)du}e^{-\int_s^t q_1(u)du}dt \\ \mathbb{P}(\tau_1 > t, \tau_2 \in dt) &= q_2(t)e^{-\int_0^t l(u)du}dt \\ \mathbb{P}(\tau_1 \in dt, \tau_2 > t) &= q_1(t)e^{-\int_0^t l(u)du}dt. \end{aligned} \quad (24)$$

(v) The correlation of H_t^1 and H_t^2 (default correlation at the time horizon t) is

$$\rho_d(t) = \frac{\exp\left(\int_0^t l_3(s)ds\right) - 1}{\sqrt{\left(\exp\left(\int_0^t q_1(s)ds\right) - 1\right)\left(\exp\left(\int_0^t q_2(s)ds\right) - 1\right)}}. \quad (25)$$

Remark 3.3. (i) In the Markov copula terminology of Bielecki *et al.* [3], the so-called *consistency condition* is satisfied (H^1 and H^2 are $\mathbb{H}$ -Markov processes). The bi-variate model H with generator A is thus a *Markovian copula model* with marginal generators A_1 and A_2 .

(ii) The default times τ_1 and τ_2 could equivalently be defined by

$$\tau_1 = \eta_1 \wedge \eta_3, \quad \tau_2 = \eta_2 \wedge \eta_3$$

where the η_i 's are independent inhomogeneous exponential random variables with parameters $l_i(t)$'s. Thus, for every $0 \leq s, t$,

$$\mathbb{P}(\tau_1 > s, \tau_2 > t) = \mathbb{P}(\eta_1 > s)\mathbb{P}(\eta_2 > t)\mathbb{P}(\eta_3 > s \vee t). \quad (26)$$

In the special case of *homogeneous* exponential random variables with (constant) parameters l_i 's, one has further (see section 4 of Embrechts *et al.* [16] or Marshall and Olkin [23]),

$$\mathbb{P}(\tau_1 > s, \tau_2 > t) = C(\mathbb{P}(\eta_1 > s), \mathbb{P}(\eta_2 > t)) , \quad (27)$$

where the *Marshall-Olkin survival copula function* C is defined by, for $p, q \in [0, 1]$,

$$C(p, q) = pq \min(p^{-\alpha_1}, q^{-\alpha_2}) \quad (28)$$

with $\alpha_i = \frac{l_3}{l_i + l_3}$. Our model is thus an extension of the classical Marshall-Olkin copula model in which *inhomogeneous* exponential random variables are used as model inputs, and where, more importantly, a *dynamic perspective* is shed on the random times τ_1 and τ_2 by introducing the model filtration $\mathbb{H}$.

3.2 Pricing

We use the notation of Proposition 2.2, which applies here since we are in the special case $\mathbb{F} = \mathbb{H}$. Recall in particular $\Pi_t = \Pi(t, H_t) = (1 - H_t^1)(1 - H_t^2)u(t)$, for a *pricing function* $\Pi(t, 0, 0) = u(t)$, as well as the identities (13), (15), (16). We assume henceforth for simplicity that:

- The discount factor writes $\beta_t = \exp(-\int_0^t r(s)ds)$, for a deterministic *short-term interest-rate* function r ,
- The recovery rates R_1 and R_2 are constant.

Proposition 3.2. *The pricing function u of the risky CDS is given by*

$$\beta_t u(t) = \int_t^T \beta_s e^{-\int_t^s l(u)du} \pi(s) ds \quad (29)$$

with

$$\pi(s) = (1 - R_1)[l_1(s) + R_2 l_3(s)] + l_2(s)[R_2 \tilde{\chi}(s)^+ - \tilde{\chi}(s)^-] - \kappa . \quad (30)$$

The function u satisfies the following ODE:

$$\begin{cases} u(T) = 0 \\ \frac{du}{dt}(t) - (r(t) + l(t))u(t) + \pi(t) = 0, \quad t \in [0, T] . \end{cases} \quad (31)$$

Proof. Recall (12):

$$u(t) = \frac{\mathbb{E}[\pi_T(t)]}{\mathbb{P}(\tau_1 \wedge \tau_2 > t)} ,$$

where the denominator can be calculated using Proposition 3.1(iv). For computing the numerator, one rewrites the expressions for the cumulative discounted Fee,

Protection and Close-out cash flows in terms of integrals with respect to $H^{(1)}$, $H^{(2)}$ and $H^{(1,2)}$, as follows:

$$\begin{aligned}
 \text{Fees Cash Flow} &= \kappa \int_0^T \beta_s(1 - H_s^1)(1 - H_s^2)ds \\
 \text{Protection Cash Flow} &= (1 - R_1) \int_0^T \beta_s(1 - H_{s-}^2)dH_s^{(1)} + R_2(1 - R_1) \int_0^T \beta_s dH_s^{(1,2)} \\
 &= (1 - R_1) \int_0^T \beta_s(1 - H_{s-}^2)dM_s^{(1)} \\
 &\quad + (1 - R_1) \int_0^T \beta_s(1 - H_s^2)q_{(1)}(s, H_s)ds \\
 &\quad + R_2(1 - R_1) \int_0^T \beta_s dM_s^{(1,2)} + R_2(1 - R_1) \int_0^T \beta_s q_{(1,2)}(s, H_s)ds \\
 \text{Close-out Cash Flow} &= \int_0^T \beta_s[R_2\tilde{\chi}(s)^+ - \tilde{\chi}(s)^-](1 - H_{s-}^1)dH_s^{(2)} \\
 &= \int_0^T \beta_s[R_2\tilde{\chi}(s)^+ - \tilde{\chi}(s)^-](1 - H_{s-}^1)dM_s^{(2)} \\
 &\quad + \int_0^T \beta_s[R_2\tilde{\chi}(s)^+ - \tilde{\chi}(s)^-](1 - H_s^1)q_{(2)}(s, H_s)ds
 \end{aligned}$$

Making use of the martingale property of $M^{(1)}$, $M^{(2)}$ and $M^{(1,2)}$ and the fact that the integrals of bounded predictable processes with respect to these martingales are indeed martingales, we thus have

$$\mathbb{E}(\pi_T(t)) = \mathbb{E}(\tilde{\pi}_T(t)) \quad (32)$$

with

$$\begin{aligned}
 \beta_t \tilde{\pi}_T(t) &= -\kappa \int_t^T \beta_s(1 - H_s^1)(1 - H_s^2)ds \\
 &\quad + (1 - R_1) \int_t^T \beta_s(1 - H_s^2)q_{(1)}(s, H_s)ds \\
 &\quad + R_2(1 - R_1) \int_t^T \beta_s q_{(1,2)}(s, H_s)ds \\
 &\quad + \int_t^T \beta_s[R_2\tilde{\chi}(s)^+ - \tilde{\chi}(s)^-](1 - H_s^1)q_{(2)}(s, H_s)ds.
 \end{aligned}$$

Moreover, in view of the expressions for $q_{(1)}$ and $q_{(2)}$ in (21), one has

$$\begin{aligned} (1 - H_s^2)q_{(1)}(s, H_s) &= (1 - H_s^1)(1 - H_s^2)l_1(s) , \\ (1 - H_s^1)q_{(2)}(s, H_s) &= (1 - H_s^1)(1 - H_s^2)l_2(s) . \end{aligned} \quad (33)$$

Plugging this into (32) and using (24), it follows that,

$$\begin{aligned} \beta_t \mathbb{E}[\pi_T(t)] &= \mathbb{E} \left[\int_t^T \beta_s (1 - H_s^1)(1 - H_s^2) \pi(s) ds \right] \\ &= \int_t^T \beta_s \mathbb{E} \left[(1 - H_s^1)(1 - H_s^2) \right] \pi(s) ds = \int_t^T \beta_s e^{-\int_0^s l(x) dx} \pi(s) ds \end{aligned}$$

where π is given by (30). One can now check by inspection that the function u satisfies the ODE (31). $\square$

Remark 3.4. The equation (31) can also be interpreted as the Kolmogorov backward equation related to the valuation of a risky CDS in our set-up. This ODE can in fact be derived directly and independently by an application of the Itô formula to the martingale $\Pi(t, H_t^1, H_t^2)$, which results in an alternative proof of Proposition 3.2.

Remark 3.5. In the set-up of the Markov chain copula model, the identity (when-ever assumed) $\chi_{(\tau_2)} = \Pi_{\tau_2-}$ (see Remark 2.1) is thus equivalent to

$$\chi_{(\tau_2)} = \Pi_{\tau_2-} = \lim_{t \rightarrow \tau_2-} u(t) = u(\tau_2) ,$$

by continuity of u . This case thus corresponds to the case where the function $\tilde{\chi}$ in Proposition 2.2(iv) is in fact given by the function u (case $\tilde{\chi} = u$). In this case the positive and negative parts of u , i.e., u^+ and u^- are sitting in the expression for π in (30). One thus deals with a non-linear valuation ODE (31), and the formula (29) is not explicit anymore, since u is ‘hidden’ in π in the right hand side of this formula. However one can still compute u by numerical solution of (31).

Proposition 3.3. *The price of a risk-free CDS with spread κ on the firm admits the representation:*

$$P_t = P(t, H_t^1) , \quad (34)$$

for a function P of the form $P(t, e_1) = (1 - e_1)v(t)$. The pricing function v is given by

$$\beta_t v(t) = \int_t^T \beta_s e^{-\int_t^s q_1(x) dx} p(s) ds$$

with

$$p(s) = (1 - R_1)q_1(s) - \kappa. \quad (35)$$

The pricing function v thus solves the following pricing ODE:

$$\begin{cases} v(T) = 0 \\ \frac{dv}{dt}(t) - (r(t) + q_1(t))v(t) + p(t) = 0, \quad t \in [0, T]. \end{cases}$$

Proof. One has

$$\begin{aligned} \beta_t p_T(t) &= -\kappa \int_t^T \beta_s(1 - H_s^1)ds + (1 - R_1) \int_t^T \beta_s dH_s^1 \\ &= -\kappa \int_t^T \beta_s(1 - H_s^1)ds + (1 - R_1) \int_t^T \beta_s dM_s^1 \\ &\quad + (1 - R_1) \int_t^T \beta_s q_1(s)(1 - H_s^1)ds. \end{aligned}$$

As M^1 is an $\mathbb{H}$ -martingale and β a bounded continuous function, thus

$$\beta_t \mathbb{E}_t[p_T(t)] = \mathbb{E}_t \left[\int_t^T \beta_s(1 - H_s^1)p(s)ds \right] = \int_t^T \beta_s \mathbb{E}_t[1 - H_s^1]p(s)ds, \quad (36)$$

with $p(t)$ defined by (35), and where in virtue of Proposition 3.1(iii) and Proposition 2.2(ii) (Key Lemma), one has for $t < s$,

$$\mathbb{E}_t[1 - H_s^1] = \mathbb{E}[1 - H_s^1 | H_t^1] = (1 - H_t^1) \frac{\mathbb{P}(\tau_1 > s)}{\mathbb{P}(\tau_1 > t)} = (1 - H_t^1) e^{-\int_t^s q_1(x)dx}. \quad \square$$

Proposition 3.4. *One has, for $t \in [0, T]$, (cf. (13), (15), (16)),*

$$\text{EPE}(t) = \left((1 - R_2)(1 - R_1) \frac{l_3(t)}{q_2(t)} \right. \quad (37)$$

$$\left. + (v(t) - (R_2 \widetilde{\chi}^+(t) - \widetilde{\chi}^-(t))) \frac{l_2(t)}{q_2(t)} \right) e^{-\int_0^t l_1(x)dx}$$

$$\text{CVA}(t) = \int_t^T \beta_s ((1 - R_2)(1 - R_1)l_3(s) \quad (38)$$

$$+ (v(s) - (R_2 \widetilde{\chi}^+(s) - \widetilde{\chi}^-(s))) l_2(s)) e^{-\int_t^s l(x)dx} ds$$

which in the special case where $\chi_{(\tau_2)} = P_{\tau_2}$, $\widetilde{\chi} = v$ reduce to

$$\text{EPE}(t) = \text{EPE}^0(t) := \quad (39)$$

$$(1 - R_2) \left((1 - R_1) \frac{l_3(t)}{q_2(t)} + v^+(t) \frac{l_2(t)}{q_2(t)} \right) e^{-\int_0^t l_1(x) dx}$$

$$\text{CVA}(t) = \text{CVA}^0(t) := \quad (40)$$

$$\int_t^T (1 - R_2) \beta_s \left((1 - R_1) l_3(s) + v^+(s) l_2(s) \right) e^{-\int_t^s l(x) dx} ds$$

Proof. Set

$$\Phi(\tau_2) = \mathbb{E}(\mathbb{1}_{\tau_1=\tau_2 \leq T} | \tau_2) , \quad \Psi(\tau_2) = \mathbb{E}(\mathbb{1}_{\tau_2 < \tau_1, \tau_2 \leq T} | \tau_2) ,$$

which are characterized by

$$\mathbb{E}(\Phi(\tau_2) f(\tau_2)) = \mathbb{E}(f(\tau_2) \mathbb{1}_{\tau_1=\tau_2 \leq T}) , \quad (41)$$

$$\mathbb{E}(\Psi(\tau_2) f(\tau_2)) = \mathbb{E}(f(\tau_2) \mathbb{1}_{\tau_2 < \tau_1, \tau_2 \leq T}) ,$$

for every Borel function f . In particular we take $f(x) = \mathbb{1}_{x \leq t}$ for some $t \in (0, T]$.

Now using the law of τ_2 , the left-hand sides of (41) are given by

$$\mathbb{E}(\Phi(\tau_2) \mathbb{1}_{\tau_2 \leq t}) = \int_0^t \Phi(s) q_2(s) e^{-\int_0^s q_2(x) dx} ds$$

$$\mathbb{E}(\Psi(\tau_2) \mathbb{1}_{\tau_2 \leq t}) = \int_0^t \Psi(s) q_2(s) e^{-\int_0^s q_2(x) dx} ds$$

As for the right-hand-sides of (41), thanks to Proposition 3.1(i) and (iv), one has

$$\begin{aligned} \mathbb{E}(\mathbb{1}_{\tau_2 \leq t} \mathbb{1}_{\tau_1=\tau_2 \leq T}) &= \mathbb{E}\left(\int_0^t dH_s^{\{1,2\}}\right) \\ &= \int_0^t \mathbb{E}((1 - H_s^1)(1 - H_s^2)) l_3(s) ds = \int_0^t e^{-\int_0^s l(x) dx} l_3(s) ds , \end{aligned}$$

and

$$\begin{aligned} \mathbb{E}(\mathbb{1}_{\tau_2 \leq t} \mathbb{1}_{\tau_2 < \tau_1, \tau_2 \leq T}) &= \mathbb{E}\left(\int_0^t \mathbb{1}_{s \leq \tau_1 \wedge T} dH_s^{\{2\}}\right) = \mathbb{E}\left(\int_0^t \mathbb{1}_{s \leq \tau_1} q_{\{2\}}(s, H_s) ds\right) \\ &= \mathbb{E}\left(\int_0^t (1 - H_s^1)(1 - H_s^2) l_2(s) ds\right) = \int_0^t e^{-\int_0^s l(x) dx} l_2(s) ds , \end{aligned}$$

where the second identity in the first line uses that $H^{\{2\}}$ does not jump at τ_1 .

Thus for $f(x) = \mathbb{1}_{x \leq t}$ the identities in (41) can be rewritten as

$$\begin{aligned} \int_0^t \Phi(s) q_2(s) e^{-\int_0^s q_2(x) dx} ds &= \int_0^t l_3(s) e^{-\int_0^s l(x) dx} ds, \\ \int_0^t \Psi(s) q_2(s) e^{-\int_0^s q_2(x) dx} ds &= \int_0^t l_2(s) e^{-\int_0^s l(x) dx} ds. \end{aligned}$$

Taking derivative with respect to t of these last equations leads us to

$$\Phi(t) = \frac{l_3(t) e^{-\int_0^t l(x) dx}}{q_2(t)} e^{\int_0^t q_2(x) dx}, \quad \Psi(t) = \frac{l_2(t) e^{-\int_0^t l(x) dx}}{q_2(t)} e^{\int_0^t q_2(x) dx}$$

and (37) follows.

Using (16), one then has for $t \in [0, T]$,

$$\begin{aligned} \beta_t \text{CVA}(t) &= \int_t^T \beta_s \text{EPE}(s) e^{\int_0^s l(x) dx} e^{-\int_0^s q_2(x) dx} q_2(s) e^{-\int_t^s l(x) dx} ds \\ &= \int_t^T \beta_s \text{EPE}(s) e^{\int_0^s l_1(x) dx} q_2(s) e^{-\int_t^s l(x) dx} ds. \end{aligned}$$

Hence (38) follows from (37). $\square$

Remark 3.6. In view of the option-theoretic interpretation of the CVA, the CVA valuation formula (38) can also be established directly, without passing by the EPE, much like formula (29) in Proposition 3.2 above (using a probabilistic computation, or resorting to the related Kolmogorov pricing ODE).

3.3 Hedging

We now give few preliminary results about hedging the risky CDS. We shall mainly consider the issue of delta-hedging, at least partially, the risky CDS, by a risk-free CDS which would also be available on the market (CDS on the firm with the same characteristics, except for the counterparty credit risk). Another perspective on the counterparty credit risk of the risky CDS can thus be given by assessing to which extent the risky CDS could, in principle, be hedged by the risk-free CDS.

3.3.1 Price dynamics

Let $\widehat{\Pi}$ denote the discounted cum-dividend price of the risky CDS, that is, the local martingale

$$\widehat{\Pi}_t = \beta_t \Pi_t + \pi_t(0).$$

The Itô formula applied to $\Pi_t = \Pi(t, H_t)$ yields, on $[0, \tau_1 \wedge \tau_2 \wedge T]$,

$$d\widehat{\Pi}_t = \beta_t (\delta \Pi_{\{1\}}(t) dM_t^{[1]} + \delta \Pi_{\{2\}}(t) dM_t^{[2]} + \delta \Pi_{\{1,2\}}(t) dM_t^{[1,2]}) \quad (42)$$

with

$$\begin{aligned}\delta\Pi_{[1]}(t) &= 1 - R_1 - u(t), \quad \delta\Pi_{[2]}(t) = R_2\widetilde{\chi}^+(t) - \widetilde{\chi}^-(t) - u(t), \\ \delta\Pi_{[1,2]}(t) &= R_2(1 - R_1) - u(t).\end{aligned}$$

Similarly, setting

$$\widehat{P}_t = \beta_t P_t + p_t(0),$$

it follows that,

$$d\widehat{P}_t = \beta_t \delta P_1(t) dM_t^1 \quad (43)$$

with

$$\delta P_1(t) = 1 - R_1 - v(t).$$

3.3.2 Min-variance hedging

Let us denote by ψ a (self-financing) strategy in the risk-free CDS with price process P (and the savings account β_t^{-1}) for tentatively hedging the risky CDS with price process Π .

Recall that $\mathbb{P}$ is the risk neutral probability chosen by market. So the discounted cum-dividend price process $\widehat{P}$ is a $\mathbb{P}$ -local martingale (actually in view of (43) $\widehat{P}$ is here a $\mathbb{P}$ -martingale). As a result of the Galtchouk-Kunita-Watanabe decomposition, the hedging strategy ψ^{va} which minimizes the $\mathbb{P}$ -variance of the hedging error, or *min-variance hedging strategy*, is given by

$$\psi_t^{va} = \frac{d\langle \widehat{\Pi}, \widehat{P} \rangle_t}{d\langle \widehat{P} \rangle_t}.$$

Remark 3.7. Note that we only deal with minimization of the risk-neutral variance of the hedging error, here, as opposed to the more difficult problem of minimizing the variance of the hedging error under the historical probability measure.

In view of the price dynamics (42)–(43), one has, for $t \leq \tau_1 \wedge \tau_2$,

$$\frac{d\langle \widehat{\Pi}, \widehat{P} \rangle_t}{d\langle \widehat{P} \rangle_t} = \frac{l_1(t)(\delta\Pi_{[1]}(t))(\delta P_1(t)) + l_3(t)(\delta\Pi_{[1,2]}(t))(\delta P_1(t))}{q_1(t)(\delta P_1(t))^2}.$$

So

$$\psi_t^{va} = \frac{l_1(t)}{q_1(t)} \frac{1 - R_1 - u(t)}{1 - R_1 - v(t)} + \frac{l_3(t)}{q_1(t)} \frac{R_2(1 - R_1) - u(t)}{1 - R_1 - v(t)}$$

on $[0, \tau^1 \wedge \tau^2 \wedge T]$ (and $\psi^{va} = 0$ on $(\tau^1 \wedge \tau^2 \wedge T, T]$). The related min-variance hedging reduction factor writes:

$$\begin{aligned}\frac{\text{Var}(\widehat{\Pi}_T)}{\text{Var}(\widehat{\Pi}_T - \int_0^T \psi_t^{va} d\widehat{P}_t)} &= \\ &= \frac{\text{Var}(\widehat{\Pi}_T)}{\text{Var}(\widehat{\Pi}_T) + \text{Var}(\int_0^T \psi_t^{va} d\widehat{P}_t) - 2\text{Cov}(\widehat{\Pi}_T, \int_0^T \psi_t^{va} d\widehat{P}_t)},\end{aligned} \quad (44)$$

where:

$$\begin{aligned}
 \mathbb{V}\text{ar}(\widehat{\Pi}_T) &= \mathbb{E}\langle \widehat{\Pi} \rangle_T = \mathbb{E} \int_0^{\tau_1 \wedge \tau_2 \wedge T} \left((l_1(t)(\delta\Pi_{\{1\}}(t))^2 \right. \\
 &\quad \left. + l_2(t)(\delta\Pi_{\{2\}}(t))^2 + l_3(t)(\delta\Pi_{\{1,2\}}(t))^2 \right) dt , \\
 \mathbb{V}\text{ar}\left(\int_0^T \psi_t^{va} d\widehat{P}_t\right) &= \mathbb{E}\left\langle \int_0^T \psi_t^{va} d\widehat{P}_t \right\rangle_T \\
 &= \mathbb{E} \int_0^{\tau_1 \wedge \tau_2 \wedge T} q_1(t)(\psi_t^{va} \delta P_1(t))^2 dt , \\
 \text{Cov}(\widehat{\Pi}_T, \int_0^T \psi_t^{va} d\widehat{P}_t) &= \mathbb{E}\langle \widehat{\Pi}, \int_0^T \psi_t^{va} d\widehat{P}_t \rangle_T \\
 &= \mathbb{E} \int_0^{\tau_1 \wedge \tau_2 \wedge T} (l_1(t)\delta\Pi_{\{1\}}(t) + l_3(t)\delta\Pi_{\{1,2\}}(t)) \psi_t^{va} \delta P_1(t) dt .
 \end{aligned} \tag{45}$$

The various quantities that arise in (45), and therefore the hedging reduction factor given by (44), can be computed by Monte Carlo simulation.

Remark 3.8. The previous min-variance hedging strategy can be easily extended to multi-instrument hedging schemes. In case three non-redundant hedging instruments are available, then, in view of (42), the risky CDS can be perfectly replicated.

4. Implementation

4.1 Affine Intensities Model Specification

Note that the Markov chain copula model primitives are the marginal pre-default intensity functions q_1 and q_2 as well as the ‘dependence intensity function’ l_3 in $A(t)$ (cf. (17)).

Let us specify, for constants a_i ’s and b_i ’s,

$$q_i(t) = a_i + b_i t , \quad l_3(t) = a_3 + b_3 t , \tag{46}$$

with

$$a_3 = \alpha \min\{a_1, a_2\} , \quad b_3 = \alpha \min\{b_1, b_2\} ,$$

for a *model dependence parameter* $\alpha \in [0, 1]$ (for the sake of Assumption 3.2).

Remark 4.1. Such an affine specification of intensities was already used by Bi-
elecki *et al.* [2] in a context of CDO modeling.

It is immediate to check that under (46), the spread κ_i of a risk-free CDS on

name i is given by

$$\kappa_i = (1 - R_i) \frac{\int_0^T \beta_t (a_i + b_i t) \exp(-a_i t - \frac{b_i}{2} t^2) dt}{\int_0^T \beta_t \exp(-a_i t - \frac{b_i}{2} t^2) dt}. \quad (47)$$

Also note that one has, by Proposition 3.1(v),

$$\rho_d := \rho_d(T) = \frac{e^{a_3 T + b_3 T^2/2} - 1}{\sqrt{(e^{a_1 T + b_1 T^2/2} - 1)(e^{a_2 T + b_2 T^2/2} - 1)}}, \quad (48)$$

or, equivalently,

$$\alpha = \frac{\ln \left(1 + \rho_d \sqrt{(e^{a_1 T + b_1 T^2/2} - 1)(e^{a_2 T + b_2 T^2/2} - 1)} \right)}{aT + bT^2/2} \quad (49)$$

where $a = \min\{a_1, a_2\}$ and $b = \min\{b_1, b_2\}$.

4.1.1 Calibration issues

Using (47), the a_i 's and b_i 's can be calibrated independently in a straightforward way to the market CDS curves of the firm and the counterpart, respectively. Note in this regard that market CDS curves can be considered as 'risk-free CDS curves'.

As for the model dependence parameter α , in case the market price of an instrument sensitive to the dependence structure of default times (basket credit instrument on the firm and the counterpart) is available, one can use it to calibrate α . Admittedly however, this situation is an exception rather than the rule. It is thus important to devise a practical way of setting α in case such a market data is not available. A possible procedure¹ thus consists in 'calibrating' α to a target value for the model probability $p_{1,2}(T) = \mathbb{P}(H_T^1 = H_T^2 = 1)$. A target value for $p_{1,2}(T)$ can be obtained by plugging a standard static Gaussian copula *asset correlation* ρ into a bivariate normal distribution function, so

$$p_{1,2}(T) = \mathcal{N}_2^\rho \left(\mathcal{N}_1^{-1}(p_1(T)), \mathcal{N}_1^{-1}(p_2(T)) \right), \quad (50)$$

where:

- $\mathcal{N}_1$ denotes the standard Gaussian c.d.f.,
- $\mathcal{N}_2^\rho$ denotes a bivariate centered Gaussian c.d.f. with unit variances and correlation coefficients ρ ,

¹We thank J.-P. Lardy for suggesting this procedure.

- $p_i(T) = \mathbb{P}(H_T^i = 1)$ for $i = 1, 2$.

Regulatory capital requirements being based on the Vasicek formula, such a static copula correlation ρ can be retrieved from the Basel II correlations per asset class (cf. [1]).

4.1.2 Special case of constant intensities

We now look at a particular case in which $b_1 = b_2 = b_3 = 0$. This case will be referred to henceforth as the case of *constant intensities*, as opposed to the more general case of *affine intensities* introduced in subsection 4.1. In the case of constant intensities, one has,

$$q_1(t) = a_1, \quad q_2(t) = a_2, \quad l_3(t) = a_3.$$

The correlation coefficient ρ_d in (48) simplifies to

$$\rho_d = \frac{e^{a_3 T} - 1}{\sqrt{(e^{a_1 T} - 1)(e^{a_2 T} - 1)}}$$

from which a_3 can be calculated as

$$a_3 = \frac{1}{T} \ln \left(1 + \rho_d \sqrt{(e^{a_1 T} - 1)(e^{a_2 T} - 1)} \right).$$

As is well known, the price of a risk-free CDS in a constant intensity model is null, i.e., $v(t) \equiv 0$ when $b_1 = 0$. So the EPE formula (37) simplifies to

$$\text{EPE}(t) = (1 - R_1)(1 - R_2) \frac{a_3}{a_2} e^{-(a_1 - a_3)t}.$$

Also in this case, the pricing formula (29) for the risky CDS reduces to (assuming here $r(t) = r$),

$$u(t) = -(1 - R_1)(1 - R_2)a_3 \frac{1 - e^{-(r + a_1 + a_2 - a_3)(T-t)}}{r + a_1 + a_2 - a_3}.$$

Finally, from Proposition 2.1, one gets,

$$\text{CVA}(t) = -u(t).$$

In particular, for low values of the coefficients,

$$\text{CVA}(0) \simeq (1 - R_1)(1 - R_2)a_3 T = (1 - R_1)(1 - R_2) \ln \left[1 + \rho_d \sqrt{(e^{a_1 T} - 1)(e^{a_2 T} - 1)} \right],$$

so, finally,

$$\text{CVA}(0) \simeq (1 - R_1)(1 - R_2) \sqrt{a_1 a_2} T \rho_d. \quad (51)$$

4.2 Numerical Results

Our aim is to assess by means of numerical experiments the impact of ρ (the asset correlation between the firm and the counterpart, cf. (50)) on one hand, and of κ_2 (the risk-free CDS fair spread of the counterparty as of (47)) on the other hand, on the counterparty risk exposure of the investor.

Towards this end we fix the general data of Table 4.2 (case with affine intensities) or 4.2 (case with constant intensities, all b 's equal to 0), and we further consider twelve alternative sets of values for a_2 , b_2 , and ρ given in columns one, two and four of Table 4.2 (case with affine intensities), resp. for a_2 and ρ given in columns one and three of Table 4.2 (case with constant intensities).

Table 1. Fixed Data — Affine Intensities.

r	R_1	R_2	T	a_1	b_1	κ_1
5%	40%	40%	10 years	.0095	.0010	84 bp

In the case of affine intensities the corresponding spreads κ_2 at time 0, default correlation ρ_d , model dependence parameter α and joint default probabilities $p_{1,2} = \mathbb{P}(H_T^1 = H_T^2 = 1)$ are displayed respectively in the third, fifth, sixth and seventh column of Table 4.2, whereas the last column of Table 4.2 (which will be commented later in the text) gives the corresponding CVA's at time 0. The risky and risk-free CDS pricing functions u and v corresponding to each of our twelve sets of parameters are displayed in Figures 4.2 and 4.2. On each graph three curves are represented (see Remark 3.5):

- $v(t)$ (dashed blue curve),
- $u(t)$ with $\widetilde{\chi} = v$ therein, denoted by $u^0(t)$ (doted red curve),
- $u(t)$ with $\widetilde{\chi} = u$ therein, denoted by $u^1(t)$ (black curve).

The analogous results in the case of constant intensities are displayed in Table 4.2 and Figures 4.2 and 4.2. Note that on each graph in Figures 4.2 and 4.2 the function v is equal to 0, as must be in the case of constant intensities.

In all the cases u^0 and u^1 are rather close to each other, and one can check numerically that using either one makes little difference regarding the related EPES and CVAs. We present henceforth the results for $u = u^0$.

Figures 4.2, 4.2 and 4.2 show the graphs of the Expected Positive Exposure as a function of time, of the Credit valuation Adjustment as a function of time, and of the Credit Valuation Adjustment at time 0 as a function of ρ , in the cases of affine (left graphs) or constant (right graphs) intensities.

Table 2. Variable Data — Affine Intensities.

a_2	b_2	κ_2	ρ	ρ_d	α	$p_{1,2}$	CVA(0)
.0056	.0006	50 bp	10%	.0378	.0520	.0147	.0013
.0085	.0009	75 bp	10%	.0418	.0472	.0211	.0018
.0122	.0010	100 bp	10%	.0444	.0522	.0269	.0021
.0189	.0014	150 bp	10%	.0476	.0702	.0376	.0028
.0056	.0006	50 bp	40%	.1859	.2531	.0286	.0056
.0085	.0009	75 bp	40%	.1998	.2230	.0388	.0074
.0122	.0010	100 bp	40%	.2074	.2406	.0472	.0087
.0189	.0014	150 bp	40%	.2145	.3107	.0616	.0110
.0056	.0006	50 bp	70%	.4020	.5406	.0489	.0119
.0085	.0009	75 bp	70%	.4256	.4673	.0640	.0153
.0122	.0010	100 bp	70%	.4336	.4937	.0754	.0178
.0189	.0014	150 bp	70%	.4306	.6100	.0925	.0214

Table 3. Fixed Data — Constant Intensities.

r	R_1	R_2	T	a_1	κ_1
5%	40%	40%	10 years	.0140	84 bp

Table 4. Variable Data — Constant Intensities.

a_2	κ_2	ρ	ρ_d	α	$p_{1,2}$	CVA(0)
.0083	50 bp	10%	.0372	.0510	.0138	.0011
.0125	75 bp	10%	.0411	.0464	.0198	.0015
.0167	100 bp	10%	.0438	.0515	.0254	.0018
.0250	150 bp	10%	.0470	.0690	.0355	.0023
.0083	50 bp	40%	.1839	.2501	.0272	.0054
.0125	75 bp	40%	.1977	.2207	.0368	.0070
.0167	100 bp	40%	.2056	.2387	.0451	.0084
.0250	150 bp	40%	.2128	.3073	.0587	.0104
.0083	50 bp	70%	.3998	.5372	.0469	.0117
.0125	75 bp	70%	.4231	.4650	.0613	.0150
.0167	100 bp	70%	.4315	.4921	.0726	.0175
.0250	150 bp	70%	.4288	.6063	.0889	.0210

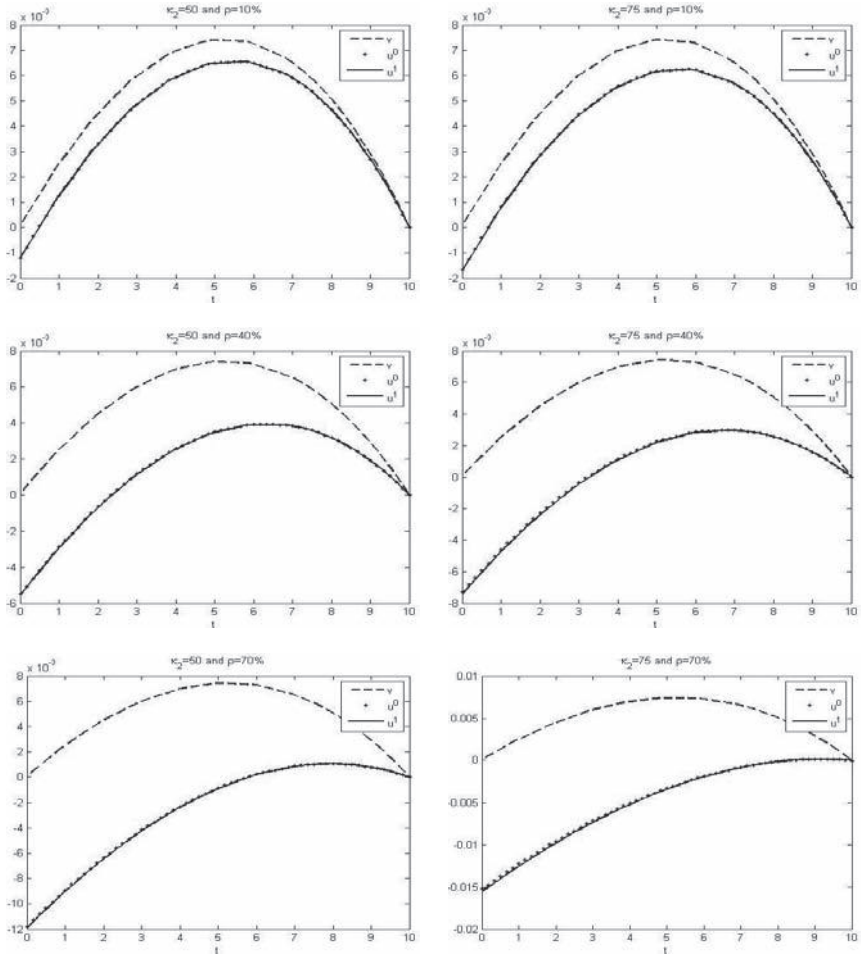


Figure 1. Pricing functions in the case of affine intensities — $v(t)$ (dashed curve), $u^0(t)$ (dotted curve) and $u^1(t)$ (line).

One can see on Figure 4.2 the impact on the counterparty risk exposure of the investor of the default risk (as measured by the risk-free spread κ_2) of the counterparty. On each graph the asset correlation ρ is fixed, with from top to down $\rho = 10\%$, 40% and 70% . The four curves on each graph of Figure 4.2 correspond to EPE(t) for $\kappa_2 = 50, 75, 100$ and 150 bps. Observe that as κ_2 decreases the counterparty risk exposure increases. This is in line with the stylized features and the financial intuition regarding the EPE: EPE(t) is the expectation of the investor's loss, given the default of the counterparty at time t . A default of

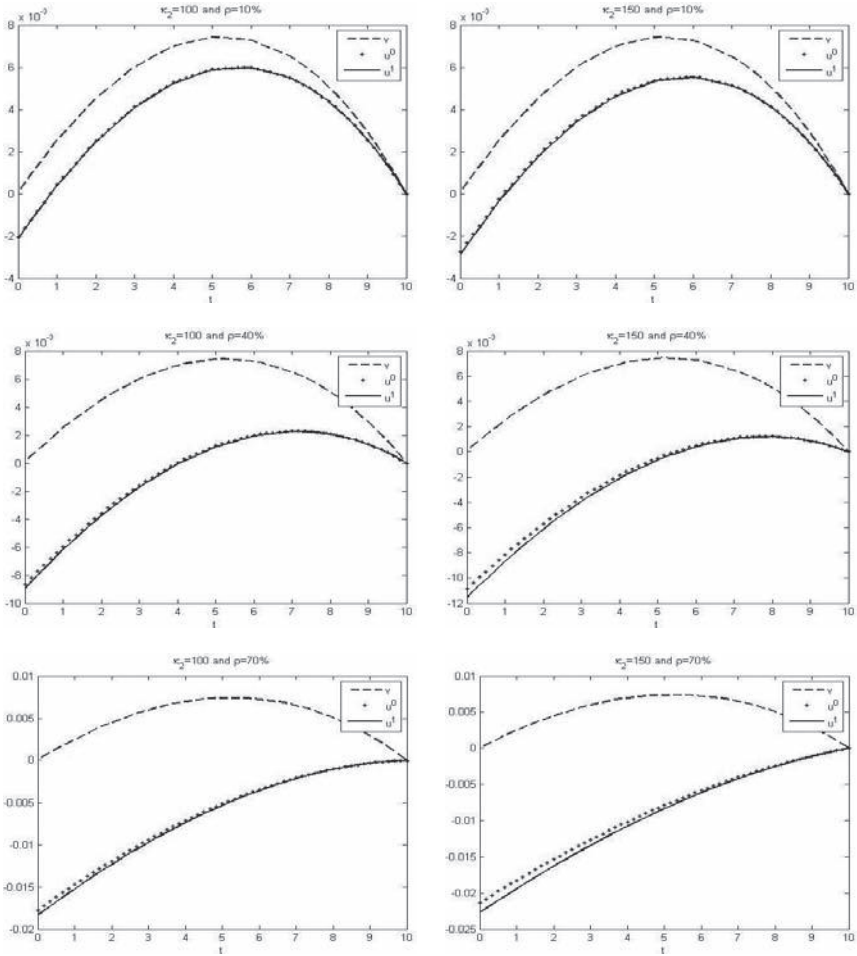


Figure 2. Pricing functions in the case of affine intensities — $v(t)$ (dashed curve), $u^0(t)$ (dotted curve) and $u^1(t)$ (black curve).

a counterpart with a lower spread is interpreted by the markets as a worse news than a default of a counterpart with a higher spread. The related EPE is thus larger.

Figure 4.2 shows the graphs of the Credit Valuation Adjustment as a function of time, for affine (left column) or constant (right column) intensities. One can thus see the impact of κ_2 on the CVA. In each graph the asset correlation ρ is fixed, with from top to down $\rho = 10\%$, 40% and 70% . The four curves on each graph of Figure 4.2 correspond to CVA(t) for $\kappa_2 = 50, 75, 100$ and 150 bps. Observe that

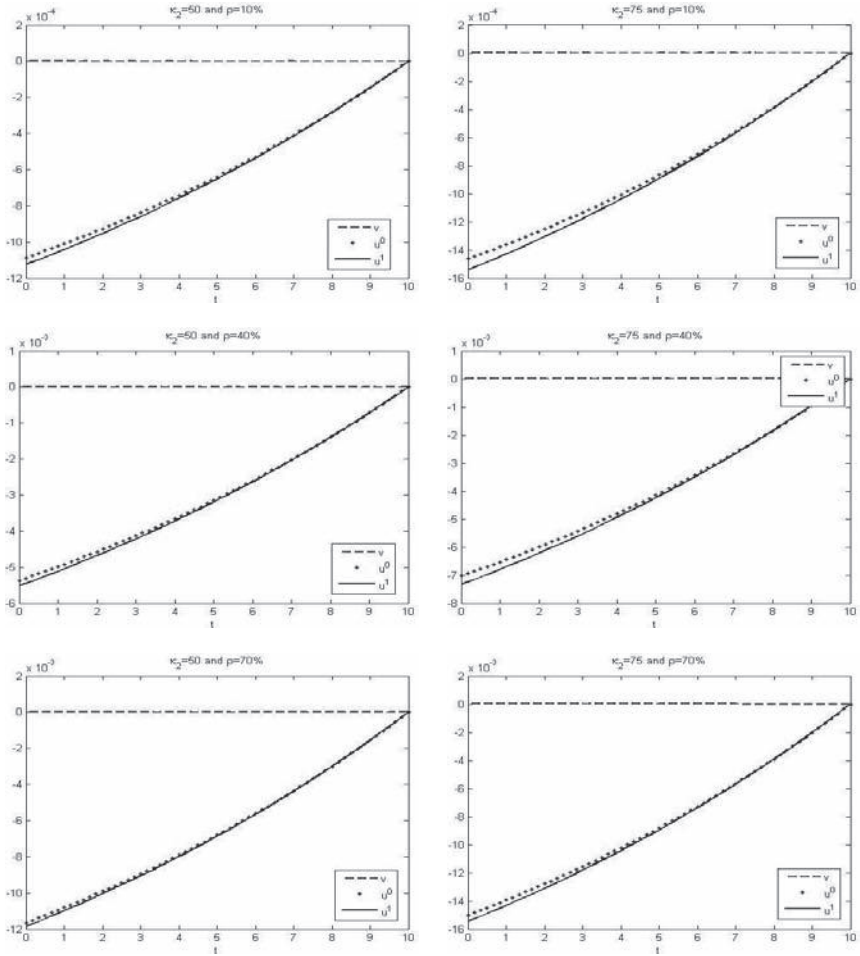


Figure 3. Pricing functions in the case of constant intensities — $v(t)$ (dashed curve), $u^0(t)$ (dotted curve) and $u^1(t)$ (line).

as opposed to the EPE, the CVA is increasing in κ_2 , in line with stylized features. Also note that the CVA is a decreasing function of time, in accordance again with expected features: less time to maturity, less risk.

Finally Figure 4.2 represents the graphs of $CVA(0)$ as a function of the asset correlation ρ for $\kappa_2 = 50, 75, 100$ and 150 bps. Note for comparison that $CVA(0)$ grows essentially linearly in the default correlation ρ_d , at least in the case of constant coefficients (cf. formula (51)).

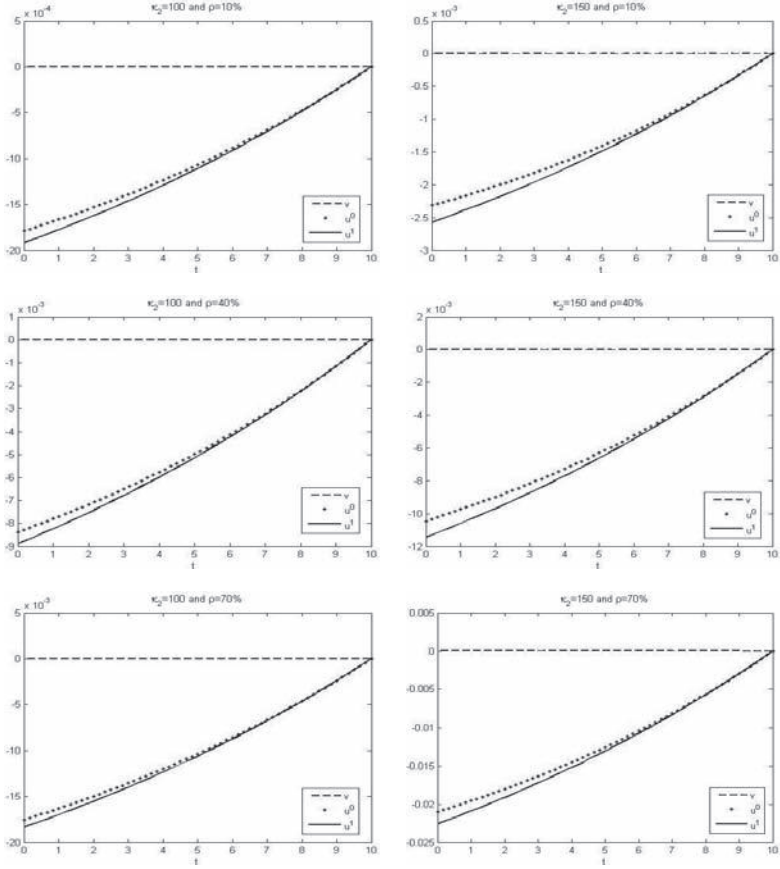


Figure 4. Pricing functions in the case of constant intensities — $v(t)$ (dashed curve), $u^0(t)$ (dotted curve) and $u^1(t)$ (line).

5. Concluding Remarks and Perspectives

In this article we propose a model of CDS with counterparty credit risk, with the following desirable properties:

- Adequacy of the behavior of EPE and CVA in the model with expected features (see Section 4.2),
- Wrong way risk (via joint defaults, specifically),
- Simplicity, since the model is a four-state Markov chain of two credit names, with one-name marginals automatically calibrated to the individual CDS curves,

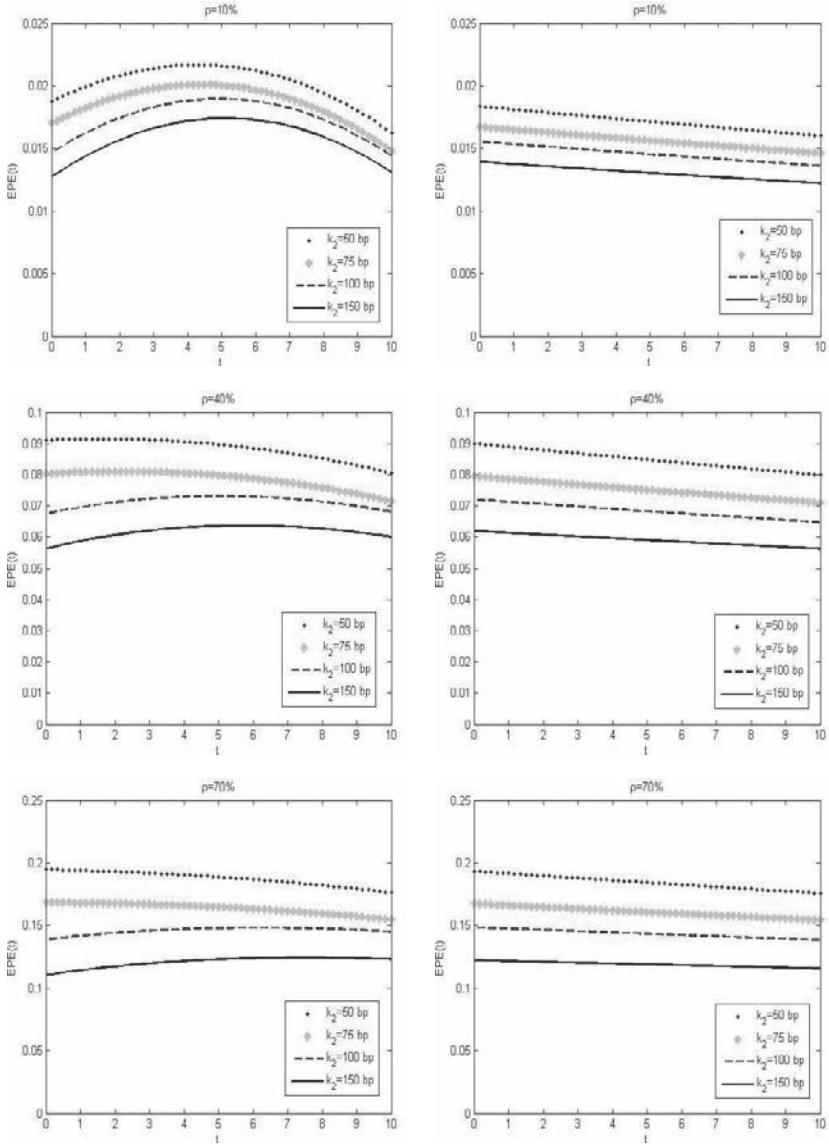


Figure 5. $EPE(t)$ ($\tilde{\chi} = v, u = u^0$). In each graph ρ is fixed. From top to down $\rho = 10\%$, $\rho = 40\%$ and $\rho = 70\%$. Left column: affine intensities. Right column: constant intensities.

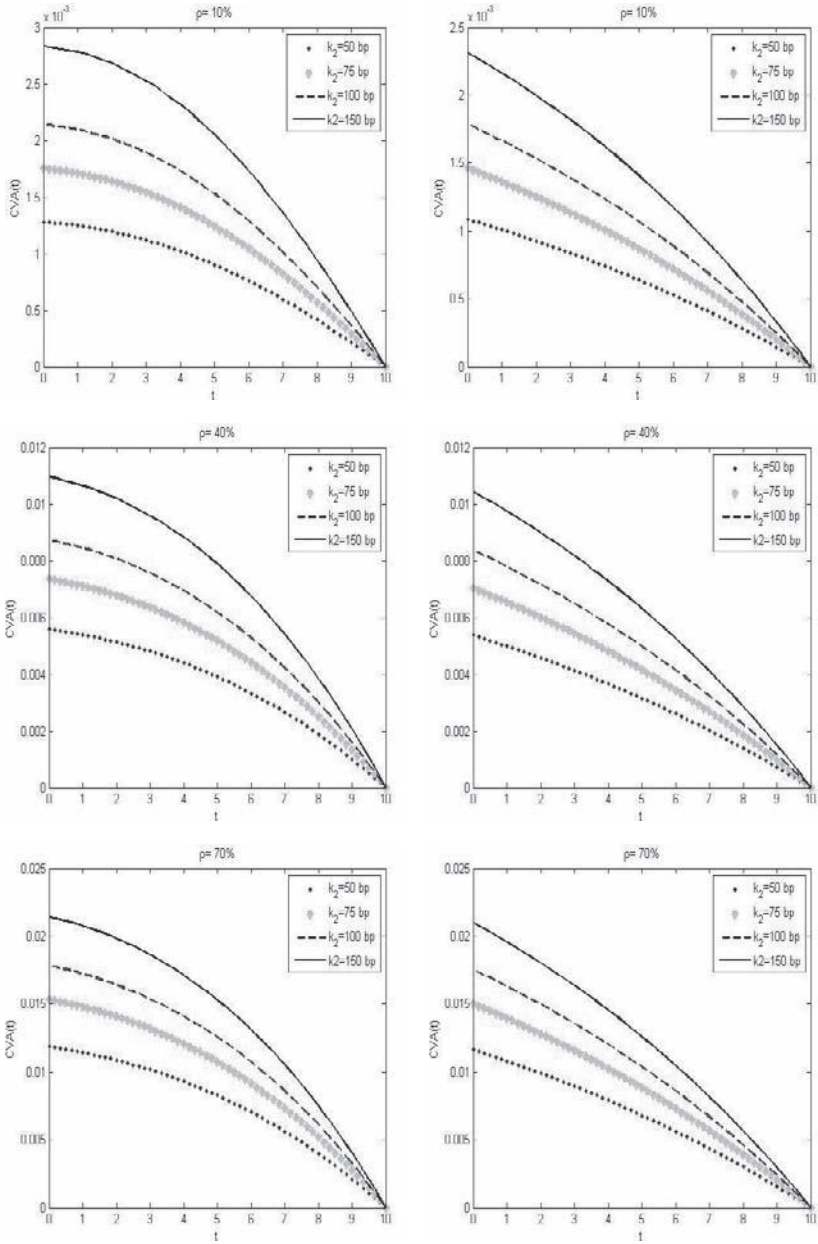


Figure 6. $CVA(t)$ ($\tilde{\chi} = v, u = u^0$). In each graph ρ is fixed. From top to down $\rho = 10\%$, $\rho = 40\%$ and $\rho = 70\%$. Left column: affine intensities. Right column: constant intensities.

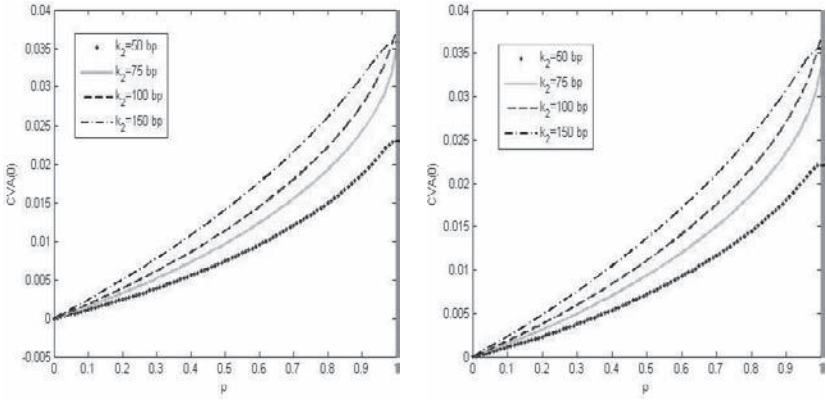


Figure 7. CVA(0) as a function of ρ for $\kappa_2 = 50$ bp, 75 bp, 100 bp and 150 bp ($\bar{\chi} = v, u = u^0$). Left: Affine intensities. Right: Constant intensities.

- Fact, related to the previous one, that the model ‘takes the right inputs to generate the right outputs’, namely it takes as basic inputs the individual default probabilities (individual CDS curves), which correspond to the more reliable information on the market, which are then ‘coupled’ in a suitable way,
- Consistency, in the sense that it is a dynamic model with replication-based valuation and hedging arguments.

The present work might be extended in at least three directions.

First, it would be desirable to add credit spread volatility into the model. This could be achieved by adding a *reference filtration* $\widetilde{\mathbb{F}}$ so that the model filtration $\mathbb{F}$ be given as $\widetilde{\mathbb{F}} \vee \mathbb{H}$, and the intensities l, q are non-negative $\widetilde{\mathbb{F}}$ -adapted processes. A second related issue is that of merging the CDS-CVA pricing tool of this paper into a more general, real-life CVA engine, including the following features:

- Netting, that is, aggregation in a suitable way of all the contracts (as opposed to only one CDS in this paper) relative to a given counterpart,
- Market (other than credit) risk factors,
- Margin agreements.

Finally, at the stage of implementation (see, e.g., Zhu and Pykhtin [26]), such real-life CVA engines pose interesting challenges from the numerical point of view of Monte Carlo simulations.

Appendix. Proof of Proposition 3.1

We shall need the following (essentially classic) Lemma.

Lemma 5.1. *Let X be a right-continuous process with a finite state space $\mathcal{E}$ and adapted to some filtration $\mathbb{F}$. Condition (i), (ii) or (iii) below are necessary and sufficient conditions for X to be an $\mathbb{F}$ – Markov chain with infinitesimal generator $\mathcal{A}(t) = \mathcal{A}_t = [\mathcal{A}_t^{i,j}]_{i,j \in \mathcal{E}}$:*

(i) *For every function h over $\mathcal{E}$,*

$$\mathcal{M}_t^h = h(X_t) - \int_0^t (\mathcal{A}_s h)(X_s) ds \quad (52)$$

is an $\mathbb{F}$ – local martingale;

(ii) *For every $j \in \mathcal{E}$, the process $\mathcal{M}^j$ defined by*

$$\mathcal{M}_t^j = \mathbb{1}_{X_t=j} - \int_0^t \mathcal{A}_s^{X_s,j} ds$$

is an $\mathbb{F}$ – local martingale;

(iii) *For every $i, j \in \mathcal{E}$ the process $\mathcal{M}^{i,j}$ given by*

$$\mathcal{M}_t^{i,j} = \mathbb{1}_{X_t=i, X_t=j} - \int_0^t \mathbb{1}_{X_s=i} \mathcal{A}_s^{i,j} ds$$

is an $\mathbb{F}$ – local martingale.

Proof. (i) is the usual local martingale characterization of Markov chains (see, e.g., Proposition 11.2.2 in Bielecki and Rutkowski [4]).

(ii) Since $\mathcal{E}$ is finite, the set of the indicator functions $\mathbb{1}_{\cdot=j}$ spans linearly the set of all functions over $\mathcal{E}$. The condition of part (ii) is thus equivalent to that of (i).

(iii) Necessity follows by combination of Proposition 11.2.2 and Lemma 11.2.3 in [4]. As for sufficiency, note that the $\mathcal{M}^{i,j}$'s being $\mathbb{F}$ – local martingales implies the same property for the $\mathcal{M}^j$'s in (ii), by summation over i . We thus conclude by the sufficiency in part (ii). $\square$

Let us proceed with the proof of Proposition 3.1. First, note the processes H^I can also be written as

$$H_t^{(1)} = \sum_{0 < s \leq t} \mathbb{1}_{\Delta H_s = (1,0)}, \quad H_t^{(2)} = \sum_{0 < s \leq t} \mathbb{1}_{\Delta H_s = (0,1)}, \quad H_t^{(1,2)} = \sum_{0 < s \leq t} \mathbb{1}_{\Delta H_s = (1,1)}.$$

(i) Let us verify that the M^I 's in (20) are $\mathbb{H}$ – local martingales. As bounded $\mathbb{H}$ – local martingales, $M^{(1)}$, $M^{(2)}$ and $M^{(1,2)}$ will thus be $\mathbb{H}$ -martingales. For $I = \{1, 2\}$,

one has,

$$\begin{aligned}
 M_t^{(1,2)} &= H_t^{(1,2)} - \int_0^t q_{\{1,2\}}(s, H_s) ds \\
 &= \sum_{0 < s \leq t} \mathbb{1}_{\Delta H_s = (1,1)} - \int_0^t \mathbb{1}_{H_s = (0,0)} l_3(s) ds \\
 &= \sum_{0 < s \leq t} \mathbb{1}_{H_{s-} = (0,0), H_s = (1,1)} - \int_0^t \mathbb{1}_{H_s = (0,0)} l_3(s) ds .
 \end{aligned}$$

Thus Lemma 5.1 with $i = (0, 0)$ and $j = (1, 1)$, implies the local martingale property of $M^{(1,2)}$.

For $M^{(1)}$, one has,

$$\begin{aligned}
 M_t^{(1)} &= H_t^{(1)} - \int_0^t q_{\{1\}}(s, H_s) ds \\
 &= \sum_{0 < s \leq t} \mathbb{1}_{\Delta H_s = (1,0)} - \int_0^t \mathbb{1}_{H_s^1 = 0} \left[\mathbb{1}_{H_s^2 = 0} l_1(s) + \mathbb{1}_{H_s^2 = 1} q_1(s) \right] ds \\
 &= \left\{ \sum_{0 < s \leq t} \mathbb{1}_{H_{s-} = (0,0), H_s = (1,0)} - \int_0^t \mathbb{1}_{H_s = (0,0)} l_1(s) ds \right\} \\
 &\quad + \left\{ \sum_{0 < s \leq t} \mathbb{1}_{H_{s-} = (0,1), H_s = (1,1)} - \int_0^t \mathbb{1}_{H_s = (0,1)} q_1(s) ds \right\} .
 \end{aligned}$$

Now we apply Lemma 5.1 to the two terms in the last equation, with $i = (0, 0)$ and $j = (1, 0)$ for the first term and $i = (0, 1)$ and $j = (1, 1)$ for the second term. Thus $M^{(1)}$ being the sum of two $\mathbb{H}$ – local martingales is an $\mathbb{H}$ – local martingale. In the same way, $M^{(2)}$ is an $\mathbb{H}$ – local martingale. As bounded $\mathbb{H}$ – local martingales, $M^{(1)}$, $M^{(2)}$ and $M^{(1,2)}$ are thus $\mathbb{H}$ –martingales.

(ii) As $q_i = l_i + l_3$ and $H^i = H^{(i)} + H^{(1,2)}$, one has $M^i = M^{(i)} + M^{(1,2)}$, so the M^i 's are in turn $\mathbb{H}$ -martingales.

(iii) Since the M^i 's are $\mathbb{H}$ -martingales, this follows easily from the sufficiency in Lemma 5.1(ii).

(iv) Formulas (24) follow directly from (23), in which we shall now show the first identity. One has for $t > s$ (see the end of the proof of Proposition 3.3),

$$\mathbb{P}(\tau_2 > t | \mathcal{H}_s) = \mathbb{P}(\tau_2 > t | H_s^2) = (1 - H_s^2) e^{-\int_s^t q_2(u) du} .$$

Thus

$$\begin{aligned}\mathbb{P}(\tau_1 > s, \tau_2 > t) &= \mathbb{E}(\mathbb{1}_{\tau_1 > s} \mathbb{E}(\mathbb{1}_{\tau_2 > t} | \mathcal{H}_s)) \\ &= \mathbb{E} \left\{ (1 - H_s^1)(1 - H_s^2) e^{-\int_s^t q_2(u) du} \right\},\end{aligned}$$

and the result follows.

(v) Since H_t^i is a Bernoulli random variable with (cf. Proposition 3.1(iv))

$$\mathbb{P}(H_t^i = 1) = \mathbb{P}(\tau_i \leq t) = 1 - \exp\left(-\int_0^t q_i(s) ds\right) := p_i(t),$$

one has

$$\mathbb{V}\text{ar}(H_t^i) = p_i(t)(1 - p_i(t))$$

Also

$$\begin{aligned}\mathbb{C}\text{ov}(H_t^1, H_t^2) &= \mathbb{C}\text{ov}(1 - H_t^1, 1 - H_t^2) \\ &= \mathbb{E} \left[(1 - H_t^1)(1 - H_t^2) \right] - \mathbb{E}(1 - H_t^1) \mathbb{E}(1 - H_t^2) \\ &= \mathbb{P}(\tau_1 > t, \tau_2 > t) - \mathbb{P}(\tau_1 > t) \mathbb{P}(\tau_2 > t) \\ &= \exp\left(-\int_0^t l(s) ds\right) - \exp\left(-\int_0^t q_1(s) ds\right) \exp\left(-\int_0^t q_2(s) ds\right).\end{aligned}$$

Thus, after some algebraic simplifications,

$$\rho_d(t) = \frac{\mathbb{C}\text{ov}(H_t^1, H_t^2)}{\sqrt{\mathbb{V}\text{ar}(H_t^1) \mathbb{V}\text{ar}(H_t^2)}} = \frac{\exp\left(\int_0^t l_3(s) ds\right) - 1}{\sqrt{\left(\exp\left(\int_0^t q_1(s) ds\right) - 1\right) \left(\exp\left(\int_0^t q_2(s) ds\right) - 1\right)}}.$$

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Portfolio Efficiency Under Heterogeneous Beliefs*

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In the standard mean variance (MV) capital asset pricing model (CAPM) with homogeneous beliefs, the optimal portfolios of investors are MV efficient. It is expected that this is no longer true in general when investors have heterogeneous beliefs in the means and variances/covariances of asset returns. This paper extends the standard Black's zero-beta CAPM to incorporate heterogeneous beliefs and verifies that the subjectively optimal portfolios of heterogeneous investors are MV inefficient in general. The paper then demonstrates that the traditional geometric relation of the mean variance frontiers with and without the riskless asset under homogeneous beliefs does not hold in general under heterogeneous beliefs. The paper further examines the impact of biased beliefs among investors on the MV efficiency of their optimal portfolios. The results provide some explanations on the risk premium puzzle, Miller's hypothesis, and under-performance of managed funds.

Keywords: Asset prices, mean-variance efficiency, heterogeneous beliefs, zero-beta CAPM.

1. Introduction

The Capital Asset Pricing Model (CAPM) developed by Sharpe [28], Lintner [22] and Mossin [25] is perhaps the most influential equilibrium model in modern

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finance. It is based on the assumptions that investors have homogeneous beliefs in the means and variances/covariances of risky assets and there is unrestricted borrowing and lending of a risk-free asset. To relax these unrealistic assumptions, Lintner [23] extends the CAPM by incorporating heterogeneous beliefs among investors. To provide a theoretical explanation on the early empirical tests of the CAPM, Black [2] removes the risk-free asset and develops the well-known zero-beta CAPM. Since then, equilibrium models have been developed in the literature to examine the impact of heterogeneity amongst investors on market equilibrium¹. Assuming that investors are bounded rational, heterogeneity may be caused by difference in information or difference in opinion².

In the mean-variance (MV) literature, the impact of heterogeneous beliefs is mostly studied for the case of a portfolio of one risky asset and one risk-free asset. Lintner [23] is the first considering the CAPM with heterogeneous beliefs and without risk-free asset and shows that heterogeneity does not change the structure of capital asset prices in any significant way, and removing risk-free asset is just a mere extension of the case with a risk-free asset. Surprisingly, this significant contribution from Lintner has not been paid much attention until recent years³. The main obstacle in dealing with heterogeneity is the complexity and heavy notation involved when the number of assets and the dimension of the heterogeneity increase. It might be due to this notational obstacle that makes the paper of Lintner hard to follow, and renders rather complicated analysis of the impact of heterogeneity on the market equilibrium prices. Recently, Sun and Yang [30] provide conditions for the existence of the market equilibrium and have shown that the zero-beta CAPM still holds under heterogeneous beliefs within the MV framework. However, they do not provide the market equilibrium price and examine the impact of heterogeneity on the market equilibrium price, including MV efficiency of the optimal portfolios of heterogeneous investors. When investors have heterogeneous beliefs in the means and variances/covariances of asset returns, it is expected in general that the subjectively optimal portfolios are no longer MV efficient. If we treat managed funds as subjectively optimal portfolios, the MV

¹Some have considered the problem in discrete time (for example, see Lintner [23], Rubinstein [27], Fan [13], Sun and Yang [30], Chiarella *et al.* [7] and Sharpe [29]) and others in continuous time (for example, see Williams [32], Detemple and Murthy [10] and Zapatero [33]), and more recently Jouni and Napp ([19], [18], [20]), Hara [14] and Brown and Rogers [5]. Some models are in the MV framework (see, Lintner [23], Williams [32] and Sun and Yang [30]), others are in the Arrow-Debreu contingent claims economy (see, for example Rubinstein [27] and Abel ([1])).

²In the first case, investors may update their beliefs as new information become available, Bayesian updating rule is often used (see, for example, Williams [32] and Zapatero [33]). In the second case, investors agree to disagree and may revise their portfolio strategies as their views of the market change over time (see, for example, Lintner [23], Rubinstein [26] and Brown and Rogers [5]). For a discussion on the difference of the two cases, we refer the reader to a survey paper by Kurz [21].

³See, for example, Wenzelburger [31], Böhm and Chiarella [3], Böhm and Wenzelburger [4], and Chiarella *et al.* ([6], [7], [8], [9]).

inefficiency would imply the under-performance of the managed funds. This paper is devoted to present an explicit equilibrium price formula, and to examine the impacts of the heterogeneous beliefs on the MV efficiency of the optimal portfolios and on the market equilibrium in general.

In their recent work, Chiarella *et al.* [7] introduce a concept of *consensus belief* and show that, when there is a riskless asset, the market consensus belief can be constructed explicitly as a weighted average of the heterogeneous beliefs. They show that the market equilibrium prices is a weighted average of the equilibrium prices perceived by each investor. They also establish a CAPM-like relation under heterogeneous beliefs. In this paper, we first extend their analysis to the case when there is no riskless asset and obtain a zero-beta CAPM-like relation under heterogeneous beliefs. It is well known that the geometric tangency relation of traditional portfolio theory plays a very important role to the establishment of the CAPM⁴. We demonstrate that this geometric relationship does not hold under heterogeneous beliefs.

The current paper is related to Jouni and Napp ([19], [18]) who investigate the impact of beliefs heterogeneity on the consumption CAPM and the risk free rate by constructing a consensus belief and consumer. They show how pessimism and doubt at the aggregate level result from pessimism and doubt at the individual level. The construction of the consensus belief in this paper shares some similarity (in a much simpler and explicit way within the MV framework) to that in Jouni and Napp, however our focus is on the portfolio analysis and MV efficiency of the subjectively optimal portfolios, rather than on the risk premium. In other words, the focus of Jouni and Napp is on the impact of the aggregation of heterogeneous beliefs on the market, while we focus on the impact of the aggregation on the MV efficiency of individuals' optimal portfolio. Also, we compare the market MV frontiers with and without riskless asset and focus on the impact of the heterogeneous beliefs on the geometric relation of the frontiers. Interestingly, a similar result on the MV efficiency of the optimal portfolio of heterogeneous beliefs is found in Easley and O'Hara [12] where the heterogeneous beliefs are due to the information asymmetry⁵. With a rational expectations equilibrium model, they show that the average market portfolio is MV efficient, but not necessarily for the investors with different information. In our setup, investors are bounded rational and the market consensus belief is endogenously determined by all market participants and we show that the market portfolio is always MV efficiency (by the

⁴The market portfolio remains the same and MV efficient with or without the existence of a riskless security

⁵The heterogeneity can be due to either asymmetric information or different interpretation about the same information among investors in general. In the first case, certain structures on information and learning (such as Bayesian updating and learning) are imposed, while in the second case, the heterogeneous beliefs are associated with certain trading strategies used in financial markets (such as the momentum and contrarian strategies).

construction of the consensus belief) and the subjectively optimal portfolios of investors are inefficient in general.

The paper is structured as follows. In Section 2, we introduce and construct the market consensus belief linking the heterogeneous market with an equivalent homogeneous market, and present an explicit market equilibrium price formula. Consequently, a zero-beta CAPM under the heterogeneous beliefs is derived. In Section 3, we examine the impacts of different aspects of heterogeneous beliefs on the market equilibrium. Through some numerical examples, Section 4 examines the implications of heterogeneity on the MV efficiency of the optimal portfolios of heterogeneous investors and the geometric tangency relation of the portfolios with and without riskless asset. Section 5 extends numerical analysis to a market with many investors and examines the impact of heterogeneity on the MV efficiency of optimal portfolios and on the market when the belief dispersions are characterized by mean-preserving spreads. Section 6 summarizes and concludes the paper. The proofs and details of an numerical example are provided in the appendices. An earlier version considering the beliefs in both payoff and return setups can be found in He and Shi[16].

2. MV Equilibrium Asset Prices Under Heterogeneous Beliefs

When a financial market consists of investors with different views on the future movement of the market, it is important to understand how market equilibrium is obtained and the roles played by different investors. Within the standard MV framework, in this section, we first introduce heterogeneous beliefs among investors and a concept of market consensus belief to reflect the market belief when market is in equilibrium. By constructing the consensus belief explicitly, we characterize the equilibrium asset prices. Consequently we obtain a zero-beta CAPM-like relation under heterogeneous beliefs.

2.1 Heterogeneous Beliefs

Following Lintner [23] and Black [2], we extend the static MV model with homogeneous belief and consider a market in which there are many risky assets but there is no risk-free asset and investors have heterogeneous beliefs of the future returns of risky assets. Similar to Chiarella *et al.* [7], asset returns are measured in the payoff in capital.

Consider a market with N risky assets, indexed by $j = 1, 2, \dots, N$ and I investors indexed by $i = 1, 2, \dots, I$. Let $\tilde{\mathbf{x}} = (\tilde{x}_1, \dots, \tilde{x}_N)^T$ be the random payoff vector of the risky assets. Assume that each investor has his/her own set of beliefs about the market in terms of means, variances and covariances of the payoffs of the assets, denoted by

$$y_{i,j} = E_i[\tilde{x}_j], \quad \sigma_{i,jk} = \text{Cov}_i(\tilde{x}_j, \tilde{x}_k) \quad \text{for } 1 \leq i \leq I, \quad 1 \leq j, k \leq N. \quad (1)$$

For investor i , we define the mean vector and variance/covariance matrix of the

payoffs of N assets as follows, $\mathbf{y}_i = \mathbb{E}_i(\tilde{\mathbf{x}}) = (y_{i,1}, y_{i,2}, \dots, y_{i,N})^T$ and $\Omega_i = (\sigma_{i,jk})_{N \times N}$, which is positive definite. Denote $\mathcal{B}_i = (\mathbb{E}_i(\tilde{\mathbf{x}}), \Omega_i)$ the set of subjective beliefs of investor i . Let $\mathbf{z}_i = (z_{i,1}, z_{i,2}, \dots, z_{i,N})^T$ be the portfolio in the risky assets (in quantity) and $W_{i,o}$ be the initial wealth of investor i . Then the end-of-period portfolio wealth of investor i is given by $\tilde{W}_i = \tilde{\mathbf{x}}^T \mathbf{z}_i$. Under the belief $\mathcal{B}_i$, the mean and variance of the portfolio wealth $\tilde{W}_i$ of investor i are given, respectively, by

$$\mathbb{E}_i(\tilde{W}_i) = \mathbf{y}_i^T \mathbf{z}_i, \quad \sigma_i^2(\tilde{W}_i) = \mathbf{z}_i^T \Omega_i \mathbf{z}_i. \quad (2)$$

As in the standard MV framework, we assume that investor i has a constant absolute risk aversion (CARA) utility function $U_i(w) = -e^{-\theta_i w}$, where θ_i is the CARA coefficient, and the end-of-period wealth $\tilde{W}_i$ of investor i is normally distributed. Under these assumptions, maximizing investor i 's expected utility of wealth is equivalent to maximizing his/her certainty equivalent end-of-period wealth $\max_{\mathbf{z}_i} Q_i(\mathbf{z}_i)$ subject to the wealth constraint

$$\mathbf{p}_0^T \mathbf{z}_i = W_{i,o}, \quad (3)$$

where

$$Q_i(\mathbf{z}_i) := \mathbb{E}_i(\tilde{W}_i) - \frac{\theta_i}{2} \sigma_i^2(\tilde{W}_i) = \mathbf{y}_i^T \mathbf{z}_i - \frac{\theta_i}{2} \mathbf{z}_i^T \Omega_i \mathbf{z}_i$$

and $\mathbf{p}_0$ is the market price vector of the risky assets. Applying the first order conditions, we obtain the following lemma on the optimal portfolio of the investor.

Lemma 2.1. *For given market price vector $\mathbf{p}_0$ of risky assets, the optimal risky portfolio $\mathbf{z}_i^*$ of investor i is uniquely determined by*

$$\mathbf{z}_i^* = \theta_i^{-1} \Omega_i^{-1} [\mathbf{y}_i - \lambda_i^* \mathbf{p}_0], \quad (4)$$

where

$$\lambda_i^* = \frac{\mathbf{p}_0^T \Omega_i^{-1} \mathbf{y}_i - \theta_i W_{i,o}}{\mathbf{p}_0^T \Omega_i^{-1} \mathbf{p}_0}. \quad (5)$$

Lemma 2.1 implies that the optimal demand of investor i depends on his/her absolute risk aversion (ARA) coefficient (θ_i), the expected payoffs and variance/covariance matrix of the risky asset payoffs, the Lagrange multiplier (λ_i^*), as well as the market price of the risky assets. Following Lintner [23], λ_i^* is a shadow price, measuring the *marginal real (riskless) certainty-equivalent of investor i 's end-of-period wealth*. In fact, applying the first order condition, we obtain $\frac{\partial Q_i(\mathbf{z}_i^*)}{\partial \mathbf{z}_i} = \lambda_i^* \mathbf{p}_0$, which leads to

$$\lambda_i^* = \frac{1}{p_{oj}} \frac{\partial Q_i(\mathbf{z}_i^*)}{\partial z_{ij}} \quad \text{for all } j = 1, 2, \dots, N. \quad (6)$$

More precisely, equation (6) indicates that λ_i^* actually measures *investor i 's optimal marginal certainty equivalent end-of-period wealth per unit of asset j relative to its market price* and it is a constant across all assets. In general, the shadow price is not necessary the same for all investors, however, it becomes the same when there exists a risk-free asset in the market. In fact, let the current price of the risk-free asset f be 1 and its payoff be $R_f = 1 + r_f$. Applying (6) to the risk-free asset leads to $\lambda_i^* = R_f$ for all investors, that is, the shadow price is equal to the payoff of the risk-free asset.

2.2 Consensus Belief and Equilibrium Asset Prices

We define the market equilibrium asset price vector $\mathbf{p}_o$ of the risky assets as the price vector under which individual's optimal demands (4) satisfy the market aggregation condition

$$\sum_{i=1}^I \mathbf{z}_i^* = \sum_{i=1}^I \bar{\mathbf{z}}_i := \mathbf{z}_m, \quad (7)$$

where $\bar{\mathbf{z}}_i$ is the endowment portfolio of investor i . Correspondingly, $\mathbf{z}_m$ is the market portfolio of the risky assets. It then follows from (7) and (4) that the market equilibrium price $\mathbf{p}_o$ is given, in terms of the heterogeneous beliefs of the investors, by

$$\mathbf{p}_o = \left(\sum_{i=1}^I \theta_i^{-1} \lambda_i^* \Omega_i^{-1} \right)^{-1} \left[\left(\sum_{i=1}^I \theta_i^{-1} \Omega_i^{-1} \mathbf{y}_i \right) - \mathbf{z}_m \right]. \quad (8)$$

This expression defines the market equilibrium price $\mathbf{p}_o$ implicitly since λ_i^* depends on $\mathbf{p}_o$ as well. For the existence of the market equilibrium price in general, we refer to Sun and Yang [30] and the references cited there. The concept of consensus belief has been used to characterize the market when investors are heterogeneous in different context (such as Jouini and Napp ([19], [18]) and Chiarella *et al.* [7]). It is closely related to but significantly different from the concept of representative investor in the classical finance literature. It is endogenously determined through the market aggregation and reflects a weight average of heterogeneous beliefs. We now introduce the concept of consensus belief for the market with the heterogeneous beliefs.

Definition 2.1. A belief $\mathcal{B}_a = (\mathbb{E}_a(\tilde{\mathbf{x}}), \Omega_a)$, defined by the expected payoff of the risky assets $\mathbb{E}_a(\tilde{\mathbf{x}})$ and the covariance matrix of the risky asset payoffs Ω_a , is called a market **consensus belief** if the market equilibrium price under the heterogeneous beliefs is also the market equilibrium price under the homogeneous belief $\mathcal{B}_a$.

When a consensus belief exists, the market with heterogeneous beliefs can be treated as a market with homogeneous consensus belief and then the classical

Markowitz portfolio analysis can be applied. Due to the complexity of heterogeneity, the existence and finding of such consensus belief is a difficult task in the literature. This obstacle makes the examination of the impact of the heterogeneity difficult. In the following, we construct the consensus belief explicitly, from which market equilibrium prices $\mathbf{p}_0$ can be determined explicitly in terms of the consensus belief. It is the explicit construction of the consensus belief that makes it easy to examine the role of heterogeneous beliefs played in determining the market equilibrium price and to derive the zero-beta CAPM relation.

Proposition 2.1. *Let*

$$\theta_a := \left(\frac{1}{I} \sum_{i=1}^I \theta_i^{-1} \right)^{-1}, \quad \lambda_a^* := \frac{1}{I} \theta_a \sum_{i=1}^I \theta_i^{-1} \lambda_i^*.$$

Then

(i) *the consensus belief $\mathcal{B}_a = (\mathbb{E}_a(\tilde{\mathbf{x}}), \Omega_a)$ is given by*

$$\Omega_a = \theta_a^{-1} \lambda_a^* \left(\frac{1}{I} \sum_{i=1}^I \lambda_i^* \theta_i^{-1} \Omega_i^{-1} \right)^{-1}, \quad (9)$$

$$\mathbf{y}_a := \mathbb{E}_a(\tilde{\mathbf{x}}) = \theta_a \Omega_a \left(\frac{1}{I} \sum_{i=1}^I \theta_i^{-1} \Omega_i^{-1} \mathbb{E}_i(\tilde{\mathbf{x}}) \right); \quad (10)$$

(ii) *the market equilibrium price $\mathbf{p}_0$ is determined by*

$$\mathbf{p}_0 = \frac{1}{\lambda_a^*} \left[\mathbf{y}_a - \frac{1}{I} \theta_a \Omega_a \mathbf{z}_m \right]; \quad (11)$$

(iii) *the equilibrium optimal portfolio of investor i is given by*

$$\mathbf{z}_i^* = \theta_i^{-1} \Omega_i^{-1} \left[\left(\mathbf{y}_i - \frac{\lambda_i^*}{\lambda_a^*} \mathbf{y}_a \right) + \frac{\lambda_i^*}{I \lambda_a^*} \theta_a \Omega_a \mathbf{z}_m \right]. \quad (12)$$

Proposition 2.1 shows how the consensus belief can be constructed explicitly from the heterogeneous beliefs. Under the consensus belief, the market equilibrium prices of the risky assets are determined in the standard way with no risk-free asset. Intuitively Proposition 2.1 indicates that the market consensus belief is a weighted average of the heterogeneous beliefs. More precisely, the market risk tolerance ($1/\theta_a$) is simply an average of the risk tolerance of the heterogeneous investors, according to Huang and Lizenberger [17], $\theta_a/I = (\sum_{i=1}^I \theta_i^{-1})^{-1}$ is called the *aggregate absolute risk aversion* and consequently $\theta_a W_{m0}/I$ is referred to as the *aggregate relative risk aversion*. The weighted average behaviour can also be

viewed in the following way. Let $\tau_i = 1/\theta_i$ be the risk tolerance of investor i and $\tau_a = \sum_{i=1}^I \tau_i$ be the market aggregate risk tolerance. Then

$$\lambda_a^* = \sum_{i=1}^I \frac{\tau_i}{\tau_a} \lambda_i^*, \quad \Omega_a^{-1} = \sum_{i=1}^I \frac{\tau_i \lambda_i}{\tau_a \lambda_a} \Omega_i^{-1}, \quad \mathbb{E}_a(\tilde{\mathbf{x}}) = \Omega_a \sum_{i=1}^I \frac{\tau_i}{\tau_a} \Omega_i^{-1} \mathbb{E}_i(\tilde{\mathbf{x}}).$$

Hence the precision matrix (Ω_a^{-1}) for the market reflects an weighted average of the precision matrices of all investors and the market expected payoff is a weighted average of the expected payoffs of the investors. The market equilibrium prices are determined such that each investor can choose their optimal portfolio subjectively and the market is cleared. It follows from (4) in Lemma 2.1 that $\mathbf{p}_0 = \frac{1}{\lambda_i^*}(\mathbf{y}_i - \frac{1}{\tau_i} \Omega_i \mathbf{z}_i^*)$ for $i = 1, \dots, I$. However, if the entire market acts as an aggregate investor, then for the market to clear, the prices must be determined by the consensus belief as in (11) or equivalently as $\mathbf{p}_0 = \frac{1}{\lambda_a^*}(\mathbf{y}_a - \frac{1}{\tau_a} \Omega_a \mathbf{z}_m)$. This suggests that the consensus belief $\mathcal{B}_a$ must correspond to the belief of the aggregate market such that the market portfolio is an optimal portfolio. The expressions in Proposition 2.1 provide explicit relationships between the heterogeneous belief and the market consensus belief under the market aggregation. Their usefulness will be revealed when we derive a zero-beta CAPM-like relation and examine the impacts of the heterogeneity on the market equilibrium in the following subsection.

2.3 The Zero-Beta CAPM Under Heterogeneous Beliefs

As a corollary of Proposition 2.1, we show now that a zero-beta CAPM-like relation holds under the constructed consensus belief with no risk-free asset.

Let the future payoff of the market portfolio $\mathbf{z}_m$ be given by $\tilde{W}_m = \tilde{\mathbf{x}}^T \mathbf{z}_m$ and its current market value is $W_{m,o} = \mathbf{z}_m^T \mathbf{p}_0 = \sum_{i=1}^I W_{i,o}$. Hence under the consensus belief $\mathcal{B}_a$, $\mathbb{E}_a(\tilde{W}_m) = \mathbf{y}_a^T \mathbf{z}_m$ and $\sigma_a^2(\tilde{W}_m) = \mathbf{z}_m^T \Omega_a \mathbf{z}_m$. Define the return vector $\tilde{\mathbf{r}} = (\tilde{r}_1, \dots, \tilde{r}_N)^T$ with $\tilde{r}_j = \tilde{x}_j/p_{j,o} - 1$ and $\tilde{r}_m = \tilde{W}_m/W_{m,o} - 1$. Under the market consensus belief $\mathcal{B}_a$, we set

$$\mathbb{E}_a(\tilde{r}_j) = \frac{\mathbb{E}_a(\tilde{x}_j)}{p_{j,o}} - 1, \quad \mathbb{E}_a(\tilde{r}_m) = \frac{\mathbb{E}_a(\tilde{W}_m)}{W_{m,o}} - 1, \quad \sigma_a^2(\tilde{r}_m) = \frac{\sigma_a^2(\tilde{W}_m)}{W_{m,o}^2}$$

and

$$\text{Cov}_a(\tilde{r}_j, \tilde{r}_m) = \frac{1}{p_{j,o} W_{m,o}} \text{Cov}_a(\tilde{x}_j, \tilde{W}_m), \quad \text{Cov}_a(\tilde{r}_j, \tilde{r}_k) = \frac{1}{p_{j,o} p_{k,o}} \text{Cov}_a(\tilde{x}_j, \tilde{x}_k).$$

Then we have the following result.

Corollary 2.1. *In market equilibrium, the relation between expected return and risk under the heterogeneous beliefs can be expressed as*

$$\mathbb{E}_a[\tilde{\mathbf{r}}] - (\lambda_a^* - 1)\mathbf{1} = \boldsymbol{\beta}[\mathbb{E}_a(\tilde{r}_m) - (\lambda_a^* - 1)], \quad (13)$$

where

$$\lambda_a^* = \frac{\mathbf{z}_m^T \mathbf{y}_a - \theta_a \mathbf{z}_m^T \Omega_a \mathbf{z}_m / I}{W_{m,o}}, \quad (14)$$

$$\mathbb{E}_a(\tilde{r}_m) - (\lambda_a^* - 1) = \frac{\theta_a \mathbf{z}_m^T \Omega_a \mathbf{z}_m / I}{W_{m0}} = \frac{1}{\tau_a} W_{m,o} \sigma_a^2(\tilde{r}_m) > 0 \quad (15)$$

and $\beta = (\beta_1, \beta_2, \dots, \beta_N)^T$ with

$$\beta_j = \frac{\text{Cov}_a(\tilde{r}_m, \tilde{r}_j)}{\sigma_a^2(\tilde{r}_m)} = \frac{W_{m0}}{p_{0j}} \frac{\text{Cov}_a(\tilde{x}_j, \tilde{W}_m)}{\sigma_{a,m}^2}, \quad j = 1, \dots, N.$$

The equilibrium relation (13) is the standard Zero-Beta CAPM except that the mean and variance/covariance are calculated based on the consensus belief $\mathcal{B}_a$. We refer it as the *Zero-beta Heterogeneous Capital Asset Pricing Model* (ZHCAPM). For risky assets, relation (13) is equivalent to

$$\mathbb{E}_a[\tilde{r}_j] - (\lambda_a^* - 1) = \beta_j [\mathbb{E}_a(\tilde{r}_m) - (\lambda_a^* - 1)], \quad \text{for } j = 1, \dots, N. \quad (16)$$

The zero-beta rate, $\lambda_a^* - 1$, corresponds to the expected return of the zero-beta portfolio of the market portfolio, where λ_a^* is the market shadow price. As in the standard case, the market risk premium, given by equation (15) is positively proportional to the aggregate relative risk aversion $W_{m,o}/\tau_a$ and the variance of the market portfolio returns $\sigma_a^2(\tilde{r}_m)$. The market price of risk under the consensus belief is given by $\phi = (\mathbb{E}_a(\tilde{r}_m) - (\lambda_a^* - 1))/\sigma_a(\tilde{r}_m) = W_{m0}\sigma(\tilde{r}_m)/\tau_a$, which is proportional to the level of volatility of the market and the aggregate relative risk aversion.

As discussed earlier, investor i 's shadow price becomes R_f across all investors when there exists a risk-free asset in the market. That is, $\lambda_i^* = \lambda_a^* = R_f$. Substituting this into Proposition 2.1 and Corollary 2.1 leads to the main results in Chiarella *et al.* [7].

3. The Impact of Heterogeneity

In this section, we use Proposition 2.1 and Corollary 2.1 to examine the impact of the heterogeneous beliefs on the market consensus belief and equilibrium price. To simplify the analysis, we focus on some special cases.

3.1 The Shadow Prices and the Aggregation Property

We first examine the relationship between individual shadow prices and the market consensus shadow price. Following (2.1), let $\lambda_a^* = f(\lambda_1^*, \lambda_2^*, \dots, \lambda_I^*; \theta_1, \theta_2, \dots, \theta_I)$. Then it is easy to see that $\frac{\partial f}{\partial \lambda_i^*} = \frac{\theta_i \theta_i^{-1}}{I} > 0$, showing that the market consensus shadow price increases as the shadow price of investor i increases, and the rate of increase depends on θ_i . It follows from

$\frac{\partial^2 f}{\partial \lambda_i^* \partial \theta_i} = \frac{1}{I} \theta_i^{-3} \theta_a (\frac{1}{I} \theta_a - \theta_i)$ and $I \theta_i^{-1} > \theta_a^{-1}$ that $\frac{\partial^2 f}{\partial \lambda_i^* \partial \theta_i} < 0$. Therefore the market consensus shadow price is more sensitive to the change of the shadow price of investor who is less risk-averse.

According to Huang and Litzenberger [17], when investors have homogeneous belief, time-additive and state independent utility functions with linear risk tolerance and a common cautiousness coefficient, the market equilibrium prices are independent of the distribution of the initial wealth among investors and, if this is the case, we say that the market satisfies the aggregation property. In a general two-period economy without specifying the type of utility function for any investors, Fan [13] shows the *Second Welfare Theorem* holds. The theorem states that investors with large capital endowments would have lower marginal utilities of capital endowments and a stronger influence on the market equilibrium. In our case, the utility is measured by $Q_i(\mathbf{z})$. From (6), the marginal utility of investor i is represented by the shadow price (λ_i^*) . It then follows from (5) that a large initial wealth or capital endowment leads to a lower marginal utility. Also, from the expression of the equilibrium price vector in (8), it can be seen that (λ_i^*) is inversely related to the price vector. This suggests that an investor with a lower shadow price or marginal utility has a stronger impact on the market equilibrium prices, and hence an investor with a larger capital is more influential in the market. This is consistent with the *Second Welfare Theorem*. In other words, the aggregation property does not hold in our case in general. However, if there is a risk-free asset in the market, then the shadow prices or marginal utilities is a constant across all investors. Correspondingly the market prices are independent of the initial wealth distribution. Summarizing the above analysis, we have the following corollary.

Corollary 3.1. *With the heterogeneous beliefs and no risk-free asset, the aggregation property does not hold. Furthermore investors with lower shadow prices or marginal utilities have a stronger impact on the market equilibrium prices, and hence investors with larger capital are more influential in the market. However, if there is a risk-free asset, the aggregation property holds.*

3.2 The Impact of Heterogeneous ARA Coefficients

Proposition 2.1 indicates that the heterogeneous ARA coefficients or risk tolerance have complicated impact on the market consensus belief and equilibrium price. To illustrate such impact, we consider a special case when investors are homogeneous in the expected payoffs and covariance matrix but heterogeneous in ARA, that is, $\Omega_i = \Omega_a := \Omega_o$, $\mathbf{y}_i = \mathbf{y}_a := \mathbf{y}_o$ for all i . Accordingly the equilibrium price vector can be written as

$$\mathbf{p}_0 = \frac{1}{\lambda_a^*} \left[\mathbf{y}_o - \frac{1}{I} \theta_a \Omega_o \mathbf{z}_m \right], \quad \lambda_a^* = \frac{\mathbf{z}_m^T \mathbf{y}_o - \theta_a \mathbf{z}_m^T \Omega_o \mathbf{z}_m / I}{W_{m0}}. \quad (17)$$

Equation (17) implies that, when the risk aversion coefficient is the only source of heterogeneity, the market equilibrium prices are independent of the initial wealth

distribution amongst individuals and hence the aggregation property holds. For any risky asset j , (17) becomes

$$p_{0,j} = \frac{1}{\lambda_a^*} \left[y_{0,j} - \frac{1}{I} \theta_a \text{Cov}(\tilde{x}_j, \tilde{W}_m) \right].$$

This, together with the market shadow price in equation (17), leads to $\frac{\partial p_{0,j}}{\partial \theta_a} = \frac{\sigma^2(\tilde{r}_m)(1-\beta_j)}{I\lambda_a^*}$. In the presence of a risk-free asset with payoff R_f , this becomes $\frac{\partial p_{0,j}}{\partial \theta_a} = -\frac{\sigma^2(\tilde{r}_m)\beta_j}{IR_f}$. Noting that, in this case, the equilibrium prices and expected returns are inversely related since the expected payoff is given. Together with the fact that $\frac{\partial \theta_a}{\partial \theta_i} = (\theta_a^{-1}\theta_i^{-1})^2/I > 0$ and $\frac{\partial^2 \theta_a}{\partial \theta_i^2} = -2\frac{\partial \theta_a}{\partial \theta_i}(\frac{\partial \theta_a}{\partial \theta_i}\theta_a^{-1} + \theta_i^{-1}) < 0$, this analysis leads to the following corollary.

Corollary 3.2. *In a market with homogeneous beliefs and no risk-free assets,*

$$(\beta_j - 1) \frac{\partial p_{0,j}}{\partial \theta_i} < 0, \quad (\beta_j - 1) \frac{\partial \mathbb{E}_o(\tilde{r}_j)}{\partial \theta_i} > 0$$

for $\beta_j \neq 1$ and $\frac{\partial p_{0,j}}{\partial \theta_i} = \frac{\partial \mathbb{E}_o(\tilde{r}_j)}{\partial \theta_i} = 0$ for $\beta_j = 1$. If there exists a risk-free asset, then

$$\beta_j \frac{\partial p_{0,j}}{\partial \theta_i} < 0, \quad \beta_j \frac{\partial \mathbb{E}_o(\tilde{r}_j)}{\partial \theta_i} > 0$$

for $\beta_j \neq 0$ and $\frac{\partial p_{0,j}}{\partial \theta_i} = \frac{\partial \mathbb{E}_o(\tilde{r}_j)}{\partial \theta_i} = 0$ for $\beta_j = 0$. The rate of change for both the equilibrium price and expected return is greater when investor is less risk averse.

Corollary 3.2 indicates that the impact of ARA on the market equilibrium depends on the beta of the asset. When there is no risk-free asset, if an asset is riskier than the market ($\beta_j > 1$), an increase in ARA for any investor increases the price and decreases the expected future return of the asset, and vice versa for a less risky asset. However, if there is a risk-free asset, the changes depend on the return correlation of the asset with the market. If the returns of the asset and market are positive correlated, an increase (decrease) in ARA of any investor leads to lower (higher) market equilibrium price and higher (lower) expected return for the asset. In addition, changing ARA of less risk averse investor has more significant impact on market equilibrium price and expected return. The market is dominated by less risk averse investors, because the market average risk aversion coefficient θ_a is a harmonic mean of θ_i s, it aggravates the impact of the small θ_i s. This suggests that, when there is no risk-free asset in the market and when the risk aversion coefficients of the investors becomes more divergent with a given average, the aggregate ARA would be reduced, resulting lower (higher) equilibrium price and higher (lower) expected return for assets with betas are below (above) the market

level. However, when there is a risk-free asset, the reduction of the market aggregate risk aversion leads to lower (higher) equilibrium price and higher (lower) expected return for assets that are negatively (positively) correlated with the market.

3.3 The Impact of Heterogeneous Expected Payoffs

We now assume that investors agree on the variances and covariances of asset payoffs, say $\Omega_i = \Omega_o$, but disagree on the expected future payoffs of the assets. Consequently $\Omega_a = \Omega_o$ and the equilibrium price for asset j becomes

$$p_{0,j} = \frac{1}{\lambda_a^*} \left[y_{a,j} - \frac{1}{\tau_a} \text{Cov}_o(\tilde{x}_j, \tilde{W}_m) \right], \quad (18)$$

where $\lambda_a^* = [\mathbf{z}_m^T \mathbf{y}_a - \mathbf{z}_m^T \Omega_o \mathbf{z}_m / \tau_a] / W_{m0}$ and $y_{a,j} = \sum_{i=1}^I (\tau_i / \tau_a) y_{i,j}$. This, together with (18), leads to

$$\frac{\partial p_{0,j}}{\partial y_{a,j}} = \frac{1 - \alpha_j}{\lambda_a^*}, \quad (19)$$

where $\alpha_j = p_{0,j} z_{m,j} / W_{m0}$ is the market share of asset j in wealth. If there is a risk-free asset in the market with payoff R_f , then (19) simply becomes $\partial p_{0,j} / \partial y_{a,j} = 1 / R_f$. Note that

$$\frac{\partial y_{a,j}}{\partial y_{i,j}} = \frac{1}{I} \frac{\theta_a}{\theta_i} y_{i,j} > 0, \quad \frac{\partial^2 y_{a,j}}{\partial y_{i,j} \partial \theta_i} = \theta_a \theta_i^{-3} \frac{y_{i,j}}{I} \left(\frac{\theta_a}{I} - \theta_i \right) < 0. \quad (20)$$

Because of $\alpha \in [0, 1]$, equations (19) and (20) indicates that investor i 's subjective belief in the expected payoff of asset j is positively related to its equilibrium price. This is also true when there is a risk-free asset in the market. The positive correlation between the subjective beliefs in the expected payoff and the equilibrium price for asset j does not necessarily lead to a negative correlation between the subjective beliefs in the expected payoff and the market expected return for asset j . To see the exact relation, we have from $\mathbb{E}_a(\tilde{r}_j) = y_{a,j} / p_{0,j} - 1$ that

$$\frac{\partial \mathbb{E}_a(\tilde{r}_j)}{\partial y_{a,j}} = \frac{p_{0,j} - (1 - \alpha_j) y_{a,j} / \lambda_a^*}{p_{0,j}^2}. \quad (21)$$

This expression is negative if and only if $(1 + \mathbb{E}_a(\tilde{r}_j))(1 - \alpha_j) > \lambda_a^*$. When this condition holds, the expected return decreases when the expected payoff increases for asset j . When there is a risk-free asset, $\lambda_a^* = R_f$ and equation (21) becomes $\frac{\partial \mathbb{E}_a(\tilde{r}_j)}{\partial y_{a,j}} = \frac{p_{0,j} - y_{a,j} / R_f}{p_{0,j}^2}$, which is negative if and only if $\mathbb{E}_a(\tilde{r}_j) > r_f$. When this condition holds, the expected return decreases when the heterogeneous belief in the expected payoff increases for asset j . Summarizing the above analysis, we obtain the following corollary.

Corollary 3.3. *In a market with homogeneous beliefs in covariance matrix and no risk-free assets, if*

$$(1 + \mathbb{E}_a(\tilde{r}_j))(1 - \alpha_j) > \lambda_a^* \quad (22)$$

for asset j , then the market expected payoff increases and the expected return decreases when the heterogeneous belief in the expected payoff of any investor increases for asset j . When there is a risk-free asset, the condition (22) becomes $\mathbb{E}_a(\tilde{r}_j) > r_f$.

The following discussion is devoted to Miller's hypothesis (Miller [24]) that assets with high dispersion in beliefs have higher market prices and lower expected future returns than otherwise similar stocks. Empirical tests performed in Diether, Malloy and Scherbina [11] support Miller's hypothesis. Intuitively, optimistic investors would increase the price of the asset and then reduce its expected future return. We now provide an explanation on this hypothesis. Let us consider a market in which investors have homogeneous beliefs in the covariance matrix but heterogeneous beliefs in the expected payoffs of two risky assets j and j' . Let the expected payoffs be $\mathbf{y}_j = (y_{1,j}, y_{2,j}, \dots, y_{I,j})^T$ and $\mathbf{y}_{j'} = (y_{1,j'}, y_{2,j'}, \dots, y_{I,j'})^T$ for asset j and j' , respectively. Assume $y_{i,j'} = y_{i,j} + \epsilon_{i,j}$, where $\{\epsilon_{j,1}, \epsilon_{j,2}, \dots, \epsilon_{j,I}\}$ is a set of real numbers such that $\sum_{i=1}^I \epsilon_{i,j} = 0$ and $\frac{1}{I} \sum_{i=1}^I (y_{i,j'} - \bar{y})^2 \geq \frac{1}{I} \sum_{i=1}^I (y_{i,j} - \bar{y})^2$, where $\bar{y} = (1/I) \sum_{i=1}^I y_{i,j}$. This condition implies that investors have more divergence of opinions in the expected payoff for asset j' than asset j . According to Miller's hypothesis, asset j' would have higher market price and lower expected future return than asset j . To see if this is true, we consider the following simple example when $I = 2$.

Example 3.1. Let $I = 2$. Given $\epsilon > 0$, consider two assets j and k with $y_{2,j} < y_{1,j}$, and $y_{1,k} = y_{1,j} + \epsilon$ and $y_{2,k} = y_{2,j} - \epsilon$. This specification indicates that the divergence of opinion about the asset's expected payoff is greater for asset k than for asset j . Then $y_{a,j} = \frac{\theta_1^{-1}}{\theta_1^{-1} + \theta_2^{-1}} y_{1,j} + \frac{\theta_1^{-1}}{\theta_1^{-1} + \theta_2^{-1}} y_{2,j}$ and $y_{a,k} = \frac{\theta_1^{-1}}{\theta_1^{-1} + \theta_2^{-1}} (y_{1,j} + \epsilon) + \frac{\theta_1^{-1}}{\theta_1^{-1} + \theta_2^{-1}} (y_{2,j} - \epsilon)$. Hence $y_{a,j} - y_{a,k} = \frac{\epsilon}{\theta_1^{-1} + \theta_2^{-1}} (\theta_2^{-1} - \theta_1^{-1})$. Accordingly, $y_{a,j} < y_{a,k}$ if and only if $\theta_1 < \theta_2$. This implies that if investor who is optimistic about the asset expected payoff is less risk averse, then a divergence of opinion among the two investors for the expected payoff for asset k leads to high expected payoff for the asset in equilibrium. This suggests that divergence of opinion on the asset expected payoffs generates higher market expected payoff if belief of assets' expected future payoffs is negatively correlated to risk aversion for any investor i . It then follows from Corollary 3.3 that, when both assets j and k satisfy the condition (22), the divergence of opinion on the asset expected payoffs generates lower expected future return for the asset.

To summarize, if our model is to be consistent with Miller's hypothesis that divergence of opinion causes asset price to increase and expected return to decrease, we need the investor with an optimistic view of the asset future payoff to

be less risk-averse comparing with the relative pessimistic investor, also the asset return satisfies condition (22).

4. MV Efficiency and Geometric Relationship of MV Frontiers

In this section, we examine the MV efficiency of the optimal portfolios of investors and the geometric relationship of the MV frontiers with and without riskless asset in market equilibrium. Following the standard Markowitz method, we can construct the MV portfolio frontier based on the consensus belief. Because the consensus belief reflects the market belief when it is in equilibrium, we call this frontier the market equilibrium MV frontier. A portfolio is MV efficient if it is located on the market equilibrium MV frontier. When investors are homogeneous in their beliefs, it is well known that the optimal portfolios of investors are always MV efficient and also the market portfolio is the unique tangency portfolio between the MV frontiers with and without a riskless asset. When investors' beliefs are heterogeneous, we would expect that the subjectively optimal portfolios of investors are MV inefficient. However, it is not clear if the geometric relationships of the MV frontiers with and without riskless asset is still hold. In this section, we first demonstrate the MV inefficiency of the subjectively optimal portfolios and then show that the geometric relationship breaks down under the heterogeneous beliefs.

4.1 MV Efficiency of the Optimal Portfolios Under Heterogeneous Beliefs

In the market we set up in Section 2, investors are bounded rational in the sense that they make their optimal decisions based on their beliefs. Based on investors' subjective beliefs, we can construct the MV frontiers (in the standard deviation and expected return space) by using the standard Markowitz method. Of course, the optimal portfolios of the investors will be located on the efficient MV frontiers under their subjective beliefs. Similarly, based on the consensus belief, the market equilibrium MV frontier can be constructed. By the market clearing condition and frontier construction, the market portfolio is always located on the market equilibrium frontier, hence always efficient. The question is whether the optimal portfolios of individual investors are MV efficient. This is a very important question both theoretically and empirically. If the answer to the question is yes, then the optimal portfolio of the bounded rational heterogeneous investors are MV efficient under market aggregation. Otherwise, market fails to provide the MV efficiency for the investors. If we refer to heterogeneous investors as fund managers and the market portfolio as the market index, the MV efficiency of the optimal portfolios will have important implications on whether fund managers can out perform the market index based on the MV criteria.

To answer this question, we consider a consensus investor with the market consensus beliefs $\mathcal{B}_o$, risk aversion coefficient θ_i and initial wealth $W_{i,o}$. Then the

optimal portfolio of the investor given by equation (12) becomes

$$\mathbf{z}_i^* = \left(1 - \frac{\lambda_i^*}{\lambda_a^*}\right) \theta_i^{-1} \Omega_a^{-1} \mathbf{y}_a + \frac{1}{I} \theta_i^{-1} \theta_a \frac{\lambda_i^*}{\lambda_a^*} \mathbf{z}_m. \quad (23)$$

Equation (23) shows that any consensus investor will divide his/her investment into two portfolios, namely, $\Omega_a^{-1} \mathbf{y}_a$ and the market portfolio $\mathbf{z}_m$, which is consistent with the *Two Fund Separation Theorem* (see Huang and Lizenberger [17] Chapter 4, page 83) and such portfolios must be MV efficient due to the construction, which means that the portfolios $\Omega_a^{-1} \mathbf{y}_a$ and $\mathbf{z}_m$ must be the MV frontier portfolios. It is easy to verify from (23) that the aggregate position of the portfolio $\Omega_a^{-1} \mathbf{y}_a$ of all investors is $\sum_i (1 - \frac{\lambda_i^*}{\lambda_a^*}) \theta_i^{-1} \Omega_a^{-1} \mathbf{y}_a = \mathbf{0}$ when the market clearing condition (7) is satisfied.

However, when investor i 's subjective belief ($\mathcal{B}_i$) differs from the market belief ($\mathcal{B}_a$), the optimal portfolio of investor i can be expressed as

$$\mathbf{z}_i^* = \theta_i^{-1} \Omega_i^{-1} (\mathbf{y}_i - \mathbf{y}_a) + \frac{1}{I} \theta_i^{-1} \theta_a \frac{\lambda_i^*}{\lambda_a^*} \Omega_i^{-1} \Omega_a \mathbf{z}_m. \quad (24)$$

Then the composition of the portfolio depends also on the belief error $\mathbf{y}_i - \mathbf{y}_a$ and $\Omega_i^{-1} \Omega_a$ of the investor i from the market. Analytically it is not easy to see if the optimal portfolio of investor i lies on the market equilibrium MV frontier. However, through Example D.1 in Appendix D, we can show that the optimal portfolios of investors are not located on the market equilibrium MV frontier in general. In this example, we consider a market with two investors and three risky assets. Given individuals' risk aversion coefficients, subjective beliefs and initial wealth, we first form the consensus belief and calculate the equilibrium price vector. Using the equilibrium price, we convert the consensus belief in asset payoffs to the consensus belief in asset returns and obtain the market expected returns and variances/covariances of asset returns. With the information provided in Table 3 in Appendix D, we can construct the portfolio frontiers for each investor and for the market equilibrium frontier in the mean-standard deviation space, and locate the optimal portfolios for individual investors as well as the market portfolio. Figure 1 exhibits the resulting graph.

Figure 1 shows two interesting and important features. Firstly, the market equilibrium MV frontier is located between two individual's MV frontiers. However, it is closer to that of investor 2. Intuitively, this is due to the fact that investor 2 is less risk averse and more optimistic about the market in the sense that he/she perceives higher expected payoffs and smaller standard deviations on the asset payoffs and hence dominates the market. Secondly, it is verified that the optimal portfolios of the two investors are always located on their MV efficient frontiers based on their own beliefs and the market portfolio is located on the market MV efficient frontier under the consensus belief. However, in market equilibrium, the

MVS without a risk-free asset

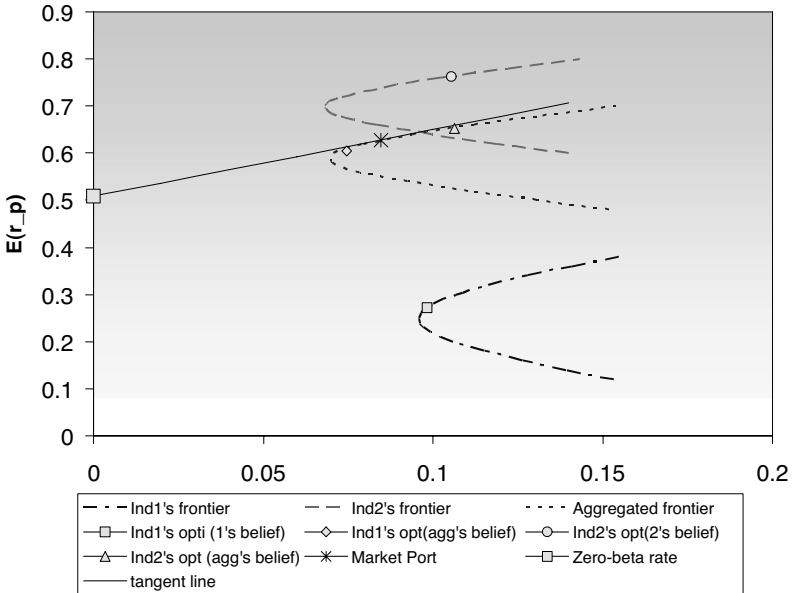


Figure 1. The mean variance frontiers under the heterogeneous beliefs and the market equilibrium consensus beliefs. The tangency line corresponding to the consensus belief has the market portfolio as the tangency portfolio and the expected return of the zero-beta portfolio of the market as the intercept with the expected return axis.

optimal portfolios of the two investors are strictly below the market equilibrium MV frontier. This may be hard to view in Figure 1. We provide a zoom-in version in Figure 2 to verify this observation.

Figure 2 clearly shows that the optimal portfolios of the two investors are not located on the MV frontier, though they are very close to it, and hence are MV inefficient. Intuitively, because of the bounded rationality and the fact that the market consensus belief is jointly determined by all market participants, no investor has knowledge about the “correct” market belief. Therefore, both investors made “wrong guesses” about the market, investor 1 being pessimistic and investor 2 being optimistic, their optimal portfolios suffer from those “wrong guesses” in terms of MV efficiency.

Sharpe [29] simulates market trading using his latest program APSIM, the program assumes a risk-free asset and there is a true probability distribution of future states of the market. Although not directly compatible due to the different setups,

Zoom in on the aggregate market frontier

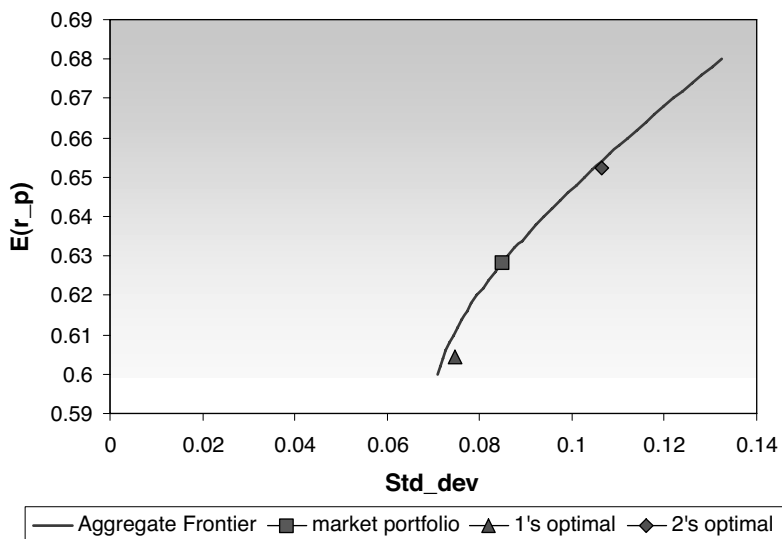


Figure 2. Close-up of the locations of individuals' optimal portfolios and the market portfolio relative to the market frontier when the market is in equilibrium.

Sharpe's findings⁶ are consistent with ours. That is the market portfolio outperforms most of other portfolios in terms of MV efficiency, superior fund managers can only at their best perform as well as the market portfolio. In our model, the "true" probability distribution of the future depends on the heterogeneous beliefs of the investors. Sharpe [29] explains the inferior performance of the active fund managers in the long run compared to the index funds by higher cost of active managers. To add to this, we suggest that it might be simply because it is very difficult for active managers to consistently make correct predictions about the future, while index funds tracks the market portfolio, which is always MV efficient. We conclude this numerical example by amending Sharpe's *Index Fund Premise* (IFP) to the following:

⁶Sharpe shows in Chapter 6, case 18, that superior fund manager who makes the "correct guesses" about the future of the market (meaning their probability assessment of the future coincide with the hypothetical true probability assessment) has a Sharpe Ratio (of 0.367) slightly above the market's value (of 0.366), other investors who make the "wrong guess" (meaning that their probability assessment of the future differs the hypothetical true probability assessment) are mostly penalized in terms of efficiency (with the lowest Sharpe Ratio of 0.237). However a lucky investor still has the same Sharpe Ratio (of 0.367) as the superior fund managers.

IFPa. Few of us are as smart as all of us.

IFPb. Few of us are as smart as all of us, and it is hard to identify such people in advance.

IFPc. Few of us are as smart as all of us, and it is hard to identify such people in advance, and they **definitely**⁷ charge more than they are worth.

4.2 The Geometric Relation of the Equilibrium MV Frontiers with and without Risk-Free Asset

To examine the *tangency relationship* of the traditional portfolio theory with heterogeneous beliefs, we consider the situation under which a riskless asset exists with future payoff R_f . Under the homogeneous belief, the classic portfolio theory tells us that the efficient portfolio frontier collapses to a straight line when a risk-free asset is added to the market. This straight line has one tangency point with the original frontier without a risk-free asset. This *tangency portfolio* is exactly the market portfolio when both the risk-free and equity markets are in equilibrium. We now examine this equilibrium tangency relationship under heterogeneous beliefs through the following example.

Example 4.1. Consider the case with $I = 2$ investors with beliefs $\mathcal{B}_i = (\Omega_i, \mathbf{y}_i)$ for $i = 1, 2$. There are $N = 3$ risky assets and a risk-free asset with payoff R_f . Let the absolute risk aversion coefficients $(\theta_1, \theta_2) = (5, 1)$, investors' initial wealth $W_{1,o} = W_{2,o} = \$10$, market endowment of risky assets $\mathbf{z}_m = (1, 1, 1)^T$, and $\mathbf{y}_o = (6.59, 9.34, 9.78)^T$, $\mathbf{1} = (1, 1, 1)^T$ and $\Omega_o = D_o C D_o$ where

$$D_o = \begin{pmatrix} 0.7933 & 0 & 0 \\ 0 & 0.8770 & 0 \\ 0 & 0 & 1.4622 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0.2233 & 0.1950 \\ 0.2233 & 1 & 0.1163 \\ 0.1950 & 0.1163 & 1 \end{pmatrix},$$

in which D_o corresponds to the standard deviation matrix and C is the correlation matrix. Assume that investors' beliefs are given by $\mathbf{y}_i = (1 + \delta_i)\mathbf{y}_o$ and $\Omega_i = D_i C D_i$, where $D_i = (1 + \epsilon_i)D_o$ for $i = 1, 2$. This implies that investors agree on the correlation of asset payoffs, but disagree about the volatilities and expected payoffs. Next we aggregate individuals' beliefs according to Proposition 2.1, first without a risk-free asset, then with a risk-free asset. The risk-free payoff R_f is determined such that the risk-free asset is in net-zero supply in equilibrium. To examine the tangency relationship, we plot the MV frontiers and optimal portfolios under the market consensus belief with and without risk-free asset for different values of δ_i and ϵ_i . Plots are shown in Figure 3.

⁷It reads "may" in Sharpe's book. Because no optimal portfolio is MV efficient unless the individual's belief coincides to the consensus belief, hence no one can beat the market portfolio when the market is in equilibrium.

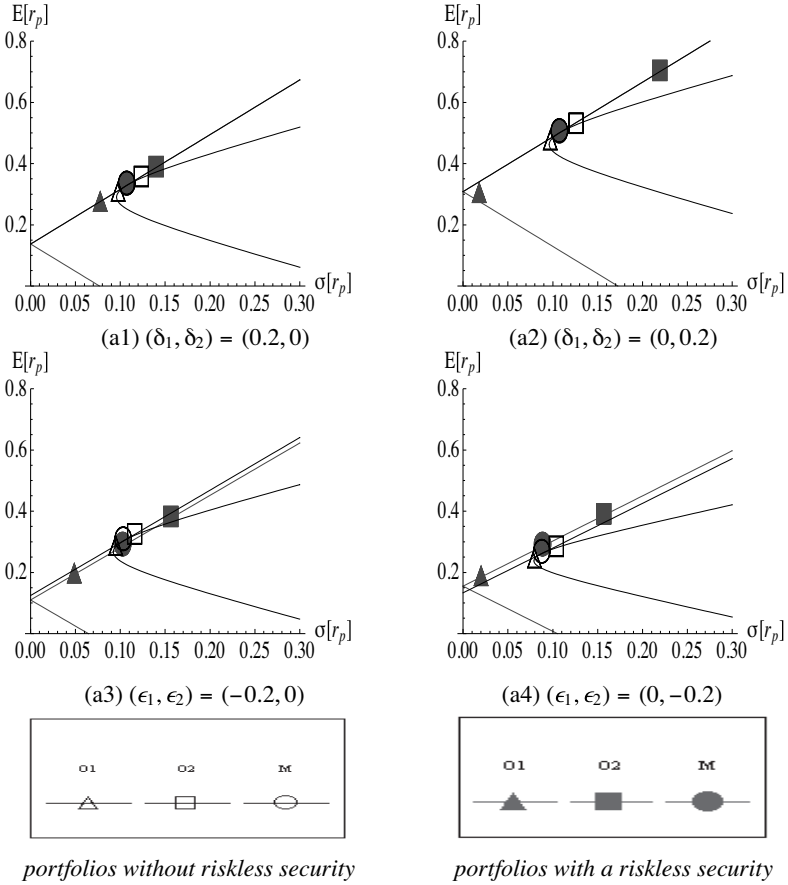


Figure 3. Compare the geometric relationships between market MV frontiers with and without a risk-free asset, when the risk-free asset is in net-zero supply. In (a1) and (a2), $y_1 \neq y_2, \Omega_1 = \Omega_2$; in (a3) and (a4), $y_1 = y_2, \Omega_1 \neq \Omega_2$.

When investors are homogeneous about the variances and covariances but heterogeneous about the expected payoffs of the risky asset, Figure 3(a1) and (a2) show that the tangency relation still holds. This is not surprising. Because of the homogeneous belief of the variance-covariance matrix $\Omega_i = \Omega_o$, the consensus variance-covariance matrix is given by $\Omega_a = \Omega_o$. From the construction of the consensus belief, the expected payoff y_a is a risk tolerance weighted average of the heterogeneous beliefs in the expected payoffs. Therefore, the consensus belief $\mathcal{B}_a$ remains the same when a risk-free asset is added to the market. Furthermore, since the risk-free asset is in net-zero supply, it follows from equation (11) in

Proposition 2.1 that

$$W_{m,o} = \mathbf{z}_m^T \mathbf{p}_0 = \frac{1}{\lambda_a^*} \left[\mathbf{y}_a^T \mathbf{z}_m - \frac{1}{I} \theta_a \mathbf{z}_m^T \Omega_a \mathbf{z}_m \right] = \frac{1}{R_f} \left[\mathbf{y}_a^T \mathbf{z}_m - \frac{1}{I} \theta_a \mathbf{z}_m^T \Omega_a \mathbf{z}_m \right].$$

Consequently, the riskless payoff R_f must equal to the zero-beta payoff λ_a^* . This implies that both the market's optimal marginal certainty equivalent wealth (CEW) and the equilibrium prices do not change when a risk-free asset is added to the market. Therefore the tangency relationship of the two market equilibrium frontiers with and without a risk-free asset holds with the market portfolio as the tangency portfolio. However, the efficiency of the optimal portfolios of the two investors depends on their expectations and risk aversion coefficients. On the one hand, when the more risk averse investor is optimistic and the less risk averse investor is pessimistic about the expected payoffs, Figure 3(a1) indicates that the optimal portfolios of both investors are located closer to the market portfolio and market MV frontiers. On the other hand, when the more risk averse investor is pessimistic and the less risk averse investor is optimistic about the expected payoffs, Figure 3(a2) indicates that the optimal portfolios of both investors are located far away from the market portfolio and the equilibrium market MV frontier, in particular, the optimal portfolio of the pessimistic investor may become even more inefficient when the risk-free asset is available. This means that adding a risk-free asset in this situation may help investor 2 to achieve a higher expected return for his optimal portfolio by sacrificing the MV efficiency of the optimal portfolio of investor 1.

When investors are heterogeneous in the variances of the asset payoffs but homogeneous in their expected payoffs, Figure 3(a3) and (a4) illustrate that the tangency relation breaks down. The risk-free payoff is no longer guaranteed to equal to the zero-beta payoff, which results in a change in the market's optimal CEW and also the equilibrium prices. In particular, when the relative less risk averse investor, investor 2 in this case, is more confident (measured by the smaller variance), Figure 3(a4) indicates that the existence of a risk-free asset actually pushes up the MV frontier, leading to higher expected return for the market portfolio. If one would believe that it is more likely that the less risk averse investor would be more confident in general, this implies that adding a risk-free asset would be more likely to push the portfolio frontier line above the tangency line of the frontier without the risk-free asset, leading to a higher market expected return. This observation would help us to explain the risk premium puzzle⁸. However, when the relative more risk averse investor, investor 1 in this case, is more confident, Figure 3(a3) implies that the existence of a risk-free asset actually pushes down the MV

⁸A detailed analysis on the conditions under which the market generates higher market risk premium and lower riskfree rate in market equilibrium when there are two assets and two beliefs can be found in He and Shi [15].

frontier, lowering the expected return of the market portfolio. This is an unexpected and surprising result. In the standard homogeneous case, the expected return of the market portfolio is independent of the existence of the risk-free asset which is in zero-net supply. The above analysis demonstrates that this is no longer the case when investors are heterogeneous. Based on Figure 3(a4), we observe that a restriction to the access of the risk-free asset may lead to a lower market expected return, a phenomena we experienced in the current financial crisis, and we leave further development along this line to future research.

5. The Impact of Heterogeneity on the Market with Many Investors

From the traditional portfolio theory under homogeneous beliefs, we know that all investors hold portfolios located on the market's MV efficient frontier, which is a hyperbola when there is no riskless asset and a straight line connecting the risk-free asset and the market portfolio, which is called the *Capital Market Line* (CML), when there is a risk-free asset. When investors are heterogeneous, we have shown through numerical examples of three assets and two investors in the previous section that investors no longer held portfolios on the equilibrium market MV frontier unless investors held the market consensus belief. Essentially, this is due to the fact that the consensus belief is determined endogenously by the heterogeneous beliefs and no individual knows the consensus belief in advance. In this section, we extend the analysis in the previous section to a market consisting of many different investors and we want to see whether those features observed for the market with two investors also hold for the market with many investors. We use mean-preserving spreads to characterize the beliefs of the investors and the spreads can be either univariate or multivariate. We use numerical examples to examine the MV efficiency of the optimal portfolios of investors and their relative position to the CML when the heterogeneity is either in the expected payoffs or the variances of the payoffs.

Example 5.1. Let the number of investors $I = 50$, number of risky assets $N = 3$, and market portfolio of risky assets is given by $\mathbf{z}_m = (25, 25, 25)^T$ (so that the average number of each stock per investor stays at 0.5 as in the previous example). Assume that there is a risk-free asset with payoff $R_f = 1.05$. Investors' initial wealth $W_{0,i} = \$10$, the ARA coefficients $\theta_i \sim \mathcal{N}(\theta_o, \sigma_\theta^2)$ with $\theta_o = 3$ and $\sigma_\theta = 0.3$ for $i = 1, 2, \dots, I$. Consider two types of probability distributions for investors' beliefs;

- (i) $\mathbf{y}_i = (1 + \delta_i)\mathbf{y}_o$ and $\Omega_i = D_i C D_i$, $D_i = (1 + \epsilon_i)D_o$ for $i = 1, \dots, 50$, where C , $\mathbf{y}_o$ and D_o are defined in Example 4.1 and $\delta_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_\delta^2)$ and $\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_\epsilon^2)$;
- (ii) $\mathbf{y}_i = \delta_i + \mathbf{y}_o$ and $\Omega_i = D_i C D_i$, $D_i = \text{Diag}[\epsilon_i + (0.7933, 0.8770, 1.4622)^T]$ for $i = 1, \dots, 50$, where $\delta_i \stackrel{iid}{\sim} \mathcal{MN}(\mathbf{0}, \Sigma_\delta)$ and $\epsilon_i \stackrel{iid}{\sim} \mathcal{MN}(\mathbf{0}, \Sigma_\epsilon)$ and $\Sigma_\delta = \sigma_\delta \text{Diag}[\mathbf{1}]$ and $\Sigma_\epsilon = \sigma_\epsilon \text{Diag}[\mathbf{1}]$.

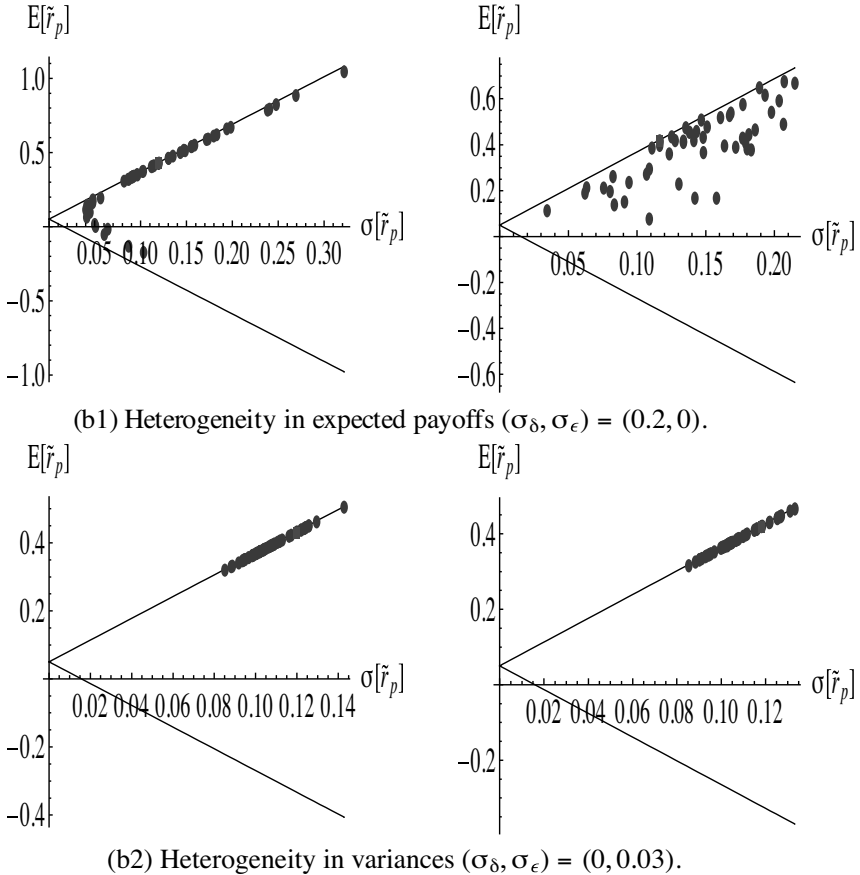


Figure 4. The optimal portfolios of all the 50 investors and their relative positions to the CML when investors' beliefs is homogeneous in variances and heterogeneous in expected payoffs in (b1) or homogeneous in expected payoffs and heterogeneous in variances in (b2). The left (right) panels correspond to univariate (multivariate) distribution in beliefs.

Symbols $\mathcal{N}$ and $\mathcal{MN}$ stand for (truncated, if necessary) normal distribution and multivariate normal distribution respectively.

The results for the two cases are plotted in Figure 4 in which the optimal portfolios of all 50 investors and their relative position to the CML are plotted. Figure 4(b1) illustrates the case when investors are homogeneous in variances but heterogeneous in the expected payoffs, while Figure 4(b2) illustrates the case other way around. The left panels correspond to the case with univariate belief dispersion and the right panels correspond to the case with multivariate dispersions. Figure 4

leads to the following interesting observations. (i) The optimal portfolios of all investors are almost on the CML when investors are heterogeneous in variances but homogeneous in the expected payoffs, illustrated by Figure 4(b2). The same effect is observed for univariate spread (left panel) and multivariate spreads (right panel). This shows that heterogeneity in covariances, characterized by mean-preserving spreads, plays insignificant role for the MV efficiency of the optimal portfolios of investors. (ii) The heterogeneity in expected payoff has significant impact on the MV efficiency of the optimal portfolios of the investors, illustrated in Figure 4(b1). The optimal portfolios become less MV efficient, in particular, when the belief dispersions are multivariate normally distributed (the right panel). Some optimal portfolios are far below the CML, even have lower expected return than the risk free rate (the left panel). In addition, when the belief dispersions are univariate, the optimal portfolios seem to form a hyperbolic curve below the equilibrium market MV efficient frontier (the left panel). However, when the divergence of opinions are not the same for each asset, optimal portfolios are scattered under the MV frontier without any significant pattern. This example shows that heterogeneity in expected payoff has more significant impact on the MV efficiency of optimal portfolios of investors than the heterogeneity in variances.

Based on the example with many investors, we find that the impact on the MV efficiency of the optimal portfolios of investors is significant for the heterogeneity in expected payoffs, but insignificant for the heterogeneity in variances and different mean preserved spreads in beliefs have different impact. However, based on the analysis in the previous section, the impact on the geometric relation of the frontiers with and without riskless asset is insignificant for the heterogeneity in expected payoffs, but significantly for the heterogeneity in variance. Therefore the heterogeneities have different impact on the MV efficiency and the geometric relation of the portfolio frontiers. Overall, we can see that, due to the heterogeneous beliefs, the market fails to provide investors with MV efficient portfolio, this generic feature is not what we would expect in the market with homogeneous belief. It shows that heterogeneous investors can never beat the market when the performance is measured by the MV efficiency.

6. Conclusion

Within the MV framework, by assuming that investors are heterogeneous, this paper examines the impact of the heterogeneity on the market equilibrium prices and equilibrium MV frontier in a market with many risky assets and no riskless asset. The heterogeneity is measured by the risk aversion coefficients, expected payoffs, and variance/covariance matrices of risky assets of heterogeneous investors. Investors are bounded rational in the sense that, based on their beliefs, they make their optimal portfolio decisions. To characterize the market equilibrium prices of the risky assets, we introduce the concept of consensus belief of the market and show how the consensus or market belief can be constructed from

the heterogeneous beliefs. Basically, under the market aggregation, the consensus belief is a weighted average of the heterogeneous beliefs. Explicit formula for the market equilibrium prices of the risky assets are derived. As a by-product of the consensus belief and equilibrium price formula, we show that the standard Black's zero-beta CAPM still holds with heterogeneous beliefs. The impact of the heterogeneity on the market equilibrium, mean variance frontier and the MV efficiency of the optimal portfolios of the investors are analyzed. In particular, through some numerical examples, we show that, in market equilibrium, the biased belief (from the market belief) of an investor makes his/her optimal portfolio below the equilibrium market MV frontier (although they may be very close to the MV efficient frontier). This demonstrates that bounded rational investors may never achieve their MV efficiency in market equilibrium. If we refer the heterogeneous investors as fund managers and the market portfolio as a market index, then our result offers an explanation on the empirical finding that, according to the MV criteria, managed funds under-perform the market indices on average. We also offer an explanation on Miller's proposition that "divergence of opinion corresponds to lower future asset returns" and the subsequent empirical findings on this. Furthermore, we show that the well known *tangency relation* of the frontiers with and without risk-free asset under the homogeneous beliefs breaks down under the heterogeneous beliefs, in particular when investors are heterogeneous in variances. Adding a risk-free asset to the market with many risky assets can have very complicated effect on the market in general. In the homogeneous market, the expected return of the market portfolio is independent of the existence of the risk-free asset. However, in the heterogeneous market, adding a risk-free asset to the market with many risky assets can have different impact on the expected return for the market portfolio in equilibrium. This result can be used to explain the risk premium puzzle and financial market crisis. In addition, the heterogeneity in the expected payoffs has significant impact on the MV efficiency of the subjectively optimal portfolios but insignificantly for the geometric relation. However, this is other way around for the heterogeneity in variance.

The implication of the heterogeneity on the market under different market conditions is far more complicated than it seems and it deserves further study. It would be interesting to extend the current static framework to a dynamic setting in which the heterogeneous beliefs are characterized by some trading strategies used in financial markets so that the MV efficiency of different trading strategies can be examined. It is also interesting to allow investors to learn overtime from the market through various learning mechanisms, such as the Bayesian updating rule, and adaptive learning mechanisms so that the expectation feedback (see Chiarella *et al.* [9]) and the MV efficiency under the learning can be examined. These extension will give us a richer modeling environment and hopefully lead to a better understanding of the phenomenons in our financial market. We leave these issues to the future research.

Appendices

A. Proof of Lemma 2.1

Let λ_i be the Lagrange multiplier and set

$$L(\mathbf{z}_i, \lambda_i) := \mathbf{y}_i^T \mathbf{z}_i - \frac{\theta_i}{2} \mathbf{z}_i^T \Omega_i \mathbf{z}_i + \lambda_i [\mathbf{p}_0^T \mathbf{z}_i - W_0^i]. \quad (25)$$

Then the optimal portfolio of agent i is determined by the first order condition

$$\frac{\partial L}{\partial \mathbf{z}_i} = \mathbf{0} \quad \Rightarrow \quad \mathbf{z}_i = \theta_i^{-1} \Omega_i^{-1} [\mathbf{y}_i - \lambda_i \mathbf{p}_0]. \quad (26)$$

Substituting (26) into (3) yields (5).

B. Proof of Proposition 2.1

From Definition 2.1, if the consensus belief $\mathcal{B}_a = (\mathbb{E}_a(\tilde{\mathbf{x}}), \Omega_a)$ exists, then

$$\mathbf{z}_i^* = \theta_a^{-1} \Omega_a^{-1} [\mathbf{y}_a - \lambda_a^* \mathbf{p}_0]. \quad (27)$$

Applying the market equilibrium condition to (27), we must have

$$\mathbf{z}_m = \sum_{i=1}^I \mathbf{z}_i^* = I \left[\theta_a^{-1} \Omega_a^{-1} [\mathbf{y}_a - \lambda_a^* \mathbf{p}_0] \right]. \quad (28)$$

This leads to the equilibrium price (11). On the other hand, it follows from the individuals demand (4) and the market clearing condition (7) that, under the heterogeneous beliefs,

$$\mathbf{z}_m = \sum_{i=1}^I \mathbf{z}_i^* = \sum_{i=1}^I \theta_i^{-1} \Omega_i^{-1} [\mathbf{y}_i - \lambda_i^* \mathbf{p}_0]. \quad (29)$$

Under the definitions (9) and (10), we can re-write equation (29) as

$$\mathbf{z}_m = \sum_{i=1}^I \theta_i^{-1} \Omega_i^{-1} \mathbf{y}_i - \left(\sum_{i=1}^I \theta_i^{-1} \lambda_i^* \Omega_i^{-1} \right) \mathbf{p}_0 = I \theta_a^{-1} \Omega_a^{-1} \mathbf{y}_a - I \theta_a^{-1} \lambda_a^* \Omega_a^{-1} \mathbf{p}_0, \quad (30)$$

which leads to the same market equilibrium price (11). This shows that $\mathcal{B}_a = \{\Omega_a, \mathbf{y}_a\}$ defined in (9) and (10) is the consensus belief. Inserting (11) into (4) give the equilibrium optimal portfolio (12) of investor i .

C. Proof of Corollary 2.1

The equilibrium price vector in (11) can be re-written to express the price of each asset

$$p_{0,j} = \frac{1}{\lambda_a^*} (y_{a,j} - \theta_a / I \sum_{k=1}^N \sigma_{j,k} z_{m,k}) = \frac{1}{\lambda_a^*} [y_{a,j} - \frac{\theta_a}{I} \text{Cov}_a(\tilde{x}_j, \tilde{W}_m)]. \quad (31)$$

It follows from (31) that $y_{a,j} - \lambda_a^* p_{0,j} = \frac{\theta_a}{I} \text{Cov}_a(\tilde{x}_j, \tilde{W}_m)$ and hence

$$\frac{y_{a,j}}{p_{0,j}} - \lambda_a^* = \frac{1}{p_{0,j}} \frac{\theta_a}{I} \text{Cov}_a(\tilde{x}_j, \tilde{W}_m).$$

Therefore

$$\mathbb{E}_a(\tilde{r}_j) - (\lambda_a^* - 1) = \frac{1}{p_{0,j}} \frac{\theta_a}{I} \text{Cov}_a(\tilde{x}_j, \tilde{W}_m). \quad (32)$$

It follows from $W_{m0} = \frac{1}{\lambda_a^*} \mathbf{z}_m^T (\mathbf{y}_a - \theta_a \Omega_a \mathbf{z}_m / I)$ that

$$\lambda_a^* = \frac{\mathbf{z}_m^T \mathbf{y}_a - \theta_a \mathbf{z}_m^T \Omega_a \mathbf{z}_m / I}{W_{m0}}. \quad (33)$$

Using the definition of λ_a^* in (33), we obtain

$$\mathbb{E}_a(\tilde{r}_m) - (\lambda_a^* - 1) = \frac{\mathbf{y}_a^T \mathbf{z}_m}{\mathbf{z}_m^T \mathbf{p}_0} - \lambda_a^* = \frac{\mathbf{y}_a^T \mathbf{z}_m}{W_{m0}} - \frac{\mathbf{z}_m^T \mathbf{y}_a - \theta_a \mathbf{z}_m^T \Omega_a \mathbf{z}_m / I}{W_{m0}}.$$

Thus

$$\mathbb{E}_a(\tilde{r}_m) - (\lambda_a^* - 1) = \frac{\theta_a \mathbf{z}_m^T \Omega_a \mathbf{z}_m / I}{W_{m0}} \neq 0. \quad (34)$$

Dividing (32) by (34) leads to

$$\begin{aligned} \frac{\mathbb{E}_a(\tilde{r}_j) - (\lambda_a^* - 1)}{\mathbb{E}_a(\tilde{r}_m) - (\lambda_a^* - 1)} &= \frac{(\frac{1}{p_{0,j}} \frac{\theta_a}{I} \text{Cov}_a(\tilde{x}_j, \tilde{W}_m))}{(\frac{\theta_a \mathbf{z}_m^T \Omega_a \mathbf{z}_m / I}{W_{m0}})} = \frac{\frac{1}{p_{0,j}} \text{Cov}_a(\tilde{x}_j, \tilde{W}_m)}{\frac{\sigma_{a,m}^2}{W_{m0}}} \\ &= \frac{\text{Cov}_a(\frac{\tilde{x}_j}{p_{0,j}}, \frac{\tilde{W}_m}{W_{m0}})}{\frac{\sigma_{a,m}^2}{W_{m0}^2}} = \frac{\text{Cov}_a(\tilde{r}_j, \tilde{r}_m)}{\sigma_a^2(\tilde{r}_m)} = \beta_j, \end{aligned} \quad (35)$$

leading to the CAPM-like relation in (13).

Table 1. Market specifications and heterogeneous beliefs.

Initial Wealth	Risk Aversion	Expected payoffs	Variance/Covariance of payoffs
$W_0^1 = 10$	$\theta_1 = 5$	$\mathbf{y}_1 = \begin{pmatrix} 6.60 \\ 9.35 \\ 9.78 \end{pmatrix}$	$\Omega_1 = \begin{pmatrix} 0.6292 & 0.1553 & 0.2262 \\ & 0.7692 & 0.1492 \\ & & 2.1381 \end{pmatrix}$
$W_0^2 = 10$	$\theta_2 = 1$	$\mathbf{y}_2 = \begin{pmatrix} 9.60 \\ 12.35 \\ 12.78 \end{pmatrix}$	$\Omega_2 = \begin{pmatrix} 0.4292 & -0.0447 & 0.0262 \\ & 0.5692 & -0.0508 \\ & & 1.7381 \end{pmatrix}$

D. A Numerical Example

Example D.1. Let $I = 2$ and $N = 3$. Consider the set up in Table 1. Assuming there is one share available for each asset, that is, $\mathbf{z}_m = (1, 1, 1)^T$. Based on the information in Table 1, we use equation (8) and Excel Solver to solve for the equilibrium price vector and obtain the market equilibrium price $\mathbf{p}_0 = (5.6436, 7.4328, 6.9236)^T$. The optimal portfolios and shadow prices of the investors are given by $\mathbf{z}_1^* = (0.380, 0.768, 0.310)^T$, $\lambda_1^* = 0.7894$ for investor 1 and $\mathbf{z}_2^* = (0.620, 0.232, 0.690)^T$ and $\lambda_2^* = 1.6520$ for investor 2. Using Proposition 2.1, we construct the consensus belief $\mathcal{B}_a$, the aggregate risk aversion coefficient θ_a , and the aggregate shadow price λ_a^* , and obtain the result in Table 2.

We then use the market equilibrium price to convert the consensus belief from payoffs to returns as follows. Let $P_0 = \text{diag}[\mathbf{p}_0] = \text{diag}(5.6436, 7.4328, 6.9236)$ and

$$\begin{aligned}
\mathbb{E}_i(\tilde{\mathbf{r}}) &:= P_0^{-1} \mathbf{y}_i - \mathbf{1}, & V_i(\tilde{\mathbf{r}}) &:= P_0^{-1} \Omega_i P_0^{-1}, & i &= 1, 2, a; \\
\mathbf{w}_i^* &:= \frac{1}{W_{i,o}} P_0 \mathbf{z}_i^*, & \mathbb{E}_i(\tilde{r}_{ip}^*) &:= \mathbb{E}_i(\tilde{\mathbf{r}})^T \mathbf{w}_i^*, & \sigma_{ip}^* &= (\mathbf{w}_i^{*T} V_i(\tilde{\mathbf{r}}) \mathbf{w}_i^*)^{1/2}, & i &= 1, 2; \\
\mathbb{E}_a(\tilde{r}_{ip}^*) &:= \mathbb{E}_a(\tilde{\mathbf{r}})^T \mathbf{w}_i^*, & \sigma_{ip}^a &= (\mathbf{w}_i^{*T} V_a(\tilde{\mathbf{r}}) \mathbf{w}_i^*)^{1/2}, & i &= 1, 2; \\
\mathbf{w}_m &:= \frac{1}{W_{m,o}} P_0 \mathbf{z}_m, & \mathbb{E}_a(\tilde{r}_m) &:= \mathbb{E}_a(\tilde{\mathbf{r}})^T \mathbf{w}_m, \\
\sigma_{a,m} &:= (\mathbf{w}_m^T V_a(\tilde{\mathbf{r}}) \mathbf{w}_m)^{1/2}, & \beta &:= \frac{V_a(\tilde{\mathbf{r}}) \mathbf{w}_m}{\sigma_{a,m}^2}.
\end{aligned}$$

We then obtain the results in Table 3.

In the above definitions, $\mathbb{E}_i(\mathbf{r})$ and $V_i(\mathbf{r})$ are the expected return vectors and covariance matrices in terms of asset returns for each investor. Subsequently, $\mathbf{w}_i^*$ are the individuals' optimal portfolio weights, $\mathbb{E}_i(\tilde{r}_{ip}^*)$ and σ_{ip}^* are the expected return and standard deviations of the optimal portfolios of investors, respectively, under their subjective beliefs, $\mathcal{B}_i = (\mathbb{E}_i(\tilde{r}_{ip}^*), \sigma_{ip}^a)$. Similarly, under the market

Table 2. The market consensus beliefs, shadow price and ARA.

Initial Market Wealth	Shadow Price	Risk Aversion	Expected payoffs	Variance/Covariance of payoffs
$W_{m0} = 20$	$\lambda_a^* = 1.5083$	$\theta_a = 1.6667$	$\mathbf{y}_a = \begin{pmatrix} 8.88 \\ 11.63 \\ 12.06 \end{pmatrix}$	$\Omega_a = \begin{pmatrix} 0.4383 & -0.0356 & 0.0352 \\ & 0.5783 & -0.0417 \\ & & 1.9472 \end{pmatrix}$

Table 3. Heterogeneous beliefs and the consensus belief, the individual optimal and market portfolios in equilibrium, and the means and standard deviations of these portfolios under heterogeneous and consensus beliefs, respectively.

Expected returns	Variance/Covariance of returns	Portfolio weights	Portfolio Return/SD
$\mathbb{E}_1(\tilde{\mathbf{r}}) = \begin{pmatrix} .1690 \\ .2577 \\ .4126 \end{pmatrix}$	$V_1 = \begin{pmatrix} .0198 & .0037 & .0058 \\ & .0139 & .0029 \\ & & .0446 \end{pmatrix}$	$\mathbf{w}_1^* = \begin{pmatrix} .2144 \\ .5711 \\ .2145 \end{pmatrix}$	$\mathbb{E}_1(r_{1p}^*) = .2719$ $\sigma_{1p}^* = .09824$ $\mathbb{E}_a(r_{1p}^*) = .6043$ $\sigma_{1p}^a = .0748$
$\mathbb{E}_2(\tilde{\mathbf{r}}) = \begin{pmatrix} .7006 \\ .6613 \\ .8459 \end{pmatrix}$	$V_2 = \begin{pmatrix} .0135 & -.0011 & .0007 \\ & .0103 & -.0010 \\ & & .0404 \end{pmatrix}$	$\mathbf{w}_2^* = \begin{pmatrix} .3499 \\ .1722 \\ .4778 \end{pmatrix}$	$\mathbb{E}_2(r_{2p}^*) = .7633$ $\sigma_{2p}^* = .1054$ $\mathbb{E}_a(r_{2p}^*) = .6522$ $\sigma_{2p}^a = .1065$
$\mathbb{E}_a(\tilde{\mathbf{r}}) = \begin{pmatrix} .5729 \\ .5644 \\ .7418 \end{pmatrix}$	$V_a = \begin{pmatrix} .0138 & -.0008 & .0009 \\ & .0105 & -.0008 \\ & & .0406 \end{pmatrix}$	$\mathbf{w}_m = \begin{pmatrix} .2822 \\ .3716 \\ .3462 \end{pmatrix}$	$\mathbb{E}_a(r_m) = .6283$ $\sigma_{a,m} = .0848$

$$\boldsymbol{\beta} = (0.5390 \ 0.4681 \ 1.9468)^T$$

belief $\mathcal{B}_a$, $\mathbf{w}_m$ is the market portfolio weight vector, $\mathbb{E}_a(\tilde{r}_m)$ and $\sigma_{a,m}$ are the market return and volatility under the market belief respectively. Finally $\boldsymbol{\beta}$ is the vector of beta coefficients. According to these definitions, we obtain results in Table 3.

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Security Pricing with Information-Sensitive Discounting*

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In this paper, incomplete-information models are developed for the pricing of securities in a stochastic interest rate setting. In particular, we consider credit-risky assets that may include random recovery upon default. The market filtration is generated by a collection of information processes associated with economic factors, on which interest rates depend, and information processes associated with market factors used to model the cash flows of the securities. We use information-sensitive pricing kernels to give rise to stochastic interest rates. Semi-analytical expressions for the price of credit-risky bonds are derived, and a number of recovery models are constructed which take into account the perceived state of the economy at the time of default. The price of a European-style call bond option is deduced, and it is shown how examples of hybrid securities, like inflation-linked credit-risky bonds, can be valued. Finally, a cumulative information process is employed to develop pricing kernels that respond to the amount of aggregate debt of an economy.

Keywords: Asset pricing, incomplete information, stochastic interest rates, credit risk, recovery models, credit-inflation hybrid securities, information-sensitive pricing kernels.

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1. Introduction

The information-based framework developed by Brody et al. in [7] and [8] is a method to price assets based on incomplete information available to market participants about the cash flows of traded assets. In this approach the value of a number of different types of assets can be derived by modelling the random cash flows defining the asset, and by explicitly constructing the market filtration that is generated by the incomplete information about independent market factors that build the cash flows. This principle has been used in [7] to derive the price processes of credit-risky securities, in [8] to value equity-type assets with various dividend structures, in [9] to price insurance and reinsurance products, and in [6] to price assets in a market with asymmetric information. However, for simplicity, in this framework it is typically assumed that interest rates are deterministic.

One of the earliest generalizations of the models developed in [7] to include stochastic interest rates can be found in [19]. Here, it is assumed that the filtration is generated jointly by the information processes associated with the future random cash flows of a defaultable bond and by an independent Brownian motion that drives the stochastic discount factor.

Pricing kernel models for interest rates have been studied by the authors of [10], [15] and [18], among others. In such models, the price P_{tT} at time t of a sovereign bond with maturity T and unit payoff, is given by the formula

$$P_{tT} = \frac{\mathbb{E}^{\mathbb{P}}[\pi_T | \mathcal{F}_t]}{\pi_t}, \quad (1)$$

where $\{\pi_t\}_{t \geq 0}$ is the $\{\mathcal{F}_t\}$ -adapted pricing kernel process and $\mathbb{P}$ denotes the real probability measure. Given the filtration $\{\mathcal{F}_t\}_{t \geq 0}$, arbitrage-free interest rate models can be obtained by specifying the dynamics of the pricing kernel. In particular, term structure models with positive interest rates are generated by requiring that $\{\pi_t\}$ is a positive supermartingale. A more recent approach to constructing interest rate models in an information-based setting, presented in [14], develops the notion of an information-sensitive pricing kernel. The pricing kernel is modelled by a function of time and information processes that are observed by market participants and that over time reveal genuine information about economic factors at a certain rate. In order to obtain positive interest rate models, this function must be chosen so that the pricing kernel has the supermartingale property. A scheme for generating appropriate functions to construct such pricing kernels in an information-based approach is considered in [2]. Incomplete information about economic factors that is available to investors is modelled in [2] by using time-inhomogeneous Markov processes. The Brownian bridge information process considered in [14] and, more generally, the subclass of the continuous Lévy random bridges, recently introduced in [12], are examples of time-inhomogeneous Markov processes.

In this paper we describe how credit-risky securities can be priced within the framework considered in [7] while including a stochastic discount factor by use of information-sensitive pricing kernels. To this end, we proceed in Section 2 to recap briefly the theory for the pricing of fixed-income securities in an information-based framework described in [14]. In Section 3 we recall the result in [2] that can be used to obtain the explicit dynamics of the pricing kernel by use of so-called “weighted heat kernels” with time-inhomogeneous Markov processes. In Section 4, we derive the price process of a defaultable discount bond and compute the yield spreads between digital bonds and sovereign bonds. Section 5 considers a number of random recovery models for defaultable bonds, and in the following section we derive a semi-analytical formula for the price of a European option on a credit-risky bond. In Section 7 we demonstrate how to price credit-inflation securities as an example of a hybrid structure. We investigate the valuation of credit-risky coupon bonds in Section 8 and conclude by considering a pricing kernel that reacts to the level of debt accumulated in a country over a finite period of time.

2. Information-Sensitive Pricing Kernels

We define the probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$, where $\mathbb{P}$ denotes the real probability measure. We fix two dates T and U , where $T < U$, and introduce a macroeconomic random variable X_U , the value of which is revealed at time U . Noisy information about the economic factor available to market participants is modelled by the information process $\{\xi_{tU}\}_{0 \leq t \leq U}$ given by

$$\xi_{tU} = \sigma t X_U + \beta_{tU}. \quad (2)$$

Here the parameter σ represents the information flow rate at which the true value of X_U is revealed as time progresses, and the noise component $\{\beta_{tU}\}_{0 \leq t \leq U}$ is a Brownian bridge that is taken to be independent of X_U . We assume that the market filtration $\{\mathcal{F}_t\}_{t \geq 0}$ is generated by $\{\xi_{tU}\}$, and note that it is shown in, e.g., [7] that $\{\xi_{tU}\}$ is a Markov process with respect to its natural filtration. We consider pricing kernels $\{\pi_t\}$ that are of the form

$$\pi_t = M_t f(t, \xi_{tU}), \quad (3)$$

where $\{M_t\}_{0 \leq t < U}$ is the density martingale associated with a change of measure from $\mathbb{P}$ to the so-called “bridge measure” $\mathbb{B}$ under which the information process has the law of a Brownian bridge. It is proven in [7], that $\{M_t\}$ satisfies the differential equation

$$dM_t = -\sigma \frac{U}{U-t} \mathbb{E}^{\mathbb{P}}[X_U | \xi_{tU}] M_t dW_t, \quad (4)$$

where $\{W_t\}_{0 \leq t < U}$ is an $(\{\mathcal{F}_t\}, \mathbb{P})$ -Brownian motion given by

$$W_t = \xi_{tU} + \int_0^t \frac{1}{U-s} \xi_{sU} ds - \sigma \int_0^t \frac{U}{U-s} \mathbb{E}^{\mathbb{P}}[X_U | \xi_{sU}] ds. \quad (5)$$

By applying Bayes change-of-measure formula to equation (1), we can express the price P_{tT} at time t of a sovereign discount bond with maturity T by

$$P_{tT} = \frac{\mathbb{E}^{\mathbb{B}}[f(T, \xi_{TU}) | \xi_{tU}]}{f(t, \xi_{tU})}. \quad (6)$$

Next we introduce the random variable Y_{tT} defined by

$$Y_{tT} = \xi_{TU} - \frac{U - T}{U - t} \xi_{tU}, \quad (7)$$

and observe that under the measure $\mathbb{B}$, Y_{tT} is a Gaussian random variable with zero mean and variance given by

$$v_{tT}^2 = \frac{(T - t)(U - T)}{U - t}. \quad (8)$$

It can be verified that Y_{tT} is independent of ξ_{tU} under $\mathbb{B}$, see [14]. Next, we introduce a Gaussian random variable Y , with zero mean and unit variance; this allows us to write $Y_{tT} = v_{tT}Y$. Since ξ_{tU} is $\mathcal{F}_t$ -measurable and Y is independent of ξ_{tU} , we can express the price of a sovereign bond by the following Gaussian integral:

$$P_{tT} = \frac{1}{f(t, \xi_{tU})} \int_{-\infty}^{\infty} f\left(T, v_{tT}y + \frac{U - T}{U - t} \xi_{tU}\right) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y^2\right) dy. \quad (9)$$

Interest rate models of various types can therefore be constructed in this framework by specifying the function $f(t, x)$. However, pricing kernels constructed by the relation (3) are not automatically $(\{\mathcal{F}_t\}, \mathbb{P})$ -supermartingales. In particular, to guarantee positive interest rates, it is a requirement that the function $f(t, x)$ satisfies the following differential inequality, see [14]:

$$\frac{x}{U - t} \frac{\partial}{\partial x} f(t, x) - \frac{1}{2} \frac{\partial^2}{\partial x^2} f(t, x) - \frac{\partial}{\partial t} f(t, x) > 0. \quad (10)$$

We emphasize that finding a function which satisfies relation (10) is equivalent to finding a process $\{f(t, \xi_{tU})\}_{0 \leq t < U}$ that is a positive $(\{\mathcal{F}_t\}, \mathbb{B})$ -supermartingale. Hence the pricing kernel $\{\pi_t\}_{0 \leq t < U}$ is a positive $(\{\mathcal{F}_t\}, \mathbb{P})$ -supermartingale since

$$\mathbb{E}^{\mathbb{P}}[\pi_T | \mathcal{F}_t] = M_t \mathbb{E}^{\mathbb{B}}[f(T, \xi_{TU}) | \xi_{tU}] \leq M_t f(t, \xi_{tU}) = \pi_t. \quad (11)$$

We now proceed to construct such positive $(\{\mathcal{F}_t\}, \mathbb{B})$ -supermartingales using a technique known as the “weighted heat kernel approach”, presented in [1] and adapted for time-inhomogeneous Markov processes in [2].

3. Weighted Heat Kernel Models

We consider the filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$ where the filtration $\{\mathcal{F}_t\}_{t \geq 0}$ is generated by the information process $\{\xi_{tU}\}$. We recall that the martingale $\{M_t\}$ satisfying equation (4), induces a change of measure from $\mathbb{P}$ to the bridge measure $\mathbb{B}$, and that the information process $\{\xi_{tU}\}$ is a Brownian bridge under $\mathbb{B}$. The Brownian bridge is a time-inhomogeneous Markov process with respect to its own filtration.

Let $w : \mathbb{R}_0^+ \times \mathbb{R}_0^+ \rightarrow \mathbb{R}^+$ be a weight function that satisfies

$$w(t, u - s) \leq w(t - s, u) \quad (12)$$

for arbitrary $t, u \in \mathbb{R}_0^+$ and $s \leq t \wedge u$. Then, for $t < U$ and a positive integrable function $F(x)$, the process $\{f(t, \xi_{tU})\}$ given by

$$f(t, \xi_{tU}) = \int_0^{U-t} \mathbb{E}^{\mathbb{B}}[F(\xi_{t+u,U}) | \xi_{tU}] w(t, u) du \quad (13)$$

is a positive supermartingale.

The proof of this result goes as follows. For $f(t, x)$ an integrable function, the process $\{f(t, \xi_{tU})\}$ is a supermartingale for $0 \leq s \leq t < U$ if

$$\mathbb{E}^{\mathbb{B}}[f(t, \xi_{tU}) | \xi_{sU}] \leq f(s, \xi_{sU}) \quad (14)$$

is satisfied. We define the process $\{p(t, u, \xi_{tU})\}$ by

$$p(t, u, \xi_{tU}) = \mathbb{E}^{\mathbb{B}}[F(\xi_{t+u,U}) | \xi_{tU}], \quad (15)$$

where $0 \leq u \leq U - t$. Then we have:

$$\begin{aligned} \mathbb{E}^{\mathbb{B}}[f(t, \xi_{tU}) | \xi_{sU}] &= \int_0^{U-t} \mathbb{E}^{\mathbb{B}}[p(t, u, \xi_{tU}) | \xi_{sU}] w(t, u) du \\ &= \int_0^{U-t} p(s, u + t - s, \xi_{sU}) w(t, u) du \\ &= \int_{t-s}^{U-s} p(s, v, \xi_{sU}) w(t, v - t + s) dv. \end{aligned} \quad (16)$$

Here we have used the tower rule of conditional expectation and the Markov property of $\{\xi_{tU}\}$. Next we make use of the relation (12) to obtain

$$\begin{aligned} \mathbb{E}^{\mathbb{B}}[f(t, \xi_{tU}) | \xi_{sU}] &\leq \int_{t-s}^{U-s} p(s, v, \xi_{sU}) w(t - (t - s), v) dv \\ &\leq \int_0^{U-s} p(s, v, \xi_{sU}) w(s, v) dv \\ &= f(s, \xi_{sU}). \end{aligned} \quad (17)$$

Thus, $\{f(t, \xi_{tU})\}$ is a positive $(\{\mathcal{F}_t\}, \mathbb{B})$ -supermartingale if $F(x)$ is positive.

The method based on equation (13) provides one with a convenient way to generate positive pricing kernels driven by the information process $\{\xi_{tU}\}$. These models can be used to generate information-sensitive dynamics of positive interest rates. In particular, the functions $f(t, x)$ underlying such interest rate models satisfy inequality (10).

4. Credit-Risky Discount Bonds

We introduce two dates T and U , where $T < U$, and attach two independent factors X_T and X_U to these dates respectively. We assume that X_T is a discrete random variable that takes values in $\{x_0, x_1, \dots, x_n\}$ with a priori probabilities $\{p_0, p_1, \dots, p_n\}$, where $1 \geq x_n > x_{n-1} > \dots > x_1 > x_0 \geq 0$. We take X_T to be the random variable by which the future payoff of a credit-risky bond issued by a firm is modelled. The second random variable X_U is assumed to be continuous and represents a macroeconomic factor. For instance, one might consider the GDP level at time U of an economy in which the bond is issued. With the two X -factors, we associate the independent information processes $\{\xi_{tT}\}_{0 \leq t \leq T}$ and $\{\xi_{tU}\}_{0 \leq t \leq U}$ given by

$$\xi_{tU} = \sigma_1 t X_U + \beta_{tU}, \quad \xi_{tT} = \sigma_2 t X_T + \beta_{tT}. \quad (18)$$

The market filtration $\{\mathcal{F}_t\}$ is generated by both information processes $\{\xi_{tT}\}$ and $\{\xi_{tU}\}$. The price B_{tT} at $t \leq T$ of a defaultable discount bond with payoff H_T at $T < U$ can be written in the form

$$B_{tT} = \frac{\mathbb{E}^{\mathbb{P}}[\pi_T H_T | \mathcal{F}_t]}{\pi_t} \quad (19)$$

where $\{\pi_t\}$ is the pricing kernel. We consider the positive martingale $\{M_t\}_{0 \leq t \leq U}$ that satisfies

$$dM_t = -\sigma_1 \frac{U}{U-t} \mathbb{E}^{\mathbb{P}}[X_U | \xi_{tU}] M_t dW_t, \quad (20)$$

and introduce the pricing kernel $\{\pi_t\}$ given by

$$\pi_t = M_t f(t, \xi_{tU}). \quad (21)$$

The dependence of the pricing kernel on $\{\xi_{tU}\}$ implies that interest rates fluctuate due to the information flow in the market about the likely value of the macroeconomic factor X_U at time U . Since the information processes are Markovian, the price of the defaultable discount bond can be expressed by

$$B_{tT} = \frac{\mathbb{E}^{\mathbb{P}}[M_T f(T, \xi_{TU}) H_T | \xi_{tT}, \xi_{tU}]}{M_t f(t, \xi_{tU})}, \quad (22)$$

where H_T is the bond payoff at maturity T . We now suppose that the payoff of the credit-risky bond is a function of X_T and the value of the information process associated with X_U at the bond's maturity T , that is

$$H_T = H(X_T, \xi_{TU}). \quad (23)$$

Due to the independence property of the information processes, the price of the credit-risky discount bond can be written as follows:

$$B_{tT} = \frac{\mathbb{E}^{\mathbb{P}} \left[\mathbb{E}^{\mathbb{P}} \left[M_T f(T, \xi_{TU}) H(X_T, \xi_{TU}) \mid \xi_{tT} \right] \mid \xi_{tU} \right]}{M_t f(t, \xi_{tU})}. \quad (24)$$

By applying the conditional form of Bayes formula, we change the measure to the bridge measure $\mathbb{B}$ with respect to which the outer expectation is taken:

$$B_{tT} = \frac{\mathbb{E}^{\mathbb{B}} \left[\mathbb{E}^{\mathbb{B}} \left[f(T, \xi_{TU}) H(X_T, \xi_{TU}) \mid \xi_{tT} \right] \mid \xi_{tU} \right]}{f(t, \xi_{tU})}. \quad (25)$$

At this stage, we define a random variable Y_{tT} by

$$Y_{tT} = \xi_{tU} - \frac{U-t}{U-t} \xi_{tU}. \quad (26)$$

Since $\{\xi_{tU}\}$ is a Brownian bridge under $\mathbb{B}$, we know that Y_{tT} is a Gaussian random variable with zero mean and variance

$$\text{Var}^{\mathbb{B}}[Y_{tT}] = \frac{(T-t)(U-t)}{(U-t)}. \quad (27)$$

Next we introduce a standard Gaussian random variable Y and write $Y_{tT} = \nu_{tT} Y$, where $\nu_{tT}^2 = \text{Var}^{\mathbb{B}}[Y_{tT}]$. We can now express the price of the defaultable discount bond in terms of Y as

$$B_{tT} = \frac{\mathbb{E}^{\mathbb{B}} \left[\mathbb{E}^{\mathbb{B}} \left[f \left(T, \nu_{tT} Y + \frac{U-t}{U-t} \xi_{tU} \right) H \left(X_T, \nu_{tT} Y + \frac{U-t}{U-t} \xi_{tU} \right) \mid \xi_{tT} \right] \mid \xi_{tU} \right]}{f(t, \xi_{tU})}. \quad (28)$$

Since $f(T, Y, \xi_{tU})$ in the numerator does not depend on ξ_{tT} , we can write

$$B_{tT} = \frac{\mathbb{E}^{\mathbb{B}} \left[f \left(T, \nu_{tT} Y + \frac{U-t}{U-t} \xi_{tU} \right) \mathbb{E}^{\mathbb{B}} \left[H \left(X_T, \nu_{tT} Y + \frac{U-t}{U-t} \xi_{tU} \right) \mid \xi_{tT} \right] \mid \xi_{tU} \right]}{f(t, \xi_{tU})}. \quad (29)$$

Because both Y and ξ_{tU} are independent of ξ_{tT} , the inner conditional expectation in this expression can be carried out explicitly. We obtain

$$B_{tT} = \frac{\mathbb{E}^{\mathbb{B}} \left[f \left(T, \nu_{tT} Y + \frac{U-t}{U-t} \xi_{tU} \right) \sum_{i=0}^n \pi_{it} H \left(x_i, \nu_{tT} Y + \frac{U-t}{U-t} \xi_{tU} \right) \mid \xi_{tU} \right]}{f(t, \xi_{tU})}, \quad (30)$$

where π_{it} denotes the conditional density of X_T , given by

$$\pi_{it} = \mathbb{P} \left[X_T = x_i \mid \xi_{iT} \right] = \frac{p_i \exp \left[\frac{T}{T-t} \left(\sigma_2 x_i \xi_{iT} - \frac{1}{2} \sigma_2^2 x_i^2 t \right) \right]}{\sum_{i=0}^n p_i \exp \left[\frac{T}{T-t} \left(\sigma_2 x_i \xi_{iT} - \frac{1}{2} \sigma_2^2 x_i^2 t \right) \right]}. \quad (31)$$

Finally, since the random variable ξ_{tU} , appearing in the arguments of $f(T, Y, \xi_{tU})$ and of $H(Y, \xi_{tU})$ in (30), is measurable at time t and Y is independent of the conditioning random variable ξ_{tU} , the conditional expectation reduces to a Gaussian integral over the range of the random variable Y :

$$\begin{aligned} B_{tT} &= \frac{1}{f(t, \xi_{tU})} \sum_{i=0}^n \pi_{it} \int_{-\infty}^{\infty} f \left(T, v_{tTY} + \frac{U-T}{U-t} \xi_{tU} \right) H \left(x_i, v_{tTY} + \frac{U-T}{U-t} \xi_{tU} \right) \\ &\quad \times \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{1}{2} y^2 \right) dy. \end{aligned} \quad (32)$$

In the case where the payoff is $H_T = X_T$, by using the expression for the sovereign bond given by equation (9), we can write the price of the defaultable bond as:

$$B_{tT} = P_{tT} \sum_{i=0}^n x_i \pi_{it}, \quad (33)$$

where π_{it} is defined by equation (31). For $n = 1$, the defaultable bond pays a principal of x_1 units of currency, if there is no default, and x_0 units of currency in the event of default; we call such an instrument a “binary bond”. In particular, if $x_0 = 0$ and $x_1 = 1$, we call such a bond a “digital bond”. The price of the digital bond is

$$B_{tT} = P_{tT} \pi_{1t}. \quad (34)$$

We can generalize the above situation slightly by considering a pricing kernel $\{\pi_t\}$ of the form

$$\pi_t = M_t f(t, \xi_{tT}, \xi_{tU}). \quad (35)$$

By following the technique in equations (22) to (32), and by using the fact that $\xi_{TT} = \sigma_2 X_T T$, we can show that

$$\begin{aligned} B_{tT} &= \frac{1}{f(t, \xi_{tT}, \xi_{tU})} \sum_{i=0}^n \pi_{it} \int_{-\infty}^{\infty} f \left(T, \sigma_2 x_i T, v_{tTY} + \frac{U-T}{U-t} \xi_{tU} \right) \\ &\quad \times H \left(x_i, v_{tTY} + \frac{U-T}{U-t} \xi_{tU} \right) \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{1}{2} y^2 \right) dy. \end{aligned} \quad (36)$$

Here we model the situation in which the pricing kernel in the economy is not only a function of information at that time about the macroeconomic variable, but is also dependent on noisy information about potential default of the firm leaked in the market through $\{\xi_{tT}\}$. This is relevant in light of events occurring in financial markets where defaults by big companies can affect interest rates and the market price of risk.

A measure for the excess return provided by a defaultable bond over the return on a sovereign bond with the same maturity, is the bond yield spread. This measure is given by the difference between the yields-to-maturity on the defaultable bond and the sovereign bond, see for example [3]. That is:

$$s_{tT} = y_{tT}^d - y_{tT} \quad (37)$$

for $t < T$, where y_{tT} and y_{tT}^d are the yields associated with the sovereign bond and the credit-risky bond, respectively. We have:

$$s_{tT} = \frac{1}{T-t} (\ln P_{tT} - \ln B_{tT}). \quad (38)$$

In particular, the bond yield spread between a digital bond and the sovereign bond is given by

$$s_{tT} = -\frac{1}{T-t} \ln \pi_{1t}. \quad (39)$$

For bonds with payoff $H_T = X_T$, we see that the information related to the macroeconomic factor X_U does not influence the spread. Thus for $0 \leq t < T$, the spread at time t depends only on the information concerning potential default. In this case, the bond yield spread between the defaultable discount bond and the sovereign bond with stochastic interest rates is of the form of that in the deterministic interest rate setting treated in [7].

Figure 1 shows the bond yield spreads between a digital bond, with all trajectories conditional on the outcome that the bond does not default, and a sovereign bond. The maturities of the bonds are taken to be $T = 2$ years and the a priori probability of default is assumed to be $p_0 = 0.2$. The effect of different values of the information flow parameter is shown by setting $\sigma_2 = 0.04$, $\sigma_2 = 0.2$ and $\sigma_2 = 1$, $\sigma_2 = 5$. Since the paths of the digital bond are conditional on the outcome that default does not occur, we observe that the bond yield spreads must eventually drop to zero. The parameter σ_2 controls the magnitude of genuine information about potential default that is available to bondholders. For low values of σ_2 , the bondholder is, so to speak, “in the dark” about the outcome until very close to maturity; while for higher values of σ_2 , the bondholder is better informed. As σ_2 increases, the noisiness in the bond yield spreads, which is indicative of the bondholder’s uncertainty of the outcome, becomes less pronounced near maturity.

Furthermore, if the bondholders in the market were well-informed, they would require a smaller premium for buying the credit-risky bond since its behaviour would be similar to that of the sovereign bond; this is illustrated in Figure 1. It is worth noting that in the information-based asset pricing approach, an increased level of genuine information available to investors about their exposure, is manifestly equivalent to a sort of “securitisation” of the risky investments.

The case for which the paths of the digital bond are conditional on default can also be simulated. Here, the effect of increasing the information flow rate parameter σ_2 is similar. However, the bondholder now requires an infinitely high reward for buying a bond that will be worthless at maturity. Thus the bond-yield spread grows to infinity at maturity.

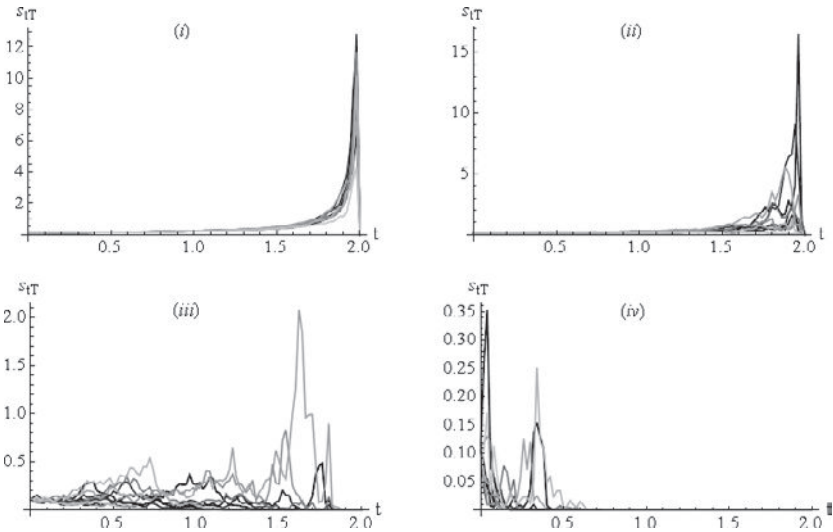


Figure 1. Bond yield spread between a digital bond (with all trajectories conditional on no default) and a sovereign bond. The bonds have maturity $T = 2$ years. The a priori probability of default is taken to be $p_0 = 0.2$. We use (i) $\sigma_2 = 0.04$, (ii) $\sigma_2 = 0.2$, (iii) $\sigma_2 = 1$, and (iv) $\sigma_2 = 5$.

5. Credit-Risky Bonds with Continuous Market-Dependent Recovery

Let us consider the case in which the credit-risky bond pays $H_T = X_T$ where X_T is a discrete random variable which takes values $\{x_0, x_1, \dots, x_n\} \in [0, 1]$ with a priori probabilities $\{p_0, p_1, \dots, p_n\}$, where $x_n > x_{n-1} > \dots > x_1 > x_0$. Such a payoff spectrum is a model for random recovery where at bond maturity one out of a discrete number of recovery levels may be realised. We can also consider credit-risky bonds with continuous random recovery in the event of default. In

doing so, we introduce the notion of “market-dependent recovery”. Suppose that the payoff of the defaultable bond is given by

$$H_T = X_T + (1 - X_T) R(\xi_{TU}), \quad (40)$$

where X_T takes the values $\{0, 1\}$ with a priori probabilities $\{p_0, p_1\}$. The recovery level $R : \mathbb{R} \rightarrow [0, 1]$ is dependent on the information at time T about the macroeconomic factor X_U . In this case, if the credit-risky bond defaults at maturity T , the recovery level of the bond depends on the state of the economy at time U that is perceived in the market at time T . In other words, if the sentiment in the market at time T is that the economy will have good times ahead, then a firm in a state of default at T may have better chances to raise more capital from liquidation (or restructuring), thus increasing the level of recovery of the issued bond. We can price the cash flow (40) by applying equation (32), with $n = 1$, $x_0 = 0$ and $x_1 = 1$. The result is:

$$\begin{aligned} B_{tT} = P_{tT} \pi_{1t} + \pi_{0t} \frac{1}{f(t, \xi_{tU})} \int_{-\infty}^{\infty} f\left(T, v_{tT}y + \frac{U-T}{U-t} \xi_{tU}\right) R\left(v_{tT}y + \frac{U-T}{U-t} \xi_{tU}\right) \\ \times \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y^2\right) dy, \end{aligned} \quad (41)$$

where P_{tT} is given by equation (9). As an example, suppose that we choose the recovery function to be of the form $R(z) = 1 - \exp(-z^2)$. In this case, it is possible to have zero recovery when the value of the information process at time t is $\xi_{tU} = -(U-t)/(U-T) v_{tT}Y$, thereby capturing the worst-case scenario in which bondholders lose their entire investment in the event of default.

The latter consideration is apt in the situation where the extent of recovery is determined by how difficult it is for the firm to raise capital by liquidating its assets, i.e. the exposure of the firm to the general economic environment. However, this model does not say much about how the quality of the management of the firm may influence recovery in the event of default. This observation brings us to another model of recovery. Default of a firm may be triggered by poor internal practices and (or) tough economic conditions. We now structure recovery by specifying the payoff of the credit risky bond by

$$H_T = X_C [X_E + (1 - X_E)R_E] + (1 - X_C) [X_E R_C + (1 - X_E)R_{CE}], \quad (42)$$

where X_C and X_E are random variables taking values in $\{0, 1\}$ with a priori probabilities $\{p_0^C, p_1^C\}$ and $\{p_0^E, p_1^E\}$, respectively. We define X_C and X_E to be indicators of good management of the company and a strong economy, respectively. We set R_C to be a continuous random variable assuming values in the interval $[0, 1]$. We take R_E to be a function of ξ_{TU} , and R_{CE} to be a function of ξ_{TU} and R_C , where both R_E and R_{CE} assume values in the interval $[0, 1]$.

The payoff in equation (42) covers the following situations: First, we suppose that despite good overall management of the firm, default is triggered as a result of a depressed economy. Here, $X_C = 1$ and $X_E = 0$ which implies that $H_T = R_E$. Therefore the recovery is dependent on the state of the economy at time T and thus, how difficult it has been for the firm to raise funds. It is also possible that a firm can default in otherwise favourable economic conditions, perhaps due to the management's negligence. In this case we have $X_E = 1$ and $X_C = 0$. Thus $H_T = R_C$ and the amount recovered is dependent on the level of mismanagement of the firm. Finally we have the worst case in which a firm is poorly managed, $X_C = 0$, and difficult economic times prevail, $X_E = 0$. Recovery is given by the amount $H_T = R_{CE}$, which is dependent on both, the extent of mismanagement of the firm and how much capital the firm can raise in the face of an economic downturn. The particular payoff structure (42) is used in [16] to model the dependence structure between two credit-risky discount bonds that share market factors in common. Further investigation may include the situation where one models such dependence structures for bonds subject to stochastic interest rates and featuring recovery functions of the form (42).

6. Call Option Price Process

Let $\{C_{st}\}_{0 \leq s \leq t < T}$ be the price process of a European-style call option with maturity t and strike K , written on a defaultable bond with price process $\{B_{tT}\}$. The price of such an option at time s is given by

$$C_{st} = \frac{1}{\pi_s} \mathbb{E}^{\mathbb{P}} [\pi_t (B_{tT} - K)^+ | \mathcal{F}_s]. \quad (43)$$

We recall that if the payoff of the credit-risky bond is $H_T = X_T$, then the price of the bond at time t is

$$B_{tT} = P_{tT} \sum_{i=0}^n \pi_{it} x_i, \quad (44)$$

where P_{tT} is given by equation (9) and the conditional density π_{it} is defined in equation (31). The filtration $\{\mathcal{F}_t\}$ is generated by the information processes $\{\xi_{tT}\}$ and $\{\xi_{tU}\}$, and the pricing kernel $\{\pi_t\}$ is of the form

$$\pi_t = M_t f(t, \xi_{tU}), \quad (45)$$

with $\{M_t\}$ satisfying equation (20). Then the price of the option at time s is expressed by

$$C_{st} = \frac{1}{M_s f(s, \xi_{sU})} \mathbb{E}^{\mathbb{P}} [M_t f(t, \xi_{tU}) (B_{tT} - K)^+ | \xi_{sT}, \xi_{sU}]. \quad (46)$$

We recall that the two information processes are independent, and use the martingale $\{M_t\}$ to change the measure as follows:

$$C_{st} = \frac{1}{f(s, \xi_{sU})} \mathbb{E}^{\mathbb{B}_U} \left[f(t, \xi_{tU}) \mathbb{E}^{\mathbb{P}} \left[\left(P_{tT} \sum_{i=0}^n x_i \pi_{it} - K \right)^+ \middle| \xi_{sT} \right] \middle| \xi_{sU} \right]. \quad (47)$$

We first simplify the inner conditional expectation by following an analogous calculation to that in [7], Section 9. The difference is that the discount factor $\{P_{tT}\}$ in (47) is stochastic. However since $\{P_{tT}\}$ is driven by $\{\xi_{tU}\}$, it is unaffected by the conditioning of the inner expectation, allowing us to use the result in [7]. Let us introduce $\{\Phi_t\}$ by

$$\Phi_t = \sum_{i=0}^n p_{it}, \quad (48)$$

where $p_{it} = p_i \exp \left[\frac{T}{T-t} \left(\sigma_2 x_i \xi_{tT} - \frac{1}{2} \sigma_2^2 x_i^2 t \right) \right]$. We write the inner expectation as

$$\mathbb{E}^{\mathbb{P}} \left[\left(P_{tT} \sum_{i=0}^n x_i \pi_{it} - K \right)^+ \middle| \xi_{sT} \right] = \mathbb{E}^{\mathbb{P}} \left[\frac{1}{\Phi_t} \left(\sum_{i=0}^n (P_{tT} x_i - K) p_{it} \right)^+ \middle| \xi_{sT} \right]. \quad (49)$$

The process $\{\Phi_t^{-1}\}$ induces a change of measure from $\mathbb{P}$ to the bridge measure $\mathbb{B}_T$, under which $\{\xi_{tT}\}$ is a Brownian bridge; this allows us to use Bayes formula to express the expectation as follows:

$$\mathbb{E}^{\mathbb{P}} \left[\frac{1}{\Phi_t} \left(\sum_{i=0}^n (P_{tT} x_i - K) p_{it} \right)^+ \middle| \xi_{sT} \right] = \frac{1}{\Phi_s} \mathbb{E}^{\mathbb{B}_T} \left[\left(\sum_{i=0}^n (P_{tT} x_i - K) p_{it} \right)^+ \middle| \xi_{sT} \right]. \quad (50)$$

In order to compute the expectation we introduce the Gaussian random variable Z_{st} , defined by

$$Z_{st} = \frac{\xi_{tT}}{T-t} - \frac{\xi_{sT}}{T-s}, \quad (51)$$

which is independent of $\{\xi_{uT}\}_{0 \leq u \leq s}$. It is possible to find the critical value, for which the argument of the expectation vanishes, in closed form if it is assumed that the defaultable bond is binary. So, for $n = 1$, the critical value z^* is given by

$$z^* = \frac{\ln \left[\frac{\pi_{0s}(K - x_0 P_{tT})}{\pi_{1s}(x_1 P_{tT} - K)} \right] + \frac{1}{2} \sigma_2^2 (x_1^2 - x_0^2) \alpha_{st}^2 T^2}{\sigma_2 (x_1 - x_0) \alpha_{st} T}, \quad (52)$$

where $\alpha_{st}^2 = \text{Var}^{\mathbb{B}_T}[Z_{st}]$. The computation of the expectation amounts to two Gaussian integrals reducing to cumulative normal distribution functions, which we de-

note by $N[x]$. We obtain the following:

$$\mathbb{E}^{\mathbb{P}} \left[\left(P_{tT} \sum_{i=0}^1 x_i \pi_{it} - K \right)^+ \middle| \xi_{sT} \right] = \pi_{1s}(P_{tT}x_1 - K)N[d_s^+] - \pi_{0s}(K - P_{tT}x_0)N[d_s^-], \quad (53)$$

where

$$d_s^{\pm} = \frac{\ln \left[\frac{\pi_{1s}(x_1 P_{tT} - K)}{\pi_{0s}(K - x_0 P_{tT})} \right] \pm \frac{1}{2} \sigma_2^2 (x_1 - x_0)^2 \alpha_{st}^2 T^2}{\sigma_2 (x_1 - x_0) \alpha_{st} T}. \quad (54)$$

We can now insert this intermediate result into equation (47) for $n = 1$; we have

$$C_{st} = \frac{1}{f(s, \xi_{sU})} \mathbb{E}^{\mathbb{B}_U} [f(t, \xi_{tU}) [\pi_{1s}(P_{tT}x_1 - K)N[d_s^+] - \pi_{0s}(K - P_{tT}x_0)N[d_s^-]] | \xi_{sU}]. \quad (55)$$

We emphasize that $\{P_{tT}\}$ is given by a function $P(t, T, \xi_{tU})$ and thus is affected by the conditioning with respect to ξ_{sU} . To compute the expectation in equation (55), we use the same technique as in Section 4 and introduce the Gaussian random variable Y_{st} , defined by

$$Y_{st} = \xi_{tU} - \frac{U-t}{U-s} \xi_{sU}, \quad (56)$$

with mean zero and variance $v_{st}^2 = \text{Var}^{\mathbb{B}_U}[Y_{st}]$. Thus, as shown in the previous sections, the outer conditional expectation reduces to a Gaussian integral:

$$\begin{aligned} C_{st} = \frac{1}{f(s, \xi_{sU})} \int_{-\infty}^{\infty} f\left(t, v_{st}y + \frac{U-t}{U-s} \xi_{sU}\right) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y^2\right) \\ \times \left[\pi_{1s} \left(P\left(t, T, v_{st}y + \frac{U-t}{U-s} \xi_{sU}\right) x_1 - K \right) N[d_s^+(y)] \right. \\ \left. - \pi_{0s} \left(K - P\left(t, T, v_{st}y + \frac{U-t}{U-s} \xi_{sU}\right) x_0 \right) N[d_s^-(y)] \right] dy. \end{aligned} \quad (57)$$

Therefore we obtain a semi-analytical pricing formula for a call option on a defaultable bond in a stochastic interest rate setting. The integral in equation (57) can be evaluated using numerical methods once the function $f(t, x)$ is specified.

7. Hybrid Securities

So far we have focused on the pricing of credit-risky bonds with stochastic discounting. The formalism presented in the above sections can also be applied to price other types of securities. In particular, as an example of a hybrid security, we show how to price an inflation-linked credit-risky discount bond. While such a security has inherent credit risk, it offers bondholders protection against inflation.

This application also gives us the opportunity to extend the thus far presented pricing models to the case where n independent information processes are employed. We shall call such models, “multi-dimensional pricing models”.

In what follows, we consider three independent information processes, $\{\xi_{tT}\}$, $\{\xi_{tU_1}\}$ and $\{\xi_{tU_2}\}$, defined by

$$\xi_{tT} = \sigma t X_T + \beta_{tT}, \quad \xi_{tU_1} = \sigma_1 t X_{U_1} + \beta_{tU_1}, \quad \xi_{tU_2} = \sigma_2 t X_{U_2} + \beta_{tU_2}, \quad (58)$$

where $0 \leq t \leq T < U_1 \leq U_2$. The positive random variable X_T is discrete, while X_{U_1} , X_{U_2} are assumed to be continuous. The market filtration $\{\mathcal{F}_t\}$ is generated jointly by the three information processes. Let $\{C_t\}_{t \geq 0}$ be a price level process, e.g., the process of the consumer price index. The price Q_{tT} , at time t , of an inflation-linked discount bond that pays C_T units of a currency at maturity T , is

$$Q_{tT} = \frac{\mathbb{E}^{\mathbb{P}} [\pi_T C_T | \mathcal{F}_t]}{\pi_t}. \quad (59)$$

We now make use of the “foreign exchange analogy” (see, e.g., [4], [5], [11], [13], [17]) in which the nominal pricing kernel $\{\pi_t\}$, and the real pricing kernel $\{\pi_t^R\}$, are viewed as being associated with “domestic” and “foreign” economies respectively, with the price level process $\{C_t\}$, acting as an “exchange rate”. The process $\{C_t\}$ is expressed by the following ratio:

$$C_t = \frac{\pi_t^R}{\pi_t}. \quad (60)$$

For further details about the modelling of the real and the nominal pricing kernels, and the pricing of inflation-linked assets, we refer to [14]. In what follows, we make use of the method proposed in [14] to price an example of an inflation-linked credit-risky discount bond (ILCR) that, at maturity T , pays a cash flow $H_T = C_T H(X_T, \xi_{TU_1}, \xi_{TU_2})$. The price H_{tT} at time $t \leq T$ of such a bond is

$$H_{tT} = \frac{1}{\pi_t} \mathbb{E}^{\mathbb{P}} [\pi_T^R H(X_T, \xi_{TU_1}, \xi_{TU_2}) | \mathcal{F}_t], \quad (61)$$

where we have used relation (60). We choose to model the real and the nominal pricing kernels by

$$\pi_t = M_t^{(1)} M_t^{(2)} f(t, \xi_{tU_1}, \xi_{tU_2}) \quad \text{and} \quad \pi_t^R = M_t^{(1)} M_t^{(2)} g(t, \xi_{tU_1}, \xi_{tU_2}), \quad (62)$$

where $f(t, x, y)$ and $g(t, x, y)$ are two functions of three variables. The process $\{M_t^{(i)}\}_{0 \leq t \leq T < U_i}$ for $i = 1, 2$ is a martingale that induces a change of measure to the bridge measure $\mathbb{B}_i$. We recall that the information process $\{\xi_{tU_i}\}$ has the law of a Brownian bridge under the measure $\mathbb{B}_i$. In order to work out the expectation

in (61) with the pricing kernel models introduced in (62), we can also define a process $\{M_t\}$ by

$$M_t = M_t^{(1)} M_t^{(2)}, \quad (63)$$

where $0 \leq t \leq T < U_1 \leq U_2$. Since the information processes $\{\xi_{tU_1}\}$ and $\{\xi_{tU_2}\}$ are independent, $\{M_t\}$ is itself an $(\{\mathcal{F}_t\}, \mathbb{P})$ -martingale, with $M_0 = 1$ and $\mathbb{E}^\mathbb{P}[M_t] = 1$. Thus $\{M_t\}$ can be used to effect a change of measure from $\mathbb{P}$ to a bridge measure $\mathbb{B}$, under which the random variables ξ_{tU_1} and ξ_{tU_2} have the distribution of a Brownian bridge for $0 \leq t \leq T < U_1$. This can be verified as follows: $\{\xi_{tU_1}\}$ is a Gaussian process with mean

$$\mathbb{E}^\mathbb{B}[\xi_{tU_1}] = \mathbb{E}^{\mathbb{B}_1} \left[\frac{M_t}{M_t^{(1)}} \xi_{tU_1} \right] = \mathbb{E}^{\mathbb{B}_1} [M_t^{(2)}] \mathbb{E}^{\mathbb{B}_1}[\xi_{tU_1}] = 0, \quad (64)$$

due to the independence property of $\{\xi_{tU_1}\}$ and $\{\xi_{tU_2}\}$. Moreover, for $0 \leq s \leq t \leq T < U_1$, the covariance is given by

$$\mathbb{E}^\mathbb{B}[\xi_{sU_1} \xi_{tU_1}] = \mathbb{E}^{\mathbb{B}_1} [M_t^{(2)}] \mathbb{E}^{\mathbb{B}_1}[\xi_{sU_1} \xi_{tU_1}] = \mathbb{E}^\mathbb{P}[M_t] \mathbb{E}^{\mathbb{B}_1}[\xi_{sU_1} \xi_{tU_1}] = \frac{s(U_1 - t)}{U_1}. \quad (65)$$

The same can be shown for $\{\xi_{tU_2}\}$.

By the definition of $\{M_t\}$ and by use of Bayes formula and the fact that $\{\xi_{tT}\}$, $\{\xi_{tU_1}\}$ and $\{\xi_{tU_2}\}$ are $\{\mathcal{F}_t\}$ -Markov processes, equation (61) reduces to

$$H_{tT} = \frac{1}{f(t, \xi_{tU_1}, \xi_{tU_2})} \mathbb{E}^\mathbb{B} \left[\mathbb{E}^\mathbb{P} \left[g(T, \xi_{TU_1}, \xi_{TU_2}) H(X_T, \xi_{TU_1}, \xi_{TU_2}) \mid \xi_{tT} \right] \mid \xi_{tU_1}, \xi_{tU_2} \right]. \quad (66)$$

Next we repeat an analogous calculation to the one leading from equation (25) to expression (32). For the ILCR discount bond under consideration, we obtain

$$H_{tT} = \frac{1}{f(t, \xi_{tU_1}, \xi_{tU_2})} \sum_{i=0}^n \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(T, z(y_1), z(y_2)) H(x_i, z(y_1), z(y_2)) \pi_{it} \\ \times \frac{1}{2\pi} \exp \left[-\frac{1}{2} (y_1^2 + y_2^2) \right] dy_1 dy_2. \quad (67)$$

Here the conditional density π_{it} is given by an expression analogous to the one in equation (31) and, for $k = 1, 2$, $z(y_k)$ is defined by

$$z(y_k) = v_{tT}^{(k)} y_k + \frac{U_k - T}{U_k - t} \xi_{tU_k}, \quad \text{where} \quad v_{tT}^{(k)} = \sqrt{\frac{(T-t)(U_k - T)}{U_k - t}}. \quad (68)$$

In the special case where $H_T = X_T$, the expression for the price at time t of the ILCR discount bond simplifies to

$$H_{tT} = Q_{tT} \sum_{i=0}^n \pi_{it} x_i. \quad (69)$$

Here Q_{tT} is the price of an inflation-linked discount bond that depends on the information processes $\{\xi_{tU_1}\}$ and $\{\xi_{tU_2}\}$. In particular, a formula similar to (57) can be derived for the price of a European-style call option written on an ILCR bond with price process given by (69) with $n = 1$. We note here that similar pricing formulae can be derived for credit-risky discount bonds traded in a foreign currency. In this case the real pricing kernel, and thus the real interest rate, is associated with the pricing kernel denominated in the foreign currency. On the other hand, the nominal pricing kernel is associated with the domestic currency, thus giving rise to the domestic interest rate.

8. Credit-Risky Coupon Bonds

Let $\{T_k\}_{k=1,\dots,n}$ be a collection of fixed dates where $0 \leq t \leq T_1 \leq \dots \leq T_n$. We consider the valuation of a credit-risky bond with coupon payment H_{T_k} at time T_k and maturity T_n . The bond is in a state of default as soon as the first coupon payment does not occur. We denote the price process of the coupon bond by $\{B_{tT_n}\}$ and introduce n independent random variables $X_{T_1}, \dots, X_{T_n}$ that are applied to construct the cash flows H_{T_k} given by

$$H_{T_k} = \mathbf{c} \prod_{j=1}^k X_{T_j}, \quad (70)$$

for $k = 1, \dots, n-1$, and for $k = n$ by

$$H_{T_n} = (\mathbf{c} + \mathbf{p}) \prod_{j=1}^n X_{T_j}. \quad (71)$$

Here $\mathbf{c}$ and $\mathbf{p}$ denote the coupon and principal payment, respectively, and the random variables $\{X_{T_k}\}_{k=1,\dots,n}$ take values in $\{0, 1\}$. With each factor X_{T_k} we associate an information process $\{\xi_{tT_k}\}$ defined by

$$\xi_{tT_k} = \sigma_k t X_{T_k} + \beta_{tT_k}. \quad (72)$$

Furthermore we introduce another information process $\{\xi_{tU}\}$ given by

$$\xi_{tU} = \sigma t X_U + \beta_{tU} \quad (0 \leq t \leq T_n < U) \quad (73)$$

that we reserve for the modelling of the pricing kernel. The market filtration $\{\mathcal{F}_t\}$ is generated jointly by the $n+1$ information processes, that is $\{\xi_{tT_k}\}_{k=1,\dots,n}$ and $\{\xi_{tU}\}$. Following the method in Section 4, we model the pricing kernel $\{\pi_t\}$ by

$$\pi_t = M_t f(t, \xi_{tU}), \quad (74)$$

where the density martingale $\{M_t\}$ which induces a change of measure to the bridge measure satisfies equation (4). Armed with these ingredients we are now in

the position to write down the formula for the price B_{tT_n} at time t of the credit-risky coupon bond:

$$\begin{aligned} B_{tT_n} &= \frac{1}{\pi_t} \sum_{k=1}^n \mathbb{E}^{\mathbb{P}} \left[\pi_{T_k} H_{T_k} \mid \xi_{tT_1}, \dots, \xi_{tT_k}, \xi_{tU} \right], \\ &= \frac{1}{M_t f(t, \xi_{tU})} \sum_{k=1}^n \mathbb{E}^{\mathbb{P}} \left[M_{T_k} f(T_k, \xi_{T_k U}) \mathbf{c} \prod_{j=1}^k X_{T_j} \mid \xi_{tT_1}, \dots, \xi_{tT_k}, \xi_{tU} \right] \\ &\quad + \frac{1}{M_t f(t, \xi_{tU})} \mathbb{E}^{\mathbb{P}} \left[M_{T_n} f(T_n, \xi_{T_n U}) \mathbf{p} \prod_{j=1}^n X_{T_j} \mid \xi_{tT_1}, \dots, \xi_{tT_n}, \xi_{tU} \right]. \end{aligned} \quad (75)$$

To compute the expectation, we use the approach presented in Section 4. Since the pricing kernel and the cash flow random variables H_{T_k} , $k = 1, \dots, n$, are independent, we conclude that the expression for the bond price B_{tT_n} reduces to

$$B_{tT_n} = \mathbf{c} \sum_{k=1}^n P_{tT_k} \mathbb{E}^{\mathbb{P}} \left[\prod_{j=1}^k X_{T_j} \mid \xi_{tT_1}, \dots, \xi_{tT_k} \right] + \mathbf{p} P_{tT_n} \mathbb{E}^{\mathbb{P}} \left[\prod_{j=1}^n X_{T_j} \mid \xi_{tT_1}, \dots, \xi_{tT_n} \right], \quad (76)$$

where the discount bond system $\{P_{tT_k}\}$ is given by

$$P_{tT_k} = \frac{1}{f(t, \xi_{tU})} \int_{-\infty}^{\infty} f\left(T_k, \nu_{tT_k} y_k + \frac{U - T_k}{U - t} \xi_{tU}\right) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} y_k^2\right) dy_k, \quad (77)$$

and $\nu_{tT_k}^2 = (T_k - t)(U - T_k)/(U - t)$. We note that formula (75) can be simplified further since the expectations therein can be worked out explicitly due to the independence property of the information processes. We have,

$$\mathbb{E}^{\mathbb{P}} \left[\prod_{j=1}^k X_{T_j} \mid \xi_{tT_1}, \dots, \xi_{tT_k} \right] = \prod_{j=1}^k \pi_{1t}^{(j)}, \quad (78)$$

where the conditional density $\pi_{1t}^{(j)}$ at time t that the random variable X_{T_j} takes value one is given by

$$\pi_{1t}^{(j)} = \frac{p_1^{(j)} \exp\left[\frac{T_j}{T_j - t} \left(\sigma_j \xi_{tT_j} - \frac{1}{2} \sigma_j^2 t\right)\right]}{p_0^{(j)} + p_1^{(j)} \exp\left[\frac{T_j}{T_j - t} \left(\sigma_j \xi_{tT_j} - \frac{1}{2} \sigma_j^2 t\right)\right]}. \quad (79)$$

Here $p_1^{(j)} = \mathbb{P}[X_{T_j} = 1]$. Thus, the price B_{tT_n} at time t of the credit-risky coupon bond is given by

$$B_{tT_n} = \sum_{k=1}^n \mathbf{c} P_{tT_k} \prod_{j=1}^k \pi_{1t}^{(j)} + \mathbf{p} P_{tT_n} \prod_{j=1}^n \pi_{1t}^{(j)}. \quad (80)$$

At this stage, we observe that the price of a credit-risky coupon bond has been derived for the case in which the cash flow functions H_{T_k} , $k = 1, \dots, n$, do not depend on the information available at time T_k about the macroeconomic factor X_U , thereby leading to independence between the discount bond system and the credit-risky component of the bond. This is generalized in a straightforward manner by considering cash flow functions of the form

$$H_{T_k} = H(X_{T_1}, \dots, X_{T_k}, \xi_{T_k U}), \quad (81)$$

for $k = 1, \dots, n$. The valuation of such cash flows at time t may include the case treated in (32), however endowed with coupon payments.

As an illustration we consider the situation in which the bond pays a coupon $\mathbf{c}$ at T_k , $k = 1, \dots, n$, and the principal amount $\mathbf{p}$ at T_n . Upon default, market-dependent recovery given by $R_k(\xi_{T_k U})$ (as a percentage of coupon plus principal) is paid at T_k . For simplicity, we consider $n = 2$. In this case, the random cash flows of the bond are given by

$$\begin{aligned} H_{T_1} &= \mathbf{c} X_{T_1} + (\mathbf{c} + \mathbf{p}) R_1(\xi_{T_1 U})(1 - X_{T_1}), \\ H_{T_2} &= (\mathbf{c} + \mathbf{p}) X_{T_1} [X_{T_2} + R_2(\xi_{T_2 U})(1 - X_{T_2})]. \end{aligned}$$

By making use of the technique presented in Section 5, we can express the price of the credit-risky coupon bond by

$$\begin{aligned} B_{tT_2} &= \mathbf{c} P_{tT_1} \pi_{1t}^{(1)} + (\mathbf{c} + \mathbf{p}) P_{tT_2} \pi_{1t}^{(1)} \pi_{1t}^{(2)} \\ &+ (\mathbf{c} + \mathbf{p}) \pi_{0t}^{(1)} \frac{1}{f(t, \xi_{tU})} \int_{-\infty}^{\infty} f(T_1, m(y_1)) R_1(m(y_1)) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y_1^2\right) dy_1 \\ &+ (\mathbf{c} + \mathbf{p}) \pi_{1t}^{(1)} \pi_{0t}^{(2)} \frac{1}{f(t, \xi_{tU})} \int_{-\infty}^{\infty} f(T_2, m(y_2)) R_2(m(y_2)) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y_2^2\right) dy_2, \end{aligned} \quad (82)$$

where, for $k = 1, 2$, we define

$$m(y_k) = v_{tT_k} y_k + \frac{U - T_k}{U - t} \xi_{tU}, \quad v_{tT_k} = \sqrt{\frac{(T_k - t)(U - T_k)}{U - t}}. \quad (83)$$

9. Credit-Sensitive Pricing Kernels

We fix the dates T_1 and T_2 , where $T_1 \leq T_2$, to which we associate the economic factors X_{T_1} and X_{T_2} respectively. The first factor is identified with a debt payment at time T_1 . For example X_{T_1} could be a coupon payment that a country is obliged to make at time T_1 . The second factor, X_{T_2} , could be identified with the measured growth (possibly negative) in the employment level in the same country at time T_2 since the last published figure. In such an economy, with two random factors

only, it is plausible that the prices of the treasuries fluctuate according to the noisy information market participants will have about the outcome of X_{T_1} and X_{T_2} . Thus the price of a sovereign bond with maturity T , where $0 \leq t \leq T < T_1 \leq T_2$, is given by:

$$P_{tT} = \frac{1}{f(t, \xi_{tT_1}, \xi_{tT_2})} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f\left(T, v_{tT}^{(1)} y_1 + \frac{T_1 - T}{T_1 - t} \xi_{tT_1}, v_{tT}^{(2)} y_2 + \frac{T_2 - T}{T_2 - t} \xi_{tT_2}\right) \times \frac{1}{2\pi} \exp\left[-\frac{1}{2}(y_1^2 + y_2^2)\right] dy_2 dy_1. \quad (84)$$

In particular, the resulting interest rate process in this model is subject to the information processes $\{\xi_{tT_1}\}$ and $\{\xi_{tT_2}\}$ making it fluctuate according to information (both genuine and misleading) about the economy's factors X_{T_1} and X_{T_2} .

We now ask the following question: *What type of model should one consider if the goal is to model a pricing kernel that is sensitive to an accumulation of losses? Or in other words, how should one model the nominal short rate of interest and the market price of risk processes if both react to the amount of debt accumulated by a country over a finite period of time?*

To treat this question we need to introduce a model for an accumulation process. We shall adopt the method developed in [9], where the idea of a gamma bridge information process is introduced. It turns out that the use of such a cumulative process is suitable to provide an answer to the question above. In fact, if in the example above, the factor X_{T_1} is identified with the total accumulated debt at time T_1 , then the gamma bridge information process $\{\xi_{tT_1}^\gamma\}$, defined by

$$\xi_{tT_1}^\gamma = X_{T_1} \gamma_{tT_1} \quad (85)$$

where $\{\gamma_{tT_1}\}_{0 \leq t \leq T_1}$ is a gamma bridge process that is independent of X_{T_1} , measures the level of the accumulated debt as of time t , $0 \leq t \leq T_1$. If the market filtration is generated, among other information processes, also by the debt accumulation process, then asset prices that are calculated by use of this filtration, will fluctuate according to the updated information about the level of the accumulated debt of a country. We now work out the price of a sovereign bond for which the price process reacts both to Brownian and gamma information.

We consider the time line $0 \leq t \leq T < T_1 \leq T_2 < \infty$. Time T is the maturity date of a sovereign bond with unit payoff and price process $\{P_{tT}\}_{0 \leq t \leq T}$. With the date T_1 we associate the factor X_{T_1} and with the date T_2 the factor X_{T_2} . The positive random variable X_{T_1} is independent of X_{T_2} , and both may be discrete or continuous random variables. Then we introduce the following information processes:

$$\xi_{tT_1}^\gamma = X_{T_1} \gamma_{tT_1}, \quad \xi_{tT_2} = \sigma t X_{T_2} + \beta_{tT_2}. \quad (86)$$

The process $\{\xi_{tT_1}^\gamma\}$ is a gamma bridge information process, and it is taken to be independent of $\{\xi_{tT_2}\}$. The properties of the gamma bridge process $\{\gamma_{tT_1}\}$ are described in great detail in [9]. We assume that the market filtration $\{\mathcal{F}_t\}_{t \geq 0}$ is generated jointly by $\{\xi_{tT_1}^\gamma\}$ and $\{\xi_{tT_2}\}$.

In this setting, the pricing kernel reacts to the updated information about the level of accumulated debt and, for the sake of example, also to noisy information about the likely level of employment growth at T_2 . Thus we propose the following model for the pricing kernel:

$$\pi_t = M_t f(t, \xi_{tT_1}^\gamma, \xi_{tT_2}) \quad (87)$$

where the process $\{M_t\}$ is the change-of-measure martingale from the probability measure $\mathbb{P}$ to the Brownian bridge measure $\mathbb{B}$, satisfying

$$dM_t = -\sigma \frac{T_2}{T_2 - t} \mathbb{E}[X_{T_2} | \xi_{tT_2}] M_t dW_t. \quad (88)$$

Here $\{W_t\}$ is an $(\{\mathcal{F}_t\}, \mathbb{P})$ -Brownian motion. The formula for the price of the sovereign bond is given by

$$P_{tT} = \frac{\mathbb{E}^\mathbb{P} [M_T f(T, \xi_{TT_1}^\gamma, \xi_{TT_2}) | \xi_{tT_1}^\gamma, \xi_{tT_2}]}{M_t f(t, \xi_{tT_1}^\gamma, \xi_{tT_2})}. \quad (89)$$

We make use of the Markov property and the independence property of the information processes, together with the change of measure to express the bond price by

$$P_{tT} = \frac{\mathbb{E}_\gamma^\mathbb{P} [\mathbb{E}^\mathbb{B} [f(T, \xi_{TT_1}^\gamma, \xi_{TT_2}) | \xi_{tT_2}] | \xi_{tT_1}^\gamma]}{f(t, \xi_{tT_1}^\gamma, \xi_{tT_2})}. \quad (90)$$

Here, the expectations $\mathbb{E}_\gamma^\mathbb{P}$ and $\mathbb{E}^\mathbb{B}$ are operators that apply according to the dependence of their argument on the random variables $\xi_{TT_1}^\gamma$ and ξ_{TT_2} respectively. This is a direct consequence of the independence of $\{\xi_{tT_1}^\gamma\}$ and $\{\xi_{tT_2}\}$. We now use the technique adopted in the preceding sections, where we introduce the Gaussian random variable Y_{tT} with mean zero and variance $v_{tT}^2 = (T - t)(T_2 - T)/(T_2 - t)$, and the standard Gaussian random variable Y . By following the approach taken in Section 4, we can compute the inner expectation explicitly since the conditional expectation reduces to a Gaussian integral over the range of the random variable Y . Thus we obtain:

$$P_{tT} = \int_{-\infty}^{\infty} \frac{\mathbb{E}_\gamma^\mathbb{P} [f(T, \xi_{TT_1}^\gamma, v_{tT} Y + \frac{T_2 - T}{T_2 - t} \xi_{tT_2}) | \xi_{tT_1}^\gamma]}{f(t, \xi_{tT_1}^\gamma, \xi_{tT_2})} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y^2\right) dy. \quad (91)$$

The feature of this model which sets it apart from those considered in preceding sections, is the fact that we have to calculate a gamma expectation $\mathbb{E}_\gamma^\mathbb{P}$. In this case, we cannot adopt the “usual” change-of-measure method we have used thus far. To this end we refer to [9], where the price process of the Arrow-Debreu security for the case that it is driven by a gamma bridge information process is derived. We use this result and obtain for the Arrow-Debreu density process $\{A_{tT}\}$ the following expression:

$$A_{tT}(y_\gamma) = \mathbb{E}^\mathbb{P} \left[\delta(\xi_{TT_1}^\gamma - y_\gamma) \middle| \xi_{tT_1}^\gamma \right] \quad (92)$$

$$= \frac{\mathbb{1}_{\{y_\gamma > \xi_{tT_1}^\gamma\}} (y_\gamma - \xi_{tT_1}^\gamma)^{m(T-t)-1} \int_{y_\gamma}^\infty p(x) x^{1-mT_1} (x - y_\gamma)^{m(T_1-T)-1} dx}{B[m(T-t), m(T_1-T)] \int_{\xi_{tT_1}^\gamma}^\infty p(z) z^{1-mT_1} (z - \xi_{tT_1}^\gamma)^{m(T_1-t)-1} dz}, \quad (93)$$

where $\delta(y)$ is the Dirac distribution and $p(x)$ is the *a priori* probability density of X_{T_1} . Here $B[a, b]$ is the beta function. Following [16], Section 3.4, we consider a function $h(\xi_{TT_1}^\gamma)$ of the random variable $\xi_{TT_1}^\gamma$ and note that for a suitable function h we may write:

$$\mathbb{E}_\gamma^\mathbb{P} \left[h(\xi_{TT_1}^\gamma) \middle| \xi_{tT_1}^\gamma \right] = \int_{-\infty}^\infty \mathbb{E}_\gamma^\mathbb{P} \left[\delta(\xi_{TT_1}^\gamma - y_\gamma) \middle| \xi_{tT_1}^\gamma \right] h(y_\gamma) dy_\gamma. \quad (94)$$

Next we see that the conditional expectation under the integral is the Arrow-Debreu density (92) for which there is the closed-form expression (93). We go back to equation (91) and observe that the conditional expectation under the integral is of the form $\mathbb{E}_\gamma^\mathbb{P} \left[h(\xi_{TT_1}^\gamma) \middle| \xi_{tT_1}^\gamma \right]$. Thus we can use (94) to calculate the gamma expectation in (91). We write:

$$\begin{aligned} \mathbb{E}_\gamma^\mathbb{P} \left[f \left(T, \xi_{TT_1}^\gamma, v_{tT} y + \frac{T_2 - T}{T_2 - t} \xi_{tT_2} \right) \middle| \xi_{tT_1}^\gamma \right] \\ = \int_{-\infty}^\infty A_{tT}(y_\gamma) f \left(T, y_\gamma, v_{tT} y + \frac{T_2 - T}{T_2 - t} \xi_{tT_2} \right) dy_\gamma. \end{aligned} \quad (95)$$

We are now in the position to write down the bond price (91) in explicit form by using equation (95). We thus obtain:

$$P_{tT} = \int_{-\infty}^\infty \int_{-\infty}^\infty \frac{A_{tT}(y_\gamma) f \left(T, y_\gamma, v_{tT} y + \frac{T_2 - T}{T_2 - t} \xi_{tT_2} \right)}{f(t, \xi_{tT_1}^\gamma, \xi_{tT_2})} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y^2\right) dy_\gamma dy. \quad (96)$$

The bond price can be written more concisely by defining

$$\begin{aligned} \tilde{f}(T, t, \xi_{tT_1}^\gamma, \xi_{tT_2}) &= \int_{-\infty}^\infty \int_{-\infty}^\infty A_{tT}(y_\gamma) f \left(T, y_\gamma, v_{tT} y + \frac{T_2 - T}{T_2 - t} \xi_{tT_2} \right) \\ &\quad \times \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y^2\right) dy_\gamma dy. \end{aligned} \quad (97)$$

We thus have:

$$P_{tT} = \frac{\tilde{f}(T, t, \xi_{tT_1}^\gamma, \xi_{tT_2})}{f(t, \xi_{tT_1}^\gamma, \xi_{tT_2})}. \quad (98)$$

Future investigation in this line of research incorporates the constructions of processes $\{f(t, \xi_{tT_1}^\gamma, \xi_{tT_2})\}$ such that the resulting pricing kernel (87) is an $(\{\mathcal{F}_t\}, \mathbb{P})$ -supermartingale. The appropriate choice of $f(t, x, y)$ depends also on a suitable description of the economic interplay of the information flows modelled by $\{\xi_{tT_1}^\gamma\}$ and $\{\xi_{tT_2}\}$. One might begin with looking at the situation in which the price of the bond depreciates due to a rising debt level and a higher level of employment. We conclude by observing that the gamma bridge information process may also be considered for the modelling of credit-risky bonds, where default is triggered by the firm's accumulated debt exceeding a specified threshold at bond maturity. Random recovery models may be constructed using the technique in Section 5.

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On Statistical Aspects in Calibrating a Geometric Skewed Stable Asset Price Model*

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Estimation of an asset price process under the physical measure can be regarded as the first step of the calibration problem, hence is of practical importance. In this article, supposing that a log-price process is expressed by a possibly skewed stable driven model and that a high-frequency dataset over a fixed period is available, we provide practical procedures of estimating the dominating parameters. Especially, the scale parameter may be time-varying and possibly random as long as it is independent of the driving skewed stable Lévy process. By means of the scaling property and realized bipower variations, it is possible to estimate the index and positivity (skewness) parameters without specific information of the scale process. When the target scale parameter is constant, our estimators are asymptotically normally distributed, the rate of convergence being $\sqrt{n}$. When the scale is actually time-varying, we focus on estimation of the integrated scale, which is an analogue to the integrated volatility in the Brownian-semimartingale framework. In this case we show that estimation of the integrated scale exhibits a kind of asymptotic singularity with respect to the unknown index parameter, with the rate of convergence being the slower $\sqrt{n}/\log n$.

Keywords: High-frequency sampling, parameter estimation, skewed stable Lévy process.

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1. Introduction

Nowadays there exists a vast amount of option-pricing theories for many kinds of underlying asset price processes, which depend on either finite- or infinite-dimensional unknown parameters. Typically, we are first given an underlying asset price process whose law is governed by a physical measure (real world), and then construct a risk-neutral measure under which a price formula is provided through a change of measure. To apply the theories in practice, we are inevitably forced to calibrate the model in question. Then, the first key step would be to estimate the structure of the underlying asset price process based on observed return data.

In this article, we address the estimation problem for a class of asset price models driven by a possibly skewed stable Lévy process. Specifically, we provide simple recipes for estimating the parameters governing the law of the log-price process $X = \log S$, where S denotes a univariate asset price process: recall that for a semimartingale X without continuous local martingale part it follows from Itô's formula that

$$dS_t = S_{t-} \{dX_t + (e^{\Delta X_t} - 1 - \Delta X_t)\}$$

with some positive initial variable S_0 , where $\Delta X_t := X_t - X_{t-}$ denotes the jump of X at time t . We model X as a stochastic integral of a positive process σ independent of the integrator Z , a skewed stable Lévy process with finite mean. Our model includes the so-called geometric stable Lévy process, where σ is constant. Undoubtedly, Lévy processes, which form *the* continuous-time counterpart of discrete-time random walks, serve as a building block for continuous-time modelling of financial data. We refer the reader to, among others, Bertoin [5] and Sato [16] for systematic accounts of Lévy processes. Recently, Miyahara and Moriwaki [15] (see also Fujiwara and Miyahara [9]) introduced an option-pricing model based on the geometric stable Lévy process and the minimal entropy martingale measure, and shown its usability to, e.g., reproduce the volatility smile/smirk properties.

Our estimation procedure utilizes empirical-sign statistics and realized multi-power variations (MPV for short), and its implementation is pretty simple and requires no hard numerical optimization, hence preferable in practice. Using MPVs essentially amounts to the classical method of moments with possibly random targets. Some authors have studied asymptotic behaviors concerning MPVs for estimating integrated- σ quantities: Barndorff-Nielsen and Shephard [4] for centered and symmetric stable Z , Woerner [18, 20] for general Z admitting a nearly symmetric Lévy density near the origin. The independence between σ and Z was crucial in these papers. On the other hand, Corcuera *et al.* [7] treated realized power variation for general strictly stable Z with σ not necessarily independent of Z .

Concerning joint estimation of the stable Lévy processes based on high-frequency data, Masuda [13] considered a joint estimation of the index, scale, and location parameters in case of symmetric Lévy density. There it was shown that the sample median based estimator of the location combined with a variant

of the central limit theorem led to full-joint estimators, which are asymptotically normal with finite and nondegenerate asymptotic covariance matrices. In particular, the sample median based estimator turned out to be rate-efficient. Our model setup in this article does not contain the drift parameter (presupposed to be zero), but instead allows possible skewness.

This article is organized as follows. Our model setup and objectives are described in Section 2. Section 3 presents our estimation procedures. Small simulation results are reported in Section 4. Concluding remarks are given in Section 5.

2. Setup

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0,1]}, P)$ be an underlying probability space, which is supposed to be rich enough to carry all the random variables and processes appearing below, and to make all the random processes adapted. We denote by E the expectation operator. For convenience, we start with describing some basic facts concerning the stable distributions and stable Lévy processes.

Denote by $S_\alpha(\rho, \sigma)$ the possibly skewed stable distribution without drift, the characteristic function of which is given by

$$u \mapsto \exp \left\{ -\sigma |u|^\alpha \left(1 - i \operatorname{sgn}(u) \tan \{ \alpha \pi (\rho - 1/2) \} \right) \right\}, \quad u \in \mathbb{R}. \quad (1)$$

The dominating parameters α , ρ , and σ correspond to:

- the *stable-index parameter* $\alpha \in (1, 2)$;
- the *positivity parameter* ρ fulfilling that $1 - 1/\alpha < \rho < 1/\alpha$; and
- the *scale parameter* $\sigma > 0$.

We here rule out the “infinite-mean” case (i.e. $\alpha \in (0, 1]$), and also the case of “one-sided jumps” (i.e. either $\rho = 1 - 1/\alpha$ or $1/\alpha$) from our scope; in many cases, this restriction is non-fatal for realistic modelling in finance.

Let ζ stand for a random variable such that $\mathcal{L}(\zeta) = S_\alpha(\rho, \sigma)$. Here and in the sequel, for a random variable ξ we denote its law by $\mathcal{L}(\xi)$. The name of “positivity parameter” of ρ comes from the fact that $P[\zeta \geq 0] = \rho$; trivially, the symmetric case corresponds to $\rho = 1/2$. Note that the positivity parameter of $\mathcal{L}(c\zeta)$ is again ρ whatever $c > 0$ is. For future reference, we mention the closed-form expressions of absolute and signed-absolute moments (cf. Kuruoğlu [11]): for any $r \in (-1, \alpha)$ and $r' \in (-2, -1) \cup (-1, \alpha)$,

$$E[|\zeta|^r] = \frac{\Gamma(1 - r/\alpha) \cos(r\xi/\alpha)}{\Gamma(1 - r) \cos(r\pi/2)} \frac{\sigma^{r/\alpha}}{|\cos(\xi)|^{r/\alpha}}, \quad (2)$$

$$E[|\zeta|^{r'} \operatorname{sgn}(\zeta)] = \frac{\Gamma(1 - r'/\alpha) \sin(r'\xi/\alpha)}{\Gamma(1 - r') \sin(r'\pi/2)} \frac{\sigma^{r'/\alpha}}{|\cos(\xi)|^{r'/\alpha}}, \quad (3)$$

where we wrote

$$\xi = \alpha\pi(\rho - 1/2)$$

and the symbol $\text{sgn}(u)$ expresses 1, 0, -1 according as $u > 0$, $= 0$, < 0 , respectively. We write

$$\mu_r = \sigma^{-r/\alpha} E[|\zeta|^r] \quad \text{and} \quad \nu_{r'} = \sigma^{-r'/\alpha} E[|\zeta|^{r'} \text{sgn}(\zeta)],$$

the r th absolute and r' th signed-absolute moments associated with $S_\alpha(\rho, 1)$, respectively.

The most familiar parametrization of the stable distribution would be, instead of (1),

$$u \mapsto \exp \left\{ -(\sigma|u|)^\alpha \left(1 - i\beta \text{sgn}(u) \tan \frac{\alpha\pi}{2} \right) \right\},$$

where the skewness parameter fulfils $\beta \in (-1, 1)$, the symmetric case corresponding to $\beta = 0$; as such, ρ and β have the one-to-one relation

$$\tan \left\{ \alpha\pi \left(\rho - \frac{1}{2} \right) \right\} = \beta \tan \frac{\alpha\pi}{2}.$$

Also, regarding ρ as a function of β (for any fixed $\alpha \in (1, 2)$), it can be seen that ρ is monotonically decreasing on $(-1, 1)$. Hence $\rho - 1/2$ and β have opposite signs for $\alpha \in (1, 2)$, which is not the case for $\alpha \in (0, 1)$; Figure 1 illustrates this point, where also included just for comparison is the case of $\alpha = 0.8$. Interested readers can consult Zolotarev [21] for more details concerning one-dimensional stable distributions; see also Borak *et al.* [6].

The reason why we have chosen the parametrization (1) is that, as is expected from Figure 1, estimation performance of β based on the empirical sign is destabilized for α close to 2. That is to say, a “small” change of the empirical-sign quantity (see Section 3.1.1) leads to a “big” diremption of the estimate of β from the true value; this point can be seen from Figure 1, where the curve is gentler for α closer to 2.

Denote by $Z = (Z_t)_{t \in [0,1]}$ a univariate Lévy process starting from the origin such that

$$\mathcal{L}(Z_t) = S_\alpha(\rho, t), \quad t \in [0, 1]. \quad (4)$$

The image measure of the process Z is completely characterized by the two parameter α and ρ . Figure 2 shows two simulated sample paths of Z .

For the stable Lévy processes, the (tail-)index α also corresponds to the Blumenthal-Gettoor activity index (see, e.g., Sato [16]). In view of (4), we see that the time parameter t directly serves as the scale in the parametrization (1).

The process Z itself does not accommodate the scale parametrization. Now we introduce a possibly time-varying scale process. Let $\sigma = (\sigma_t)_{t \in [0,1]}$ be a positive

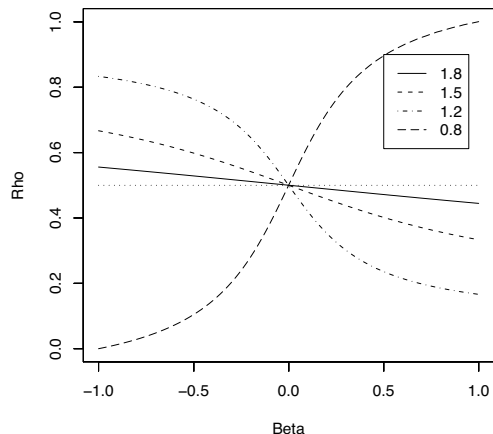


Figure 1. Plots of ρ as a function of β for the values $\alpha = 0.8, 1.2, 1.5$, and 1.8 .

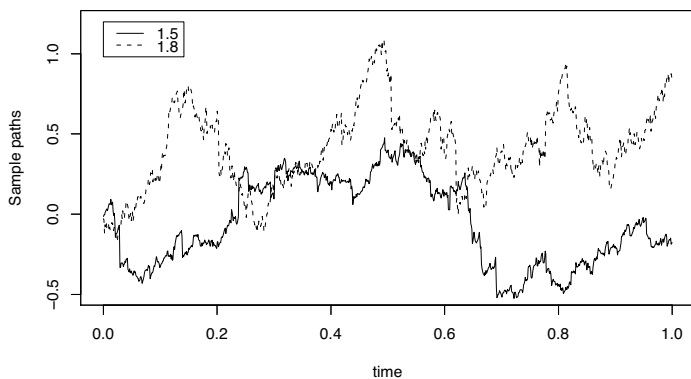


Figure 2. Two simulated sample paths of Z of (4) for $\alpha = 1.5$ and 1.8 , with $\beta = -0.5$ and $\sigma_t \equiv 1$; although we drew solid and dashed lines for clarity, they are actually of pure jump in theory.

càdlàg process (right-continuous and having left-hand side limits) independent of Z , such that

$$P\left[\int_0^1 \sigma_s^2 ds < \infty\right] = 1. \quad (5)$$

Then we consider the process $X = (X_t)_{t \in [0,1]}$ given by

$$X_t = \int_0^t \sigma_s dZ_s$$

as a model of a univariate log-price process under physical measure; without loss of generality, we have set $X_0 = 0$. The condition (5) is sufficient in order to make the stochastic integral well-defined; see, e.g., Applebaum [1] for a general account of stochastic integration. Additionally, for a technical reason, we impose the following structure on σ^α (the α th power process of σ) which is borrowed from Barndorff-Nielsen *et al.* [3] (see also Barndorff-Nielsen *et al.* [2]):

$$\begin{aligned}\sigma_t^\alpha &= \sigma_0^\alpha + \int_0^t a_s ds + \int_0^t b_{s-} dw_s \\ &\quad + \int_0^t \int h \circ c(s-, z)(\mu - \nu)(ds, dz) + \int_0^t \int (c - h \circ c)(s-, z)\mu(ds, dz).\end{aligned}$$

Here the ingredients are as follows: w is a standard Wiener process; μ is a Poisson random measure having the intensity measure $\nu(ds, dz) = dsF(dz)$, where F is a σ -finite measure on $(0, \infty) \times \mathbb{R}$; a and b are real-valued càdlàg processes; $c : \Omega \times [0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}$ is a càdlàg process satisfying that (i) $c(s, z) = c(\omega; s, z)$ is $\mathcal{F}_s \otimes \mathcal{B}(\mathbb{R})$ -measurable for each s , and that (ii) $\sup_{\omega \in \Omega, s < S_k(\omega)} |\delta(\omega; s, z)| \leq \psi_k(z)$ for some nonrandom functions $\psi_k(z)$ fulfilling that $\int_{\mathbb{R}} \{1 \wedge \psi_k(z)^2\} F(dz) < \infty$ and stopping times S_k such that $S_k \rightarrow \infty$ a.s.; finally, h is a continuous function on $\mathbb{R}$ with compact support such that $h(x) = x$ near the origin.

Such σ s constitute a broad class of the so-called Itô's semimartingales, including diffusions with jumps.

Remark 2.1. Extending the present time period $[0, 1]$ to $[0, \infty)$, we may equivalently set $X_t = Z_{\int_0^t \sigma_s^\alpha ds}$ provided that the “clock” process $\int_0^t \sigma_s^\alpha ds \rightarrow \infty$ a.s. for $t \rightarrow \infty$. This time-change representation is known to be inherent in the case of stable-Lévy integrators among general Lévy ones; see Kallsen and Shiryaev [10] for details.

Remark 2.2. We have set the target period is $[0, 1]$ from the very beginning. However, this point is a matter of no importance: enlarging the length of the period is reflected in making $\int_0^1 \sigma_s^\alpha ds$ larger through σ .

Suppose that we have a discrete-time data with sampling mesh $1/n$ over the target period $[0, 1]$, where n denotes the sample size; namely, we observe the sequence of log prices

$$X_{1/n}, X_{2/n}, \dots, X_{(n-1)/n}, X_1.$$

The log-price model described above is governed by the parameter (ρ, α, σ) unknown to observers. Nevertheless, note that (ρ, α, σ) is possibly infinite-dimensional. Hopefully we will be able to estimate σ_t for each $t \in [0, 1]$, but this is beyond the scope of this article; to the best of author's knowledge, no such result has been obtained in the non-Gaussian stable driven case. Instead, we are going to confine our objective to the following:

- (A) estimation of (ρ, α, σ) when $\sigma_t \equiv \sigma$ for a positive constant σ .
- (B) estimation of $(\rho, \alpha, \int_0^1 \sigma_s^\alpha ds)$ when σ is actually time-varying (possibly random).

Our goal is to provide an explicit recipe of interval estimation based on the available high-frequency data (i.e., for $n \rightarrow \infty$). To this end we are going to derive asymptotic (mixed) normality with specific asymptotic covariance matrix as well as rate of convergence.

Of course the case (A) is formally included in the case (B), however, we need a separate argument to consider the latter. In both cases, we first construct a simple estimator of (ρ, α) with leaving σ unknown. Then, using the estimates, we provide a estimator of σ or $\int_0^1 \sigma_s^\alpha ds$. Estimation of integrated quantities such as $\int_0^1 \sigma_s^\alpha ds$ is already known to be possible in the light of recently developed theory of MPV for pure-jump processes; see Woerner [20] and references therein. However, to implement the procedure, as a matter of fact we need estimates of α and ρ beforehand. We can avoid this inconvenience since, in our estimation procedure, an estimator of (ρ, α) is first provided without using information of σ . This is a great advantage of our estimation procedure.

According to the scaling property of the strictly stable distributions and the independence between σ and Z , we have

$$\mathcal{L}(X_1|\sigma) = S_\alpha \left(\rho, \int_0^1 \sigma_s^\alpha ds \right) \quad (6)$$

in the case (B). It seems natural to target at the integrated scale $\int_0^1 \sigma_s^\alpha ds$; the major estimation target in the familiar Brownian semimartingale framework (e.g., Barndorff-Nielsen *et al.* [2, 3] as well as their references) is the integrated volatility $\int_0^1 \sigma_s^2 ds$. The author expects that the pricing strategy of Miyahara and Moriwaki [15] for the geometric stable Lévy process remains valid even for the cases of time-varying scale, as long as the option in question is of European type, in which only an expectation of the “terminal” variable (namely, X_1 in our framework) is concerned: this is just because, as specified in (6), the $\mathcal{L}(X_1|\sigma)$ is exactly stable.

3. Description of Estimation Procedure

3.1 Preliminaries

Write the increments of successive observations as

$$\Delta_i X = X_{i/n} - X_{(i-1)/n}, \quad i \leq n.$$

Conditional on the process σ , the random variables $\Delta_i X$ are mutually independent and for each $n \in \mathbb{N}$ and $i \leq n$

$$\mathcal{L}(\Delta_i X|\sigma) = S_\alpha \left(\rho, \int_{(i-1)/n}^{i/n} \sigma_s^\alpha ds \right).$$

Before proceeding let us remind two fundamental facts, which are several times used in the sequel without notice.

- Since we are concerned here with the weak property, we may set

$$\Delta_i X = (\bar{\sigma}_i/n)^{1/\alpha} \zeta_i \quad \text{a.s.,}$$

where $\bar{\sigma}_i := n \int_{(i-1)/n}^{i/n} \sigma_s^\alpha ds$ and (ζ_i) is an i.i.d. sequence with common law $S_\alpha(\rho, 1)$.

- Let Λ_n be a sequence of essentially bounded functionals on the product space of the path spaces of Z and σ , and let $\lambda_n(\sigma) := \int \Lambda_n(\sigma, z) P^Z(dz)$, where P^ξ denotes the image measure of a variable ξ . Suppose $\lambda_n(\sigma) \rightarrow^P \lambda_0(\sigma)$ for some functional λ_0 on the path spaces of σ , where $\rightarrow^P$ denotes the convergence in probability. In view of the independence between Z and σ , a disintegration argument gives $\lambda_n(\sigma) = E[\Lambda_n(\sigma, Z) | \sigma]$ a.s., moreover, the boundedness of $\{\lambda_n(\sigma)\}_{n \in \mathbb{N}}$ yields convergence of moments, namely, $E[\Lambda_n(\sigma, Z)] = \int \lambda_n(\sigma) P^\sigma(d\sigma) \rightarrow \int \lambda_0(\sigma) P^\sigma(d\sigma)$. That is to say, we may actually treat σ a nonrandom process in the process of deriving weak limit theorems. In particular, if some functionals $S_n(\sigma_0, Z)$ with fixed σ_0 are asymptotically centered normal with covariance matrix $V(\sigma_0)$, then it automatically follows that the limit distribution of $S_n(\sigma, Z)$ has the characteristic function $u \mapsto \int \exp\{-u^\top V(\sigma)u/2\} P^\sigma(d\sigma)$, a mixed normal if σ is random.

These are trivial, but crucial in our study.¹

As mentioned before, first we construct concrete estimators of ρ and α in this order without any further information of the scale process σ . (Section 3.2), and then, using the estimates of ρ and α so obtained, we give estimators of the remaining σ or $\int_0^1 \sigma_s^\alpha ds$ according as the cases (A) or (B), respectively (Sections 3.3 and 3.4). For later use, in the rest of this subsection we give some background information on the empirical-sign statistics and MPVs.

3.1.1 Expression of empirical-sign statistics

Let $H_n := n^{-1} \sum_{i=1}^n \text{sgn}(\Delta_i X)$, then $H_n = n^{-1} \sum_{i=1}^n \text{sgn}(\zeta_i) \rightarrow^P E[\text{sgn}(\zeta_1)] = 2\rho - 1$. Hence

$$\hat{\rho}_n := \frac{1}{2}(H_n + 1) \quad (7)$$

serves as a consistent estimator of ρ . Since

$$\sqrt{n}(\hat{\rho}_n - \rho) = \sum_{i=1}^n \frac{1}{2\sqrt{n}} \{\text{sgn}(\zeta_i) - (2\rho - 1)\}, \quad (8)$$

¹Moreover, if necessary in the proof, we may suppose that $(\sigma_t)_{t \in [0,1]}$ is bounded from above and bounded away from zero without loss of generality: this follows from the localization arguments as in Barndorff-Nielsen *et al.* [3].

we easily deduce the asymptotic normality $\sqrt{n}(\rho_n - \rho) \rightarrow^d \mathcal{N}_1(0, \rho(1 - \rho))$, where the symbol $\rightarrow^d$ stands for the weak convergence. It is nice that the asymptotic variance only depends on ρ as it directly enables us to provide a confidence interval of ρ . Despite of its simplicity, it exhibits unexpectedly good finite-sample performances; see Section 4.

Perhaps the simplest possible estimator of ρ is not (7) but $n^{-1} \sum_{i=1}^n I(\Delta_i X \geq 0)$, where $I(A)$ denotes the indicator function of an event A . The reason why we chose (7) is that, thanks to (3), it directly leads to an explicit asymptotic covariance between the estimator of the remaining parameters. Moreover, the asymptotic variance of $n^{-1} \sum_{i=1}^n I(\Delta_i X \geq 0)$ is $\rho(1 - \rho)$, which is the same as that of (7). See Section 3.2 for details.

Remark 3.1. There are other possible ways to construct an estimate of ρ , for example, the method of moments based on $E[|\zeta|^q]$ together with $E[\zeta^{(q)}]$, where $\mathcal{L}(\zeta) = S_\alpha(\rho, 1)$ (see Kuruoğlu [11]). However, in this case the asymptotic variance of the resulting estimator must depend on the true value of α .

Remark 3.2. It may be expected that there is no other Lévy process than the stable one, for which we can consistently estimate the “degree of skewness” in such a simple way. For instance, the familiar generalized hyperbolic Lévy process has the skewness parameter, but it can be consistently estimated only when we target the long-term asymptotics; see, e.g., Woerner [19].

3.1.2 Expression of normalized MPV

Fix an $m \in \mathbb{N}$, and let $r = (r_l)_{l=1}^m$ be such that $r_l \geq 0$, $r_+ := \sum_{l=1}^m r_l > 0$, and $\max_{l \leq m} r_l < \alpha/2$. Then we define the r th MPV as

$$M_n(r) := \frac{1}{n} \sum_{i=1}^{n-m+1} \prod_{l=1}^m |n^{1/\alpha} \Delta_{i+l-1} X|^{r_l}. \quad (9)$$

By the equivalent expression of $(\Delta_i X)$, we may replace “ $|n^{1/\alpha} \Delta_{i+l-1} X|^{r_l}$ ” in the right-hand side of (9) by “ $\sigma_{i+l-1}^{r_l/\alpha} |\zeta_{i+l-1}|^{r_l}$ ”. Let

$$\sigma_q^* := \int_0^1 \sigma_s^q ds$$

for $q > 0$ and $\mu(r) := \prod_{l=1}^m \mu_{r_l}$. Here we prepare a first-order stochastic expansion useful for our goal.

Observe that

$$\sqrt{n} \{M_n(r) - \mu(r) \sigma_{r_+}^*\} = \sum_{i=1}^{n-m+1} \frac{1}{\sqrt{n}} \chi'_{ni}(r) + R_n(r),$$

where

$$\begin{aligned}\chi'_{ni}(r) &:= \left(\prod_{l=1}^m \bar{\sigma}_{i+l-1}^{r_l/\alpha} \right) \left(\prod_{l=1}^m |\zeta_{i+l-1}|^{r_l} - \mu(r) \right), \\ R_n(r) &:= \mu(r) \left\{ \sum_{i=1}^{n-m+1} \frac{1}{\sqrt{n}} \left(\prod_{l=1}^m \bar{\sigma}_{i+l-1}^{r_l/\alpha} - \sigma_{(i-1)/n}^{r_+} \right) \right. \\ &\quad \left. + \sum_{i=1}^{n-m+1} \sqrt{n} \int_{(i-1)/n}^{i/n} (\sigma_{(i-1)/n}^{r_+} - \sigma_s^{r_+}) ds \right\} + O\left(\frac{1}{\sqrt{n}}\right).\end{aligned}$$

From the same argument as in Woerner [20] together with Barndorff-Nielsen *et al.* [3] (see also Masuda [14]), we can deduce that $R_n(r) \rightarrow^p 0$. Similarly, straightforward but rather messy computations lead to

$$\sum_{i=1}^{n-m+1} \frac{1}{\sqrt{n}} \chi'_{ni}(r) = \sum_{i=m}^n \frac{1}{\sqrt{n}} \chi_{ni}(r) + o_p(1),$$

where

$$\chi_{ni}(r) := \left(\prod_{l=1}^m \bar{\sigma}_{i-m+l}^{r_l/\alpha} \right) \sum_{q=1}^m \left(\prod_{l=1}^{q-1} |\zeta_{i+l-q}|^{r_l} \right) \left(\prod_{l=q+1}^m \mu_{r_l} \right) (|\zeta_i|^{r_q} - \mu_{r_q}).$$

In summary, we have

$$\sqrt{n} \{M_n(r) - \mu(r) \sigma_{r_+}^*\} = \sum_{i=m}^n \frac{1}{\sqrt{n}} \chi_{ni}(r) + o_p(1). \quad (10)$$

3.1.3 A basic limit result

Building on the arguments above, we now derive a basic distributional result.

Let $r = (r_l)_{l=1}^m$ be as before, and also let $r' = (r'_l)_{l=1}^m$ be another vector fulfilling the same conditions as r . In what follows we set

$$r_+ = r'_+ = p \quad (11)$$

for some $p > 0$; this setting is enough for both (A) and (B). We here derive the limit distribution (normal conditional on σ) of the random vectors

$$S_n(r, r') := \sqrt{n} \begin{pmatrix} H_n & - (2\rho - 1) \\ M_n(r) & - \mu(r) \sigma_p^* \\ M_n(r') & - \mu(r') \sigma_p^* \end{pmatrix},$$

which serves as a basic tool for our purpose.

In view of (8) and (10), it follows that $S_n(r, r')$ admits the stochastic expansion

$$S_n(r, r') = \sum_{i=m}^n \frac{1}{\sqrt{n}} \begin{pmatrix} \text{sgn}(\xi_i) - (2\rho - 1) \\ \chi_{ni}(r) \\ \chi_{ni}(r') \end{pmatrix} + o_p(1) =: \sum_{i=m}^n \frac{1}{\sqrt{n}} \gamma_{ni} + o_p(1).$$

For the leading term $\sum_{i=m}^n n^{-1/2} \gamma_{ni}$, we can apply a central limit theorem either for finite-order dependent arrays or for martingale difference arrays. Here we formally use the latter, where the underlying filtration may be taken as $\{\mathcal{G}_{ni}\}_{i \leq n}$ with $\mathcal{G}_{ni} := \sigma(\xi_j : j \leq i)$; recall that we are now regarding σ a nonrandom process. The Lindeberg condition readily follows from the condition

$$\max_{l \leq m} (r_l \vee r'_l) < \frac{\alpha}{2},$$

hence it suffices to compute the quadratic variation. Therefore we are left to finding the limits in probability of $n^{-1} \sum_{i=m}^n E[\gamma_{ni} \gamma_{ni}^\top | \mathcal{G}_{n,i-1}]$. After lengthy computation, it turns out that, under the regularity conditions imposed on σ ,

$$\frac{1}{n} \sum_{i=m}^n E[\gamma_{ni} \gamma_{ni}^\top | \mathcal{G}_{n,i-1}] \rightarrow^p \Sigma(\rho, \alpha, \sigma) := \begin{pmatrix} 4\rho(1-\rho) & A(r)\sigma_{r_+}^* & A(r')\sigma_{r'_+}^* \\ & B(r, r)\sigma_{2r_+}^* & B(r, r')\sigma_{r_++r'_+}^* \\ \text{sym.} & & B(r', r')\sigma_{2r'_+}^* \end{pmatrix},$$

where we conveniently wrote

$$\begin{aligned} A(r) &= \sum_{q=1}^m \left(\prod_{1 \leq l \leq m, l \neq q} \mu_{r_l} \right) \{v_{r_q} - (2\rho - 1)\mu_{r_q}\}, \\ B(r, r') &= \prod_{l=1}^m \mu_{r_l+r'_l} - (2m-1) \prod_{l=1}^m \mu_{r_l} \mu_{r'_l} \\ &\quad + \sum_{q=1}^{m-1} \left\{ \left(\prod_{l=1}^{m-q} \mu_{r'_l} \right) \left(\prod_{l=m-q+1}^m \mu_{r'_l+r_{l-m+q}} \right) \left(\prod_{l=q+1}^m \mu_{r_l} \right) \right. \\ &\quad \left. + \left(\prod_{l=1}^{m-q} \mu_{r_l} \right) \left(\prod_{l=m-q+1}^m \mu_{r_l+r'_{l-m+q}} \right) \left(\prod_{l=q+1}^m \mu_{r'_l} \right) \right\}, \end{aligned}$$

with obvious analogues $A(r')$ and $B(r, r)$, and $B(r', r')$. Thus we arrive at

$$S_n(r, r') \rightarrow^d \mathcal{N}_3(0, \Sigma(\rho, \alpha, \sigma)), \quad (12)$$

which implies that the limit distribution of $S_n(r, r')$ is a normal scale mixture conditional on σ with conditional covariance matrix $\Sigma(\rho, \alpha, \sigma)$. Here we note that $\Sigma(\rho, \alpha, \sigma)$ depends on the process σ only through the integrated quantities $\sigma_{r_+}^*$, $\sigma_{r'_+}^*$, $\sigma_{2r_+}^*$, $\sigma_{2r'_+}^*$, and $\sigma_{r_++r'_+}^*$.

Having the basic convergence (12) in hand, we now turn to our main objectives, (A) and (B) mentioned in Section 2.

3.2 Joint Asymptotic (Mixed) Normality

Given a $p > 0$ and (r, r') (remind that we are assuming (11)), we write $(\hat{\rho}_n, \hat{\alpha}_{p,n}, \hat{\sigma}_{p,n}^*)$ for the random root of

$$\begin{pmatrix} H_n & -(2\rho - 1) \\ M_n(r) - \mu(r)\sigma_p^* \\ M_n(r') - \mu(r')\sigma_p^* \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \quad (13)$$

For a moment we suppose that such a root indeed exists. We introduce the function

$$F(\rho, \alpha, s) := \begin{pmatrix} 2\rho - 1 \\ \mu(r)s \\ \mu(r')s \end{pmatrix}.$$

Now let us recall (2) with $\sigma = 1$. As we are assuming that $\alpha \in (1, 2)$ and $1 - 1/\alpha < \rho < 1/\alpha$, we have $\xi \in (-\pi/2, \pi/2)$, so that $\cos(\xi) > 0$. Hence the quantities $\mu(r)$ and $\mu(r')$ are continuously differentiable with respect to (ρ, α) . Let $D_\rho(r) := \frac{\partial}{\partial \rho}\mu(r)$ and $D_\alpha(r) := \frac{\partial}{\partial \alpha}\mu(r)$: here, the variable “ s ” is supposed to be independent of (ρ, α) . Trivially,

$$\nabla F(\rho, \alpha, s) = \begin{pmatrix} 2 & 0 & 0 \\ sD_\rho(r) & sD_\alpha(r) & \mu(r) \\ sD_\rho(r') & sD_\alpha(r') & \mu(r') \end{pmatrix},$$

which is nonsingular for each $s > 0$ as soon as

$$\mu(r')D_\alpha(r) \neq \mu(r)D_\alpha(r'). \quad (14)$$

Again let us recall that we may proceed as if σ is nonrandom. The classical delta method (e.g., van der Vaart [17]) yields that, if (14) holds true, then

$$\sqrt{n} \begin{pmatrix} \hat{\rho}_n - \rho \\ \hat{\alpha}_{p,n} - \alpha \\ \hat{\sigma}_{p,n}^* - \sigma_p^* \end{pmatrix} \rightarrow^d \mathcal{N}_3(0, V(\rho, \alpha, \sigma)), \quad (15)$$

where

$$V(\rho, \alpha, \sigma) := \{\nabla F(\rho, \alpha, \sigma_p^*)\}^{-1} \Sigma(\rho, \alpha, \sigma) \{\nabla F(\rho, \alpha, \sigma_p^*)\}^{-1, \top}.$$

We see that $\Sigma(\rho, \alpha, \sigma)$ here depends on σ only through σ_p^* and σ_{2p}^* ; hence, more specifically we may write $\Sigma(\rho, \alpha, \sigma) = \Sigma(\rho, \alpha, \sigma_p^*, \sigma_{2p}^*)$, and accordingly, $V(\rho, \alpha, \sigma) = V(\rho, \alpha, \sigma_p^*, \sigma_{2p}^*)$. We should note that the function $V(\rho, \alpha, \sigma_p^*, \sigma_{2p}^*)$ is fully explicit as a function of its four arguments.

Now we set $m = 2$ and consider $r = (2q, 0)$ and $r' = (q, q)$ for a $q > 0$ (hence $p = 2q$). In order to make (12) valid, we need $q < \alpha/4$: as we are assuming that $\alpha \in (1, 2)$, a naive choice is $q = 1/4$ (see Remark 3.3 below).

Let us mention the computation of the solution to (13). We already have a closed-form solution $\hat{\rho}_n$ in (7). As for $\hat{\alpha}_n$, we can conveniently utilize the second and third arguments of (13): write $\hat{\mu}(\cdot)$ for the $\mu(\cdot)$ with (ρ, α) replaced by $(\hat{\rho}_n, \hat{\alpha}_{p,n})$, and then consider the estimating equation $M_n(q, q)/M_n(2q, 0) = \hat{\mu}(q, q)/\hat{\mu}(2q)$, which can be rewritten as

$$\frac{\sum_{i=1}^{n-1} |\Delta_i X|^q |\Delta_{i+1} X|^q}{\sum_{i=1}^n |\Delta_i X|^{2q}} = C_1(q) C_2(q, \hat{\rho}_n) \frac{\{\Gamma(1 - q/\hat{\alpha}_{p,n})\}^2}{\Gamma(1 - 2q/\hat{\alpha}_{p,n})}, \quad (16)$$

where, having $\hat{\rho}_n$ beforehand, we can regard

$$C_1(q) := \frac{\Gamma(1 - 2q) \cos(q\pi)}{\{\Gamma(1 - q) \cos(q\pi/2)\}^2} \quad \text{and} \quad C_2(q, \hat{\rho}_n) := \frac{[\cos\{q\pi(\hat{\rho}_n - 1/2)\}]^2}{\cos\{2q\pi(\hat{\rho}_n - 1/2)\}}$$

as constants. Since the function

$$\alpha \mapsto \frac{\{\Gamma(1 - q/\alpha)\}^2}{\Gamma(1 - 2q/\alpha)} \quad (17)$$

is strictly monotone on $(1, 2)$, it is easy to search the root $\hat{\alpha}_{p,n}$. Clearly, the root does uniquely exist with probability tending to one.

Remark 3.3. We see that the range of the function (17) becomes narrower for smaller q , so that the root $\hat{\alpha}_{p,n}$ becomes too sensitive for a small change of the sample quantity in the left-hand side of (16). This implies that the law of large numbers for the sample quantity should be in force with high degree of accuracy for smaller q .

Thus, given a $p = 2q > 0$, we could get the estimates $\hat{\rho}_n$ and $\hat{\alpha}_{p,n}$ without special information of σ , which may be time-varying and random as long as the regularity conditions on σ imposed on Section 2 hold true. It is important here that we have used the bipower variation in part; the procedure using the first and second empirical moments as in Masuda [13] is valid only when σ is constant.

The present asymptotic covariance matrix is $V(\rho, \alpha, \sigma_{2q}^*, \sigma_{4q}^*)$, for which we want to provide a consistent estimator. We only need to give consistent estimators of σ_{2q}^* and σ_{4q}^* ; recall that we need

$$4q < \alpha$$

in order to make the distributional result (15) with $p = 2q$ valid. For instance, we can proceed as follows. First, (15) with $p = 2q$ implies that $M_n(2p, 0) \rightarrow^P \mu(2q, 0) \sigma_{2q}^*$. Using the estimates $(\hat{\rho}_n, \hat{\alpha}_{p,n})$ and the continuous mapping theorem, we deduce that $M_n(2q, 0)/\hat{\mu}(2q, 0)$ is a consistent estimator of σ_{2q}^* . We should notice the dependence of $M_n(2q, 0)$ on α (recall (9)): $M_n(2q, 0) = n^{2q/\alpha-1} \sum_{i=1}^n |\Delta_i X|^{2q}$. Nevertheless, as in Masuda [13], we see that the α can be

replaced by $\hat{\alpha}_{p,n}$ since we already know that $\sqrt{n}(\hat{\alpha}_{p,n} - \alpha) = O_p(1)$. Therefore,

$$\hat{\sigma}_{2q,n}^* := \frac{n^{2q/\hat{\alpha}_{p,n}-1}}{\hat{\mu}(2q, 0)} \sum_{i=1}^n |\Delta_i X|^{2q} \rightarrow^p \sigma_{2q}^*. \quad (18)$$

Once again, let us remind that $\hat{\mu}(2q, 0)$ can be easily computed in view of (2) with $\sigma = 1$. By the same token, we could deduce that (still under $4q < \alpha$, of course)

$$\hat{\sigma}_{4q,n}^* := \frac{n^{4q/\hat{\alpha}_{p,n}-1}}{\hat{\mu}(2q, 2q)} \sum_{i=1}^{n-1} |\Delta_i X|^{2q} |\Delta_{i+1} X|^{2q} \rightarrow^p \sigma_{4q}^*.$$

After all, $V(\hat{\rho}_n, \hat{\alpha}_{p,n}, \hat{\sigma}_{2q,n}^*, \hat{\sigma}_{4q,n}^*)$ can serve as a desired consistent estimator.

Now we are in a position to complete our main objectives (A) and (B).

3.3 Case (A): Geometric Skewed Stable Lévy Process

When $\sigma_t \equiv \sigma > 0$, our model reduces to the geometric skewed stable Lévy process. In this case we can perform a full-joint interval estimation concerning the dominating (three-dimensional) parameter (ρ, α, σ) at rate $\sqrt{n}$.

We keep using the framework of the last subsection. It directly follows from (15) that

$$\sqrt{n} \begin{pmatrix} \hat{\rho}_n - \rho \\ \hat{\alpha}_{p,n} - \alpha \\ (\hat{\sigma}_{p,n})^p - \sigma^p \end{pmatrix} \rightarrow^d \mathcal{N}_3(0, V(\rho, \alpha, \sigma)), \quad (19)$$

where $V(\rho, \alpha, \sigma)$ explicitly depends on the three-dimensional parameter (ρ, α, σ) ; recall that $p = 2q < \alpha/2$. Applying the delta method to (19) in order to convert $(\hat{\sigma}_{p,n})^p$ to $\hat{\sigma}_{p,n}$ in (19), we readily get the asymptotic normality of $\sqrt{n}(\hat{\rho}_n - \rho, \hat{\alpha}_{p,n} - \alpha, \hat{\sigma}_{p,n} - \sigma)$; we omit the details. Our first objective (A) is thus achieved.

In summary, we may proceed with the choice $q = 1/4$ (so $p = 1/2$) as follows.

1. Compute the estimate $\hat{\rho}_n$ of ρ by (7).
2. Using the $\hat{\rho}_n$, find the root $\hat{\alpha}_{1/2,n}$ of (16).
3. Using $(\hat{\rho}_n, \hat{\alpha}_{1/2,n})$ thus obtained, an estimate of σ is provided by, e.g. (recall (18)),

$$\hat{\sigma}_{1/2,n} := \left\{ \frac{n^{1/(2\hat{\alpha}_{p,n})-1}}{\hat{\mu}(1/2, 0)} \sum_{i=1}^n \sqrt{|\Delta_i X|} \right\}^2.$$

3.4 Case (B): Time-Varying Scale Process

Now we turn to the case (B). Again by means of the argument give in Section 3.2, it remains to construct an estimator of $\sigma_\alpha^* = \int_0^1 \sigma_s^\alpha ds$. The point here is that, different from the case (A), a direct use of (15) is not sufficient to deduce the

distributional result concerning estimating σ_α^* because the dependence of (r, r') on α is not allowed there. In order to utilize $M_n(r)$ with r depending on α , we need some additional arguments.

Extracting the second row of (12), we have

$$\sqrt{n}\{M_n(r) - \mu(r)\sigma_{r_+}^*\} \rightarrow^d \mathcal{N}(0, B(r, r)\sigma_{2r_+}^*). \quad (20)$$

In view of the condition $\max_{l \leq m} r_l < \alpha/2$, we need (at least) a tripower variation for setting $r_+ = \alpha$. For simplicity, we set $m = 3$ and

$$r = r(\alpha) = \left(\frac{\alpha}{3}, \frac{\alpha}{3}, \frac{\alpha}{3}\right).$$

With this choice, we are going to provide an estimator of σ_α^* with specifying its rate of convergence and limiting distribution.

Let $M_n^*(\alpha) := M_n(\alpha/3, \alpha/3, \alpha/3)$. In this case the normalizing factor is $n^{r_+/\alpha-1} \equiv 1$, so that

$$M_n^*(\alpha) = \sum_{i=1}^{n-2} \prod_{l=1}^3 |\Delta_{i+l-1} X|^{\alpha/3},$$

which is computable as soon as we have an estimate of α . We have already obtained the estimator $\hat{\alpha}_{p,n}$, hence want to use $M_n^*(\hat{\alpha}_{p,n})$. For this, we have to look at the asymptotic behavior of the gap

$$\sqrt{n}\{M_n^*(r(\alpha)) - \mu(r(\alpha))\sigma_\alpha^*\} - \sqrt{n}\{M_n^*(\hat{\alpha}_{p,n}) - \mu(r(\hat{\alpha}_{p,n}))\sigma_\alpha^*\},$$

namely, the effect of “plugging in $\hat{\alpha}_{p,n}$ ”.

By means of Taylor’s formula

$$a^x = a^y + (\log a)^y(x-y) + (\log a)^2 \int_0^1 (1-u) a^{y+u(x-y)} du (x-y)^2$$

applied to the function $x \mapsto a^x$ ($x, y, a > 0$), we get

$$\begin{aligned} & \sqrt{n}\left\{M_n^*(\hat{\alpha}_{p,n}) - \mu\left(\frac{\alpha}{3}, \frac{\alpha}{3}, \frac{\alpha}{3}\right)\sigma_\alpha^*\right\} \\ &= \sqrt{n}\left\{M_n^*(\alpha) - \mu\left(\frac{\alpha}{3}, \frac{\alpha}{3}, \frac{\alpha}{3}\right)\sigma_\alpha^*\right\} + \frac{1}{3}\sqrt{n}(\hat{\alpha}_{p,n} - \alpha) \sum_{i=1}^{n-2} x_i^{\alpha/3} \log x_i \\ & \quad + \left\{\frac{1}{3}\sqrt{n}(\hat{\alpha}_{p,n} - \alpha)\right\}^2 \frac{1}{\sqrt{n}} \sum_{i=1}^{n-2} (\log x_i)^2 \int_0^1 (1-u) x_i^{\{\alpha+u(\hat{\alpha}_{p,n}-\alpha)\}/3} du, \quad (21) \end{aligned}$$

where we wrote $x_i = \prod_{l=1}^3 |\Delta_{i+l-1} X|$. We look at the right-hand side of (21) termwise. Let $y_i := \prod_{l=1}^3 |n^{1/\alpha} \Delta_{i+l-1} X|$.

- The first term is $O_p(1)$, as is evident from (20).
- Concerning the second term, we have

$$\begin{aligned}
 \sum_{i=1}^{n-2} x_i^{\alpha/3} \log x_i &= \frac{1}{n} \sum_{i=1}^{n-2} y_i^{\alpha/3} \log y_i - \frac{3}{\alpha} (\log n) \frac{1}{n} \sum_{i=1}^{n-2} y_i^{\alpha/3} \\
 &= O_p(1) - (\log n) \frac{3}{\alpha} \left\{ \mu \left(\frac{\alpha}{3}, \frac{\alpha}{3}, \frac{\alpha}{3} \right) \sigma_{\alpha}^* + O_p \left(\frac{1}{\sqrt{n}} \right) \right\} \\
 &= O_p(1) - (\log n) \frac{3}{\alpha} \mu \left(\frac{\alpha}{3}, \frac{\alpha}{3}, \frac{\alpha}{3} \right).
 \end{aligned}$$

- Write the third term as $\{\sqrt{n}(\hat{\alpha}_{p,n} - \alpha)/3\}^2 T_n$, and let us show that $T_n = o_p(1)$. Fix any $\varepsilon > 0$ and $\varepsilon_0 \in (0, \alpha/2)$ in the sequel. Then

$$P[|T_n| > \varepsilon] \leq P[|\hat{\alpha}_{p,n} - \alpha| > \varepsilon_0] + P[|T_n| > \varepsilon, |\hat{\alpha}_{p,n} - \alpha| \leq \varepsilon_0] =: p'_n + p''_n.$$

Clearly $p'_n \rightarrow 0$ by the $\sqrt{n}$ -consistency of $\hat{\alpha}_{p,n}$. As for p''_n , we first note that

$$\inf_{u \in [0,1]} \frac{1}{\alpha} \{\alpha + u(\hat{\alpha}_{p,n} - \alpha)\} \geq 1 - \frac{\varepsilon_0}{\alpha} > 0$$

on the event $\{|\hat{\alpha}_{p,n} - \alpha| \leq \varepsilon_0\}$. We estimate p''_n as follows:

$$\begin{aligned}
 p''_n &= P \left[|\hat{\alpha}_{p,n} - \alpha| \leq \varepsilon_0, \right. \\
 &\quad \left. \frac{1}{\sqrt{n}} \sum_{i=1}^{n-2} (\log x_i)^2 \int_0^1 (1-u) y_i^{\{\alpha + u(\hat{\alpha}_{p,n} - \alpha)\}/3} n^{-\{\alpha + u(\hat{\alpha}_{p,n} - \alpha)\}/\alpha} du > \varepsilon \right] \\
 &\leq P \left[|\hat{\alpha}_{p,n} - \alpha| \leq \varepsilon_0, n^{\varepsilon_0/\alpha - 1/2} \frac{1}{n} \sum_{i=1}^{n-2} (\log x_i)^2 \int_0^1 (1-u) y_i^{\{\alpha + u(\hat{\alpha}_{p,n} - \alpha)\}/3} du > \varepsilon \right] \\
 &\leq P \left[n^{\varepsilon_0/\alpha - 1/2} \frac{1}{n} \sum_{i=1}^{n-2} \{(\log n)^2 + (\log y_i)^2\} (1 + y_i)^{(\alpha + \varepsilon_0)/3} > C\varepsilon \right] \\
 &\leq \frac{1}{C\varepsilon} n^{\varepsilon_0/\alpha - 1/2} (\log n)^2 \rightarrow 0
 \end{aligned}$$

for some constant $C > 0$. Here we used Markov's inequality in the last inequality; note that $(\alpha + \varepsilon_0)/3 < \alpha/2$, hence the moment does exist.

Piecing together these three items and (21), we arrive at the asymptotic relation:

$$\frac{\sqrt{n}}{\log n} \left\{ M_n^*(\hat{\alpha}_{p,n}) - \mu \left(\frac{\alpha}{3}, \frac{\alpha}{3}, \frac{\alpha}{3} \right) \sigma_{\alpha}^* \right\} = -\frac{1}{\alpha} \mu \left(\frac{\alpha}{3}, \frac{\alpha}{3}, \frac{\alpha}{3} \right) \sigma_{\alpha}^* \sqrt{n}(\hat{\alpha}_{p,n} - \alpha) + O_p \left(\frac{1}{\log n} \right). \quad (22)$$

Now, recalling (2) we note that the quantity $\mu(\alpha/3, \alpha/3, \alpha/3)$ is a continuously differentiable function of (ρ, α) . Write $\bar{\mu}(\rho, \alpha) = \mu(\alpha/3, \alpha/3, \alpha/3)$. In view of the $\sqrt{n}$ -consistency of $(\hat{\rho}_n, \hat{\alpha}_{p,n})$ and the delta method, we obtain

$$\bar{\mu}(\rho, \alpha) = \bar{\mu}(\hat{\rho}_n, \hat{\alpha}_{p,n}) + O_p\left(\frac{1}{\sqrt{n}}\right). \quad (23)$$

Substituting (23) in (22) we end up with

$$\frac{\sqrt{n}}{\log n} \left\{ \frac{M_n^*(\hat{\alpha}_{p,n})}{\bar{\mu}(\hat{\rho}_n, \hat{\alpha}_{p,n})} - \sigma_\alpha^* \right\} = -\frac{1}{\alpha} \sigma_\alpha^* \sqrt{n}(\hat{\alpha}_{p,n} - \alpha) + O_p\left(\frac{1}{\log n}\right), \quad (24)$$

which implies that

$$\hat{\sigma}_{\alpha,n}^* := \frac{M_n^*(\hat{\alpha}_{p,n})}{\bar{\mu}(\hat{\rho}_n, \hat{\alpha}_{p,n})} \quad (25)$$

serves as $(\sqrt{n}/\log n)$ -consistent estimator of σ_α^* . Its asymptotic distribution is the centered normal scale mixture with limiting variance being

$$v(\rho, \alpha, \sigma_\alpha^*, \sigma_p^*, \sigma_{2p}^*) := \left(\frac{\sigma_\alpha^*}{\alpha} \right)^2 V_{22}(\rho, \alpha, \sigma_p^*, \sigma_{2p}^*),$$

where V_{22} denotes the (2,2)th entry of V ; recall that p is a parameter-free constant (see Section 3.2). A consistent estimator of $v(\rho, \alpha, \sigma_\alpha^*, \sigma_p^*, \sigma_{2p}^*)$ can be constructed by plugging in the estimators of its arguments.

The stochastic expansion (24) indicates an asymptotic linear dependence of $\sqrt{n}(\hat{\alpha}_{p,n} - \alpha)$ and $(\sqrt{n}/\log n)(\hat{\sigma}_{\alpha,n}^* - \sigma_\alpha^*)$. Of course, this occurs even for constant σ , if we try to estimate (α, σ^α) instead of (α, σ) . The point is that, plugging in a $\sqrt{n}$ -consistent estimator of α into the index r of the MPV $M_n(r)$ slows down estimation of σ_α^* from $\sqrt{n}$ to $\sqrt{n}/(\log n)$. It is beyond the scope of this article to explore a better alternative estimator of σ_α^* .

4. Simulation Experiments

Based on the discussion above, let us briefly observe finite-sample performance of our estimators. For simplicity, we here focus on nonrandom σ .

4.1 Case (A)

First, let σ is a positive constant, so that X is the geometric skewed stable Lévy process and the parameter to be estimated is (ρ, α, σ) .

As a simulation design, we set $\alpha = 1.3, 1.5, 1.7$, and 1.9 with common $\beta = -0.5$ and $\sigma = 1$; hence $(\alpha, \rho) = (1.2, 0.7638), (1.5, 0.5984), (1.7, 0.5467)$, and $(1.9, 0.5132)$. The sample size are taken as $n = 500, 1000, 2000$, and 5000 . In

all cases, the tuning parameter q is set to be $1/4$, and 1000 independent sample paths of X are generated. Empirical means and empirical s.d.'s are given with the 1000 independent estimates obtained. The results are reported in Table 4.1. We see that estimation of (ρ, α) is, despite of its simplicity, quite reliable. On the other hand, estimation variance of σ is relatively large compared with those of ρ and α . Nevertheless, it is clear that the bias is small. Moreover, as α gets close to 2, the performance of $\hat{\sigma}_n$ becomes better, while that of $(\hat{\rho}_n, \hat{\alpha}_{p,n})$ is seemingly unchanged.

In the unreported simulation results, we have observed that a change of q within its admissible region does not lead to a drastic change unless it is too small (see Remark 3.3).

Table 1. Estimation results for the true parameters $(\rho, \alpha, \sigma) = (0.7638, 1.2, 1)$, $(0.5984, 1.5, 1)$, $(0.5467, 1.7, 1)$, and $(0.5132, 1.9, 1)$ with the geometric stable Lévy processes. In each case, the empirical mean and standard deviation (in parenthesis) are given.

$\alpha = 1.2$						
n	ρ		α		σ	
500	0.7627	(0.0186)	1.2026	(0.0790)	1.1021	(0.8717)
1000	0.7634	(0.0137)	1.2031	(0.0575)	1.0450	(0.4643)
2000	0.7645	(0.0096)	1.2031	(0.0437)	1.0253	(0.5102)
5000	0.7636	(0.0061)	1.2023	(0.0313)	1.0123	(0.2854)
$\alpha = 1.5$						
n	ρ		α		σ	
500	0.5988	(0.0222)	1.4929	(0.1030)	1.0751	(0.4066)
1000	0.5981	(0.0162)	1.5010	(0.0757)	1.0289	(0.2549)
2000	0.5986	(0.0106)	1.4986	(0.0564)	1.0284	(0.2355)
5000	0.5984	(0.0073)	1.4983	(0.0364)	1.0169	(0.1516)
$\alpha = 1.7$						
n	ρ		α		σ	
500	0.5476	(0.0219)	1.6810	(0.1103)	1.0633	(0.2359)
1000	0.5474	(0.0158)	1.6830	(0.0823)	1.0567	(0.1948)
2000	0.5472	(0.0113)	1.6930	(0.0625)	1.0308	(0.1611)
5000	0.5466	(0.0070)	1.6977	(0.0375)	1.0126	(0.1022)
$\alpha = 1.9$						
n	ρ		α		σ	
500	0.5129	(0.0224)	1.8553	(0.1026)	1.0821	(0.1767)
1000	0.5133	(0.0164)	1.8767	(0.0808)	1.0535	(0.1568)
2000	0.5131	(0.0109)	1.8870	(0.0579)	1.0330	(0.1111)
5000	0.5128	(0.0073)	1.8971	(0.0401)	1.0097	(0.0809)

4.2 Case (B)

Next we observe a case of time-varying but nonrandom scale. We set

$$\sigma_t^\alpha = \frac{2}{5} \left\{ \cos(2\pi t) + \frac{3}{2} \right\}, \quad (26)$$

so that $\sigma_\alpha^* = 0.6$.

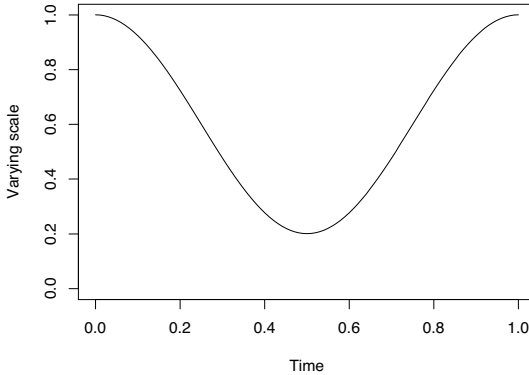


Figure 3. The plot of the function $t \mapsto \sigma_t^\alpha$ given by (26).

With the same choices of (ρ, α) , q , and n as in the previous case, we obtain the result in Tables 4.2; the estimator of σ_α^* here is based on (25). There we can observe a quite similar tendency as in the previous case.

5. Concluding Remarks

We have studied some statistical aspects in the calibration problem of a geometric skewed stable asset price models. Estimation of stable asset price models with possibly time-varying scale can be done easily by means of the simple empirical-sign statistics and MPVs. Especially, we could estimate integrated scale, which is a natural quantity as in the integrate variance in the framework of Brownian semimartingales, with multistep estimating procedure: we estimate ρ , α , and σ (or σ_α^*) one by one in this order. Our simulation results say that finite-sample performance of our estimators are unexpectedly good despite of their simplicity, except for a relatively bigger variance in estimating σ (or σ_α^*).

We close with mentioning some possible future issues.

- Throughout we supposed the independence between the scale process σ and the driving skewed stable Lévy process Z . This may be disappointing

Table 2. Estimation results for the true parameters $(\rho, \alpha) = (0.7638, 1.2)$, $(0.5984, 1.5)$, $(0.5467, 1.7)$, and $(0.5132, 1.9)$ with $\sigma_\alpha^* = 0.6$ in common under (26). In each case, the empirical mean and standard deviation (in parenthesis) are given.

$\alpha = 1.2$					
n	ρ		α		σ_α^*
500	0.7632	(0.0179)	1.1951	(0.0794)	0.6730 (0.3857)
1000	0.7636	(0.0139)	1.2042	(0.0619)	0.6274 (0.3094)
2000	0.7638	(0.0098)	1.2044	(0.0472)	0.6105 (0.2323)
5000	0.7641	(0.0059)	1.2025	(0.0305)	0.6029 (0.1521)
$\alpha = 1.5$					
n	ρ		α		σ_α^*
500	0.5978	(0.0220)	1.4877	(0.1023)	0.6697 (0.3031)
1000	0.5981	(0.0159)	1.4908	(0.0733)	0.6551 (0.2488)
2000	0.5985	(0.0111)	1.4960	(0.0573)	0.6349 (0.2033)
5000	0.5987	(0.0069)	1.4990	(0.0376)	0.6151 (0.1414)
$\alpha = 1.7$					
n	ρ		α		σ_α^*
500	0.5460	(0.0216)	1.6727	(0.1038)	0.6832 (0.2465)
1000	0.5465	(0.0160)	1.6801	(0.0820)	0.6714 (0.2280)
2000	0.5468	(0.0113)	1.6931	(0.0600)	0.6318 (0.1607)
5000	0.5465	(0.0071)	1.6988	(0.0393)	0.6116 (0.1135)
$\alpha = 1.9$					
n	ρ		α		σ_α^*
500	0.5130	(0.0229)	1.8440	(0.1039)	0.7196 (0.2233)
1000	0.5131	(0.0159)	1.8703	(0.0823)	0.6762 (0.1897)
2000	0.5138	(0.0114)	1.8851	(0.0588)	0.6412 (0.1349)
5000	0.5135	(0.0068)	1.8956	(0.0411)	0.6168 (0.0998)

as it excludes accommodating the leverage effect, however, the simple constructions of our estimators (especially, $\hat{\rho}_n$) break down if they are allowed to be dependent. We may be able to deal with correlated σ and Z if we have an extension of the power-variation results obtained in Corcuera *et al.* [7] to the MPV version. To the best of author's knowledge, such an extension does not seem to have been explicitly mentioned as yet.

- Assuming that σ is indeed time-varying and possibly random, estimation of “spot” scales σ_t is an open problem. Needless to say, this is much more difficult and delicate to deal with than the integrated scale. We know several results for Brownian-semimartingale cases (see, among others, Fan and Wang [8] and Malliavin and Mancino [12]), however, yet no general result for the case of pure-jump Z .

- Finally, it might be interesting to derive an option-pricing formula for the case of time-varying scale, which seems more realistic than the mere geometric skewed stable Lévy processes.

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A Note on a Statistical Hypothesis Testing for Removing Noise by the Random Matrix Theory and Its Application to Co-Volatility Matrices

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It is well known that the bias called market microstructure noise will arise, when estimating realized co-volatility matrix which is calculated as a sum of cross products of intraday high-frequency returns. An existing conventional technique for removing such a market microstructure noise is to perform eigenvalue decomposition of the sum of cross products matrix and to identify the elements corresponding to the decomposed values which are smaller than the maximum eigenvalue of the random matrix as noises. Although the maximum eigenvalue of a random matrix follows asymptotically Tracy-Widom distribution, the existing technique does not take this asymptotic nature into consideration, but only the convergence value is used for it. Therefore, it cannot evaluate quantitatively such a risk that regards accidentally essential volatility as a noise. In this paper, we propose a statistical hypothesis test for removing noise in co-volatility matrix based on the nature in which the maximum eigenvalue of a random matrix follows Tracy-Widom distribution asymptotically.

Keywords: Realized volatility, market microstructure noise, random matrix theory.

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1. Introduction

In recent years, we can easily obtain “high frequency data in finance”, so we may estimate and forecast (co)-volatility more correctly than before using by Realized Volatility (RV) which is a series of the sum of intraday squared log return and Realized Co-volatility (RC) which is a series of the sum of cross-product of two log returns, see [2] or [1]. However, it is well known that when forecasting volatility, RV and RC are contaminated by large biases, so called micro structural noise which is progressively increased as sampling frequency becomes higher, see [7]. Thus, the research considers a statistical method of removing such a noise in RV and RC by using random matrix theory. Doing eigenvalue decomposition of cross product matrix, we consider that noises in a covolatility matrix are elements corresponding to eigenvalues smaller than the maximum eigenvalue of the random matrix. It is known that the maximum eigenvalue of a random matrix will follow Tracy-Widom distribution asymptotically. However, existing methods haven't taken into consideration a distribution of the maximum eigenvalue of a random matrix, but have used only the maximum eigenvalue itself, for example, see [9]. Therefore, they cannot evaluate quantitatively a risk of considering accidentally that essential volatility is a noise.

Therefore, we propose a statistical hypothesis test for removing noise in covolatility matrix based on the nature in which the maximum eigenvalue of a random matrix follows Tracy-Widom distribution asymptotically.

This paper is organized as follows. Section 2 describes theoretical background of this study and gives brief explanation of random matrix theory and our proposal. Section 3 investigates empirical analysis. Section 4 concludes.

2. Theoretical Background

In this section, we will introduce theoretical properties of random matrix with some simulation results.

2.1 Random matrix

Random matrix is a matrix which has random variables as its elements. First, [16] and [17] developed a eigenvalue distribution of $N \times N$ real symmetric matrix $A = (a_{ij})$ with elements $\{a_{ij}\} \sim \text{i.i.d.}(0, 1/N)$. Following [16] and [17], we introduce $N \times N$ real symmetric random matrix $A = (a_{ij})$ with elements $\{a_{ij} | i \leq j\}$ which independently follows a distribution with a mean 0 and a variance $1/N$. If eigenvalues of A are $\lambda_1, \dots, \lambda_N$ and an empirical eigenvalue distribution of A is defined by

$$\rho_A(\lambda) = \frac{1}{N} \sum_{i=1}^N \delta(\lambda - \lambda_i),$$

then

$$\lim_{N \rightarrow \infty} \rho_A(\lambda) = \begin{cases} \frac{\sqrt{4-\lambda^2}}{2\pi} & (|\lambda| \leq 2) \\ 0 & (\text{otherwise}) \end{cases},$$

where $\delta(\cdot)$ is Dirac measure. Figures 1 and 2 show simulated eigenvalue distribution of A with $n = 1000$. The left panel is sampled from a Normal distribution and the right one is from uniform distribution. From these figures we can see that asymptotic behavior of eigenvalues of A is identical whatever distribution they follow.

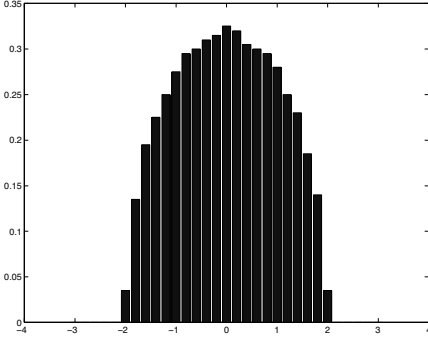


Figure 1. Sampled from normal.

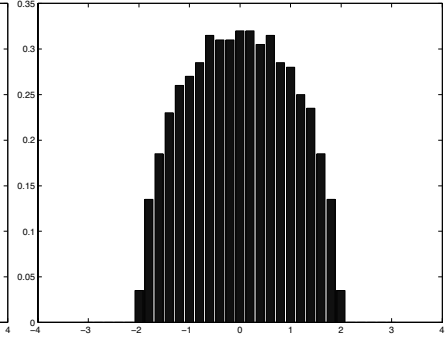


Figure 2. Sampled from uniform.

Second, we introduce Wishart distribution which plays very important role in multivariate analysis. Wishart found it for describing behavior of a sample covariance matrix $XX^\top$ in 1928. A distribution of $XX^\top$ depends on a distribution of random variables x , so we can estimate the original distribution of x from the distribution of $XX^\top$. If each column vector of $N \times p$ matrix $X = (\mathbf{x}^{(1)} \dots \mathbf{x}^{(n)})$ independently follows N dimensional Gaussian distribution, $\mathbf{x}^{(i)} \sim \mathcal{N}_N(\mathbf{0}, \Sigma)$, then $N \times N$ random matrix $XX^\top$

$$XX^\top \sim \mathcal{W}_N(p, \Sigma)$$

follows N dimensional Wishart distribution with a degree of freedom p and a covariance matrix Σ . If $N = 1$, then it follows χ^2 distribution with a degree of freedom p , and for $N = 2$, [6] found a relevant distribution.

Next, we also consider an asymptotic eigenvalue distribution of Wishart matrix. Supposed that $\Sigma = I_N$ and each element of a random matrix $X \in \mathbb{R}^{N \times p}$ independently follows $\mathcal{N}_1(0, 1^2)$,

$$XX^\top \sim \mathcal{W}_N(p, I_N).$$

For a random matrix $X \in \mathbb{R}^{N \times p}$ with $XX^\top \sim \mathcal{W}_N(p, I_N)$, if keeping proportion of

$$\alpha = p/N$$

and $N \rightarrow \infty$, then an eigenvalue distribution of $XX^\top$ converges to some function.

If $N \times p$ matrix X doesn't follow Gaussian distribution, say, $XX^\top$ isn't Wishart matrix, then the eigenvalue distribution of $XX^\top$ still converges to the same function above. This property is known as "the universality" of random matrix theory. It is very important characteristics that there is no necessity that each element of X follows Gaussian distribution. That is, it is not a necessary condition that $XX^\top$ follows Wishart distribution. Hence, we can generalize the limiting distribution of eigenvalues of $XX^\top$. We have the following theorem related to the universality by referring to [10] and [5].

Theorem 1 (Marčenko-Pastur law): Let X be an $N \times p$ matrix with independent, identically distributed entries $X_{i,j}$. We assume that $E(X_{i,j}) = 0$ and $\text{var}(X_{i,j}) = 1$. If p, N are large enough and p/N is a non-zero constant, then the distribution of eigenvalues of $XX^\top$ converges almost surely to a known density.

We set eigenvalues $\{\lambda_1, \dots, \lambda_N\}$ sampled from $XX^\top \sim \mathcal{W}_N(p, I_N)$ are scaled by

$$u_i = \lambda_i/p, \quad i = 1, \dots, N.$$

An empirical distribution of u follows

$$\delta P = \frac{1}{N} \{\delta(u_1) + \dots + \delta(u_N)\},$$

where $\delta(u)$ is Dirac measure. If $\alpha = p/N$ and $p, N \rightarrow \infty$, δP converges a.e. to $p(u)du$,

$$p(u) = \begin{cases} \frac{1}{2\pi\alpha} \frac{\sqrt{(u-u_{\min})(u_{\max}-u)}}{u} & \text{if } u_{\min} < u < u_{\max}, \\ 0 & \text{otherwise,} \end{cases}$$

$$u_{\min} = (\sqrt{\alpha} - 1)^2, \quad u_{\max} = (\sqrt{\alpha} + 1)^2.$$

The asymptotic eigenvalue distribution is given by the following formula from Marčenko-Pastur law,

$$\lambda_{\min} = (1 - \sqrt{\alpha})^2, \lambda_{\max} = (1 + \sqrt{\alpha})^2.$$

Limiting distribution of eigenvalues of $A = XX^\top$ is given by

$$\lim_{N \rightarrow \infty} \rho_A(\lambda) = \begin{cases} \frac{1}{2\pi\lambda} \sqrt{(\lambda - \lambda_{\min})(\lambda_{\max} - \lambda)} & \lambda_{\min} \leq \lambda \leq \lambda_{\max}, \\ 1 - \alpha & \lambda = 0 \text{ and } \alpha < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Figures 3 and 4 show simulated eigenvalue distribution of $XX^\top$ with $p = 1000$. The left panel is sampled from Normal distribution and the right one is from

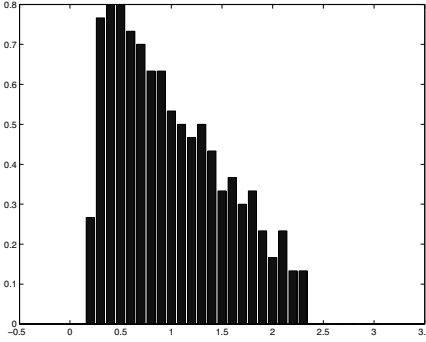


Figure 3. Sampled from normal.

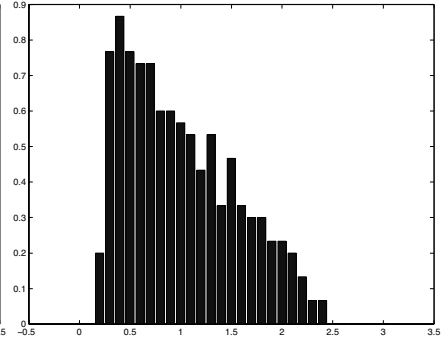


Figure 4. Sampled from uniform.

uniform distribution. From these figures we can see that asymptotic behavior of eigenvalues of XX^T is identical whatever distribution they follow just as Wigner's case.

Figures 5 and 6 show theoretical and empirical distribution of XX^T . The left panel is simulated distribution sampled from Normal distribution with $p = 1000$ and $N = 600$, $N = 1000$ and $N = 1400$. The right panel is empirical distribution sampled from individual stocks listed in Tokyo Stock Exchange. From these figures we can see that empirical distribution resembles simulated one in appearance however its scale is very different from each other.

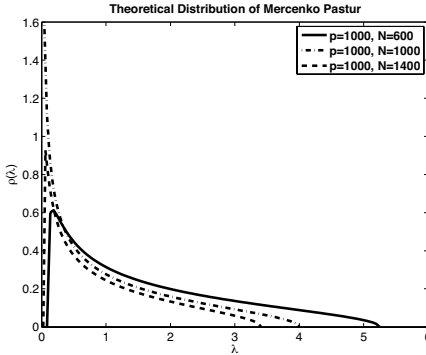


Figure 5. Theoretical distribution.

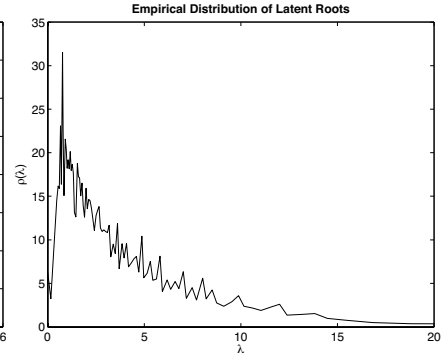


Figure 6. Empirical distribution.

2.2 Extraction of Essential Volatility

Here, we describe the technique of dividing a matrix $V = RR^T$ calculated from a standardized log return matrix R into essential parts and noise parts, following

[9]. We first obtain unit eigenvalues u_k corresponding to eigenvalues $\lambda_1, \dots, \lambda_N$ and $\lambda_k, (k = 1, \dots, N)$ of V where N denotes the number of stocks. Letting $V_k := \lambda_k u_k u_k^\top$ be the k th element of the matrix, then $V = \sum_{k=1}^N V_k$ hold, so we can divide V into N elements. Among the N elements, we can see that the elements corresponding to big eigenvalues are more essential to heavily influence on overall market. On the other hand, we can consider that the elements corresponding to small eigenvalues are less essential to be independent of overall market. To put it briefly, independent elements which are inside the maximum eigenvalue make no sense to portfolio strategy which are connected to correlation of log returns.

We schematically show the denoising method above.

- If the elements of V_k independent and identically distributed, then the corresponding eigenvalue λ_k must lie in the support of the Marčenko-Pastur law.
- If k th eigenvalue λ_k lies out of the support of the Marčenko-Pastur law, then the corresponding element V_k is not independent and identically distributed, say, can be considered to contain something other than noise.
- Thus, we can consider that the sum of elements V corresponding to a bigger eigenvalue than a threshold value θ which is the maximum eigenvalue of the matrix is so called denoised daily realized volatility¹.

$$V_+ = \sum_{k|\lambda_k > \theta} V_k.$$

As you can see, conventional studies have dichotomously distinguished noise and substantial parts in a convergence point of the maximum eigenvalue. In the existing research, the threshold value θ is determined only by the maximum eigenvalue of the matrix regardless of the asymptotic nature of the maximum eigenvalue of a random matrix. That is, they consider that the sum of the elements corresponding to eigenvalues such as $\lambda_k > \theta_1$ is denoised, and the sum of others is contaminated with noise. However, such a deterministic and “digital” method may accidentally cause an error misidentifying denoised volatility as contaminated one and vice versa since the maximum eigenvalue of a random matrix is still a random variable.

Therefore, we propose an interval estimation of eigenvalues which can distinguish noises, paying attention to the point that the maximum eigenvalue is also a random variable. We perform a statistical hypothesis testing with respect to denoised and contaminated volatility by using characteristics of which the maximum

¹No-one knows essential volatility since it is usually unobservable, so we dare to use the term “denoised” in stead of “essential”.

eigenvalue of V follows Tracy-Widom distribution which is explained in the next subsection. Here, the null hypothesis, which assumes that log returns only consist of pure noises, can be rejected by the fact that the largest eigenvalue of the sample covariance matrix does not lie in the support of the Marčenko-Pastur law. Specifically we set up a null hypothesis as contaminated with noise and alternative as denoised and vice versa, and the statistics value is obtained by a eigenvalue of V calculated from standardized log return matrix R .

2.3 Maximum Eigenvalue Density of Random Matrix

We suppose that X is $n \times p$ random matrix and $XX^\top$ is its covariance matrix. Under Gaussian assumptions, $XX^\top$ is said to have a Wishart distribution $W_p(n, \Sigma)$. If $\Sigma = I$, it is called as a white Wishart, in analogy with time series settings where a white spectrum is one with the same variance at all frequencies, see [8]. Asymptotic distribution of maximum eigenvalue of Wishart matrix $XX^\top$ with unit covariance follows the first order Tracy-Widom distribution, if $\alpha = p/n$ is constant, see [12], [13] and [14]. Moreover, even if the size of n or p is about ten, this asymptotic property is not lost, and it is known that Tracy-Widom distribution appear as a solution of Painlevé II type differential equation.

Theorem 2 (Tracy-Widom Law): Suppose that $\mathcal{W}$ is white Wishart matrix, γ is a constant, l_1 is the maximum eigenvalue, and $n/p \rightarrow \gamma \geq 1$, then

$$\frac{l_1 - \mu_{np}}{\sigma_{np}} \xrightarrow{\text{dist}} \mathcal{W} \sim F_1,$$

where the location and scale parameters are given by

$$\begin{aligned} \mu_{np} &= (\sqrt{n-1} + \sqrt{p})^2, \\ \sigma_{np} &= \sqrt{\mu_{np}} \left((n-1)^{-\frac{1}{2}} + p^{-\frac{1}{2}} \right)^{1/3}, \end{aligned}$$

where F_1 denotes the density function of the first order Tracy-Widom law.

We have to mention that the Tracy-Widom law also has “the universality” of random matrix theory. Hence, the Tracy-Widom law still holds without Gaussian assumptions, see [11], for example.

The asymptotic distribution function F_1 is a special case of the distribution family F_β . For $\beta = 1, 2, 4$, the function F_β appears as a asymptotic distribution for the maximum eigenvalue of Gaussian Orthogonal Ensemble (GOE), Gaussian Unitary Ensemble (GUE) and Gaussian Symplectic Ensemble (GSE), respectively. According to this fact, a distribution function $F_{N,\beta}(s) \stackrel{\text{def}}{=} \mathbb{P}(l_{\max}(\mathcal{A}) < s)$, $\beta = 1, 2, 4$ for the maximum eigenvalue $l_{\max}(\mathcal{A})$ of each random matrix $\mathcal{A}$ of GOE ($\beta = 1$), GUE ($\beta = 2$) or GSE ($\beta = 4$) satisfies the asymptotic law as follows:

$$F_\beta(s) = \lim_{N \rightarrow \infty} F_{N,\beta}(2\sigma\sqrt{N} + \sigma N^{-1/6}s)$$

where F_β is explicitly given by

$$\begin{aligned} F_1(s) &= \exp\left(-\frac{1}{2} \int_s^\infty sq(x)dx\right) [F_2(s)]^{\frac{1}{2}}, \\ F_2(s) &= \exp\left(-\int_s^\infty (x-s)q^2(x)dx\right), \\ F_4(2^{-\frac{2}{3}}s) &= \cosh\left(-\frac{1}{2} \int_s^\infty q(x)dx\right) [F_2(s)]^{\frac{1}{2}} \end{aligned}$$

and $q(s)$ is a unique solution for Painlevé equation type II. Again,

$$q'' = sq + 2q^3 + \alpha, \quad \alpha = 0$$

satisfies the boundary condition

$$q(s) \sim \text{Ai}(s), \quad s \rightarrow +\infty$$

where $\text{Ai}(s)$ denotes Airy function. Figures 7 and 8 show simulated maximum distribution of $XX^\top$ with $p = 1000$ which is known as Tracy-Widom distribution. The left panel is sampled from Normal distribution and the right one is from uniform distribution. From these figures we can see that asymptotic behavior of maximum eigenvalues of $XX^\top$ is identical whatever distribution they follow just as previous cases.

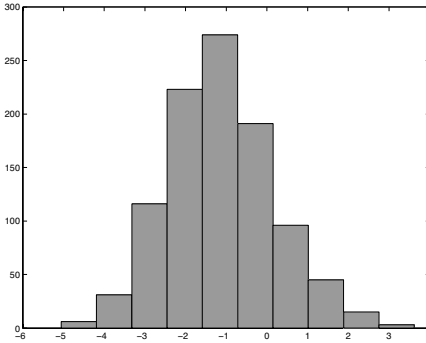


Figure 7. Sampled from normal.

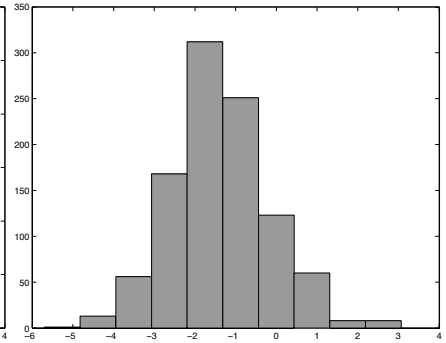


Figure 8. Sampled from uniform.

Then we can construct the following two types of hypothesis testing for noises by comparing sample eigenvalues $\lambda_k, (k = 1, \dots, N)$ with the theoretical Tracy-Widom statistic tw_α . To make it easy to understand, we illustrate these tests in Figures 9 and 10. These plots are from numerical work of [15] reporting that

the F_1 distribution has mean -1.21 and standard deviation 1.27 . The density is asymmetric and its left tail has exponential order of decay like $e^{|s|^{3/24}}$, while its right tail is of exponential order $e^{-\frac{2}{3}s^{3/2}}$, see [8]. The asymmetric feature is just the reason to propose two types of hypothesis testing for noises.²

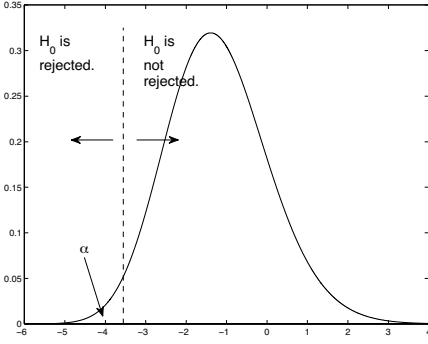


Figure 9. Illustration of Type I.

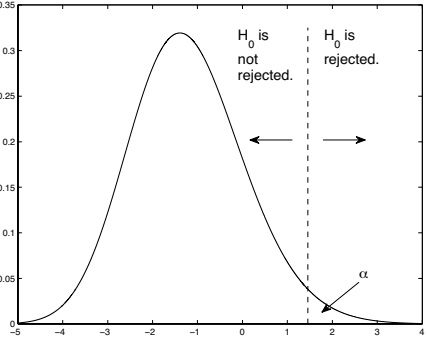


Figure 10. Illustration of Type II.

Type I: We test the probability that we take accidentally denoised parts as noises. So in this case, the null hypothesis H_0 assumes that log returns R are not pure noises:

$$H_0 : R \sim \text{not i.i.d. distributed.}$$

If $\lambda > tw_\alpha$, say, a sample eigenvalue is larger than the relevant critical value, then we fail to reject the null hypothesis.

Type II: We test the probability that we take accidentally noises as denoised parts. So in this case, the null hypothesis H_0 assumes that log returns R are pure noises:

$$H_0 : R \sim \text{i.i.d. distributed.}$$

If $\lambda_k < tw_{1-\alpha}$, say, a sample eigenvalue is smaller than the relevant critical value, then we fail to reject the null hypothesis.

Hence, Type I is a lower test for Tracy-Widom distribution and Type II an upper one. Tail probability of Tracy-Widom distribution is given by numerical computation as shown in the following Table 1, see [3] for more detailed description. Therefore, statistical hypothesis testing to the maximum eigenvalue of a covariance matrix becomes possible by using these values and significance level α .

²If the asymptotic distribution is symmetric such as a normal distribution and t distribution, then, of course, it is not necessary to consider two types of hypothesis testing.

Table 1. Probability values ($\beta = 1, 2, 4$).

$\beta \setminus \alpha$	0.995	0.975	0.95	0.05	0.025	0.005
1	-4.1505	-3.5166	-3.1808	0.9793	1.4538	2.4224
2	-3.9139	-3.4428	-3.1945	-0.2325	0.0915	0.7462
4	-4.0531	-3.6608	-3.4556	-1.0904	-0.8405	-0.3400

2.4 Realized Quantities

In recent years, we can estimate and forecast volatilities more correctly by using Realized Volatility (RV) which is a consistent estimator of Integrated Volatility (IV). We can estimate RV by the sum of intraday log return of high frequency data in finance. Realized Covariance (RC) is also important as application to finance which can be estimated by the sum of cross-product of two log returns.

We define logarithmic stock price at time t as p_t and assume that p_t follows the following diffusion process:

$$dp_t = \mu_t dt + \sigma_t dw_t,$$

where μ_t , σ_t and w_t are instantaneous drift and diffusion terms and standard Brownian motion, respectively. If $\Delta \rightarrow 0$, then

$$RV_\zeta := \sum_{\tau} r_{\zeta,\tau}^2 \rightarrow \int_{\zeta-1}^{\zeta} \sigma_s^2 ds$$

where Δ is a small time interval in each day and $r_{\zeta,\tau}$ is τ th intraday logarithmic return $p_{\zeta,\tau\Delta} - p_{\zeta,(\tau-1)\Delta}$ in ζ day. If sampling interval is small enough, then RV is a consistent estimator of IV.

Provided that τ th logarithmic returns of two stocks i, j in ζ day are defined by $r_{\zeta,\tau,i}$ and $r_{\zeta,\tau,j}$ respectively,

$$CV_{\zeta,ij} := \sum_{\tau} r_{\zeta,\tau,i} r_{\zeta,\tau,j}.$$

Unifying RV and RC, we can obtain $N \times N$ matrix:

$$V_\zeta = R_\zeta^\top R_\zeta$$

where R_ζ is $p \times N$ log return matrix, N is the number of stocks, and p is the length of time series.

3. Empirical Analysis

We use high frequency data as follows in our empirical analysis. Data that we use consists of individual stocks listed in Nikkei 225 and TOPIX ($N = 226$),

Sampling period is from January 4, 2007 to December 28, 2007 (245 days). We calculate intraday log return against $\Delta = 1, \dots, 10$ (minute) in order to evaluate denoising performance for each method. Trading time in a day is 4.5 hours in Tokyo Stock Exchange, so if $\Delta = 1[\text{min}]$ then $p = 270, \alpha = 2.67$.

However, there is a problem resulting from using high frequency data in RV and RC. It is well known as microstructure noise which may be derived from asymmetric information and the bid-ask spread and brings some bias to volatility estimates obtained by RV. Figure 11 is an example from TOPIX 100³ of microstructure noise and this is so called volatility signature plot (VSP) whose horizontal axis denotes Δ and whose vertical axis denotes volatility. Each value indicates a 245 days average value of $\text{tr}(V), \text{tr}(V_{+(m)})$ calculated by $V, V_{+(m)}, (m = 1, 2, 3)$ in each day, where $\text{tr}(\cdot)$ is the sum of diagonal elements. Here $m = 1$ denotes volatility obtained from conventional method, $m = 2$ from Type I test and $m = 4$ from Type II test, respectively. In Figure 12, solid line is $\text{tr}(V)$ obtained from raw data, dashed line is $\text{tr}(V_{+(1)})$ dotted line is $\text{tr}(V_{+(2)})$ and chained line is $\text{tr}(V_{+(3)})$. From this figure, we can see that $\text{tr}(V)$ obviously diverge when sampling frequency is small but others are stable and almost identical to each other.

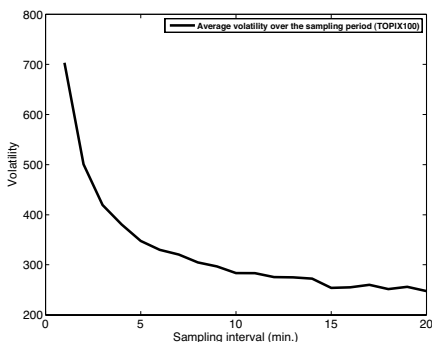


Figure 11. Microstructure noise.

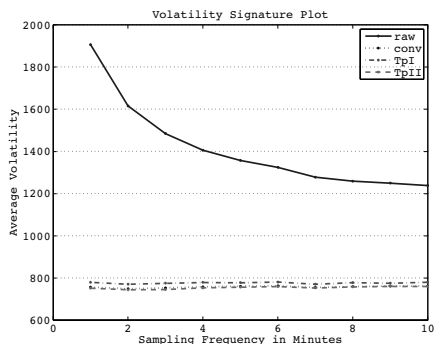


Figure 12. Average volatility.

Next we calculate minimum variance portfolio p_k without risk-free rate defined by

$$p_k = \frac{1}{Z} \sum_{i=1}^N C_{ij}^{-1}, \quad Z = \sum_{i,j=1}^N C_{ij}^{-1},$$

where $k = 1, \dots, N$ and C denotes $N \times N$ correlation matrix, see [4]. Furthermore,

³TOPIX 100 consists of 100 more liquid individual stocks from Tokyo Stock Exchange.

we compute the total variance σ_p^2 of the minimum variance portfolio given by

$$\sigma_p^2 = \mathbf{p} C \mathbf{p}^\top,$$

where $\mathbf{p}$ is a $1 \times N$ vector which contains $p_1, \dots, p_N$. If the correlation matrix C is not contaminated by noises, that is, C consists entirely of significant elements, then the total variance σ_p^2 of the minimum variance portfolio must be relatively small. Table 2 shows the estimated σ_p^2 s for each sampling interval and each method. From the table we can see that the variance of Type I is better particularly in smaller sampling intervals than 5 minutes where noise causes explosive volatility as we explained in Figure 11.

Table 2. Minimum variance portfolio without risk-free rate.

	raw	conv.	Type I	Type II
01 min.	3.5648	3.3221	3.3123	3.3214
02 min.	2.4060	2.2242	2.2130	2.2243
03 min.	2.4260	2.3802	2.3807	2.3910
04 min.	2.6374	2.3718	2.3553	2.3725
05 min.	1.5812	1.4415	1.4195	1.4431
06 min.	2.6137	2.4255	2.3724	2.4402
07 min.	2.1421	2.3312	2.3266	2.3300
08 min.	2.6004	2.2699	2.3505	2.2701
09 min.	1.5338	2.1933	2.1939	2.1935
10 min.	2.0842	1.7209	1.7280	1.7209

Furthermore, we investigate the efficient portfolio taking into account risk-free rate. We use interbank rate 0.0599 as of July 2007 for risk-free rate. Figure 13 and 14 show two remarkable examples of empirical efficient frontier. In these figures, circle denotes raw data, dotted line conventional method which means existing research, solid line Type I, and chained line Type II, respectively. Type I and Type II mean data denoised by hypothetical testing that we proposed in previous section. The right panel is efficient frontiers calculated by the data of March 15, 2007. It is interesting that dotted and chained lines which mean conventional method and Type II are placed nearer a vertical axis, which may underestimate the risk. The left panel is efficient frontiers calculated by the data of May 15, 2007. It is remarkable that circle which means raw data is situated inside others, which also may underestimate the risk. As you see from above result, efficient

frontiers differ from day to day however the outcome of Type I may seem to be stable⁴.

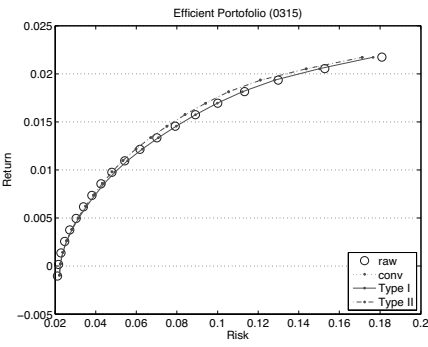


Figure 13. March 15, 2007.

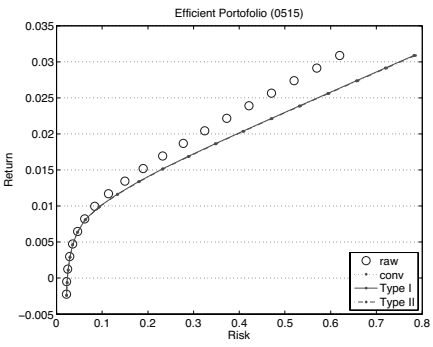


Figure 14. May 15, 2007.

Finally we present some results of empirically estimated volatility and co-volatility. Table 3 shows average values of volatility in 2007 for each sampling interval. The S.D. means standard deviation over all intervals. From the table we can see that mean volatility of Type I is relatively stable. Table 4 shows average

Table 3. Mean values of volatility.

	raw	conv.	Type I	Type II
01 min.	1.9067	0.7572	0.7803	0.7521
02 min.	1.6163	0.7495	0.7709	0.7444
03 min.	1.4847	0.7536	0.7753	0.7457
04 min.	1.4064	0.7589	0.7793	0.7540
05 min.	1.3577	0.7615	0.7780	0.7569
06 min.	1.3248	0.7630	0.7818	0.7597
07 min.	1.2785	0.7556	0.7711	0.7528
08 min.	1.2597	0.7597	0.7788	0.7582
09 min.	1.2501	0.7620	0.7757	0.7605
10 min.	1.2389	0.7625	0.7806	0.7603
S.D.	210.6982	4.3976	3.8342	5.7993

Note: S.D. is $\times 1/1000$ and others are $\times 1/100$.

⁴The efficient frontiers over the sampling period are available upon request.

values of absolute covolatility in 2007 for each sampling interval. However, we cannot find a remarkable difference in covolatility other than raw data.

Table 4. Mean values of absolute covolatility.

	raw	conv.	Type I	Type II
01 min.	4.3136	4.2221	4.2334	4.2177
02 min.	4.6013	4.5301	4.5399	4.5271
03 min.	4.7614	4.6913	4.6985	4.6877
04 min.	4.8437	4.7672	4.7773	4.7651
05 min.	4.8765	4.7950	4.8039	4.7929
06 min.	4.9548	4.8649	4.8757	4.8631
07 min.	4.8942	4.7978	4.8073	4.7947
08 min.	4.9551	4.8525	4.8664	4.8497
09 min.	4.9508	4.8429	4.8534	4.8418
10 min.	4.9850	4.8679	4.8820	4.8649
S.D.	2.1010	2.0388	2.0426	2.0457

Note: S.D. is $\times 1/1000$ and others are $\times 1/100$.

4. Concluding Remarks

We focused on denoising a covariance matrix of log-return by using the random matrix theory. Conventional researches have dichotomously distinguished noise and substantial parts in a convergence point of the maximum eigenvalue. Paying attention to the point that the maximum eigenvalue is also a random variable, we introduced an interval estimation of eigenvalues which can distinguish between noises. Here, we applied this technique to an empirical analysis of high frequency data in finance. Challenges for the future are introduction of time series structure and comparison of the forecasting ability of covolatility models.

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Quantile Hedging for Defaultable Claims

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We study the quantile hedging problem for defaultable claims in incomplete markets modeled by Itô processes, in the case where the portfolio processes are adapted to the full filtration. Using the convex duality method as in Cvitanić and Karatzas (Bernoulli, 7 (2001), 79–97) and a good structure of the class of the equivalent martingale measures, we derive a closed form solution for the problem.

Keywords: Quantile hedging, defaultable claims, convex duality, Neyman-Pearson lemma, jump processes.

1. Introduction

It is known that, in arbitrage-free, incomplete financial markets, the super-hedging cost of a contingent claim is often too high. More precisely, for any European call option in markets with transaction costs, the cheapest super-hedging is given by the buy-and-hold portfolio. This result is conjectured by Davis and Clark [11], and proved by, to name a few, Soner, Shreve and Cvitanić [24], Cvitanić, Pham and Touzi [9], Levental and Skorohod [16], and Jakubenas, Levental and Ryznar [14]. Similar results are obtained by Bellamy-Jeanblanc [2] in jump-diffusion models and by Cvitanić, Pham and Touzi [10] in stochastic volatility models.

In such a situation, it is reasonable that a hedger of a claim starts with an initial capital less than the super-hedging cost and accepts the possibility of the shortfall. One criterion for measuring this downside risk is the probability of super-hedging being successful. Optimizing this criterion is usually called quantile hedging, which is first studied by Kulldorff [15] in the context of gambling theory. Browne [5] considers the case of financial markets modeled by Itô processes with deterministic coefficients. Föllmer and Leukert [12] studies this problem for general

semimartingale financial market models. Spivak and Cvitanić [25] treats partial information market models and markets with different interest rates for borrowing and for lending. Sekine [22] analyzes the case of defaultable claims in the Brownian market models. Other criteria, such as the expected loss function or the risk measures, for the shortfall risk are also considered. See Cvitanić [6], Cvitanić and Karatzas [7], Föllmer and Leukert [13], Nakano [17, 18, 19], Pham [21], and Sekine [23], for examples.

In this paper, we consider the quantile hedging problem for defaultable claims in Brownian market models as in [23]. It investigates the case where the portfolios are adapted to the market information structure and gives closed form solutions by some reductions of the original problems to default-free ones. In our framework presented below, the portfolio processes are assumed to be adapted to the full filtration, i.e., the filtration generated by both the price and default indicator processes.

The quantile hedging problem is non-standard as a stochastic control problem and the usual dynamic programming approach cannot be applicable in a trivial way. Thus, in [12], they combine a super-hedging argument with a Neyman-Pearson lemma in the hypothesis testing to reduce the original dynamic problem to a static one. In a complete market framework, the reduced static problem is stated as the testing problem of a single null hypothesis versus a single alternative hypothesis, and so is directly solved by the classical Neyman-Pearson lemma (see [12]). However, this is not the case in our incomplete markets. To handle this issue, as in [6] and [18], we follow the convex duality approach for the generalized Neyman-Pearson lemma developed in [8].

This paper is organized as follows: In Section 2, we describe our market models. As a basic result, we give an explicit formula for the super-replication cost. Section 3 presents a solution to our quantile hedging problem for defaultable claims with zero recovery rate. In doing so, we explicitly solve the dual problem with the help of a good structure of the class of the equivalent martingale measures. Section 4 deals with the case of non-zero recovery rate.

2. Model

We consider the financial market with terminal time $T \in (0, \infty)$ consisting of one stock with price process $\{S_t\}_{0 \leq t \leq T}$ and one riskless bond with price process $\{B_t\}_{0 \leq t \leq T}$, whose dynamics are given respectively by

$$\begin{aligned} dS_t &= S_t \{b_t dt + \sigma_t dW_t\}, \quad 0 \leq t \leq T, \quad S_0 = s_0 \in (0, \infty), \\ dB_t &= r_t B_t dt, \quad 0 \leq t \leq T, \quad B_0 = 1. \end{aligned}$$

Here, $\{W_t\}_{t \geq 0}$ is a standard one-dimensional Brownian motion on a complete probability space $(\Omega, \mathcal{G}, \mathbb{P})$. The filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$ is generated by $\{W_t\}_{t \geq 0}$, augmented with $\mathbb{P}$ -null sets in $\mathcal{G}$. The processes $\{b_t\}$, $\{r_t\}$, $\{\sigma_t\}$ are all assumed to

be bounded $\mathbb{F}$ -predictable processes. Moreover we assume that $\sigma_t > 0$ for $t \in [0, T]$ a.s. and that $\{\sigma_t^{-1}\}$ is also bounded. Then, the process

$$\theta_t := \sigma_t^{-1}(b_t - r_t), \quad 0 \leq t \leq T$$

is a bounded $\mathbb{F}$ -predictable process.

Let τ be a nonnegative random variable satisfying $\mathbb{P}(\tau = 0) = 0$ and $\mathbb{P}(\tau > t) > 0$ for any $t \geq 0$, and let $\{N_t\}_{t \geq 0}$ be the counting process with respect to τ , i.e.,

$$N_t = 1_{\{\tau \leq t\}}, \quad t \geq 0.$$

Denote by $\mathbb{H} = \{\mathcal{H}_t\}_{t \geq 0}$ the filtration generated by $\{N_t\}$ and by $\mathbb{G} = \{\mathcal{G}_t\}_{t \geq 0}$ the filtration $\mathbb{F} \vee \mathbb{H}$. For simplicity we assume that $\mathcal{G} = \mathcal{G}_T$. The survival process $\{G_t\}_{t \geq 0}$ of τ with respect to $\mathbb{F}$ is then defined by

$$G_t = \mathbb{P}(\tau > t \mid \mathcal{F}_t), \quad 0 \leq t \leq T.$$

We assume that $G_t > 0$ for $t \geq 0$, and consider the hazard process $\{\Gamma_t\}_{t \geq 0}$ of τ with respect to $\mathbb{F}$ defined by $G_t = e^{-\Gamma_t}$ or $\Gamma_t = -\log G_t$ for every $t \geq 0$. We also assume that $\Gamma_t = \int_0^t \mu_s ds$, $t \geq 0$, for some nonnegative $\mathbb{F}$ -predictable process $\{\mu_t\}_{t \geq 0}$, so-called $\mathbb{F}$ -intensity of the random time τ . Then the process

$$M_t := N_t - \int_0^t \mu_s(1 - N_{s-})ds = N_t - \int_0^{t \wedge \tau} \mu_s ds, \quad t \geq 0,$$

follows a $\mathbb{G}$ -martingale (see Bielecki and Rutkowski [3]).

We now make the standing assumption that $\{W_t\}$ is a $(\mathbb{G}, \mathbb{P})$ -standard Brownian motion. Notice that if τ is independent of $\{W_t\}$ then this assumption is satisfied. Moreover, we can construct the random time τ such that $\{W_t\}$ is a $(\mathbb{G}, \mathbb{P})$ -standard Brownian motion (see, e.g., [3]).

As in the usual Brownian market models, we consider the $\mathbb{G}$ -martingale

$$Z_t^* = \exp\left(-\int_0^t \theta_s dW_s - \frac{1}{2} \int_0^t \theta_s^2 ds\right), \quad 0 \leq t \leq T.$$

Then by Girsanov's theorem, the process

$$W_t^* := W_t + \int_0^t \theta_s ds, \quad 0 \leq t \leq T$$

is a standard Brownian motion under the probability measure $\mathbb{P}^*$ defined by

$$\frac{d\mathbb{P}^*}{d\mathbb{P}} = Z_T^*.$$

In addition, we consider the process

$$Z_t^K = (1 + \kappa_\tau 1_{\{\tau \leq t\}}) \exp\left(-\int_0^{t \wedge \tau} \kappa_s \mu_s ds\right), \quad 0 \leq t \leq T,$$

where $\{\kappa_t\}_{0 \leq t \leq T}$ is taken from the class

$$\mathcal{D} = \{\{\kappa_t\}_{0 \leq t \leq T} : \text{bounded, } \mathbb{G}\text{-predictable, } \kappa_t > -1 \text{ dt} \times d\mathbb{P}\text{-a.e.}\}.$$

Then $\{Z_t^\kappa\}$, $\kappa \in \mathcal{D}$, satisfies

$$Z_t^\kappa = 1 + \int_0^t \kappa_s Z_{s-}^\kappa dM_s, \quad 0 \leq t \leq T,$$

and follows a $(\mathbb{P}, \mathbb{G})$ -martingale (see Brémaud [4] for example). Since the quadratic covariation process $[Z^*, Z^\kappa]$ is identically zero,

$$dZ_t^* Z_t^\kappa = Z_t^* Z_{t-}^\kappa (-\theta_t dW_t + \kappa_t dM_t). \quad (2.1)$$

Thus, $\{Z_t^* Z_t^\kappa\}$ is a $(\mathbb{P}, \mathbb{G})$ -positive martingale for $\kappa \in \mathcal{D}$. Each $\{Z_t^\kappa\}$ is orthogonal to $(\mathbb{P}, \mathbb{F})$ -martingales, so we can show that $\{W_t^*\}$ is also a Brownian motion under $\mathbb{Q}^\kappa$ defined by $d\mathbb{Q}^\kappa/d\mathbb{P} = Z_T^* Z_T^\kappa$. Hence $\{\mathbb{Q}^\kappa : \kappa \in \mathcal{D}\}$ defines the class of the equivalent martingale measures. We refer to [3] for details.

We consider $\mathbb{G}$ as the available information for the market participants. The portfolio process is thus defined as a $\mathbb{G}$ -predictable process $\{\pi_t\}_{0 \leq t \leq T}$ satisfying $\int_0^T |\pi_t|^2 dt < \infty$, a.s. The (self-financing) wealth process $\{X_t^{x, \pi}\}_{0 \leq t \leq T}$ for an initial wealth $x \geq 0$ and a portfolio process $\{\pi_t\}$ is then described by

$$dX_t^{x, \pi} = r_t X_t^{x, \pi} dt + \pi_t (b_t - r_t) dt + \pi_t \sigma_t dW_t, \quad X_0^{x, \pi} = x.$$

The solution to this equation is given by

$$X_t^{x, \pi} = B_t \left[x + \int_0^t B_u^{-1} \pi_u \{ (b_u - r_u) du + \sigma_u dW_u \} \right], \quad 0 \leq t \leq T.$$

We write $\mathcal{A}(x)$ for the set of all portfolio processes $\{\pi_t\}_{0 \leq t \leq T}$ such that $X_t^{x, \pi} \geq 0$, $0 \leq t \leq T$, a.s.

By Itô formula and (2.1), we get, for $\pi \in \mathcal{A}(x)$,

$$dL_t^\kappa X_t^{x, \pi} = L_{t-}^\kappa [(\pi_t \sigma_t - X_t^{x, \pi} \theta_t) dW_t + X_t^{x, \pi} \kappa_t dM_t],$$

where

$$L_t^\kappa = B_t^{-1} Z_t^* Z_t^\kappa, \quad \kappa \in \mathcal{D}.$$

This and the nonnegativity of the wealth process mean that the process $\{L_t^\kappa X_t^{x, \pi}\}$ is a supermartingale for each $\pi \in \mathcal{A}(x)$. We denote by $\mathcal{L}$ the set of all random variable L_T^κ , $\kappa \in \mathcal{D}$.

In this setting, we consider hedging problems for the defaultable claim H defined by

$$H = Y 1_{\{\tau > T\}} + \delta Y 1_{\{\tau \leq T\}}. \quad (2.2)$$

Here, Y is an $\mathcal{F}_T$ -measurable nonnegative random variable, which represents the payoff received by the holder at time T if the default does not occur in $[0, T]$. We assume that $\mathbb{E}^*[Y] < \infty$, where $\mathbb{E}^*$ stands for the expectation with respect to $\mathbb{P}^*$. The constant $\delta \in [0, 1]$ is the recovery rate of the payoff in case the default occurs in $[0, T]$.

The most conservative way of hedging the claims is the so-called superhedging, and its cost $\Pi(H)$ of H is defined by

$$\Pi(H) = \inf\{x \geq 0 : X_T^{x, \pi} \geq H \text{ a.s. for some } \pi \in \mathcal{A}(x)\}.$$

In our setting, this superhedging cost can be obtained explicitly.

Proposition 2.1. *Let H be as in (2.2) such that $\mathbb{E}^*[Y] < \infty$. Then we have*

$$\Pi(H) = \mathbb{E}^*[B_T^{-1}Y].$$

Moreover, the replicating portfolio for Y becomes a superhedging portfolio for H .

Proof. Set $\tilde{x} = \mathbb{E}^*[B_T^{-1}Y]$ and let $\tilde{\pi}$ be the replicating portfolio for Y . Then we find that $\tilde{\pi} \in \mathcal{A}(\tilde{x})$ and $X_T^{\tilde{x}, \tilde{\pi}} = Y \geq H$. Thus $\tilde{x} \geq \Pi(H)$.

On the other hand, suppose that $X_T^{x, \pi} \geq H$ for some $\pi \in \mathcal{A}(x)$. Then, from the supermartingale property of $\{L_t^\kappa X_t^{x, \pi}\}$,

$$\mathbb{E}[L_T^\kappa H] \leq \mathbb{E}[L_T^\kappa X_T^{x, \pi}] \leq x, \quad \kappa \in \mathcal{D}. \quad (2.3)$$

It follows from $H = \delta Y + (1 - \delta)Y1_{\{\tau > T\}}$ that the left-hand side in (2.3) can be written as

$$\mathbb{E}[L_T^\kappa H] = \mathbb{E}[B_T^{-1}Z_T^*Z_T^\kappa \delta Y] + \mathbb{E}[L_T^\kappa (1 - \delta)Y1_{\{\tau > T\}}]. \quad (2.4)$$

Since the quadratic covariation of $\{Z_t^\kappa\}$ and an $\mathbb{F}$ -martingale is equal to zero, the process $Z_t^\kappa \mathbb{E}[B_T^{-1}Z_T^* \delta Y | \mathcal{F}_t]$ is a local martingale. So, if Y is bounded then this process is a martingale. Therefore, by approximating Y with $Y \wedge n$ and by the monotone convergence theorem, we find that the first term in the right-hand side in (2.4) is given by $\mathbb{E}[B_T^{-1}Z_T^* \delta Y]$. From this and (2.3) we have

$$\mathbb{E}[B_T^{-1}Z_T^* \delta Y] + \sup_{\kappa \in \mathcal{D}} \mathbb{E}[L_T^\kappa (1 - \delta)Y1_{\{\tau > T\}}] \leq \Pi(H).$$

However, for any constant $\kappa > -1$,

$$\begin{aligned} \mathbb{E}[L_T^\kappa Y1_{\{\tau > T\}}] &= \mathbb{E}[B_T^{-1}Z_T^*Y(1 + \kappa 1_{\{\tau \leq T\}})e^{-\kappa \int_0^{\tau \wedge T} \mu_t dt} 1_{\{\tau > T\}}] \\ &= \mathbb{E}[B_T^{-1}Z_T^*Y1_{\{\tau > T\}}e^{-\kappa \int_0^T \mu_t dt}] = \mathbb{E}[B_T^{-1}Z_T^*Y G_T e^{-\kappa \int_0^T \mu_t dt}] \\ &= \mathbb{E}[B_T^{-1}Z_T^*Y e^{-(\kappa+1) \int_0^T \mu_t dt}]. \end{aligned}$$

Hence $\mathbb{E}[L_T^\kappa Y1_{\{\tau > T\}}] \rightarrow \mathbb{E}[B_T^{-1}Z_T^*Y]$ as $\kappa \searrow -1$. Thus the proposition follows. $\square$

3. Quantile Hedging Problem

Proposition 2.1 implies that if the hedger of the defaultable claim wants to hedge the claim almost surely then s/he needs to have the perfect hedging cost for the liability to be paid when the default does not occur. However, the price of H should reflect the possibility of default and be smaller than $\mathbb{E}^*[B_T^{-1}Y]$ since one can receive Y almost surely with this cost buying in the default-free market. In other words, an initial wealth for hedging of H may be smaller than $\mathbb{E}^*[B_T^{-1}Y]$. In such a case there is the possibility of the shortfall in hedging of H . One criterion for measuring this downside risk is the probability of super-hedging being successful. Our objective is thus to solve the following problem: for $x < \mathbb{E}^*[B_T^{-1}Y]$,

$$\max_{\pi \in \mathcal{A}(x)} \mathbb{P}(X_T^{x,\pi} \geq H). \quad (3.1)$$

Adapting an optimal portfolio for this problem as a hedging strategy for H is usually called a quantile hedging. To solve the quantile hedging problem (3.1), as in [18], we first reduce the original dynamic problem to a Neyman-Pearson type problem via a super-hedging argument, then adapt the convex duality approach to solve the Neyman-Pearson type problem.

To this end, we first introduce the class $\overline{\mathcal{L}}$ defined by the closed hull of $\mathcal{L}$ with respect to $L^1 := L^1(\Omega, \mathcal{G}, \mathbb{P})$ convergence. Since $\mathcal{L}$ is convex (see, e.g., [4]), so is $\overline{\mathcal{L}}$. Thus, $\overline{\mathcal{L}}$ is a closed convex set in L^1 .

Let us consider the Neyman-Pearson type problem defined by

$$\max_{\varphi \in \mathcal{R}} \mathbb{E}[\varphi] \quad (3.2)$$

where

$$\mathcal{R} = \{\varphi : 0 \leq \varphi \leq 1 \text{ a.s., } \sup_{L \in \overline{\mathcal{L}}} \mathbb{E}[LH\varphi] \leq x\}.$$

As in [12], our problem is reduced to the Neyman-Pearson type problem via the following proposition.

Proposition 3.1. *Suppose that there exist $A \in \mathcal{G}_T$ and $\{\hat{\pi}_t\} \in \mathcal{A}(x)$ such that 1_A solves the Neyman-Pearson type problem (3.2) and $X_T^{x,\hat{\pi}} \geq H1_A$ a.s. Then $\hat{\pi}$ is optimal for the quantile hedging problem (3.1).*

Proof. For $\pi \in \mathcal{A}(x)$,

$$\mathbb{E} \left[L_T^K 1_{\{X_T^{x,\pi} \geq H\}} H \right] \leq \mathbb{E} \left[L_T^K 1_{\{X_T^{x,\pi} \geq H\}} X_T^{x,\pi} \right] \leq \mathbb{E} [X_T^{x,\pi} L_T^K] \leq x, \quad \kappa \in \mathcal{D}.$$

Let $L \in \overline{\mathcal{L}}$. Then there exist $L_n \in \mathcal{L}$, $n = 1, 2, \dots$, such that $L = \lim_{n \rightarrow \infty} L_n$ a.s. (possibly along with a subsequence). Thus it follows from Fatou's lemma that $1_{\{X_T^{x,\pi} \geq H\}} \in \mathcal{R}$. Hence

$$\max_{\pi \in \mathcal{A}(x)} \mathbb{P}(X_T^{x,\pi} \geq H) \leq \max_{\varphi \in \mathcal{R}} \mathbb{E}[\varphi]. \quad (3.3)$$

On the other hand, denoting $\hat{X} = X_T^{x, \hat{\kappa}}$, we see

$$\mathbb{P}(\hat{X} \geq H) \geq \mathbb{P}(\hat{X} \geq H, A) = \mathbb{P}(\hat{X} \geq H 1_A, A) = \mathbb{P}(A) = \max_{\varphi \in \mathcal{R}} \mathbb{E}[\varphi].$$

Combining this with (3.3), we have the proposition. $\square$

We adapt the convex duality approach in [8] and [18]. Observe that for $\varphi \in \mathcal{R}$, $y \geq 0$, $L \in \overline{\mathcal{L}}$,

$$\mathbb{E}[\varphi] = E[\varphi(1 - yLH)] + y\mathbb{E}[LH\varphi] \leq \mathbb{E}[(1 - yLH)_+] + yx. \quad (3.4)$$

Thus the following dual problem naturally arises:

$$V(x) := \inf_{y \geq 0, L \in \overline{\mathcal{L}}} \{ \mathbb{E}[(1 - yLH)_+] + yx \}. \quad (3.5)$$

In what follows we will see that the minimization above can be completely solved in the case of zero recovery rate, i.e., in the case that H is of the following form:

$$H = Y 1_{\{\tau > T\}}. \quad (3.6)$$

Define

$$\hat{L} = B_T^{-1} Z_T^* 1_{\{\tau > T\}} e^{\int_0^T \mu_s dt}. \quad (3.7)$$

Then we have the following:

Theorem 3.2. *Suppose that H is as in (3.6) with $\mathbb{E}^*[Y] < \infty$. Then $\hat{L}$ defined by (3.7) solves*

$$\inf_{L \in \overline{\mathcal{L}}} \mathbb{E}[(1 - yLH)_+].$$

Moreover, there exists $\hat{y} > 0$ that minimizes

$$h(y) := \mathbb{E}[(1 - y\hat{L}H)_+] + yx$$

over $y \geq 0$. The pair $(\hat{y}, \hat{L})$ is optimal for the minimization problem (3.5).

Proof. First notice that $\hat{L} \in \overline{\mathcal{L}}$ since $\hat{L} = \lim_{\kappa \searrow -1} L_T^\kappa$ a.s. and in L^1 .
For $\kappa \in \mathcal{D}$,

$$\begin{aligned} \mathbb{E}[1 \wedge (yL_T^\kappa H)] &= \mathbb{E}[1 \wedge (yB_T^{-1} Z_T^* e^{-\int_0^T \kappa_s \mu_s ds} Y) 1_{\{\tau > T\}}] \\ &\leq \mathbb{E}[1 \wedge (yB_T^{-1} Z_T^* e^{\int_0^T \mu_s ds} Y) 1_{\{\tau > T\}}] \\ &= \mathbb{E}[1 \wedge (y\hat{L}H)]. \end{aligned}$$

Thus, by Fatou's lemma and $(1 - z)_+ = 1 - 1 \wedge z$, we find

$$\mathbb{E}[(1 - yLH)_+] \geq \mathbb{E}[(1 - y\hat{L}H)_+], \quad L \in \overline{\mathcal{L}}.$$

Next, we claim that there exists $y_0 > 0$ such that $h(y_0) < 1$. Suppose otherwise. Then we find that $\mathbb{E}[1 \wedge (y\hat{L}H)] \leq yx$ for every $y > 0$. Dividing by y and then letting $y \searrow 0$, we obtain $\mathbb{E}[\hat{L}H] \leq x$. However, this contradicts to the assumption $x < \mathbb{E}^*[B_T^{-1}Y]$ since

$$\mathbb{E}[\hat{L}H] = \mathbb{E}[B_T^{-1}Z_T^* e^{\int_0^T \mu_t dt} Y \mathbb{P}(\tau > T | \mathcal{F}_T)] = \mathbb{E}^*[B_T^{-1}Y].$$

The existence of the minimizer $\hat{y} > 0$ now follows from the convexity of h and the facts that $h(0) = 1$ and $h(+\infty) = +\infty$.

The optimality of the pair $(\hat{y}, \hat{L})$ for (3.5) is easy to see, so omitted. $\square$

Let $\hat{y} > 0$ be as in the previous theorem and consider the $\mathcal{F}_T$ -measurable random variable ξ defined by

$$\xi = \hat{y}B_T^{-1}Z_T^* e^{\int_0^T \mu_t dt} Y.$$

Then we can describe an optimal quantile hedging portfolio in the following way:

Theorem 3.3. *Suppose that H is as in (3.6) with $\mathbb{E}^*[Y] < \infty$ and that $\mathbb{P}(\xi = 1) = 0$. Then the perfect hedging portfolio for $Y1_{\{\xi < 1\}}$ is optimal for the quantile hedging problem (3.1).*

Proof. We follow essentially the same arguments as in [8] and [18].

The set

$$\mathcal{M} := \{(L, y) \in L^1 \times \mathbb{R} : L \in \overline{\mathcal{L}}, y \geq 0\}$$

is closed and convex in the Banach space $L^1 \times \mathbb{R}$ with the norm $\|(L, y)\| := \mathbb{E}[L] + y$. Moreover, it is straightforward to see that the functional

$$L^1 \times \mathbb{R} \ni (L, y) \mapsto U(L, y) := yx + \mathbb{E}(1 - L)_+$$

is proper, convex and lower semi-continuous on $L^1 \times \mathbb{R}$, and that

$$\inf_{(L, y) \in \mathcal{M}} U(yLH, y) = U(\hat{y}\hat{L}H, \hat{y}).$$

Denote $L^\infty(\Omega, \mathcal{G}, \mathbb{P})$ by L^∞ . Let us consider the set $\mathcal{M}^* = \{(yLH, y) : (L, y) \in \mathcal{M}\}$, the normal cone

$$\mathcal{N}(\hat{y}\hat{L}H, \hat{y}) := \{(\varphi, u) \in L^\infty \times \mathbb{R} : \mathbb{E}[\hat{y}\hat{L}H\varphi] + \hat{y}u \geq \mathbb{E}[yLH\varphi] + yu, (L, y) \in \mathcal{M}\}$$

to the set $\mathcal{M}$ at $(\hat{y}\hat{L}H, \hat{y})$, and the subdifferential

$$\begin{aligned} \partial U(\hat{y}\hat{L}H, \hat{y}) &:= \left\{ (\varphi, u) \in L^\infty \times \mathbb{R} : U(\hat{y}\hat{L}H, \hat{y}) - U(L, y) \right. \\ &\quad \left. \leq \mathbb{E}[\varphi(\hat{y}\hat{L}H - L)] + u(\hat{y} - y), \quad (L, y) \in L^1 \times \mathbb{R} \right\} \end{aligned}$$

at this point. Then, by Corollary 4.6.3 in Aubin and Ekeland [1],

$$(0, 0) \in \partial U(\hat{y}\hat{L}H, \hat{y}) + \mathcal{N}(\hat{y}\hat{L}H, \hat{y}).$$

This implies that there exists $(\hat{\phi}, \hat{u}) \in L^\infty \times \mathbb{R}$ such that $(\hat{\phi}, \hat{u}) \in \mathcal{N}(\hat{y}\hat{L}H, \hat{y})$ and $(-\hat{\phi}, -\hat{u}) \in \partial U(\hat{y}\hat{L}H, \hat{y})$. Hence we obtain

$$\mathbb{E}[H\hat{\phi}(\hat{y}\hat{L} - yL)] + (\hat{y} - y)\hat{u} \geq 0, \quad (L, y) \in \mathcal{M}, \quad (3.8)$$

$$(x + \hat{u})(\hat{y} - y) \leq \mathbb{E}[\hat{\phi}(L - \hat{y}\hat{L}H)] + \mathbb{E}(1 - L)_+ - \mathbb{E}(1 - \hat{y}\hat{L}H)_+, \quad (L, y) \in L^1 \times \mathbb{R}. \quad (3.9)$$

By letting $y \rightarrow \pm\infty$, we see that (3.9) holds only if $\hat{u} = -x$. From (3.8) with $\hat{u} = -x, y = \hat{y} \pm \delta$ ($\delta > 0$), and $L = \hat{L}$, we have

$$\mathbb{E}[H\hat{\phi}\hat{L}] = x. \quad (3.10)$$

Thus, reading (3.8) with $y = \hat{y}$, we get

$$\mathbb{E}[H\hat{\phi}L] \leq x, \quad L \in \overline{\mathcal{L}}. \quad (3.11)$$

Eq. (3.9) is now written as

$$\mathbb{E}[\hat{\phi}(L - \hat{y}\hat{L}H)] + \mathbb{E}(1 - L)_+ - \mathbb{E}(1 - \hat{y}\hat{L}H)_+ \geq 0, \quad L \in L^1. \quad (3.12)$$

Considering (3.12) for $L = \hat{y}\hat{L}H + 1_A$ with arbitrary $A \in \mathcal{G}$, we see that $0 \leq \mathbb{E}[\hat{\phi}1_A]$. Thus $\hat{\phi} \geq 0$ a.s. Similarly, considering (3.12) for $L = \hat{y}\hat{L}H - 1_A$ with arbitrary $A \in \mathcal{G}$ and using $(x + y)_+ \leq (x)_+ + (y)_+$ for $x, y \in \mathbb{R}$, we see that $0 \leq \mathbb{E}(1 - \hat{\phi})1_A$. Thus $\hat{\phi} \leq 1$ a.s. Combining with (3.11), we have $\hat{\phi} \in \mathcal{R}$.

Eq. (3.12) for $L = 1$ implies $\mathbb{E}[\hat{\phi}(1 - \hat{y}\hat{L}H)] \geq \mathbb{E}(1 - \hat{y}\hat{L}H)_+$. Thus $\hat{\phi}(1 - \hat{y}\hat{L}H) = (1 - \hat{y}\hat{L}H)_+$ a.s. From this and $\hat{\phi} \in \mathcal{R}$ we find that $\hat{\phi} = 1$ on $\{\hat{y}\hat{L}H < 1\}$ and $\hat{\phi} = 0$ on $\{\hat{y}\hat{L}H > 1\}$. Hence there must be some $[0, 1]$ -valued random variable C such that the representation

$$\hat{\phi} = 1_{\{\hat{y}\hat{L}H < 1\}} + C1_{\{\hat{y}\hat{L}H = 1\}} \quad (3.13)$$

holds. Moreover, we have $\mathbb{E}[\hat{\phi}] = \mathbb{E}[(1 - \hat{y}\hat{L}H)_+] + \hat{y}x$. This and (3.4) imply that $\hat{\phi}$ is optimal for the Neyman-Pearson type problem and that there is no duality gap. By the assumption of the theorem, $\mathbb{P}(\hat{y}\hat{L}H = 1) = \mathbb{P}(\xi = 1, \tau > T) = 0$. Moreover,

since $H\hat{\phi} = Y1_{\{\xi < 1\}}1_{\{\tau > T\}}$, we can apply Proposition 3.1 to deduce that a super-hedging portfolio for $H\hat{\phi}$ is given by the perfect hedging portfolio for $Y1_{\{\xi < 1\}}$. From this and Proposition 2.1, we arrive at the conclusion of the theorem. $\square$

Remark 3.4. The condition $\mathbb{P}(\xi = 1) = 0$ is satisfied, e.g., in the case where the following are all satisfied: τ is independent to $\{W_t\}$, the market model is described by Black-Scholes one, and Y is given by a plain vanilla option. In more general Markovian cases, Hörmander's condition in Malliavin calculus (see, e.g., Nualart [20]) can be used to check that ξ is continuously distributed.

Remark 3.5. Since the perfect hedging strategy for $Y1_{\{\xi < 1\}}$ is an $\mathbb{F}$ -predictable process, Theorem 3.3 gives an explicit solution to the quantile hedging problems with respect to $\mathbb{F}$ -predictable portfolios studied in [22] in the case of zero-recovery rate.

4. Case of Non-Zero Recovery Rate

Next we consider the case of $\delta > 0$. In this case, solving the dual problem is more difficult, and we leave it for a future study. Instead, we present a solution to our quantile hedging problem in a restricted class of portfolio processes.

Notice that the payoff of the defaultable claim satisfies

$$H = \delta Y + (1 - \delta)Y1_{\{\tau > T\}} \geq \delta Y.$$

This implies that a seller of the claim must pay at least δY at the maturity. Since δY is $\mathcal{F}_T$ -measurable, there exists a unique portfolio process $\{\pi_t^*\}_{0 \leq t \leq T}$ such that

$$X_t^{x^*, \pi^*} = \mathbb{E}^*[B_T^{-1}B_t\delta Y | \mathcal{F}_t],$$

where $x^* = \mathbb{E}^*[B_T^{-1}\delta Y]$. In view of these considerations, we impose the following capital requirements on the wealth process:

$$X_t^{x, \pi} \geq \mathbb{E}^*[B_T^{-1}B_t\delta Y | \mathcal{F}_t], \quad 0 \leq t \leq T, \text{ a.s.} \quad (4.1)$$

In particular, x must be at least x^* . We denote by $\mathcal{A}^*(x)$ all portfolio processes such that the corresponding wealth process with initial wealth x satisfies (4.1), and restrict ourselves the class of portfolio processes to $\mathcal{A}^*(x)$. Then let us consider the following quantile hedging problem

$$\max_{\pi \in \mathcal{A}^*(x)} \mathbb{P}(X_T^{x, \pi} \geq H). \quad (4.2)$$

It follows from $H = \delta Y + (1 - \delta)Y1_{\{\tau > T\}}$ that $\mathbb{P}(X_T^{x, \pi} \geq H) = \mathbb{P}(X_T^{x - x^*, \pi - \pi^*} \geq (1 - \delta)Y1_{\{\tau > T\}})$. Thus the problem (4.2) is reduced to the maximization problem

of $\mathbb{P}(X_T^{x', \pi'} \geq H')$ over all portfolio processes $\pi' \in \mathcal{A}(x')$. Here, $x' = x - x^* \geq 0$ and $H' = (1 - \delta)Y1_{\{\tau > T\}}$. Thus,

$$\max_{\pi \in \mathcal{A}^*(x)} \mathbb{P}(X_T^{x, \pi} \geq H) = \max_{\pi' \in \mathcal{A}(x')} \mathbb{P}(X_T^{x', \pi'} \geq H').$$

Therefore we can apply Theorem 3.3 to the problem (4.2) and obtain the following:

Theorem 4.1. *Suppose that $\mathbb{E}^*[B_T^{-1}Y] < \infty$, and let y' and ξ' be defined by*

$$y' = \arg \min_{y \geq 0} \{ \mathbb{E}(1 - y\hat{L}H')_+ + yx' \},$$

$$\xi' = y'B_T^{-1}Z_T^*e^{\int_0^T \mu_t dt}(1 - \delta)Y.$$

Suppose moreover that $\mathbb{P}(\xi' = 1) = 0$. Then the perfect hedging portfolio for $\delta Y + (1 - \delta)Y1_{\{\xi < 1\}}$ is optimal for the quantile hedging problem (4.2).

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New Unified Computational Algorithm in a High-Order Asymptotic Expansion Scheme*

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An asymptotic expansion scheme in finance initiated by Kunitomo and Takahashi [6] and Yoshida [29] is a widely applicable methodology for analytic approximation of the expectation of a certain functional of diffusion processes. Mathematically, this methodology is justified by Watanabe's theory ([27]) in Malliavin calculus. In practical applications, it is desirable to investigate the accuracy and stability of the method especially with expansion up to high orders in situations where the underlying processes are highly volatile as seen in the recent financial markets. Although Takahashi [17], [18] and Takahashi and Takehara [20] provided explicit formulas for the expansion up to the third order, to our best knowledge a general computation scheme for an arbitrary-order expansion has not been given yet. This paper proposes two general methods for computing the conditional expectations that are powerful especially for high order expansions. The first one, as an extension of the method introduced by the preceding papers, presents a unified scheme for computation of the conditional expectations. The second one develops a new calculation algorithm for computing the coefficients of the expansion through solving a system of ordinary differential equations that is equivalent to computing the conditional expectations. To demonstrate their effectiveness, the paper gives numerical examples of the approximation for λ -SABR model up to the fifth order and a cross-currency Libor market model with a general stochastic volatility model of the spot foreign exchange rate up to the fourth order.

Keywords: Asymptotic expansion, Malliavin calculus, approximation formula, stochastic volatility, λ -SABR model, Libor market model, currency options.

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1. Introduction

This paper presents two alternative schemes for computation in the method so-called “an asymptotic expansion approach” based on Watanabe’s theory (Watanabe [27]) in Malliavin calculus by extending the preceding papers and also by developing a new calculation algorithm.

To our best knowledge, the asymptotic expansion is first applied to finance for evaluation of an average option that is a popular derivative in commodity markets. [6] and [17] derive the approximation formulas for an average option by an asymptotic method based on log-normal approximations of an average price distribution when the underlying asset price follows a geometric Brownian motion. [29] applies a formula derived by the asymptotic expansion of certain statistical estimators for small diffusion processes. Thereafter, the asymptotic expansion have been applied to a broad class of problems in finance. See [18], [19], Kunitomo and Takahashi [7], [8], Matsuoka, Takahashi and Uchida [12], Takahashi and Yoshida [25], [26], Muroi [13], and Takahashi and Takehara [20], [21], [22]. For other asymptotic methods in finance which do not depend on Watanabe’s theory, see also Fouque, Papanicolaou and Sircar [3], [4], Henry-Labordere [10], [11], Kusuoka and Osajima [9], and Siopacha and Teichmann [16].

Recently, not only academic researchers but also many practitioners such as Antonov and Misirpashaev [1] or Andersen and Hutchings [2] have used the asymptotic expansion method based on Watanabe’s theory in their proposed techniques for a variety of financial issues, e.g. pricing or hedging complex derivatives under high-dimensional underlying stochastic environments. These methods are fully or partially based on the framework developed by [6], [17], [18] in financial literature.

In theory, this method provides us the expansion of underlying stochastic processes which has a proper meaning in the limit of some ideal situations such as cases where they become deterministic ones (for details see [27], [28] or [8]).

In practice, however, we are often interested in cases far from that situation, where the underlying processes are highly volatile, as seen in the recent financial markets especially after the crisis of 2008. Then from the viewpoint of the accuracy or stability of the techniques in practical uses, it is desirable to investigate behaviors of its estimators in such situations, especially with expansion up to high orders.

In the existing application of the asymptotic expansion based on Watanabe’s theory, they calculated certain conditional expectations which appear in their expansions and play a key role in computation, by the formulas up to the third order given explicitly in [17], [18] and [20]. In many applications, these formulas give sufficiently accurate approximation, but in some cases, for example in the cases with long maturities and/or with highly volatile underlying variables, the approximation up to the third order may not provide satisfactory accuracies. Thus, the formulas for the higher order computation are desirable. But to our knowledge,

asymptotic expansion formulas higher than the third order have not been given yet.

This paper provides the general procedures for the explicit computation of conditional expectations in the asymptotic expansion. Moreover, we develop an alternative but equivalent calculation algorithm which computes the unconditional expectations directly instead of the conditional ones and enables us to derive high order approximation formulas in an automatic manner. While these techniques can be applied to a broad class of Itô processes, for simplicity and limitation of space, in this paper we concentrate on a much simpler setting as described in Section 2. For further explanations in more general environment, see our online working paper, Takahashi, Takehara and Toda [23].

Finally, our approximation generally shows sufficient accuracy with computation of high order expansions, which is confirmed by numerical experiments in more complex cases than that in Section 2.

The organization of this paper is as follows: After Section 2 will develop our methods in the simple setting, Section 3 applies our algorithms described in the previous section to the concrete financial models, and confirms the effectiveness of the higher order expansions by numerical examples in λ -SABR model and a cross-currency Libor market model with a general stochastic volatility model of the spot foreign exchange rate. Due to limitations of space, detailed proofs, the concrete expressions of some formulas and equations are omitted in this paper, which are given in [23]. We will refer to it if necessary.

2. An Asymptotic Expansion Approach in a Black-Scholes Economy

In this section, our essential idea is explained in a simple Black-Scholes-type economy. For discussions in more general settings, refer to Section 3 and 4 of [23].

2.1 An Asymptotic Expansion Approach in a Black-Scholes Economy

Let (W, P) be a one-dimensional Wiener space. Hereafter P is considered as a risk-neutral equivalent martingale measure and a risk-free interest rate is set to be zero for simplicity. Then, the underlying economy is specified with a $(\mathbb{R}_+$ -valued) single risky asset $S^{(\epsilon)} = \{S_t^{(\epsilon)}\}$ satisfying

$$S_t^{(\epsilon)} = S_0 + \epsilon \int_0^t \sigma(S_s^{(\epsilon)}, s) dW_s \quad (1)$$

where $\epsilon \in (0, 1]$ is a constant parameter; $\sigma: \mathbb{R}_+^2 \mapsto \mathbb{R}$ satisfies some regularity conditions. We will consider the following pricing problem;

$$V(0, T) = \mathbf{E}[\Phi(S_T^{(\epsilon)})] \quad (2)$$

where Φ is a payoff function written on $S_T^{(\epsilon)}$ (for example, $\Phi(x) = \max(x - K, 0)$ for call options or $\Phi(x) = \delta_x(x)$, a delta function with mass at x for the density function) and $\mathbf{E}[\cdot]$ is an expectation operator under the probability measure P . Rigorously speaking, they are a generalized function on the Wiener functional $S^{(\epsilon)}$ and a generalized expectation defined for generalized functions respectively, whose mathematically proper definitions will be given in Section 2 of [23].

Let $A_{kt} = \frac{\partial^k S_t^{(\epsilon)}}{\partial \epsilon^k} |_{\epsilon=0}$. Here we represent A_{1t} , A_{2t} and A_{3t} explicitly by

$$A_{1t} = \int_0^t \sigma(S_s^{(0)}, s) dW_s, \quad (3)$$

$$A_{2t} = 2 \int_0^t \partial \sigma(S_s^{(0)}, s) A_{1s} dW_s, \quad (4)$$

$$A_{3t} = 3 \int_0^t \left(\partial^2 \sigma(S_s^{(0)}, s) (A_{1s})^2 + \partial \sigma(S_s^{(0)}, s) (A_{2s}) \right) dW_s \quad (5)$$

recursively and then $S_T^{(\epsilon)}$ has its asymptotic expansion

$$S_T^{(\epsilon)} = S_0 + \epsilon A_{1T} + \frac{\epsilon^2}{2!} A_{2T} + \frac{\epsilon^3}{3!} A_{3T} + o(\epsilon^3). \quad (6)$$

Note that $S_t^{(0)} = \lim_{\epsilon \downarrow 0} S_t^{(\epsilon)} = S_0$ for all t .

Next, normalize $S_T^{(\epsilon)}$ with respect to ϵ as

$$G^{(\epsilon)} = \frac{S_T^{(\epsilon)} - S_T^{(0)}}{\epsilon}$$

for $\epsilon \in (0, 1]$. Then,

$$G^{(\epsilon)} = A_{1T} + \frac{\epsilon}{2!} A_{2T} + \frac{\epsilon^2}{3!} A_{3T} + o(\epsilon^2) \quad (7)$$

in L^p for every $p > 1$. Here the following assumption is made:

$$\Sigma_T = \int_0^T \sigma^2(S_t^{(0)}, t) dt > 0. \quad (8)$$

Note that A_{1T} follows a normal distribution with mean 0 and variance Σ_T , and hence this assumption means that the distribution of A_{1T} does not degenerate. It is clear that this assumption is satisfied when $\sigma(S_t^{(0)}, t) > 0$ for some $t > 0$.

Then, the expectation of $\Phi(G^{(\epsilon)})$ is expanded around $\epsilon = 0$ up to ϵ^2 -order in the sense of Watanabe ([27], Yoshida [28]) as follows (hereafter the asymptotic

expansion of $\mathbf{E}[\Phi(G^{(\epsilon)})]$ up to the second order will be considered):

$$\begin{aligned}
\mathbf{E}[\Phi(G^{(\epsilon)})] &= \mathbf{E}[\Phi(A_{1T})] + \epsilon \mathbf{E}[\Phi^{(1)}(A_{1T})A_{2T}] \\
&\quad + \epsilon^2 \left\{ \mathbf{E}[\Phi^{(1)}(A_{1T})A_{3T}] + \frac{1}{2} \mathbf{E}[\Phi^{(2)}(A_{1T})(A_{2T})^2] \right\} + o(\epsilon^2) \\
&= \mathbf{E}[\Phi(A_{1T})] + \epsilon \mathbf{E}[\Phi^{(1)}(A_{1T})\mathbf{E}[A_{2T}|A_{1T}]] \\
&\quad + \epsilon^2 \left\{ \mathbf{E}[\Phi^{(1)}(A_{1T})\mathbf{E}[A_{3T}|A_{1T}]] + \frac{1}{2} \mathbf{E}[\Phi^{(2)}(A_{1T})\mathbf{E}[(A_{2T})^2|A_{1T}]] \right\} + o(\epsilon^2) \\
&= \int_{\mathbb{R}} \Phi(x) f_{A_{1T}}(x) dx + \epsilon \int_{\mathbb{R}} \Phi^{(1)}(x) \mathbf{E}[A_{2T}|A_{1T} = x] f_{A_{1T}}(x) dx \\
&\quad + \epsilon^2 \left\{ \int_{\mathbb{R}} \Phi^{(1)}(x) \mathbf{E}[A_{3T}|A_{1T} = x] f_{A_{1T}}(x) dx \right. \\
&\quad \left. + \frac{1}{2} \int_{\mathbb{R}} \Phi^{(2)}(x) \mathbf{E}[(A_{2T})^2|A_{1T} = x] f_{A_{1T}}(x) dx \right\} + o(\epsilon^2) \\
&= \int_{\mathbb{R}} \Phi(x) f_{A_{1T}}(x) dx + \epsilon \int_{\mathbb{R}} \Phi(x) (-1) \frac{\partial}{\partial x} \{\mathbf{E}[A_{2T}|A_{1T} = x] f_{A_{1T}}(x)\} dx \\
&\quad + \epsilon^2 \left(\int_{\mathbb{R}} \Phi(x) (-1) \frac{\partial}{\partial x} \{\mathbf{E}[A_{3T}|A_{1T} = x] f_{A_{1T}}(x)\} dx \right. \\
&\quad \left. + \frac{1}{2} \int_{\mathbb{R}} \Phi(x) (-1)^2 \frac{\partial^2}{\partial x^2} \{\mathbf{E}[(A_{2T})^2|A_{1T} = x] f_{A_{1T}}(x)\} dx \right) + o(\epsilon^2). \quad (9)
\end{aligned}$$

where $\Phi^{(m)}(x)$ is m -th order derivative of $\Phi(x)$ and $f_{A_{1T}}(x)$ is a probability density function of A_{1T} following a normal distribution;

$$f_{A_{1T}}(x) := \frac{1}{\sqrt{2\pi\Sigma_T}} \exp\left(-\frac{x^2}{2\Sigma_T}\right). \quad (10)$$

In particular, letting $\Phi = \delta_x$, we have the asymptotic expansion of the density function of $G^{(\epsilon)}$ as seen later.

Then, all we have to do to evaluate this expansion is a computation of these conditional expectations. In particular, we present two alternative approaches.

2.2 An Approach with an Expansion into Iterated Itô Integrals

In this subsection we show an approach with a further expansion of A_{2T} , A_{3T} and $(A_{2T})^2$ into iterated Itô integrals to compute the conditional expectations in (9).

Recall that we have

$$\begin{aligned} \mathbf{E}[\Phi(G^{(\epsilon)})] &= \int_{\mathbb{R}} \Phi(x) f_{A_{1T}}(x) dx + \epsilon \int_{\mathbb{R}} \Phi(x) (-1) \frac{\partial}{\partial x} \{\mathbf{E}[A_{2T} | A_{1T} = x] f_{A_{1T}}(x)\} dx \\ &\quad + \epsilon^2 \left(\int_{\mathbb{R}} \Phi(x) (-1) \frac{\partial}{\partial x} \{\mathbf{E}[A_{3T} | A_{1T} = x] f_{A_{1T}}(x)\} dx \right. \\ &\quad \left. + \frac{1}{2} \int_{\mathbb{R}} \Phi(x) (-1)^2 \frac{\partial^2}{\partial x^2} \{\mathbf{E}[(A_{2T})^2 | A_{1T} = x] f_{A_{1T}}(x)\} dx \right) + o(\epsilon^2). \end{aligned} \quad (11)$$

Next, it is shown that A_{2T} , A_{3T} , $(A_{2T})^2$ can be expressed as summations of iterated Itô integrals. First, note that A_{2T} is

$$A_{2T} = 2 \int_0^T \int_0^{t_1} \partial \sigma(S_{t_1}^{(0)}, t_1) \sigma(S_{t_2}^{(0)}, t_2) dW_{t_2} dW_{t_1} \quad (12)$$

Next, by application of Itô's formula to (5) we obtain

$$\begin{aligned} A_{3T} &= 6 \int_0^T \int_0^{t_1} \int_0^{t_2} \partial \sigma(S_{t_1}^{(0)}, t_1) \partial \sigma(S_{t_2}^{(0)}, t_2) \sigma(S_{t_3}^{(0)}, t_3) dW_{t_3} dW_{t_2} dW_{t_1} \\ &\quad + 6 \int_0^T \int_0^{t_1} \int_0^{t_2} \partial^2 \sigma(S_{t_1}^{(0)}, t_1) \sigma(S_{t_2}^{(0)}, t_2) \sigma(S_{t_3}^{(0)}, t_3) dW_{t_3} dW_{t_2} dW_{t_1} \\ &\quad + 3 \int_0^T \int_0^{t_1} \partial^2 \sigma(S_{t_1}^{(0)}, t_1) \sigma^2(S_{t_2}^{(0)}, t_2) dt_2 dW_{t_1}. \end{aligned} \quad (13)$$

Similarly,

$$\begin{aligned} (A_{2T})^2 &= 16 \int_0^T \int_0^{t_1} \int_0^{t_2} \int_0^{t_3} \partial \sigma(S_{t_1}^{(0)}, t_1) \partial \sigma(S_{t_2}^{(0)}, t_2) \sigma(S_{t_3}^{(0)}, t_3) \sigma(S_{t_4}^{(0)}, t_4) dW_{t_4} dW_{t_3} dW_{t_2} dW_{t_1} \\ &\quad + 8 \int_0^T \int_0^{t_1} \int_0^{t_2} \int_0^{t_3} \partial \sigma(S_{t_1}^{(0)}, t_1) \sigma(S_{t_2}^{(0)}, t_2) \partial \sigma(S_{t_3}^{(0)}, t_3) \sigma(S_{t_4}^{(0)}, t_4) dW_{t_4} dW_{t_3} dW_{t_2} dW_{t_1} \\ &\quad + 8 \int_0^T \int_0^{t_1} \int_0^{t_2} \partial \sigma(S_{t_1}^{(0)}, t_1) \partial \sigma(S_{t_2}^{(0)}, t_2) \sigma^2(S_{t_3}^{(0)}, t_3) dt_3 dW_{t_2} dW_{t_1} \\ &\quad + 8 \int_0^T \int_0^{t_1} \int_0^{t_2} \partial \sigma(S_{t_1}^{(0)}, t_1) \partial \sigma(S_{t_2}^{(0)}, t_2) \sigma(S_{t_2}^{(0)}, t_2) \sigma(S_{t_3}^{(0)}, t_3) dW_{t_3} dt_2 dW_{t_1} \\ &\quad + 8 \int_0^T \int_0^{t_1} \int_0^{t_2} (\partial \sigma(S_{t_1}^{(0)}, t_1))^2 \sigma(S_{t_2}^{(0)}, t_2) \sigma(S_{t_3}^{(0)}, t_3) dW_{t_3} dW_{t_2} dt_1 \\ &\quad + 4 \int_0^T \int_0^{t_1} (\partial \sigma(S_{t_1}^{(0)}, t_1))^2 \sigma^2(S_{t_2}^{(0)}, t_2) dt_2 dt_1. \end{aligned} \quad (14)$$

Then, by Proposition 1 in [23], the conditional expectations in (11) can be com-

puted as

$$\begin{aligned}
 & \mathbf{E}[A_{2T}|A_{1T} = x] \\
 &= \left(2 \int_0^T \int_0^{t_1} \partial \sigma(S_{t_1}^{(0)}, t_1) \sigma(S_{t_1}^{(0)}, t_1) \sigma^2(S_{t_2}^{(0)}, t_2) dt_2 dt_1 \right) \frac{H_2(x; \Sigma_T)}{\Sigma_T^2} \\
 &=: c_2^{2,1} H_2(x; \Sigma_T)
 \end{aligned} \tag{15}$$

$$\begin{aligned}
 & \mathbf{E}[A_{3T}|A_{1T} = x] \\
 &= \left(6 \int_0^T \int_0^{t_1} \int_0^{t_2} \partial \sigma(S_{t_1}^{(0)}, t_1) \sigma(S_{t_1}^{(0)}, t_1) \partial \sigma(S_{t_2}^{(0)}, t_2) \sigma(S_{t_2}^{(0)}, t_2) \sigma^2(S_{t_3}^{(0)}, t_3) dt_3 dt_2 dt_1 \right. \\
 &\quad \left. + 6 \int_0^T \int_0^{t_1} \int_0^{t_2} \partial^2 \sigma(S_{t_1}^{(0)}, t_1) \sigma(S_{t_1}^{(0)}, t_1) \sigma^2(S_{t_2}^{(0)}, t_2) \sigma^2(S_{t_3}^{(0)}, t_3) dt_3 dt_2 dt_1 \right) \\
 &\quad \times \frac{H_3(x; \Sigma_T)}{\Sigma_T^3} \\
 &\quad + \left(3 \int_0^T \int_0^{t_1} \partial^2 \sigma(S_{t_1}^{(0)}, t_1) \sigma(S_{t_1}^{(0)}, t_1) \sigma^2(S_{t_2}^{(0)}, t_2) dt_2 dt_1 \right) \frac{H_1(x; \Sigma_T)}{\Sigma_T} \\
 &=: c_3^{3,1} H_3(x; \Sigma_T) + c_1^{3,1} H_1(x; \Sigma_T)
 \end{aligned} \tag{16}$$

and

$$\begin{aligned}
 & \mathbf{E}[(A_{2T})^2|A_{1T} = x] \\
 &= \left(16 \int_0^T \int_0^{t_1} \int_0^{t_2} \int_0^{t_3} \partial \sigma(S_{t_1}^{(0)}, t_1) \sigma(S_{t_1}^{(0)}, t_1) \partial \sigma(S_{t_2}^{(0)}, t_2) \sigma(S_{t_2}^{(0)}, t_2) \sigma^2(S_{t_3}^{(0)}, t_3) \sigma^2(S_{t_4}^{(0)}, t_4) dt_4 dt_3 dt_2 dt_1 \right. \\
 &\quad \left. + 8 \int_0^T \int_0^{t_1} \int_0^{t_2} \int_0^{t_3} \partial \sigma(S_{t_1}^{(0)}, t_1) \sigma(S_{t_1}^{(0)}, t_1) \sigma^2(S_{t_2}^{(0)}, t_2) \partial \sigma(S_{t_3}^{(0)}, t_3) \sigma(S_{t_3}^{(0)}, t_3) \sigma^2(S_{t_4}^{(0)}, t_4) dt_4 dt_3 dt_2 dt_1 \right) \\
 &\quad \times \frac{H_4(x; \Sigma_T)}{\Sigma_T^4} \\
 &\quad + \left(16 \int_0^T \int_0^{t_1} \int_0^{t_2} \partial \sigma(S_{t_1}^{(0)}, t_1) \sigma(S_{t_1}^{(0)}, t_1) \partial \sigma(S_{t_2}^{(0)}, t_2) \sigma(S_{t_2}^{(0)}, t_2) \sigma^2(S_{t_3}^{(0)}, t_3) dt_3 dt_2 dt_1 \right. \\
 &\quad \left. + 8 \int_0^T \int_0^{t_1} \int_0^{t_2} \left(\partial \sigma(S_{t_1}^{(0)}, t_1) \right)^2 \sigma^2(S_{t_2}^{(0)}, t_2) \sigma^2(S_{t_3}^{(0)}, t_3) dt_3 dt_2 dt_1 \right) \frac{H_2(x; \Sigma_T)}{\Sigma_T^2} \\
 &\quad + \left(4 \int_0^T \int_0^{t_1} \left(\partial \sigma(S_{t_1}^{(0)}, t_1) \right)^2 \sigma^2(S_{t_2}^{(0)}, t_2) dt_2 dt_1 \right) H_0(x; \Sigma_T) \\
 &=: c_4^{2,2} H_4(x; \Sigma_T) + c_2^{2,2} H_2(x; \Sigma_T) + c_0^{2,2} H_0(x; \Sigma_T)
 \end{aligned} \tag{17}$$

where $H_n(x; \Sigma)$ is an n -th order Hermite polynomial defined by

$$H_n(x; \Sigma) := (-\Sigma)^n e^{x^2/2\Sigma} \frac{d^n}{dx^n} e^{-x^2/2\Sigma}.$$

Substituting these into (11), we have the asymptotic expansion of $\mathbf{E}[\Phi(G^{(\epsilon)})]$ up to ϵ^2 -order. Further, letting $\Phi = \delta_x$, we have the expansion of $f_{G^{(\epsilon)}}$, the density

function of $G^{(\epsilon)}$:

$$\begin{aligned}
 f_{G^{(\epsilon)}} &= f_{A_{1T}}(x) + \epsilon(-1) \frac{\partial}{\partial x} \{ \mathbf{E}[A_{2T} | A_{1T} = x] f_{A_{1T}}(x) \} \\
 &\quad + \epsilon^2 \left((-1) \frac{\partial}{\partial x} \{ \mathbf{E}[A_{3T} | A_{1T} = x] f_{A_{1T}}(x) \} + \frac{1}{2} (-1)^2 \frac{\partial^2}{\partial x^2} \{ \mathbf{E}[(A_{2T})^2 | A_{1T} = x] f_{A_{1T}}(x) \} \right) + o(\epsilon^2) \\
 &= f_{A_{1T}}(x) + \epsilon(-1) \frac{\partial}{\partial x} \{ c_2^{2,1} H_2(x; \Sigma_T) f_{A_{1T}}(x) \} \\
 &\quad + \epsilon^2 \left((-1) \frac{\partial}{\partial x} \left\{ \sum_{i=1,3} c_i^{3,1} H_i(x; \Sigma_T) f_{A_{1T}}(x) \right\} + \frac{1}{2} (-1)^2 \frac{\partial^2}{\partial x^2} \left\{ \sum_{i=0,2,4} c_i^{2,2} H_i(x; \Sigma_T) f_{A_{1T}}(x) \right\} \right) + o(\epsilon^2).
 \end{aligned} \tag{18}$$

2.3 An Alternative Approach with a System of Ordinary Differential Equations

In this subsection, we present an alternative approach in which the conditional expectations are computed through some system of ordinary differential equations. Again, the asymptotic expansion of $\mathbf{E}[\Phi(G^{(\epsilon)})]$ up to ϵ^2 -order is considered in this subsection.

Note that the expectations of A_{2T} , A_{3T} and $(A_{2T})^2$ conditional on A_{1T} are expressed by linear combinations of a finite number of Hermite polynomials as in (15), (16) and (17). Thus, by Lemma 4 in [23], we have

$$\mathbf{E}[A_{2T} | A_{1T} = x] = \sum_{n=0}^2 a_n^{2,1} H_n(x; \Sigma_T), \tag{19}$$

$$\mathbf{E}[A_{3T} | A_{1T} = x] = \sum_{n=0}^3 a_n^{3,1} H_n(x; \Sigma_T), \tag{20}$$

$$\text{and } \mathbf{E}[(A_{2T})^2 | A_{1T} = x] = \sum_{n=0}^4 a_n^{2,2} H_n(x; \Sigma_T), \tag{21}$$

where the coefficients are given by

$$\begin{aligned}
 a_n^{2,1} &= \frac{1}{n!} \frac{1}{(i\Sigma)^n} \frac{\partial^n}{\partial \xi^n} \Big|_{\xi=0} \left\{ \mathbf{E}[Z_T^{(\xi)} A_{2T}] \right\}, \\
 a_n^{3,1} [4pt] &= \frac{1}{n!} \frac{1}{(i\Sigma)^n} \frac{\partial^n}{\partial \xi^n} \Big|_{\xi=0} \left\{ \mathbf{E}[Z_T^{(\xi)} A_{3T}] \right\}, \\
 a_n^{2,2} &= \frac{1}{n!} \frac{1}{(i\Sigma)^n} \frac{\partial^n}{\partial \xi^n} \Big|_{\xi=0} \left\{ \mathbf{E}[Z_T^{(\xi)} (A_{2T})^2] \right\}, \\
 \text{and } Z_t^{(\xi)} &:= \exp \left(i\xi A_{1t} + \frac{\xi^2}{2} \Sigma_t \right).
 \end{aligned}$$

Note that $Z^{(\xi)}$ is a martingale with $Z_0^{(\xi)} = 1$. Since these conditional expectations can be represented by linear combinations of Hermite polynomials as seen in the previous subsection, the following should hold, which can be confirmed easily with results of this subsection:

$$\begin{cases} a_2^{2,1} = c_2^{2,1}; & a_1^{2,1} = a_0^{2,1} = 0; \\ a_3^{3,1} = c_3^{3,1}; & a_1^{3,1} = c_1^{3,1}; & a_2^{3,1} = a_0^{2,1} = 0; \\ a_4^{2,2} = c_4^{2,2}; & a_2^{2,2} = c_2^{2,2}; & a_0^{2,2} = c_0^{2,2}; & a_3^{2,2} = a_1^{2,2} = 0. \end{cases} \quad (22)$$

Then, computation of these conditional expectations is equivalent to that of the unconditional expectations $\mathbf{E}[Z_T^{(\xi)} A_{2T}]$, $\mathbf{E}[Z_T^{(\xi)} A_{3T}]$ and $\mathbf{E}[Z_T^{(\xi)} (A_{2T})^2]$.

First, applying Itô's formula to $(Z_t^{(\xi)} A_{2t})$ we have

$$\begin{aligned} \mathbf{E}[Z_t^{(\xi)} A_{2t}] &= \mathbf{E}\left[\int_0^t Z_s^{(\xi)} dA_{2s} + \int_0^t A_{2s} dZ_s^{(\xi)} + \langle A_2, Z^{(\xi)} \rangle_t\right] \\ &= 2(i\xi) \int_0^t \partial\sigma(S_s^{(0)}, s) \sigma(S_s^{(0)}, s) \mathbf{E}[Z_s^{(\xi)} A_{1s}] ds \end{aligned} \quad (23)$$

Then, applying Itô's formula to $(Z_t^{(\xi)} A_{1t})$ again, we also have

$$\begin{aligned} \mathbf{E}[Z_t^{(\xi)} A_{1t}] &= \mathbf{E}\left[\int_0^t Z_s^{(\xi)} dA_{1s} + \int_0^t A_{1s} dZ_s^{(\xi)} + \langle A_1, Z^{(\xi)} \rangle_t\right] \\ &= (i\xi) \int_0^t \sigma^2(S_s^{(0)}, s) \mathbf{E}[Z_s^{(\xi)}] ds \\ &= (i\xi) \int_0^t \sigma^2(S_s^{(0)}, s) ds \end{aligned} \quad (24)$$

since $\mathbf{E}[Z_t^{(\xi)}] = 1$ for all t .

Similarly, the following are obtained;

$$\begin{aligned} \mathbf{E}[Z_t^{(\xi)} A_{3t}] &= 3(i\xi) \left(\int_0^t \partial^2 \sigma(S_s^{(0)}, s) \sigma(S_s^{(0)}, s) \mathbf{E}[Z_s^{(\xi)} (A_{1s})^2] ds \right. \\ &\quad \left. + \int_0^t \partial\sigma(S_s^{(0)}, s) \sigma(S_s^{(0)}, s) \mathbf{E}[Z_s^{(\xi)} A_{2s}] ds \right) \end{aligned} \quad (25)$$

$$\begin{aligned} \mathbf{E}[Z_t^{(\xi)} (A_{1t})^2] &= \int_0^t \sigma^2(S_s^{(0)}, s) ds \\ &\quad + 2(i\xi) \int_0^t \sigma^2(S_s^{(0)}, s) \mathbf{E}[Z_s^{(\xi)} A_{1s}] ds \end{aligned} \quad (26)$$

$$\begin{aligned}\mathbf{E} \left[Z_t^{(\xi)} (A_{2t})^2 \right] &= 4 \int_0^t \left(\partial \sigma(S_s^{(0)}, s) \right)^2 \mathbf{E} \left[Z_s^{(\xi)} (A_{1s})^2 \right] ds \\ &\quad + 4(i\xi) \int_0^t \partial \sigma(S_s^{(0)}, s) \sigma(S_s^{(0)}, s) \mathbf{E} \left[Z_s^{(\xi)} A_{2s} A_{1s} \right] ds\end{aligned}\quad (27)$$

$$\begin{aligned}\mathbf{E} \left[Z_t^{(\xi)} A_{2t} A_{1t} \right] &= 2 \int_0^t \partial \sigma(S_s^{(0)}, s) \sigma(S_s^{(0)}, s) \mathbf{E} \left[Z_s^{(\xi)} A_{1s} \right] ds \\ &\quad + (i\xi) \int_0^t (\sigma(S_s^{(0)}, s))^2 \mathbf{E} \left[Z_s^{(\xi)} A_{2s} \right] ds \\ &\quad + 2(i\xi) \int_0^t \partial \sigma(S_s^{(0)}, s) \sigma(S_s^{(0)}, s) \mathbf{E} \left[Z_s^{(\xi)} (A_{1s})^2 \right] ds.\end{aligned}\quad (28)$$

Then, $\mathbf{E}[Z_T^{(\xi)} A_{2T}]$, $\mathbf{E}[Z_T^{(\xi)} A_{3T}]$ and $\mathbf{E}[Z_T^{(\xi)} (A_{2T})^2]$ can be obtained as solutions of the system of ordinary differential equations (23), (24), (25), (26), (27) and (28). In fact, since they have a grading structure that the higher-order equations depend only on the lower ones as

$$\begin{aligned}\mathbf{E} \left[Z_t^{(\xi)} A_{1t} \right] &= (i\xi) \int_0^t \sigma^2(S_s^{(0)}, s) ds \\ \mathbf{E} \left[Z_t^{(\xi)} A_{2t} \right] &= 2(i\xi) \int_0^t \partial \sigma(S_s^{(0)}, s) \sigma(S_s^{(0)}, s) \mathbf{E} \left[Z_s^{(\xi)} A_{1s} \right] ds \\ \mathbf{E} \left[Z_t^{(\xi)} (A_{1t})^2 \right] &= \int_0^t \sigma^2(S_s^{(0)}, s) ds \\ &\quad + 2(i\xi) \int_0^t \sigma^2(S_s^{(0)}, s) \mathbf{E} \left[Z_s^{(\xi)} A_{1s} \right] ds \\ \mathbf{E} \left[Z_t^{(\xi)} A_{3t} \right] &= 3(i\xi) \left(\int_0^t \partial^2 \sigma(S_s^{(0)}, s) \sigma(S_s^{(0)}, s) \mathbf{E} \left[Z_s^{(\xi)} (A_{1s})^2 \right] ds \right. \\ &\quad \left. + \int_0^t \partial \sigma(S_s^{(0)}, s) \sigma(S_s^{(0)}, s) \mathbf{E} \left[Z_s^{(\xi)} A_{2s} \right] ds \right) \\ \mathbf{E} \left[Z_t^{(\xi)} A_{2t} A_{1t} \right] &= 2 \int_0^t \partial \sigma(S_s^{(0)}, s) \sigma(S_s^{(0)}, s) \mathbf{E} \left[Z_s^{(\xi)} A_{1s} \right] ds \\ &\quad + (i\xi) \int_0^t (\sigma(S_s^{(0)}, s))^2 \mathbf{E} \left[Z_s^{(\xi)} A_{2s} \right] ds \\ &\quad + 2(i\xi) \int_0^t \partial \sigma(S_s^{(0)}, s) \sigma(S_s^{(0)}, s) \mathbf{E} \left[Z_s^{(\xi)} (A_{1s})^2 \right] ds \\ \mathbf{E} \left[Z_t^{(\xi)} (A_{2t})^2 \right] &= 4 \int_0^t \left(\partial \sigma(S_s^{(0)}, s) \right)^2 \mathbf{E} \left[Z_s^{(\xi)} (A_{1s})^2 \right] ds \\ &\quad + 4(i\xi) \int_0^t \partial \sigma(S_s^{(0)}, s) \sigma(S_s^{(0)}, s) \mathbf{E} \left[Z_s^{(\xi)} A_{2s} A_{1s} \right] ds,\end{aligned}$$

they can be easily solved with substituting each solution into the next ordinary differential equation recursively. Moreover, since these solutions are clearly the polynomial of $(i\xi)$, we can easily implement differentiations with respect to ξ in (19), (20) and (21). It is obvious that the resulting coefficients given by these solutions are equivalent to the results in the previous subsection.

Moreover, we also remark the relationship between our method and an approach presented by [18] in which the density function of $G^{(\epsilon)}$ is derived by Fourier inversion of its formally expanded characteristic function. Precisely speaking, [18] formally expanded $\Psi_{G^{(\epsilon)}}(\xi) = \mathbf{E}[e^{i\xi G^{(\epsilon)}}]$ as

$$\begin{aligned}\Psi_{G^{(\epsilon)}}(\xi) &= \mathbf{E}\left[e^{i\xi G^{(\epsilon)}}\right] = e^{-\frac{\epsilon^2}{2}\Sigma_T} \times \left\{1 + \epsilon(i\xi)\mathbf{E}\left[Z_T^{(\xi)}A_{2T}\right]\right. \\ &\quad \left.+ \epsilon^2\left((i\xi)\mathbf{E}\left[Z_T^{(\xi)}A_{3T}\right] + \frac{(i\xi)^2}{2}\mathbf{E}\left[Z_T^{(\xi)}(A_{2T})^2\right]\right)\right\} + o(\epsilon^2) \\ &= e^{-\frac{\epsilon^2}{2}\Sigma_T} \times \left\{1 + \epsilon(i\xi)\mathbf{E}\left[Z_T^{(\xi)}\mathbf{E}[A_{2T}|A_{1T}]\right]\right. \\ &\quad \left.+ \epsilon^2\left((i\xi)\mathbf{E}\left[Z_T^{(\xi)}\mathbf{E}[A_{3T}|A_{1T}]\right] + \frac{(i\xi)^2}{2}\mathbf{E}\left[Z_T^{(\xi)}\mathbf{E}[(A_{2T})^2|A_{1T}]\right]\right)\right\} + o(\epsilon^2)\end{aligned}\quad (29)$$

and computed the conditional expectations in this expansion. Then, $f_{G^{(\epsilon)}}(x)$, the density function of $G^{(\epsilon)}$, was derived by Fourier inversion of $\Psi_{G^{(\epsilon)}}(\xi)$;

$$f_{G^{(\epsilon)}}(x) = \mathcal{F}^{-1}(\Psi_{G^{(\epsilon)}}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ix\xi} \Psi_{G^{(\epsilon)}}(\xi) d\xi. \quad (30)$$

This approach is completely equivalent to our method based on Watanabe's theory as also mentioned in [18]. In fact, from (18) and (22) we obtain

$$\begin{aligned}f_{G^{(\epsilon)}}(x) &= f_{A_{1T}}(x) + \epsilon(-1)\frac{\partial}{\partial x}\{c_2^{2,1}H_2(x; \Sigma_T)f_{A_{1T}}(x)\} \\ &\quad + \epsilon^2\left\{(-1)\frac{\partial}{\partial x}\left\{\sum_{n=1,3}c_n^{3,1}H_n(x; \Sigma_T)f_{A_{1T}}(x)\right\}\right. \\ &\quad \left.+ \frac{1}{2}(-1)^2\frac{\partial^2}{\partial x^2}\left\{\sum_{n=0,2,4}c_n^{2,2}H_n(x; \Sigma_T)f_{A_{1T}}(x)\right\}\right\} + o(\epsilon^2)\end{aligned}\quad (31)$$

$$\begin{aligned}
&= \mathcal{F}^{-1} \left(e^{-\frac{\epsilon^2}{2} \Sigma_T} \right) + \epsilon c_2^{2,1} \mathcal{F}^{-1} \left((i\xi)(i\xi \Sigma_T)^2 e^{-\frac{\epsilon^2}{2} \Sigma_T} \right) \\
&\quad + \epsilon^2 \left(\sum_{n=1,3} c_n^{3,1} \mathcal{F}^{-1} \left((i\xi)(i\xi \Sigma_T)^n e^{-\frac{\epsilon^2}{2} \Sigma_T} \right) \right. \\
&\quad \left. + \frac{1}{2} \sum_{n=0,2,4} c_n^{2,2} \mathcal{F}^{-1} \left((i\xi)^2 (i\xi \Sigma_T)^n e^{-\frac{\epsilon^2}{2} \Sigma_T} \right) \right) + o(\epsilon^2) \\
&= \mathcal{F}^{-1} \left(e^{-\frac{\epsilon^2}{2} \Sigma_T} \times \left\{ 1 + \epsilon(i\xi) \sum_{n=0}^2 a_n^{2,1} (i\Sigma_T)^n \xi^n \right. \right. \\
&\quad \left. \left. + \epsilon^2 \left((i\xi) \sum_{n=0}^3 a_n^{3,1} (i\Sigma_T)^n \xi^n + \frac{(i\xi)^2}{2} \sum_{n=0}^4 a_n^{2,2} (i\Sigma_T)^n \xi^n \right) \right\} \right) + o(\epsilon^2) \\
&= \mathcal{F}^{-1} \left(e^{-\frac{\epsilon^2}{2} \Sigma_T} \times \left\{ 1 + \epsilon(i\xi) \mathbf{E}[Z_T^{(\xi)} A_{2T}] \right. \right. \\
&\quad \left. \left. + \epsilon^2 \left((i\xi) \mathbf{E}[Z_T^{(\xi)} A_{3T}] + \frac{(i\xi)^2}{2} \mathbf{E}[Z_T^{(\xi)} (A_{2T})^2] \right) \right\} \right) + o(\epsilon^2). \tag{32}
\end{aligned}$$

Then it is obvious that the inversion of the characteristic function expanded up to ϵ^2 -order (29) coincides with the density function obtained by our approach. Moreover, it can be shown that this equivalence holds at any order.

Here, at the end of this section, we state a brief summary. In the Black-Scholes-type economy, we consider the risky asset $S^{(\epsilon)}$ and evaluate some quantities, expressed as an expectation of the function of the future price, such as prices or risk sensitivities of the securities on this asset. First we expand them around the limit to $\epsilon = 0$ so that we obtain the expansion (9) which contains some conditional expectations. Then, by approaches described in Section 2.2 or 2.3, we compute these conditional expectations. Finally, substituting computation results into (9), we obtain the asymptotic expansion of those quantities. Or equivalently, one can use the formulas for these conditional expectations listed in [23].

3. Numerical Examples

In this section we apply the proposed techniques to the model more complex than Black-Scholes-type case in the previous section, to demonstrate their effectiveness. Detailed discussions in a general setting including following examples are found in Section 3 and 4 of [23].

3.1 λ -SABR Model

We first consider the European plain-vanilla call and put prices under the following λ -SABR model [10] (interest rate = 0%):

$$\begin{aligned}
dS^{(\epsilon)}(t) &= \epsilon \sigma^{(\epsilon)}(t) (S^{(\epsilon)}(t))^\beta dW_t^1, \\
d\sigma^{(\epsilon)}(t) &= \lambda(\theta - \sigma^{(\epsilon)}(t))dt + \epsilon \nu_1 \sigma^{(\epsilon)}(t) dW_t^1 + \epsilon \nu_2 \sigma^{(\epsilon)}(t) dW_t^2,
\end{aligned}$$

where $v_1 = \rho v v_2 = (\sqrt{1 - \rho^2})v$ (the correlation between S and σ is $\rho \in [-1, 1]$).

Approximated prices by the asymptotic expansion method are calculated up to the fifth order. Note that all the solutions to differential equations are obtained analytically. Benchmark values are computed by Monte Carlo simulations. ϵ is set to be one and other parameters used in the test are given in Table 1:

Table 1. Parameter specifications of the λ -SABR model for our numerical experiments.

Parameter	$S(0)$	λ	$\sigma(0)$	β	ρ	θ	v	T
i	100	0.1	3.0	0.5	-0.7	3.0	0.3	10
ii	100	0.1	0.3	1.0	-0.7	0.3	0.3	10
iii	100	0.1	0.3	1.0	-0.7	0.3	0.3	30

For the case of $\beta = 1$ (i.e. case ii and iii), we calculate approximated prices by the “log-normal asymptotic expansion method” described in Section 4.3 in [23] up to the fourth order. In Monte Carlo simulations for benchmark values, we use Euler-Maruyama scheme as a discretization scheme with 1024 time steps for case i and for case ii and iii the second order discretization scheme given by Ninomiya-Victoir [14] with 128 and 256 time steps, respectively. Each simulation contains 10^8 paths. The results are in Table 2.

From the results, in each case, the higher order asymptotic expansion or log-normal asymptotic expansion almost always improve the accuracy of approximation by the lower expansions. Improvement is significant especially in long-term cases in which the lower order asymptotic expansions cannot approximate the price well.

3.2 Currency Option under a Libor Market Model of Interest Rates and a Stochastic Volatility of a Spot Exchange Rate

In this subsection, we apply our methods to pricing options on currencies under Libor Market Models (LMMs) of interest rates and a stochastic volatility of the spot foreign exchange rate (forex). Due to limitations of space, only the structure of the stochastic differential equations of our model is described here. For details of the underlying model, see Takahashi and Takehara [20].

3.2.1 Cross-Currency Libor Market Models

Let $(\Omega, \mathcal{F}, \tilde{P}, \{\mathcal{F}_t\}_{0 \leq t \leq T^* < \infty})$ be a complete probability space with a filtration satisfying the usual conditions. We consider the following pricing problem for the call option with maturity $T \in (0, T^*]$ and strike rate $K > 0$;

$$V^C(0; T, K) = P_d(0, T) \times \mathbf{E}^P [(S(T) - K)^+] = P_d(0, T) \times \mathbf{E}^P [(F_T(T) - K)^+] \quad (33)$$

where $V^C(0; T, K)$ denotes the value of an European call option at time 0 with maturity T and strike rate K , $S(T)$ denotes the spot exchange rate at time $t \geq 0$

Table 2. Comparisons of the absolute and relative differences between the estimators by our asymptotic expansion at different order and Monte Carlo simulations. “Absolute Differences” and “Relative Differences” are given by (the approximate value by our asymptotic expansion) – (the estimator by Monte Carlo simulations) and (Absolute Differences) / (the estimator by Monte Carlo simulations).

Case	Strike (C/P)	MC	A.E. (Difference)					A.E. (Relative Difference)				
			1st	2nd	3rd	4th	5th	1st	2nd	3rd	4th	5th
i	50 Put	13.109	4.876	5.000	2.313	1.067	0.260	37.20	38.14	17.64	8.14	1.98
	60 Put	16.618	4.544	4.648	1.931	0.938	0.195	27.34	27.97	11.62	5.65	1.17
	70 Put	20.482	4.241	4.322	1.585	0.844	0.149	20.71	21.10	7.74	4.12	0.73
	80 Put	24.720	3.965	4.020	1.269	0.778	0.117	16.04	16.26	5.14	3.15	0.47
	90 Put	29.347	3.710	3.738	0.980	0.735	0.094	12.64	12.74	3.34	2.51	0.32
	100 Call	34.375	3.472	3.472	0.712	0.712	0.077	10.10	10.10	2.07	2.07	0.22
	110 Call	29.811	3.246	3.217	0.459	0.704	0.063	10.89	10.79	1.54	2.36	0.21
	120 Call	25.659	3.026	2.971	0.220	0.711	0.050	11.79	11.58	0.86	2.77	0.19
	130 Call	21.914	2.809	2.728	-0.010	0.731	0.035	12.82	12.45	-0.04	3.33	0.16
	140 Call	18.571	2.591	2.487	-0.230	0.762	0.018	13.95	13.39	-1.24	4.10	0.10
	150 Call	15.615	2.370	2.246	-0.441	0.804	-0.002	15.18	14.38	-2.83	5.15	-0.02
Case	Strike (C/P)	MC	Log Normal A.E. (Difference)				Log Normal A.E. (Relative Difference)					
			Log-Norm	1st	2nd	3rd	4th	Log-Norm	1st	2nd	3rd	4th
ii	50 Put	9.429	-0.896	0.250	0.470	-0.223	0.021	-9.51	2.65	4.99	-2.36	0.22
	60 Put	13.095	-0.187	0.168	0.449	-0.215	0.028	-1.43	1.29	3.43	-1.64	0.21
	70 Put	17.307	0.678	0.045	0.431	-0.203	0.034	3.92	0.26	2.49	-1.17	0.19
	80 Put	22.041	1.620	-0.099	0.414	-0.190	0.039	7.35	-0.45	1.88	-0.86	0.18
	90 Put	27.272	2.577	-0.253	0.397	-0.177	0.045	9.45	-0.93	1.45	-0.65	0.17
	100 Call	32.971	3.503	-0.416	0.379	-0.163	0.051	10.62	-1.26	1.15	-0.49	0.15
	110 Call	29.110	4.367	-0.589	0.360	-0.149	0.057	15.00	-2.02	1.24	-0.51	0.20
	120 Call	25.655	5.149	-0.773	0.338	-0.135	0.063	20.07	-3.01	1.32	-0.53	0.25
	130 Call	22.576	5.837	-0.972	0.315	-0.120	0.069	25.85	-4.30	1.39	-0.53	0.31
	140 Call	19.842	6.424	-1.186	0.289	-0.104	0.076	32.38	-5.98	1.46	-0.53	0.38
	150 Call	17.420	6.912	-1.416	0.261	-0.088	0.083	39.68	-8.13	1.50	-0.50	0.47
iii	50 Put	19.801	2.280	-0.889	1.143	-0.592	0.182	11.51	-4.49	5.77	-2.99	0.92
	60 Put	25.471	3.371	-1.248	1.254	-0.581	0.154	13.23	-4.90	4.93	-2.28	0.60
	70 Put	31.500	4.459	-1.594	1.351	-0.560	0.120	14.15	-5.06	4.29	-1.78	0.38
	80 Put	37.847	5.520	-1.927	1.437	-0.535	0.081	14.59	-5.09	3.80	-1.41	0.21
	90 Put	44.476	6.541	-2.246	1.515	-0.505	0.039	14.71	-5.05	3.41	-1.14	0.09
	100 Call	51.357	7.512	-2.555	1.586	-0.474	-0.005	14.63	-4.98	3.09	-0.92	-0.01
	110 Call	48.465	8.430	-2.856	1.652	-0.442	-0.051	17.39	-5.89	3.41	-0.91	-0.10
	120 Call	45.780	9.291	-3.150	1.715	-0.409	-0.098	20.30	-6.88	3.75	-0.89	-0.21
	130 Call	43.281	10.097	-3.439	1.774	-0.376	-0.147	23.33	-7.94	4.10	-0.87	-0.34
	140 Call	40.954	10.848	-3.724	1.831	-0.342	-0.197	26.49	-9.09	4.47	-0.84	-0.48
	150 Call	38.782	11.545	-4.007	1.886	-0.309	-0.248	29.77	-10.33	4.86	-0.80	-0.64

and $F_T(t)$ denotes the time t value of the forex forward rate with maturity T . Similarly, for the put option we consider

$$V^P(0; T, K) = P_d(0, T) \times \mathbf{E}^P [(K - S(T))^+] = P_d(0, T) \times \mathbf{E}^P [(K - F_T(T))^+]. \quad (34)$$

It is well known that the arbitrage-free relation between the forex spot rate and the forex forward rate are given by $F_T(t) = S(t) \frac{P_f(t, T)}{P_d(t, T)}$ where $P_d(t, T)$ and $P_f(t, T)$ denote the time t values of domestic and foreign zero coupon bonds with maturity T respectively. $\mathbf{E}^P[\cdot]$ denotes an expectation operator under EMM (Equivalent Martingale Measure) P whose associated numeraire is the domestic zero coupon bond maturing at T .

For these pricing problems, a market model and a stochastic volatility model are applied to modeling interest rates' and the spot exchange rate's dynamics respectively.

We first define domestic and foreign forward interest rates as $f_{dj}(t) = \left(\frac{P_d(t, T_j)}{P_d(t, T_{j+1})} - 1 \right) \frac{1}{\tau_j}$ and $f_{fj}(t) = \left(\frac{P_f(t, T_j)}{P_f(t, T_{j+1})} - 1 \right) \frac{1}{\tau_j}$ respectively, where $j = n(t), n(t) + 1, \dots, N$, $\tau_j = T_{j+1} - T_j$, and $P_d(t, T_j)$ and $P_f(t, T_j)$ denote the prices of domestic/foreign zero coupon bonds with maturity T_j at time $t (\leq T_j)$ respectively; $n(t) = \min\{i : t \leq T_i\}$. We also define spot interest rates to the nearest fixing date denoted by $f_{d, n(t)-1}(t)$ and $f_{f, n(t)-1}(t)$ as $f_{d, n(t)-1}(t) = \left(\frac{1}{P_d(t, T_{n(t)})} - 1 \right) \frac{1}{(T_{n(t)} - t)}$ and $f_{f, n(t)-1}(t) = \left(\frac{1}{P_f(t, T_{n(t)})} - 1 \right) \frac{1}{(T_{n(t)} - t)}$. Finally, we set $T = T_{N+1}$ and will abbreviate $F_{T_{N+1}}(t)$ to $F_{N+1}(t)$ in what follows.

Under the framework of the asymptotic expansion in the standard cross-currency Libor market model, we have to consider the following system of stochastic differential equations (henceforth called S.D.E.s) under the domestic terminal measure P to price options. For detailed arguments on the framework of these S.D.E.s see [20].

As for the domestic and foreign interest rates we assume forward market models; for $j = n(t) - 1, n(t), n(t) + 1, \dots, N$,

$$\begin{aligned} f_{dj}^{(\epsilon)}(t) &= f_{dj}(0) + \epsilon^2 \sum_{i=j+1}^N \int_0^t g_{di}^{0,(\epsilon)}(u)' \gamma_{dj}(u) f_{dj}^{(\epsilon)}(u) du \\ &\quad + \epsilon \int_0^t f_{dj}^{(\epsilon)}(u) \gamma'_{dj}(u) dW_u, \end{aligned} \quad (35)$$

$$\begin{aligned} f_{fj}^{(\epsilon)}(t) &= f_{fj}(0) - \epsilon^2 \sum_{i=0}^j \int_0^t g_{fi}^{0,(\epsilon)}(u)' \gamma_{fj}(u) f_{fj}^{(\epsilon)}(u) du \\ &\quad + \epsilon^2 \sum_{i=0}^N \int_0^t g_{di}^{0,(\epsilon)}(u)' \gamma_{fj}(u) f_{fj}^{(\epsilon)}(u) du \\ &\quad - \epsilon^2 \int_0^t \sigma^{(\epsilon)}(u) \bar{\sigma}' \gamma_{fj}(u) f_{fj}^{(\epsilon)}(u) du + \epsilon \int_0^t f_{fj}^{(\epsilon)}(u) \gamma'_{fj}(u) dW_u, \end{aligned} \quad (36)$$

where $g_{dj}^{0,(\epsilon)}(t) := \frac{-\tau_j f_{dj}^{(\epsilon)}(t)}{1+\tau_j f_{dj}^{(\epsilon)}(t)} \gamma_{dj}(t)$, $g_{fj}^{0,(\epsilon)}(t) := \frac{-\tau_j f_{fj}^{(\epsilon)}(t)}{1+\tau_j f_{fj}^{(\epsilon)}(t)} \gamma_{fj}(t)$; x' denotes the transpose of x and W is a r -dimensional standard Wiener process under the domestic terminal measure P ; $\gamma_{dj}(s)$, $\gamma_{fj}(s)$ are r -dimensional vector-valued functions of time-parameter s ; $\bar{\sigma}$ denotes a r -dimensional constant vector satisfying $\|\bar{\sigma}\| = 1$ and $\sigma^{(\epsilon)}(t)$, the volatility of the spot exchange rate, is specified to follow a $\mathbb{R}_{++}$ -valued general time-inhomogeneous Markovian process as follows:

$$\begin{aligned} \sigma^{(\epsilon)}(t) = & \sigma(0) + \int_0^t \mu(u, \sigma^{(\epsilon)}(u)) du + \epsilon^2 \sum_{j=1}^N \int_0^t g_{dj}^{0,(\epsilon)}(u)' \omega(u, \sigma^{(\epsilon)}(u)) du \\ & + \epsilon \int_0^t \omega'(u, \sigma^{(\epsilon)}(u)) dW_u, \end{aligned} \quad (37)$$

where $\mu(s, x)$ and $\omega(s, x)$ are functions of s and x .

Finally, we consider the process of the forex forward $F_{N+1}(t)$. Since $F_{N+1}(t) \equiv F_{T_{N+1}}(t)$ can be expressed as $F_{N+1}(t) = S(t) \frac{P_r(t, T_{N+1})}{P_d(t, T_{N+1})}$, we easily notice that it is a martingale under the domestic terminal measure. In particular, it satisfies the following stochastic differential equation

$$F_{N+1}^{(\epsilon)}(t) = F_{N+1}(0) + \epsilon \int_0^t \sigma_F^{(\epsilon)}(u)' F_{N+1}^{(\epsilon)}(u) dW_u \quad (38)$$

where

$$\sigma_F^{(\epsilon)}(t) := \sum_{j=0}^N \left(g_{fj}^{0,(\epsilon)}(t) - g_{dj}^{0,(\epsilon)}(t) \right) + \sigma^{(\epsilon)}(t).$$

3.2.2 Numerical Examples

Here, we specify our model and parameters, and confirm the effectiveness of our method in this cross-currency framework.

First of all, the processes of domestic and foreign forward interest rates and of the volatility of the spot exchange rate are specified. We suppose $r = 4$, that is the dimension of a Brownian motion is set to be four; it represents the uncertainty of domestic and foreign interest rates, the spot exchange rate, and its volatility. Note that in this framework correlations among all factors are allowed. We also suppose $S(0) = 100$.

Next, we specify a volatility process of the spot exchange rate in (37) with

$$\mu(s, x) = \kappa(\theta - x), \quad \omega(s, x) = \omega x, \quad (39)$$

where θ and κ represent the level and speed of its mean-reversion respectively, and ω denotes a volatility vector on the volatility. In this section the parameters are

Table 3. Initial domestic/foreign forward interest rates and their volatilities.

	f_d	γ_d^*	f_f	γ_f^*
case (i)	0.05	0.12	0.05	0.12
case (ii)	0.02	0.3	0.05	0.12
case (iii)	0.05	0.12	0.02	0.3
case (iv)	0.02	0.3	0.02	0.3

set as follows; $\epsilon = 1$, $\sigma(0) = \theta = 0.1$, and $\kappa = 0.1$; $\omega = \omega^* \bar{v}$ where $\omega^* = 0.3$ and $\bar{v}$ denotes a four dimensional constant vector given below.

We further suppose that initial term structures of domestic and foreign forward interest rates are flat, and their volatilities also have flat structures and are constant over time: that is, for all j , $f_{dj}(0) = f_d$, $f_{fj}(0) = f_f$, $\gamma_{dj}(t) = \gamma_d^* \bar{\gamma}_d 1_{\{t < T_j\}}(t)$ and $\gamma_{fj}(t) = \gamma_f^* \bar{\gamma}_f 1_{\{t < T_j\}}(t)$. Here, γ_d^* and γ_f^* are constant scalars, and $\bar{\gamma}_d$ and $\bar{\gamma}_f$ denote four dimensional constant vectors. Moreover, given a correlation matrix $\underline{C}$ among all four factors, the constant vectors $\bar{\gamma}_d$, $\bar{\gamma}_f$, $\bar{\sigma}$ and $\bar{v}$ can be determined to satisfy $\|\bar{\gamma}_d\| = \|\bar{\gamma}_f\| = \|\bar{\sigma}\| = \|\bar{v}\| = 1$ and $V'V = \underline{C}$ where $V := (\bar{\gamma}_d, \bar{\gamma}_f, \bar{\sigma}, \bar{v})$.

In this subsection, we consider four different cases for f_d , γ_d^* , f_f and γ_f^* as in Table 3. For correlations, four sets of parameters are considered: In the case “Corr. 1”, all the factors are independent: In “Corr. 2”, there exists only the correlation of -0.5 between the spot exchange rate and its volatility (i.e. $\bar{\sigma}'\bar{v} = -0.5$) while there are no correlations among the others: In “Corr. 3”, the correlation between interest rates and the spot exchange rate are allowed while there are no correlations among the others; the correlation between domestic ones and the spot forex is $0.5(\bar{\gamma}_d'\bar{\sigma} = 0.5)$ and the correlation between foreign ones and the spot forex is $-0.5(\bar{\gamma}_f'\bar{\sigma} = -0.5)$. In “Corr. 4”, more intricately correlated structure is considered; $\bar{\gamma}_d'\bar{\sigma} = 0.5$, $\bar{\gamma}_f'\bar{\sigma} = -0.5$ between interest rates and the spot forex; and $\bar{\sigma}'\bar{v} = -0.5$ between the spot forex and its volatility. It is well known that (both of exact and approximate) evaluation of the long-term options is a hard task in the case with complex structures of correlations such as in “Corr. 3” or “Corr. 4”. Lastly, we make an assumption that $\gamma_{dn(t)-1}(t)$ and $\gamma_{fn(t)-1}(t)$, volatilities of the domestic and foreign interest rates applied to the period from t to the next fixing date $T_{n(t)}$, are equal to be zero for arbitrary $t \in [t, T_{n(t)}]$.

In Figure 1, we compare our estimations of the values of call and put options by an asymptotic expansion up to the fourth order to the benchmarks estimated by 10^6 trials of Monte Carlo simulation which is discretized by Euler-Maruyama scheme with time step 0.05 and applied the Antithetic Variable Method. For the moneynesses (defined by $K/F_{N+1}(0)$) less than one, the prices of put options are shown; otherwise, the prices of call options are displayed. Detailed data obtained in this experiment, see Table 6–9 of [23].

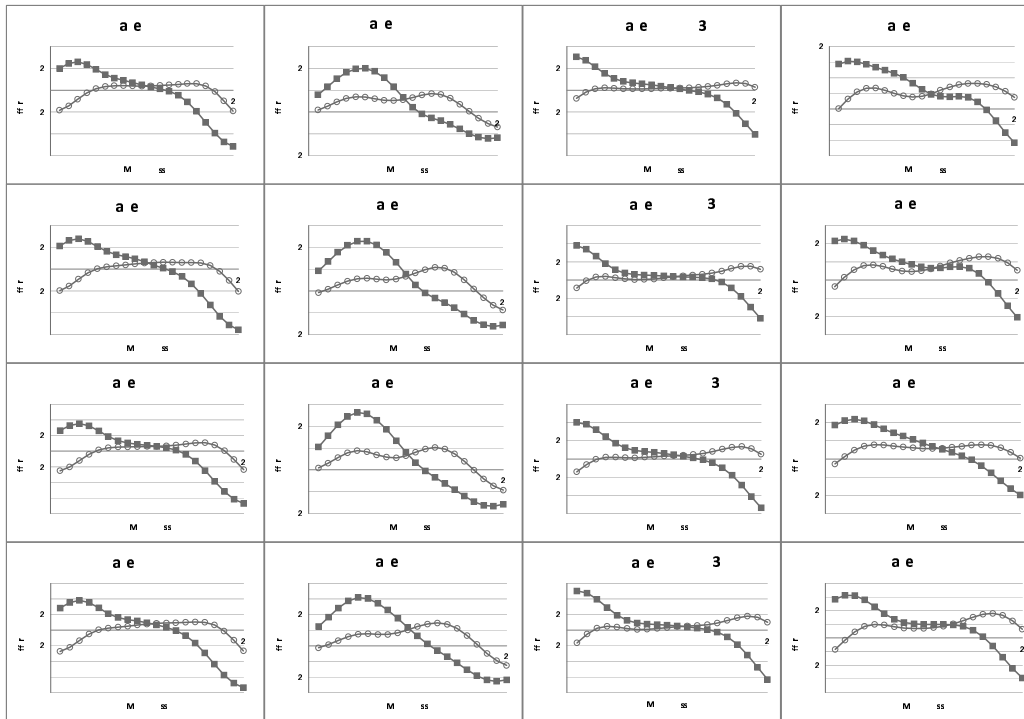


Figure 1. Graphs of comparisons of estimators by the third- and fourth-order asymptotic expansion and Monte Carlo simulations in Corr. 1–4, with a ten-year maturity. Squares denote the differences between the third-order estimators and Monte Carlo estimators; circles denote those between the fourth-order ones and Monte Carlo ones. These differences are defined by (the approximate value by our asymptotic expansion) – (the estimator by Monte Carlo simulations).

As seen in this figure, in general the estimators show more accuracy as the order of the expansion increases. Especially, for the deep OTM options the fourth order approximation performs much better and is stabler than the approximation with lower orders.

4. Concluding Remarks

In this paper, we provided the general procedures for the explicit computation of conditional expectations necessary for practical computations of the asymptotic expansion method. Moreover, the alternative but equivalent calculation algorithm which computes the unconditional expectations directly instead of the conditional ones was developed. For simplicity and limitation of space, we focused on the simple case of Black-Scholes-type economy as in Section 2, which illustrated our key ideas. For further explanations in more general environment, see [23].

Finally, we examined the accuracy of our approximation with high order expansions in the λ -SABR model and in the cross currency Libor market model with a stochastic volatility of the spot exchange rate, and confirmed satisfactory results in the both examples.

At the end of this section, we state our future plans: we will develop a similar result in the case with a jump component; we will also pursue an efficient method for the evaluation of multi-factor path-dependent or/and American derivatives. In fact, our proposed scheme can be applied to average options under a general setting of the underlying factors.

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Can Financial Synergy Motivate M&A?*

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1. Introduction

Recently, M&A has seen explosive growth. In 2007, M&A activity totaled a record \$4.38 trillion globally, up 21 percent from 2006. More and more firms are considering M&A as a firm value creation strategy instead of internal growth. M&A also has been the subject of considerable research in financial economics. Most studies have focused on positive or negative operational synergy, e.g., economies of scale, market power, and managerial benefits. However, financial synergy has rarely been analyzed.

The Modigliani-Miller (1958) theorem states that, without tax benefits and default costs, capital structure is irrelevant to total firm value. As a result, there is no financial synergy. However, in the real world with tax benefits and default costs, capital structure does matter. Therefore, adjusting capital structure through M&A may create financial synergy. Although some empirical papers relate firms' incentive of M&A to capital structure motives based on tax benefits, financial slack, wealth transfers, etc., they do not have an explicit model to analyze financial synergy realized in M&A. On the other hand, with the exception of Leland (2007) and Morellec and Zhdanov (2008), M&A related theoretical works have not focused on optimal capital structure by taking tax benefits and default costs into consideration.

This paper develops a continuous model to examine financial synergy when M&A timing is determined endogenously by equityholders. The main questions are as follows: (i) Can purely financial synergy (i.e., without operational synergy) motivate M&A? (ii) How is financial synergy distributed between equityholders and debtholders? Related to the first question, Lewellen (1971) asserts that the

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financial synergy of mergers is always positive. However, Leland (2007) suggests that financial synergy by itself is insufficient to justify M&A in many cases. Related to the second question, Scott (1977) and Shastri (1990) report that, while total firm value may increase through mergers because of lower risk, debtholders may gain at the expense of equityholders. On the other hand, Ghosh and Jain (2000) argue that equityholders can appropriate benefits from debtholders by financing M&A with debt and increasing financial leverage after M&A.

In this paper, we examine financial synergy in both scenario F and scenario E. By scenario F, we mean that the optimal capital structure after M&A is determined to maximize the total firm value. Because the M&A decision is made by equityholders, they choose scenario E rather than scenario F if there is no restriction on their behavior. By scenario E, we mean that equityholders maximize the sum of equity value and newly issued debt value to determine the optimal capital structure, ignoring the existing debt value. We find that purely financial synergy can motivate M&A in both scenarios. However, the optimal M&A timing is delayed and financial synergy is larger in scenario E.

We demonstrate that the distribution of financial synergy between equityholders and debtholders is different in the two scenarios. In scenario F, a part of the value created from exercising M&A option goes to existing debtholders, irrespective of the fact that M&A cost is fully borne by equityholders. The *ex post* wealth transfer leads equityholders to choose scenario E. This reflects the debt overhang problem discussed in Myers (1977), which may delay or prevent an investment decision to improve the total firm value because of the existing debt. In scenario E, equityholders may issue a significant amount of new debt, which results in higher default risk. Such actions that transfer wealth from existing debtholders to equityholders are similar to the risk shifting problem discussed in Jensen and Meckling (1976). Actually, scenario F corresponds to a situation where debt is issued with covenants protecting existing debtholders. On the other hand, scenario E provides a clear rationale for LBOs (leveraged buyouts), where the acquirer issues a significant amount of debt to pay for M&A and then uses the cash flows of target firm to pay off debt over time.

The main contribution of our paper is that we examine financial synergy with endogenous M&A timing. Two recent papers on synergy in M&A are related to ours: Lambrecht (2004) and Leland (2007). Lambrecht (2004) analyzes the optimal timing of mergers motivated by economies of scale. Because Lambrecht (2004) focuses on operational synergy, the tax benefits and default costs are out of consideration, which are central to our analysis of financial synergy. Leland (2007) develops a one-period model to examine the role of purely financial synergy in motivating M&A with exogenous timing given as current time. Although the focus of our paper, financial synergy in M&A, is similar to Leland (2007), our modelling differs from Leland (2007) in that we provide a continuous model with endogenous M&A timing. Our justification is as follows. Practically, M&A

is usually regarded as a value creation strategy of separate firms with initial asset in place. While Leland (2007) considers brand-new firms with no initial asset in place to start their operations at current time, we assume that two separate firms have already started their operations with optimal capital structures initially. However, their initial capital structures are not any longer optimal, because the state variable changes as time goes by. Therefore, adjusting capital structure to optimal level through M&A may create financial synergy. Due to the uncertainty in the values after M&A and the irreversibility of M&A cost, the M&A decision resembles an option exercise. It is appropriate to derive the optimal M&A timing using a real options approach. Moreover, the adjustment of capital structure through M&A rightly depends on M&A timing. By taking the optimal M&A timing into consideration endogenously, our paper analyzes the interaction between capital structures and investment decisions and obtains results different from Leland (2007).¹ That is, we find that financial synergy can motivate M&A without operational synergy when M&A timing is endogenously determined, whereas Leland (2007) concludes that financial synergy by itself is insufficient to justify M&A in many cases. Therefore, we complement the literature by demonstrating that the results derived from endogenously determined M&A timing may significantly depart from those derived from exogenously given M&A timing.

The remainder of this paper is organized as follows. Section 2 describes the model setup. Section 3 examines the adjustment of capital structure and determination of optimal M&A timing in both scenarios. Section 4 calibrates the model to measure financial synergy and provides model implications. Section 5 concludes. Some detailed proofs can be found in the Appendix.

2. Model Setup

The model is set in a continuous-time framework. Since specialization effect is more important than diversification effect nowadays, we consider M&A in the same industry.² There are two risk-neutral firms: a potential acquiring firm and a potential target firm.³ These roles are exogenously assigned and are determined by firms' specific characteristics, not modeled in this paper. Let "a" and "tar" stand for the acquiring firm and target firm, respectively. The firm j ($\in \{a, tar\}$)

¹Leland (1994) uses firstly a contingent claims approach to study the optimal capital structure in corporate finance. Dixit and Pindyck (1994) is a standard textbook on the real options approach to investment under uncertainty.

²During the 1960s and the early 1970s, most mergers were motivated by diversification effect. That is, when activities' cash flows are imperfectly correlated, risk can be lowered via mergers. However, because business circumstances become increasingly competitive, it is inefficient to manage different activities as a conglomerate. See Rhodes-Kropf and Robinson (2004) for empirical evidence that similar firms merge.

³We do not consider competition among several potential acquiring firms. See Morellec and Zhadanov (2008) for an analysis of a takeover contest with two potential acquiring firms and a potential target firm.

collects a revenue flow $Q_j X$, where Q_j is the quantity produced by firm j and X is the price. We assume that the price process $(X(t))_{t \geq 0}$ be given by the following geometric Brownian motion process:

$$dX(t) = \mu X(t)dt + \sigma X(t)dz(t),$$

where μ and $\sigma(> 0)$ are constant parameters and $z(t)$ is a standard Brownian motion. The initial value $X(0)$ is sufficiently low. As in most real options model, we suppose the discount rate $r > \mu$ for convergence. Let τ denote the corporate tax rate. Then, the unlevered firm value at time t can be calculated as

$$\Pi_j(x) := \frac{1 - \tau}{r - \mu} Q_j x, \quad (1)$$

given that $X(t) = x$.

Both the acquirer and target have already been financed optimally by equity and debt. For simplicity, we assume that the issued debt has infinite maturity and the contractual continuous coupon payment of the perpetual debt issued by firm j is c_j . The profit flow of firm j at time t before M&A is $(1 - \tau)(Q_j x - c_j)$. Although issuing debt can obtain tax benefits, it is also accompanied with default costs. As in Leland (1994), we consider a stock-based definition of default whereby equityholders inject funds in the firm as long as equity value is positive. In other words, equityholders default on their debt obligations the first time equity value is equal to zero. Let x_j^d denote the default threshold of firm j before M&A. At the default threshold, we assume that the firm value is given by $(1 - \alpha)\Pi_j(x_j^d)$, where $\alpha \in [0, 1]$ measures the loss in firm value incurred by default costs.

We suppose that firms behave in the interests of equityholders and they can only receive M&A option unexpectedly.⁴ If either the acquirer or target goes into default before M&A occurs, M&A can never be realized. If the price process $(X(t))_{t > 0}$ is sufficiently high to hit the optimal M&A threshold x_m^i before each firm's default threshold, then the acquiring equityholders exercise the M&A option by providing the stand-alone value to target equityholders and bearing the fixed M&A cost I .⁵ The M&A cost is financed by issuing new equity and new debt with coupon c_n . After M&A, the profit flow of the merged firm is $(1 - \tau)(Q_m x - c_m)$, where the subscript "m" stands for the merged firm. The unlevered firm value after M&A is

$$\Pi_m(x) := \frac{1 - \tau}{r - \mu} Q_m x. \quad (2)$$

⁴We abstract from potential agency conflicts between managers and equityholders by assuming that the incentives of these two groups are perfectly aligned. See Zwiebel (1996), Morellec (2004), Shibata and Nishihara (2010) for analysis of the relation between agency conflicts, financing decisions, and control transactions.

⁵The fixed M&A cost here refers to the due diligence cost paid to the third party.

Since our paper focuses on whether purely financial synergy can motivate M&A or not, we assume $Q_m \equiv Q_a + Q_{tar}$ and $c_m \equiv c_a + c_{tar} + c_n$. The quantity Q_m excludes the effect of operational synergy. The coupon c_m reflects the adjustment of capital structure through M&A. We assume that firms cannot call back their existing debt when exercising M&A option, consequently, $c_n \geq 0$.⁶

3. Model Analysis

In our model, acquiring equityholders make two types of interrelated decisions: M&A investment decision and financing decision. The M&A decision is characterized by an endogenously determined threshold; when the price process $(X(t))_{t \geq 0}$ reaches M&A threshold x_m^i before each firm's default threshold x_j^d , acquiring equityholders exercise M&A option. The financing decision involves the choice of newly issued debt and an endogenous default threshold. The coupon level of newly issued debt $c_n(x_m^i)$, which is characterized by a trade-off between the tax benefits and default costs of debt financing, is determined simultaneously with the M&A decision. In contrast, the default threshold $x_m^d(c_m)$, which depends on coupon level after M&A, is determined after M&A option is exercised. Note that the three endogenous variables (i.e., x_m^i , $c_n(x_m^i)$, and $x_m^d(c_m)$) form a nested structure, which is an important characteristic of this model.

We derive the equityholders' decisions using backward induction. Section 3.1 examines default threshold after M&A (step 1) and the coupon of newly issued debt (step 2), which depends on M&A timing. Section 3.2 analyzes the optimal M&A timing (step 3), taking the possibility of default before M&A into consideration.

3.1 After M&A

The first step is to derive the values after M&A and determine the default threshold for the merged firm, x_m^d . Let T_m^i and T_m^d denote the endogenously chosen times for M&A investment and default after M&A:

$$T_m^i = \inf\{t \geq 0; X(t) \geq x_m^i\}, \quad T_m^d = \inf\{t \geq T_m^i; X(t) \leq x_m^d\}.$$

According to our model setup, for $T_m^i \leq t \leq T_m^d$, the equity value after M&A can be expressed as follows:

$$E_m(x) = \mathbb{E} \left[\int_t^{T_m^d} e^{-r(s-t)} (1 - \tau)(Q_m X(s) - c_m) ds \middle| X(t) = x \right],$$

where $\mathbb{E}[\cdot | X(t) = x]$ denotes the expectation operator given that $X(t) = x$. The instantaneous change in the equity value after M&A satisfies the following ordinary

⁶Goldstein *et al.* (2001) argue that, while covenants are often in place to protect debtholders, in practice firms typically have the option to issue additional debt in the future without recalling the outstanding debt issues.

differential equation (ODE):

$$rE_m(x) = (1 - \tau)(Q_mx - c_m) + \mu x E'_m(x) + \frac{1}{2} \sigma^2 x^2 E''_m(x), \quad x \geq x_m^d. \quad (3)$$

Once process $(X(s))_{s>0}$ hits the threshold x_m^d , the merged firm defaults. The following boundary conditions ensure that the optimal default threshold is chosen by equityholders:

$$\begin{cases} E_m(x_m^d) = 0, \\ E'_m(x_m^d) = 0, \\ \lim_{x \rightarrow \infty} \frac{E_m(x)}{x} < \infty. \end{cases} \quad (4)$$

Here, the first condition is the value-matching condition. Following the stock-based definition of default, at the default threshold x_m^d , the equity value equals 0. The second condition is the smooth-pasting condition, which ensures that x_m^d is chosen to maximize the equity value. The third condition is the no-bubbles condition.

Solving the ODE (3) under these boundary conditions, we obtain the equity value after M&A as follows (see Appendix A):

$$E_m(x) = \Pi_m(x) - (1 - \tau) \frac{c_m}{r} - \left[\Pi_m(x_m^d) - (1 - \tau) \frac{c_m}{r} \right] \left(\frac{x}{x_m^d} \right)^\gamma, \quad (5)$$

where

$$x_m^d = \frac{\gamma}{\gamma - 1} \frac{r - \mu}{r} \frac{c_m}{Q_m}, \quad (6)$$

and γ is the negative root of the quadratic equation $\frac{1}{2} \sigma^2 y^2 + (\mu - \frac{1}{2} \sigma^2) y - r = 0$, i.e.,

$$\gamma = \frac{1}{\sigma^2} \left[- \left(\mu - \frac{1}{2} \sigma^2 \right) - \sqrt{\left(\mu - \frac{1}{2} \sigma^2 \right)^2 + 2 \sigma^2 r} \right] < 0. \quad (7)$$

The equity value after M&A has two components: (i) the unlevered firm value minus the present value of the contractual coupon paid to the debtholders, and plus the present value of tax benefits; (ii) the value of default option, which is the product of savings from default and the default probability, given by $(x/x_m^d)^\gamma$. Note that the default threshold x_m^d depends on the ratio c_m/Q_m .

Similarly, for $T_m^i \leq t \leq T_m^d$, the debt value after M&A can be expressed as follows:

$$D_m(x) = \mathbb{E} \left[\int_t^{T_m^d} e^{-r(s-t)} c_m ds + e^{-r(T_m^d-t)} (1 - \alpha) \Pi_m(X(T_m^d)) \middle| X(t) = x \right],$$

and we obtain the debt value as

$$D_m(x) = \frac{C_m}{r} - \left[\frac{C_m}{r} - (1 - \alpha)\Pi_m(x_m^d) \right] \left(\frac{x}{x_m^d} \right)^\gamma. \quad (8)$$

It also has two components: (i) the present value of perpetual coupon payments; (ii) the present value of the loss in default. The firm value $V_m(x)$ is the sum of equity value and debt value.

$$V_m(x) = E_m(x) + D_m(x) = \Pi_m(x) + \tau \frac{C_m}{r} - \left[\alpha \Pi_m(x_m^d) + \frac{\tau C_m}{r} \right] \left(\frac{x}{x_m^d} \right)^\gamma. \quad (9)$$

The second step is to determine the coupon of newly issued debt. Following Sundaresan and Wang (2007), we assume that the existing debt and newly issued debt have equal priority at the default threshold.⁷ Then, the existing debt value after M&A is $D_m^e(x) = [(c_a + c_{tar})/c_m]D_m(x)$ and the newly issued debt value after M&A is $D_m^n(x) = (c_n/c_m)D_m(x)$.

We consider the determination of newly issued debt in both scenario F and scenario E. In scenario F, equityholders choose c_n^* to maximize the total firm value $V_m(x)$ at the optimal M&A threshold x_m^{i*} , which is endogenously determined later. The superscript “*” stands for the solution corresponding to scenario F. In scenario E, equityholders choose c_n^{**} at the optimal M&A threshold x_m^{i**} to maximize $V_m^n(x)$, which represents the sum of equity value $E_m(x)$ and newly issued debt value $D_m^n(x)$. That is,

$$V_m^n(x) = \Pi_m(x) + \frac{\tau C_m - c_a - c_{tar}}{r} + \left[\left((1 - \alpha) \frac{c_n}{c_m} - 1 \right) \Pi_m(x_m^d) + \frac{c_a + c_{tar} - \tau C_m}{r} \right] \left(\frac{x}{x_m^d} \right)^\gamma. \quad (10)$$

The superscript “**” stands for the solution corresponding to scenario E. The distinction between $V_m(x)$ and $V_m^n(x)$ is essential, because equityholders no longer care about the existing debt value when exercising M&A option and issuing new debt. This creates the differences between the two scenarios.

The coupon of newly issued debt in scenario F is derived by taking the first-order condition of $V_m(x)$ in Eq. (9):

$$c_n^* = -c_a - c_{tar} + \frac{r}{r - \mu} \frac{\gamma - 1}{\gamma} \frac{Q_m}{h} x_m^{i*}, \quad (11)$$

where

$$h = \left[1 - \gamma \left(1 - \alpha + \frac{\alpha}{\tau} \right) \right]^{-1/\gamma} > 1, \quad (12)$$

⁷A number of papers, including Weiss (1990) and Goldstein *et al.* (2001), report that the priority of claims is frequently violated in bankruptcy. It is typical that all unsecured debt receives the same recovery rate, regardless of the issuance date.

provided that the right hand side of Eq. (11) is nonnegative. It is obvious that $dc_n^*/dx_m^{i*} > 0$. On the other hand, the coupon of the newly issued debt in scenario E is derived by taking the first-order condition of $V_m^n(x)$ in Eq. (10):

$$c_n^{**} = -c_a - c_{tar} + \frac{r}{r - \mu} \frac{\gamma - 1}{\gamma} \frac{Q_m}{h} x_m^{i**} \times \left[1 - \frac{\gamma}{\gamma - 1} \left(\frac{\tau^{-1} - \gamma(1 - \alpha + \alpha/\tau)}{1 - \gamma(1 - \alpha + \alpha/\tau)} \right) \frac{c_a + c_{tar}}{c_n^{**} + c_a + c_{tar}} \right]^{1/\gamma}, \quad (13)$$

provided that the right hand side of Eq. (13) is nonnegative. Totally differentiating Eq. (13), and then rearranging yields $dc_n^{**}/dx_m^{i**} > 0$.

Comparing c_n^* and c_n^{**} in Eq. (11) and Eq. (13), respectively, we find that the expression of c_n^* is explicit, while c_n^{**} is implicit. Moreover, both of them positively depend on M&A thresholds x_m^{i*} and x_m^{i**} , respectively, which are derived in section 3.2. It means that waiting for a better state to exercise M&A option results in issuing more new debt.

3.2 Before M&A

The third step is to determine the M&A threshold, taking the possibility of default before M&A into consideration. While the upper boundary x_m^i is determined by the acquiring equityholders, the lower boundary $\max[x_{am}^d, x_{tar}^d]$ is determined by either the acquiring equityholders (if $x_{tar}^d \leq x_{am}^d$) or the target equityholders (if $x_{tar}^d \geq x_{am}^d$). The subscript “am” differs from “a” in that it represents value with M&A option. Because default means losing M&A option in the future, equityholders may be less willing to go into default before M&A, compared to the case without M&A option. Therefore, even if $x_a^d > x_{tar}^d$, it is possible that $x_{am}^d < x_{tar}^d$.⁸

Let $\mathcal{H}(x; y, z)$ denote the present value of a claim that pays \$1 contingent on x reaching the upper threshold y before reaching the lower threshold z . In contrast, let $\mathcal{L}(x; y, z)$ denote the present value of a claim that pays \$1 contingent on x reaching the lower threshold z before reaching the upper threshold y . In Appendix B, we demonstrate that:

$$\mathcal{H}(x; y, z) = \frac{z^\gamma x^\beta - z^\beta x^\gamma}{z^\gamma y^\beta - z^\beta y^\gamma}, \quad \mathcal{L}(x; y, z) = \frac{x^\gamma y^\beta - x^\beta y^\gamma}{z^\gamma y^\beta - z^\beta y^\gamma}, \quad (14)$$

where β is the positive root of the quadratic equation $\frac{1}{2}\sigma^2 y^2 + (\mu - \frac{1}{2}\sigma^2)y - r = 0$, i.e.,

$$\beta = \frac{1}{\sigma^2} \left[-\left(\mu - \frac{1}{2}\sigma^2\right) + \sqrt{\left(\mu - \frac{1}{2}\sigma^2\right)^2 + 2\sigma^2 r} \right] > 1. \quad (15)$$

⁸Morellec and Zhdanov (2008) also jointly determine the financing strategies and the takeover timing. However, in their model, the takeover threshold is chosen by target equityholders. Furthermore, they did not explicitly consider the change in the lower boundary when M&A option is available.

We suppose that if the acquiring equityholders bear M&A cost I and provide the stand-alone value for target equityholders, the agreement on M&A can be realized. Therefore, the expression of target equity value is similar to Eq. (5):

$$E_{tar}(x) = \Pi_{tar}(x) - (1 - \tau) \frac{C_{tar}}{r} - \left[\Pi_{tar}(x_{tar}^d) - (1 - \tau) \frac{C_{tar}}{r} \right] \left(\frac{x}{x_{tar}^d} \right)^\gamma, \quad (16)$$

where

$$x_{tar}^d = \frac{\gamma}{\gamma - 1} \frac{r - \mu}{r} \frac{C_{tar}}{Q_{tar}}. \quad (17)$$

However, the target debt value with M&A option differs from the stand-alone value (i.e., $D_{tarm}(x) \neq D_{tar}(x)$). Because of the assumption that the existing debt cannot be called back when M&A occurs, the target debt value is passively affected by the acquiring equityholders' exercise of M&A option. At the upper boundary,

$$D_{tarm}(x_m^i) = \frac{C_{tar}}{C_m} D_m(x_m^i). \quad (18)$$

At the lower boundary, since M&A option is lost, $D_{tarm}(\max[x_{am}^d, x_{tar}^d]) = D_{tar}(\max[x_{am}^d, x_{tar}^d])$, which is similar to Eq. (8).

$$D_{tarm}(\max[x_{am}^d, x_{tar}^d]) = \frac{C_{tar}}{r} - \left[\frac{C_{tar}}{r} - (1 - \alpha) \Pi_{tar}(x_{tar}^d) \right] \left(\frac{\max[x_{am}^d, x_{tar}^d]}{x_{tar}^d} \right)^\gamma, \quad (19)$$

Therefore, we have the following expression for target debt value with M&A option:

$$\begin{aligned} D_{tarm}(x) = & \frac{C_{tar}}{r} + e_{tar}^i \mathcal{H}(x; x_m^i, \max[x_{am}^d, x_{tar}^d]) \\ & + e_{tar}^d \mathcal{L}(x; x_m^i, \max[x_{am}^d, x_{tar}^d]), \end{aligned} \quad (20)$$

where

$$\begin{aligned} e_{tar}^i &= \frac{C_{tar}}{C_m} D_m(x_m^i) - \frac{C_{tar}}{r}, \\ e_{tar}^d &= \begin{cases} - \left[\frac{C_{tar}}{r} - (1 - \alpha) \Pi_{tar}(x_{tar}^d) \right] \left(\frac{x_{am}^d}{x_{tar}^d} \right)^\gamma, & \text{if } x_{tar}^d < x_{am}^d, \\ - \left[\frac{C_{tar}}{r} - (1 - \alpha) \Pi_{tar}(x_{tar}^d) \right], & \text{if } x_{tar}^d \geq x_{am}^d. \end{cases} \end{aligned} \quad (21)$$

Eq. (20) has three components: (i) the present value of the contractual coupon payments; (ii) the present value when M&A option is exercised, which is given by the product of the net payoff e_{tar}^i at the upper boundary x_m^i and the present value of unit-payoff contingent claim $\mathcal{H}(x; x_m^i, \max[x_{am}^d, x_{tar}^d])$; and (iii) the present

value when default option is exercised, which is given by the product of the net payoff e_{tar}^d at the lower boundary $\max[x_{am}^d, x_{tar}^d]$ and the present value of unit-payoff contingent claim $\mathcal{L}(x; x_m^i, \max[x_{am}^d, x_{tar}^d])$. The target firm value is the sum of Eq. (16) and Eq. (20) as follows:

$$V_{tarm}(x) = E_{tar}(x) + D_{tarm}(x). \quad (22)$$

The following boundary conditions ensure that the optimal M&A threshold and default threshold of the acquirer are chosen in scenario F:

$$\begin{cases} V_{am}(x_m^i) + V_{tarm}(x_m^i) = V_m(x_m^i) - I, \\ V'_{am}(x_m^i) + V'_{tarm}(x_m^i) = V'_m(x_m^i), \\ E_{am}(x_{am}^d) = 0, \\ E'_{am}(x_{am}^d) = 0. \end{cases} \quad (23)$$

Here, the first condition is the value-matching condition at x_m^i . After M&A, the acquiring equityholders internalize the tax benefits and default costs of the merged firms. By paying the fixed cost I to exercise M&A option at x_m^i , the acquiring firm collects the surplus from the merged firm value subtracting the value paid to the target firm ($V_{tarm} = E_{tar} + D_{tarm}$). The second condition is the smooth pasting condition at x_m^i . This condition ensures that x_m^i is chosen to maximize the total firm value. The remaining two conditions are the value-matching and smooth-pasting conditions at x_{am}^d .

According to the two value-matching conditions in (23), the firm value of the acquiring firm with M&A option can be written as:

$$\begin{aligned} V_{am}(x) = & \Pi_a(x) + \tau \frac{C_a}{r} + \hat{e}_a^i \mathcal{H}(x; x_m^i, \max[x_{am}^d, x_{tar}^d]) \\ & + \hat{e}_a^d \mathcal{L}(x; x_m^i, \max[x_{am}^d, x_{tar}^d]), \end{aligned} \quad (24)$$

where

$$\begin{aligned} \hat{e}_a^i = & V_m(x_m^i) - V_{tarm}(x_m^i) - I - \left[\Pi_a(x_m^i) + \frac{\tau C_a}{r} \right], \\ \hat{e}_a^d = & \begin{cases} - \left[\alpha \Pi_a(x_{am}^d) + \frac{\tau C_a}{r} \right], & \text{if } x_{am}^d > x_{tar}^d, \\ - \left[\alpha \Pi_a(x_{tar}^d) + \frac{\tau C_a}{r} \right], & \text{if } x_{am}^d \leq x_{tar}^d \leq x_a^d, \\ - \left[\alpha \Pi_a(x_a^d) + \frac{\tau C_a}{r} \right] \left(\frac{x_{tar}^d}{x_a^d} \right)^\gamma, & \text{if } x_{am}^d \leq x_{tar}^d, x_a^d < x_{tar}^d. \end{cases} \end{aligned} \quad (25)$$

The equity value of the acquiring firm with M&A option can be written as:

$$\begin{aligned} E_{am}(x) = & \Pi_a(x) - (1 - \tau) \frac{C_a}{r} + e_a^i \mathcal{H}(x; x_m^i, \max[x_{am}^d, x_{tar}^d]) \\ & + e_a^d \mathcal{L}(x; x_m^i, \max[x_{am}^d, x_{tar}^d]), \end{aligned} \quad (26)$$

where

$$e_a^i = V_m^n(x_m^i) - E_{tar}(x_m^i) - I - \left[\Pi_a(x_m^i) - (1 - \tau) \frac{C_a}{r} \right],$$

$$e_a^d = \begin{cases} - \left[\Pi_a(x_{am}^d) - (1 - \tau) \frac{C_a}{r} \right], & \text{if } x_{am}^d > x_{tar}^d, \\ - \left[\Pi_a(x_{tar}^d) - (1 - \tau) \frac{C_a}{r} \right], & \text{if } x_{am}^d \leq x_{tar}^d \leq x_a^d, \\ - \left[\Pi_a(x_a^d) - (1 - \tau) \frac{C_a}{r} \right] \left(\frac{x_{tar}^d}{x_a^d} \right)^\gamma, & \text{if } x_{am}^d \leq x_{tar}^d, x_a^d < x_{tar}^d. \end{cases} \quad (27)$$

Note that if $x_{am}^d \leq x_{tar}^d$ (the second and third lines in Eq. (27)), then the lower boundary turns out to be x_{tar}^d . Once the price process $(X(s))_{s>0}$ hits x_{tar}^d , the acquirer loses M&A option. Moreover, if $x_{am}^d \leq x_{tar}^d \leq x_a^d$ (the second line in Eq. (27)), then the acquirer immediately goes into default at the lower boundary x_{tar}^d ; if $x_{am}^d \leq x_{tar}^d$ and $x_a^d < x_{tar}^d$ (the third line in Eq. (27)), then the acquirer continues operating the firm and goes into default optimally when the price process $(X(s))_{s>0}$ hits x_a^d .

By now, we have obtained all the value expressions appeared in boundary conditions (23). Substituting these expressions into the smooth-pasting conditions at x_m^i and $\max[x_{am}^d, x_{tar}^d]$ in (23), respectively, we obtain:

$$\nu_1 \gamma (x_m^{i*})^\gamma = \frac{(\hat{e}_a^d + e_{tar}^d)(\gamma - \beta)(x_m^{i*})^{\gamma+\beta}}{\left(\max[x_{am}^{d*}, x_{tar}^d] \right)^\gamma (x_m^{i*})^\beta - \left(\max[x_{am}^{d*}, x_{tar}^d] \right)^\beta (x_m^{i*})^\gamma}$$

$$+ \frac{(\hat{e}_a^i + e_{tar}^i) \left[\beta (x_m^{i*})^\beta \left(\max[x_{am}^{d*}, x_{tar}^d] \right)^\gamma - \gamma (x_m^{i*})^\gamma \left(\max[x_{am}^{d*}, x_{tar}^d] \right)^\beta \right]}{\left(\max[x_{am}^{d*}, x_{tar}^d] \right)^\gamma (x_m^{i*})^\beta - \left(\max[x_{am}^{d*}, x_{tar}^d] \right)^\beta (x_m^{i*})^\gamma}, \quad (28)$$

where

$$\nu_1 = - \left[\alpha \Pi_m(x_m^d) + \frac{\tau C_m}{r} \right] (x_m^d)^{-\gamma} + \left[\Pi_{tar}(x_{tar}^d) - \frac{(1 - \tau) C_{tar}}{r} \right] (x_{tar}^d)^{-\gamma},$$

and

$$\Pi_a(x_{am}^{d*}) + \frac{e_a^i(\beta - \gamma)(x_{am}^{d*})^{\beta+\gamma} + e_a^d \left[\gamma (x_{am}^{d*})^\gamma (x_m^{i*})^\beta - \beta (x_{am}^{d*})^\beta (x_m^{i*})^\gamma \right]}{\left(x_{am}^{d*} \right)^\gamma (x_m^{i*})^\beta - \left(x_{am}^{d*} \right)^\beta (x_m^{i*})^\gamma} = 0. \quad (29)$$

On the other hand, in scenario E, the value-matching and smooth-pasting conditions at x_m^i are given as follows:

$$\begin{cases} E_{am}(x_m^i) + E_{tar}(x_m^i) = V_m^n(x_m^i) - I, \\ E'_{am}(x_m^i) + E'_{tar}(x_m^i) = V_m^{n'}(x_m^i), \end{cases} \quad (30)$$

where $E_{am}(x)$ and $V_m^n(x)$ are given as Eq. (26) and Eq. (10), respectively. The value-matching and smooth-pasting conditions at the lower boundary are the same with those in scenario F. The smooth-pasting condition at x_m^i in (30) implies:

$$\begin{aligned} v_2 \gamma (x_m^{i**})^\gamma = & \frac{e_a^d (\gamma - \beta) (x_m^{i**})^{\gamma+\beta}}{\left(\max[x_{am}^{d**}, x_{tar}^d] \right)^\gamma (x_m^{i**})^\beta - \left(\max[x_{am}^{d**}, x_{tar}^d] \right)^\beta (x_m^{i**})^\gamma} \\ & + \frac{e_a^i \left[\beta (x_m^{i**})^\beta \left(\max[x_{am}^{d**}, x_{tar}^d] \right)^\gamma - \gamma (x_m^{i**})^\gamma \left(\max[x_{am}^{d**}, x_{tar}^d] \right)^\beta \right]}{\left(\max[x_{am}^{d**}, x_{tar}^d] \right)^\gamma (x_m^{i**})^\beta - \left(\max[x_{am}^{d**}, x_{tar}^d] \right)^\beta (x_m^{i**})^\gamma}, \quad (31) \end{aligned}$$

where

$$\begin{aligned} v_2 = & \left[\left((1 - \alpha) \frac{c_n}{c_m} - 1 \right) \Pi_m(x_m^d) + \frac{c_a + c_{tar} - \tau c_m}{r} \right] (x_m^d)^{-\gamma} \\ & + \left[\Pi_{tar}(x_{tar}^d) - \frac{(1 - \tau) c_{tar}}{r} \right] (x_{tar}^d)^{-\gamma}. \end{aligned}$$

Proposition 3.1. *The optimal M&A threshold, default threshold of acquirer with M&A option, and coupon level of newly issued debt, can be obtained by simultaneously solving the following equations:*

- (i) *For scenario F, the three equations that determine x_m^{i*} , x_{am}^{d*} , and c_n^* are Eq. (11), Eq. (28), and Eq. (29);*
- (ii) *For scenario E, the three equations that determine x_m^{i**} , x_{am}^{d**} and c_n^{**} are Eq. (13), Eq. (31), and Eq. (29) (x_{am}^{d**} instead of x_{am}^{d*}).*

4. Model Implications

Since the equations above are nonlinear in the thresholds, analytical solutions in closed forms are impossible. In this section, we calibrate the model to analyze the characteristics of the solutions and provide several empirical predictions. In particular, we measure financial synergy when M&A option is exercised optimally.

We use the following input parameter values for calibration: $\mu = 0.01$, $\sigma = 0.25$, $r = 0.06$, $\tau = 0.4$, $\alpha = 0.4$, $c_a = 2.5$, $c_{tar} = 3$, $Q_a = 1$, $Q_{tar} = 1.5$, $I = 10$, $x = 2.3$. The growth rate $\mu = 0.01$ and volatility $\sigma = 0.25$ of cash flows are selected to match the data of an average Standard and Poor's (S&P) 500 firms (see Strebulaev (2007)). The risk-free rate $r = 0.06$ is taken from the yield curve on Treasury bonds. The corporate tax rate $\tau = 0.4$ follows the estimation by Kemsley and Nissim (2002). The default costs parameter $\alpha = 0.4$ is chosen to be consistent with Gilson (1997), which reports that default costs are equal to 0.365 and 0.455 for the median firm in his samples. The remaining parameter values (the coupon c_j , the quantity Q_j , the fixed cost I , and the current value of state variable x) are

not essentially important, because they can be normalized. We simply set them as above to show the results clearly.

Under these parameter setting, $x_a^d = 1.09$, $x_{tar}^d = 0.89$. We can also calculate inversely that the initial value of state variable (denoted by x_j^0 , $j \in \{a, tar\}$) for acquirer and target to establish their firms are $x_a^0 = 2.74$ and $x_{tar}^0 = 2.19$, respectively, given c_a and c_{tar} are their optimal coupons at the establishment timing 0.⁹ As time goes by, their initial capital structures are not any longer optimal because the state variable changes. Since we set $x = 2.3$ at current time, the acquirer is a firm with excessive debt and the target is a firm with insufficient debt relative to their optimal capital structures now. Therefore, adjusting capital structure to optimal level through M&A may create financial synergy. We also analyze a parameter setting when $c_a = 3$, $c_{tar} = 2.5$, $Q_a = 1.5$, $Q_{tar} = 1$, with other parameters unchanged. In such a case, the acquirer is a firm with insufficient debt and the target is a firm with excessive debt relative to their optimal capital structures now. After comparing the results of the two cases (the case when the acquirer's debt is excessive and the case when the acquirer's debt is insufficient), we find that in scenario E, there is little difference between the two cases, because the existing debt value is ignored in the maximization process. On the other hand, in scenario F, M&A is delayed in the case when the acquirer's debt is excessive in comparison to the case when the acquirer's debt is insufficient, because the debt overhang problem is more serious. Except for this point, the results when acquirer's debt is insufficient are very similar to the results when acquirer's debt is excessive, which we will analyze below in detail.

4.1 Measure of Financial Synergy

Since we have assumed no operational synergy, financial synergy of M&A is measured by the difference between the value of the optimally levered merged firm, and the sum of the stand-alone acquirer value and target value. The purely financial synergy at current time is defined as:

$$FS(x) = [\Delta TB(x_m^i) - \Delta DC(x_m^i)] (x/x_m^i)^\beta, \quad (32)$$

where

$$\Delta TB(x_m^i) = \frac{\tau}{r} \left[c_m \left[1 - \left(\frac{x_m^i}{x_m^d} \right)^\gamma \right] - c_a \left[1 - \left(\frac{x_m^i}{x_a^d} \right)^\gamma \right] - c_{tar} \left[1 - \left(\frac{x_m^i}{x_{tar}^d} \right)^\gamma \right] \right], \quad (33)$$

$$\Delta DC(x_m^i) = \alpha \left[\Pi_m(x_m^d) \left(\frac{x_m^i}{x_m^d} \right)^\gamma - \Pi_a(x_a^d) \left(\frac{x_m^i}{x_a^d} \right)^\gamma - \Pi_{tar}(x_{tar}^d) \left(\frac{x_m^i}{x_{tar}^d} \right)^\gamma \right]. \quad (34)$$

⁹From Eq. (11), we know that there is a linear relationship between the optimal coupon and the initial investment threshold.

The financial synergy can be divided into two components, which are directly related to changes in financial structure through M&A. The first component ΔTB denotes the change in the present value of tax benefits from the optimally levered merged firm versus separate firms. The second component ΔDC denotes the change in the present value of default costs. The credit spread and leverage at x_m^i are defined as follows:

$$CS_j(x_m^i) = \frac{c_j}{D_j(x_m^i)} - r, \quad (35)$$

$$L_j(x_m^i) = \frac{D_j(x_m^i)}{V_j(x_m^i)}, \quad (36)$$

where $j \in \{m, a, tar\}$.

4.2 Main Results

Table 1 demonstrates the results in both scenarios.¹⁰ According to our computation, the main results are robust across a wide range of parameter values c_j , Q_j , I , and x .

Table 1. Results of scenarios F and E.

	FS	ΔTB	ΔDC	x_m^i	c_m	x_m^d	ΔE	ΔD_a	ΔD_{tar}
F	0.23	0.46	0.24	2.51	5.71	0.99	0.86	1.09	-1.72
E	1.63	5.19	3.56	5.18	15.95	2.77	7.73	-2.60	-3.50
	CS_a	CS_{tar}	CS_m	L_a	L_{tar}	L_m			
F	0.0292	0.0207	0.0253	0.739	0.654	0.705			
E	0.0105	0.0079	0.0418	0.474	0.403	0.819			

There are three interesting findings. First, consider the financial synergy and M&A threshold. We find that financial synergy can be positive in both scenarios. In other words, purely financial synergy by itself can motivate M&A. Both the tax benefits and default costs increase; however, the increase in tax benefits is much larger than that in default costs, resulting in positive financial synergy. Moreover, in comparison to scenario F, the M&A threshold is higher and the financial synergy is larger in scenario E. Because $x_m^i = 5.18$ in scenario E is much higher than $x_a^0 = 2.74$ and $x_{tar}^0 = 2.19$, the distortion of $V_a(x_m^i)$ and $V_{tar}(x_m^i)$ with initial coupons from those with optimal coupons is larger. Therefore, the financial synergy defined in Eq. (32) is larger in scenario E.

¹⁰Values are firstly calculated at x_m^i , and then multiplied by the M&A probability $(x/x_m^i)^\beta$. The credit spread and leverage are calculated at x_m^i .

Claim 4.1. When operational synergy is zero, purely financial synergy can motivate M&A in both scenario F and scenario E.

This result differs from that of Leland (2007) who assumes two separate firms with no initial asset in place. With the assumption that M&A timing is exogenously given as current time, Leland (2007) concludes that purely financial synergy by itself is insufficient to justify M&A in many cases. By contrast, we assume two separate firms with initial asset in place. By deriving M&A timing endogenously, we find that purely financial synergy can motivate M&A in both scenarios. We therefore demonstrate that financial synergy hinges in large part on whether M&A timing is exogenously given or endogenously determined.

Second, consider the changes in coupon and values. In scenario F, although the coupon after M&A increases, default threshold $x_m^d = 0.99$ lies between $x_a^d = 1.09$ and $x_{tar}^d = 0.89$. Therefore, default threshold decreases and existing debt value increases from the viewpoint of acquiring firm with excessive debt. Irrespective of the fact that M&A cost is fully borne by acquiring equityholders, a part of the increase in the total firm value accrues to existing debtholders. The wealth transfer discourages equityholders from exercising M&A option at a lower threshold in scenario F. This reflects the debt overhang problem discussed in Myers (1977) and Sundaresan and Wang (2007), which may delay or prevent an investment decision to improve the total firm value. In scenario E, default threshold increases and existing debt value decreases. The reason is that acquiring equityholders appropriate the benefits from existing debtholders by issuing a significant amount of new debt and increasing the leverage of the merged firm.¹¹ That is the so-called risk shifting problem discussed in Jensen and Meckling (1976). The equity value increases in both scenarios, which ensures the participation constraint of equityholders in M&A.

Third, consider the changes in leverage and credit spread. In scenario F, although the coupon after M&A increases a little, the default threshold is between that of the two firms before M&A. Therefore, both the leverage and credit spread are also between those of the two firms before M&A. On the other hand, in scenario E, because the coupon level increases significantly and the default threshold increases, both the leverage and credit spread increase. In fact, scenario F corresponds to a situation where debt is issued with covenants protecting the existing debtholders, while scenario E corresponds to LBOs. In LBOs, acquirers issue a significant amount of debt to pay for M&A and then use the cash flows of target firm to pay off debt over time. After LBOs, firms usually have high leverage,

¹¹Although we assumed both existing debt and newly issued debt have equal priority at the default threshold, even with seniority provisions, existing debtholders lose value when new debt is issued. Ziegler (2004) demonstrates that seniority provisions do protect existing debtholders against losing value to new debtholders; however, they do not protect existing debtholders against wealth transfers driven by changes in the timing and probability of default.

and the debt usually is below investment grade. From the perspective of existing debtholders, LBOs represent a fundamental shift in the firm's risk profile and result in a decrease in debt value.¹² However, our results demonstrate that the loss in debt values is not large enough to explain the gain in equity values. This is consistent with the empirical findings documented in Brealey *et al.* (2008).

To examine the effect of uncertainty on optimal M&A threshold, Fig. 1 plots M&A thresholds for varying volatilities of the price process. We find that in sce-

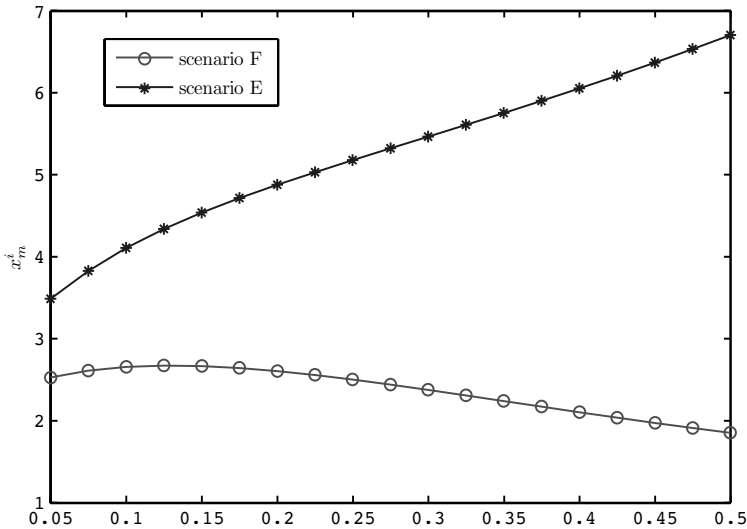


Figure 1. The effects of uncertainty on M&A threshold.

nario E, the optimal M&A threshold increases with uncertainty. By contrast, in scenario F, the optimal M&A threshold increases with uncertainty at first, and then decreases with uncertainty. The intuition is as follows. The uncertainty has two countervailing effects on the optimal M&A threshold. One is the usual positive effect explained in the standard real options model (all-equity firm without default). Higher uncertainty implies a larger option value of waiting to exercise M&A option. Therefore, M&A threshold increases with uncertainty. The other is a negative effect because of the existence of the lower default threshold before M&A. As Fig. 2 shows (with parameters $x = 2.3$, $y = 2.5$, $z = 1.8$), the present

¹²The famous LBO was that Kohlberg Kravis Roberts (KKR) acquired RJR Nabisco in the late 1980s and this illustrates the wealth transfer from the existing debtholders to equityholders.

value of claim $\mathcal{L}(x; y, z)$ in Eq. (14) (pay \$1 contingent on x reaching the lower threshold z before reaching the upper threshold y) increases with uncertainty. On the other hand, the present value of claim $\mathcal{H}(x; y, z)$ in Eq. (14) (pay \$1 contingent on x reaching the upper threshold y before reaching the lower threshold z) has little change with uncertainty. Since the probability of hitting the default threshold before M&A increases, there is an incentive for equityholders to exercise M&A earlier, which induces a lower M&A threshold. In scenario E, irrespective of the uncertainty level, the positive effect dominates the negative effect; while in scenario F, the negative effect becomes stronger as uncertainty increases and begins to dominate the positive effect when uncertainty increases at a certain degree.

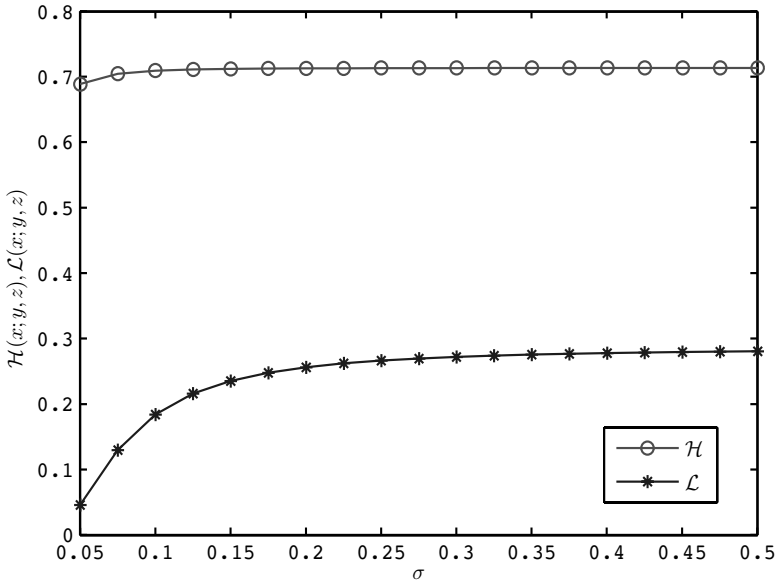


Figure 2. The effects of uncertainty on contingent claims $\mathcal{H}$ and $\mathcal{L}$.

5. Conclusions

This paper developed a continuous model to examine financial synergy when M&A timing is determined endogenously. We demonstrated that purely financial synergy can motivate M&A in both scenarios. However, the optimal M&A timing is delayed and financial synergy is larger in scenario E.

The analysis in this paper is suitable for settings where the firm receives a new growth option (like M&A) unexpectedly. Our theoretical model generates implications that are consistent with empirical evidences in corporate finance. One implication is the debt overhang problem. While total firm value increases through

M&A, a part of the value created from exercising M&A option goes to existing debtholders. This *ex post* wealth transfer discourages equityholders from exercising M&A option at the optimal timing in scenario F, because M&A cost is fully borne by equityholders. Another implication is the risk shifting problem. The existence of debtholders already in place creates an incentive for equityholders to issue a significant amount of new debt which results in higher default risk. Our results also have implications for empirical works that examine the sources of M&A synergies. Those parameters mentioned above, such as the tax rate and default costs, which can create substantial financial synergy, should be included as possible explanatory variables.

Lastly, we should point out an important but difficult topic for future research. While our paper considered the situation where firms receive M&A option unexpectedly, the analysis when firms are able to anticipate a future growth option can endogenously derive the initial capital structure to defer *ex post* inefficiency. We will consider this problem in the future.

Appendix A

The general solution of ODE (3) is:

$$E_m(x) = A_+ x^\beta + A_- x^\gamma + (1 - \tau) \frac{Q_m x}{r - \mu} - \frac{c_m}{r}, \quad (\text{A.1})$$

where β and γ are the positive and negative roots of the quadratic equation $\frac{1}{2}\sigma^2 y^2 + (\mu - \frac{1}{2}\sigma^2)y - r = 0$.

According to the no-bubbles condition, A_+ must equal zero. From the value-matching and smooth-pasting conditions, we know that:

$$\begin{cases} A_- (x_m^d)^\gamma + \frac{1-\tau}{r-\mu} Q_m x_m^d - \frac{c_m}{r} = 0, \\ A_- \gamma (x_m^d)^{\gamma-1} + \frac{1-\tau}{r-\mu} Q_m = 0. \end{cases} \quad (\text{A.2})$$

Solving the equations above yields the default threshold and equity value. The debt value can be obtained similarly.

Appendix B

Because $\mathcal{H}(x; y, z)$ is a claim that receives no dividend, we know from (A.1) that $\mathcal{H}(x; y, z)$ is of the form:

$$\mathcal{H}(x; y, z) = A_+ x^\beta + A_- x^\gamma. \quad (\text{A.3})$$

Substituting (A.3) into the boundary conditions:

$$\mathcal{H}(y; y, z) = 1, \quad \mathcal{H}(z; y, z) = 0,$$

we obtain that

$$\mathcal{H}(x; y, z) = \frac{z^\gamma x^\beta - z^\beta x^\gamma}{z^\gamma y^\beta - z^\beta y^\gamma}.$$

Similarly, $\mathcal{L}(x; y, z)$ can be derived as

$$\mathcal{L}(x; y, z) = \frac{x^\gamma y^\beta - x^\beta y^\gamma}{z^\gamma y^\beta - z^\beta y^\gamma}.$$

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